Retained cusp topology, triality morphisms, and the marked Niemeier–Leech construction

Standalone theorem note for the higher-rung search

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Abstract

For the retained modulo-840 lattice N with roots $A_5^4D_4$ and its specified index-six Leech neighbour Λ, we compute the arithmetic of the original Albert cubic, retaining its coefficient 2. Fix the Cayley–Dickson frame $(1, i, j, k, \ell, i\ell, j\ell, k\ell)$ and the explicit isometry $\mathcal{L} : N_{\mathbb{R}} \to \mathbb{O}^3$. Writing $\tau(q) = \mathrm{Re}((q_1q_2)q_3)$ and $F = \tau \circ \mathcal{L}$ on the original Niemeier span, let $\mathcal{V}_F(L)$ be the additive group generated by its values. Then

$$\mathcal{V}_F(N) = \tfrac{1}{27}\mathbb{Z} \oplus \tfrac{\sqrt{2}}{54}\mathbb{Z} \oplus \tfrac{\sqrt{3}}{18}\mathbb{Z},$$
$$\mathcal{V}_F(\Lambda) = \tfrac{1}{27}\mathbb{Z} \oplus \tfrac{\sqrt{2}}{216}\mathbb{Z} \oplus \tfrac{\sqrt{3}}{54}\mathbb{Z}, \qquad \mathcal{V}_F(\Lambda)/\mathcal{V}_F(N) \cong \mathbb{Z}/12.$$

The determinant is $\lambda^3 - \lambda(y, y) + 2F(y)$; this value-group inclusion is not a lattice inclusion $N \subset \Lambda$. The residue operator $r = \rho(361)$ cycles the three neighbours $\Lambda_j = r^j\Lambda$; all their pairwise intersections equal J, with $\det J = 81$ and $[\Lambda_j : J] = 9$. A fixed fourth isotropic plane gives the even unimodular A_2^{12} lattice M, with 72 roots, $M \cap \Lambda_j = J$ and $[M : J] = 9$. Its cubic-value group already equals that of the common lattice:

$$\mathcal{V}_F(J) = \mathcal{V}_F(M) = \tfrac{1}{27}\mathbb{Z} \oplus \tfrac{\sqrt{2}}{216}\mathbb{Z} \oplus \tfrac{\sqrt{3}}{18}\mathbb{Z}.$$

The two successive value-group inclusions have matrices $\mathrm{diag}(1, 4, 1)$ and $\mathrm{diag}(1, 1, 3)$, with quotients $\mathbb{Z}/4$ and $\mathbb{Z}/3$.

Wilson's Leech construction [31] is embedded in the same original metric by the specified isometry $x \mapsto x/\sqrt{2}$. Its cyclic fixed Gram matrix is $3G_{E_8}$, its orthogonal complement is explicitly isometric to $A_2 \otimes E_8$, and their sum has index 3^8 with quotient $\mathbb{F}_3^8$. Retaining $c(q_1, q_2, q_3) = (q_3, q_1, q_2)$, Wilson's opposite cycle is $c_{\mathrm{W}} = c^{-1}$. The literal intersections with both $\mathcal{L}(N)$ and $\mathcal{L}(\Lambda)$ equal $\{\sqrt{2}(a, a, a) : a \in D_{\mathbb{H}}\}$, where $D_{\mathbb{H}}$ is D_4 in the first four fixed Cayley coordinates. This rank-four lattice has Gram matrix $6G_{D_4}$, determinant 5184, minimum squared norm 12, and value-generated subgroup $4\sqrt{2}\mathbb{Z}$ for τ ($8\sqrt{2}\mathbb{Z}$ for 2τ). Its complete image is $2\sqrt{2}\big(18\mathbb{Z} \cup (6\mathbb{Z} + 2) \cup (6\mathbb{Z} - 2)\big)$, with an explicit integral section. Writing $a = a_0 + v$, $t = |v|$, the corresponding Albert matrix retains the scalar μ and has eigenvalues $\mu + 2\sqrt{2}a_0$ and $\mu - \sqrt{2}a_0 \pm \sqrt{6}t$. Its three labelled projectors recover v exactly. The complete orbit-family proof keeps mean, radius and Cartan angle, including the two focal projective-plane strata. The marked umbral shadow is $(5I_{12} + J)S_6$, with $J_{rs} = \mathbf{1}_{s=5r\ (12)}$; its coefficient map gives the eta quotient T_6 and $T_6(-1/(6\tau)) = 72/T_6(\tau)$, with the exact multiplier and original glue-action comparisons.

The higher target is three-directional, not required to carry a complex structure. The ordered triality products retain the Jordan factor 1/2, mixed determinant derivative 2τ, and symmetric polarization $\tau/3$, where $\tau(a, b, c) = \mathrm{Re}((ab)c)$. The spectral graph identifies $\mathbb{O}^3$ with the equal-diagonal positive determinant boundary. An explicit flag-cell chart and its inverse give the degree-+1 collapse $F_4/\mathrm{Spin}(8) \to S^{24}$; every reverse sphere map has degree zero. The angle-forgetting cusp map bounds an explicit framed $D^2 \times T^2$. The retained class $[H] = \eta_4\eta_5 \neq$

0, under the specified based $X \cong S^6$ identification, instead gives the nontrivial real three-plane reduction inside the triality carrier and the Hopf–cusp class $12\nu_6 \neq 0$ in $\pi_9(S^6)$. The complex-toroidal comparison retains its original periods, kernel, monodromy and fillings; its Euler contribution 6 prevents the unchanged wrap from closing on S^8. These comparisons do not by themselves certify the claimed complex S^6 or an interacting quantum spectrum.

1 The source construction and the toroidal comparison

The higher-rung investigation asks how a structure with three interacting directions can extend the source geometry. The present algebraic datum is the vector and two half-spin sectors of triality, together with their invariant trilinear tensor. The complex-torus extension studied first below is one specified comparison with the source construction. Its complex structure is a hypothesis of those toroidal theorems; it is not a requirement on the higher-rung target. The retained cusp-angle map and the marked $A_5^4D_4$ lattice are retained as inputs to the subsequent comparison maps.

We use the following exact input from the reconstructed candidate construction [2]. Let

$$B = \mathbb{P}^1, \qquad \Sigma = \{p_0, p_1, p_2\}, \qquad B^\circ = B \setminus \Sigma,$$

where p_0 is a cusp and p_1, p_2 have orbifold orders $3, 4$. There is a claimed compact connected complex threefold and proper holomorphic map

$$f : X \longrightarrow B, \qquad X \cong_{\text{diff}} S^6,$$

whose regular fibres are complex two-tori. In the basis (γ, u, w, δ) of $V = \mathbb{Z}^4$ the integral monodromies are generated by

$$T_1 = \begin{pmatrix} 1 & 0 & -6 & 2 \\ 0 & -1 & 1 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad T_2 = \begin{pmatrix} 1 & 6 & 0 & -3 \\ 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{1}$$

with $T_1^3 = T_2^4 = I$ and $T_0 = (T_1T_2)^{-1} = I + N$, $N^2 = 0$. On the dual lattice $\Lambda = V^*$ the monodromies are $A_j = (T_j^{-1})^{\mathsf{T}}$. The period matrix is

$$\Pi(z) = \begin{pmatrix} 6\mu(z) & \tau(z) & 1 & 0 \\ \beta(z) & \mu(z) & 0 & 1 \end{pmatrix}, \qquad F_z = \mathbb{C}^2/\Pi(z)\Lambda. \tag{2}$$

At p_0 the filling is a Mumford toric quotient with reduced irreducible normal-crossing central fibre W: its normalization is the degree-six del Pezzo surface, opposite sides of its anticanonical hexagon are identified, and $\chi(W) = 2$. At p_j, $j = 1, 2$, the filling is a logarithmic transform with

$$f^*p_1 = 3S_1, \qquad f^*p_2 = 4S_2,$$

where the S_j are smooth bielliptic surfaces and hence have Euler characteristic zero. The primitive affine translations have

$$v_1 = \widehat{\gamma} + 2\widehat{u} - 4\widehat{w}, \qquad v_2 = -\widehat{\gamma} - 3\widehat{u} + 3\widehat{w}, \qquad (\ell_0, \ell_1, \ell_2) = (0, 1, -1). \tag{3}$$

They give $\pi_1(X) = \mathbb{Z}/|12\ell_0 - 4\ell_1 - 3\ell_2| = 0$ and the integral homology of S^6.

The later marked observation uses the integral difference matrix and its fixed right inverse

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \quad u = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad C = PP^* = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad R = P^*C^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ -1 & 1 \\ -1 & -2 \end{pmatrix}. \tag{4}$$

Thus $PR = I_2$, $Pu = 0$, $u^*R = 0$, and for $X_P = (R\ u)$,

$$X_P^{-1} = \begin{pmatrix} P \\ u^*/3 \end{pmatrix}, \qquad \det X_P = 1, \qquad X_P^* X_P = \operatorname{diag}(C^{-1}, 3). \tag{5}$$

Definition 1.1 (Data for the complex-toroidal comparison)**.** A *full higher toroidal wrap* consists of all the following data and proofs. Neither its dimension nor its homeomorphism type is prescribed.

(F1) A compact connected smooth carrier obtained from the stated pieces, with its smooth and homotopy type computed from the attaching maps.

(F2) A geometric tangent structure on the whole carrier. For a literal toroidal continuation this is an integrable complex structure; a different category must state its replacement rather than inherit the word "complex".

(F3) A proper holomorphic map to the same one-dimensional base, with a complex torus of the dimension forced by total dimension as regular fibre.

(F4) Specified local spaces over the cusp and the two finite points, including their multiplicities, reduced fibres, attaching maps, and Euler characteristics.

(F5) An integral period lattice, monodromy representation, affine translations, and actual specialization maps at every filling.

(F6) The van Kampen and integral Mayer–Vietoris maps proving the claimed smooth sphere topology.

(F7) A marked observation whose domain, codomain, lattice map, and compatibility with periods and fillings are specified.

(F8) Rigidity: after all markings above are fixed, no auxiliary metric, phase section, fan identification, or gluing choice is left unrecorded, and existence and uniqueness up to the stated notion of isomorphism are proved.

This definition specifies the complex-toroidal comparison studied in the following sections. The three-sector triality construction is developed through its own explicitly defined products, bundles, boundary, and lattice maps below; the list above does not constrain it to be another complex structure.

For a rank-three successor over the regular base, require a complex-linear surjection $P_z : \mathbb{C}^3 \to \mathbb{C}^2$ with $K_{\mathbb{C},z} = \ker P_z$, period embeddings $\widetilde{\Pi}_z : \widetilde{\Lambda} \hookrightarrow \mathbb{C}^3$ and $\Pi_z : \Lambda \hookrightarrow \mathbb{C}^2$, and

$$P_z\widetilde{\Pi}_z = \Pi_z\Phi, \qquad K_{\mathbb{C},z} \cap \widetilde{\Pi}_z(\widetilde{\Lambda}) = \widetilde{\Pi}_z\kappa(K_{\mathbb{Z}}).$$

Put

$$E_z = K_{\mathbb{C},z}/\widetilde{\Pi}_z\kappa(K_{\mathbb{Z}}), \qquad \widetilde{F}_z = \mathbb{C}^3/\widetilde{\Pi}_z(\widetilde{\Lambda}), \qquad F_z = \mathbb{C}^2/\Pi_z(\Lambda).$$

The desired portion of the map diagram is then

$$\begin{array}{ccccccccc}
0 & \longrightarrow & K_{\mathbb{Z}} & \xrightarrow{\kappa} & \widetilde{\Lambda} & \xrightarrow{\Phi} & \Lambda & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \widetilde{\Pi}_z\kappa} & & \downarrow{\scriptstyle \widetilde{\Pi}_z} & & \downarrow{\scriptstyle \Pi_z} & & \\
0 & \longrightarrow & K_{\mathbb{C},z} & \longrightarrow & \mathbb{C}^3 & \xrightarrow{P_z} & \mathbb{C}^2 & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle q_K} & & \downarrow{\scriptstyle q_3} & & \downarrow{\scriptstyle q_2} & & \\
0 & \longrightarrow & E_z & \longrightarrow & \widetilde{F}_z & \xrightarrow{\overline{P}_z} & F_z & \longrightarrow & 0.
\end{array} \tag{6}$$

All three rows are exact. The maps $\overline{P}_z$ must assemble to a map $\widetilde{X}^\circ \to X^\circ$ over B°, together with commuting specialization squares at all three deleted points. Sections 3 and 4 prove exactly which parts of this diagram exist.

2 Sphere and Euler obstructions

Theorem 2.1 (The S^8 tangent obstruction with all hypotheses)**.** *Let M be a closed smooth manifold homotopy equivalent to S^8. There is no bundle endomorphism $J : TM \to TM$ satisfying $J^2 = -\mathrm{id}$. Consequently no smooth manifold in that homotopy type is a complex fourfold, and in particular no proper holomorphic complex-three-torus fibration over $\mathbb{P}^1$ can have such a total space.*

Proof. Suppose J exists and let $E = (TM, J)$, a complex rank-four bundle. Orient M by J and choose $a \in H^8(M;\mathbb{Z})$ with $\langle a, [M]\rangle = 1$. Since the cohomology below the top degree vanishes,

$$c_1(E) = c_2(E) = c_3(E) = 0.$$

The top Chern class of a complex bundle is the Euler class of its underlying oriented real bundle. Poincaré–Hopf and $\chi(M) = 2$ therefore give

$$c_4(E) = e(TM) = 2a.$$

For the underlying real bundle,

$$p_2(E_{\mathbb{R}}) = c_2(E)^2 - 2c_1(E)c_3(E) + 2c_4(E) = 4a,$$

whereas $p_1(TM) = 0$ because $H^4(M;\mathbb{Z}) = 0$. The middle cohomology of M vanishes, so $\mathrm{sign}(M) = 0$. Hirzebruch's degree-eight signature formula now gives

$$0 = \mathrm{sign}(M) = \left\langle \frac{7p_2(TM) - p_1(TM)^2}{45}, [M] \right\rangle = \frac{28}{45},$$

a contradiction. The last displayed rational number is not being asserted to be a possible signature: its nonintegrality is precisely the contradiction forced by the incompatible characteristic classes. □

Theorem 2.2 (Parity and the complete higher-sphere exclusion)**.** *Let M be a smooth homotopy m-sphere with $m > 6$. Then M admits no almost-complex structure and hence no integrable complex structure. In particular S^9 cannot be a complex carrier, and there is no different higher sphere dimension to which the literal complex part of the candidate can be moved.*

Proof. If m is odd, an endomorphism J with $J^2 = -I$ would make each real tangent space a complex vector space, whose real dimension is even. Hence m must be even. Write $m = 2n$; then $m > 6$ gives $n \geq 4$.

If $E = (TM, J)$ existed, $c_i(E) = 0$ for $i < n$ and, after using the complex orientation, $c_n(E) = 2a$ for the integral top generator a. Pull E back along a homotopy equivalence $S^{2n} \to M$. Bott integrality says that the top Chern class of every complex bundle on S^{2n} is divisible by $(n-1)!$ [7, Chapter 17, Section 7, Theorem 7.2, pp. 274–275]. Since $(n-1)! \geq 6$ does not divide 2, this contradicts $c_n(E) = 2a$. Borel and Serre prove the same conclusion for every differentiable structure on S^{2n} in [4, Section 15, Proposition 15.1 and the following remark, p. 434]; their universal reduced-power input is [4, Section 12, Propositions 12.4 and 12.6 and Lemma 12.5, p. 430]. □

Proposition 2.3 (The direct rank-three Euler mismatch)**.** *Consider the direct rank-three toroidal local model obtained by coning the periodic Kuhn triangulation of $\mathbb{R}^3$ and dividing by $\mathbb{Z}^3$. Its central fibre W_3 has*

$$\chi(W_3) = 6.$$

If a proper triangulable map $\widetilde{f} : \widetilde{X}^8 \to S^2$ has this as one special fibre, has precisely two other special fibres that are free finite quotients of T^6, and is a T^6-bundle elsewhere, then $\chi(\widetilde{X}) = 6$. Therefore the direct three-fibre pattern cannot have total space S^8 even after forgetting complex structures.

Proof. A fundamental cube of the Kuhn triangulation is divided into the six tetrahedra indexed by permutations of its three coordinate directions. The central toric fibre is stratified by algebraic tori; every positive-dimensional torus stratum has Euler characteristic zero, while the zero-dimensional strata are indexed by the maximal tetrahedra modulo $\mathbb{Z}^3$. There are exactly six, so additivity over the finite torus stratification gives $\chi(W_3) = 6$.

Let Σ_f be the three critical values. Euler integration, or a cut into the regular bundle and three filling pieces, gives

$$\chi(\widetilde{X}) = \chi(S^2 \setminus \Sigma_f)\chi(T^6) + \sum_{p \in \Sigma_f} \chi(\widetilde{f}^{-1}(p)).$$

The regular term vanishes since $\chi(T^6) = 0$. A free finite quotient of a torus has Euler characteristic zero: the covering formula multiplies its Euler characteristic by the positive degree and the covering torus has Euler characteristic zero. Only W_3 remains, giving six. Since $\chi(S^8) = 2$, the total spaces cannot be homeomorphic. □

Corollary 2.4 (The exact singular-fibre repair equation)**.** *Let $g : M \to S^2$ be a proper triangulable map which is locally trivial with fibre T^6 off a finite set Δ, and suppose the direct Kuhn cusp fibre W_3 occurs over $p_0 \in \Delta$. Then*

$$\chi(M) = 6 + \sum_{p \in \Delta \setminus \{p_0\}} \chi(g^{-1}(p)). \tag{7}$$

Consequently an S^8 carrier requires, exactly at the Euler-characteristic level,

$$\sum_{p \in \Delta \setminus \{p_0\}} \chi(g^{-1}(p)) = -4. \tag{8}$$

If there are only the original two finite values, their replacement fibres must have Euler characteristics summing to -4. If those two fibres remain free finite quotients of T^6, they still contribute zero, so the cusp fibre itself must instead be replaced by a filling with Euler characteristic 2. Without either change, topology is an output and every closed carrier of the unchanged wrap necessarily has Euler characteristic 6.

Proof. Euler integration gives $\chi(M) = \chi(S^2 \setminus \Delta)\chi(T^6) + \sum_{p\in\Delta} \chi(g^{-1}(p))$. The first term is zero and $\chi(W_3) = 6$ by proposition 2.3, giving (7). Substitution of $\chi(S^8) = 2$ gives (8). The remaining assertions are its two specializations, using the covering calculation in the proof of proposition 2.3. □

Remark 2.5 (What remains possible topologically)**.** A merely real singular map $S^8 \to S^2$ with regular fibre T^6 is not excluded by theorem 2.1; it must have special-fibre Euler sum 2. It cannot be a fibre bundle: the homotopy sequence would require the connecting map $\pi_2(S^2) = \mathbb{Z} \to \pi_1(T^6) = \mathbb{Z}^6$ to be surjective. A real $S^9 \to S^2$ model with regular fibre T^7 would likewise require genuine singularities and special-fibre Euler sum 0, but neither its carrier nor its odd-dimensional regular fibre can be complex in the required sense.

3 The exact toroidal successor over the cusp

We now retain, rather than discard, every exact map that survives the obstructions. Put

$$A_F = \mathbb{Z}\langle \widehat{w}, \widehat{\delta}\rangle, \qquad B_F = \mathbb{Z}\langle \widehat{\gamma}, \widehat{u}\rangle, \qquad B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} : B_F \to A_F.$$

In angular–deck order the candidate cusp monodromy and filling map are

$$M_F(a, b) = (a + B_0 b, b), \qquad c_F : A_F \oplus B_F \longrightarrow B_F, \quad c_F(a, b) = b. \tag{9}$$

Let $L_3 = \mathbb{Z}^3$, let $k = (1, 1, 1)^{\mathsf{T}}$, and put

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad Q = B_0^{-1} P = \begin{pmatrix} 0 & -1 & 1 \\ 1 & -1 & 0 \end{pmatrix}. \tag{10}$$

Both maps are onto, $\ker P = \ker Q = \mathbb{Z}k$, and $B_0 Q = P$. The rank-three cusp module and its specialization are

$$M_3(a, b) = (a + b, b), \qquad c_3 : L_3 \oplus L_3 \longrightarrow L_3, \quad c_3(a, b) = b. \tag{11}$$

Theorem 3.1 (Primitive lattice, monodromy, and filling quotient)**.** *The homomorphism*

$$\Phi = P \oplus Q : L_3 \oplus L_3 \longrightarrow A_F \oplus B_F, \qquad \Phi(a, b) = (Pa, Qb),$$

fits into the exact sequence

$$0 \longrightarrow \mathbb{Z}^2 \xrightarrow{\kappa} \mathbb{Z}^6 \xrightarrow{\Phi} \mathbb{Z}^4 \longrightarrow 0, \qquad \kappa(m, n) = (mk, nk), \tag{12}$$

and satisfies

$$\Phi M_3 = M_F \Phi, \qquad c_F \Phi = Q c_3. \tag{13}$$

The kernel in (12) *is primitive.*

Proof. The integer matrix

$$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

satisfies $PS = I_2$ and $QSB_0 = I_2$. Thus P and Q, and hence Φ, are integrally surjective. Their displayed kernels give $\ker \Phi = \mathbb{Z}(k,0) \oplus \mathbb{Z}(0,k)$, proving exactness. A kernel of a surjection between free abelian groups is primitive.

Directly,

$$\Phi M_3(a,b) = (P(a+b), Qb),$$

while

$$M_F \Phi(a,b) = (Pa + B_0 Qb, Qb) = (Pa + Pb, Qb).$$

The filling identity is equally literal: $c_F \Phi(a,b) = Qb = Qc_3(a,b)$. □

Let $t = e^{2\pi i s}$ on a logarithmic cusp cover. Retain the complete bounded holomorphic correction $C_2(t)$ of the candidate and define

$$Z_2 = sB_0 + C_2(t), \qquad C_3(t) = SC_2(t)Q, \qquad Z_3 = sI_3 + C_3(t). \tag{14}$$

We use the angular–deck period convention $\Omega_r = (I_r \ Z_r)$, and write $\Omega_r(t)$ when the dependence through s and $C_r(t)$ must be displayed. The manuscript convention $(Z_r \ I_r)$ is related by the displayed integral block permutation $\left(\begin{smallmatrix} 0 & I_r \\ I_r & 0 \end{smallmatrix}\right)$; no period, sign, or lattice coordinate is silently changed.

Theorem 3.2 (Holomorphic elliptic extension and its exact fibre)**.** *After shrinking the cusp disc,* $\operatorname{Im} Z_3$ *is invertible and*

$$PZ_3 = Z_2 Q, \qquad P\Omega_3 = \Omega_2 \Phi. \tag{15}$$

Consequently $P : \mathbb{C}^3 \to \mathbb{C}^2$ *descends to a short exact sequence of complex tori*

$$0 \longrightarrow E_s \longrightarrow \frac{\mathbb{C}^3}{\mathbb{Z}^3 + Z_3\mathbb{Z}^3} \xrightarrow{\overline{P}_s} \frac{\mathbb{C}^2}{\mathbb{Z}^2 + Z_2\mathbb{Z}^2} \longrightarrow 0, \qquad E_s \cong \frac{\mathbb{C}}{\mathbb{Z} + s\mathbb{Z}}. \tag{16}$$

The correction C_2 *is recoverable from its prescribed lift by the exact left inverse*

$$C_2 = PC_3SB_0. \tag{17}$$

Thus the extension neither deletes the original periods nor replaces them by an identity period matrix.

Proof. Since C_3 is bounded while $\operatorname{Im} s \to +\infty$, $\operatorname{Im} Z_3 = (\operatorname{Im} s) I_3 + \operatorname{Im} C_3$ is invertible on a sufficiently small cusp. Using $PS = I_2$ and $B_0 Q = P$,

$$PZ_3 = sP + PSC_2Q = sB_0Q + C_2Q = Z_2Q,$$

which is exactly (15). It makes the complex linear map P a homomorphism of the displayed quotient tori. It is surjective on vector spaces and its lattice image is the whole target lattice because Φ is onto; therefore its kernel is connected.

The vector-space kernel is $\mathbb{C}k$. Since $Qk = 0$, one has $C_3 k = 0$ and $Z_3 k = sk$. If a source period lies in $\mathbb{C}k$, its image under P is the zero target period. The real period map $\Omega_2 : \mathbb{R}^4 \to \mathbb{C}^2$ is an isomorphism, so (15) and $\ker \Phi = \mathbb{Z}(k,0) \oplus \mathbb{Z}(0,k)$ show that all such periods are precisely $\mathbb{Z}k + s\mathbb{Z}k$. This proves the fibre formula.

Finally $QS = B_0^{-1}$, whence

$$PC_3SB_0 = PSC_2QSB_0 = C_2.$$

□

3.1 The cusp filling map, not merely its boundary lattice

For $m \in \mathbb{Z}^3$ and a permutation $\pi = (\pi_1, \pi_2, \pi_3) \in S_3$, let

$$\sigma_{m,\pi} = \mathrm{conv}\{m, m + e_{\pi_1}, m + e_{\pi_1} + e_{\pi_2}, m + e_1 + e_2 + e_3\}. \tag{18}$$

These are the tetrahedra of the periodic Kuhn triangulation; denote that triangulation by $\mathcal{T}_3$. Let $\mathcal{T}_2$ be the periodic triangulation of $\mathbb{R}^2$ whose triangles are

$$\mathrm{conv}\{u, u + e_1, u + e_2\}, \qquad \mathrm{conv}\{u + e_1, u + e_2, u + e_1 + e_2\}, \qquad u \in \mathbb{Z}^2.$$

For $r = 2, 3$, let $\mathfrak{S}_r$ be the fan in $\mathbb{Z}^r \oplus \mathbb{Z}$ consisting of all faces of the cones over $\tau \times \{1\}$ for $\tau \in \mathcal{T}_r$, and put $Y_r = X_{\mathfrak{S}_r}$. In the target lattice write

$$n_1 = \widehat{w}, \qquad n_2 = \widehat{\delta} - \widehat{w}, \qquad n_3 = -\widehat{\delta}; \qquad n_1 + n_2 + n_3 = 0.$$

The map P in (10) is exactly $P(e_i) = n_i$.

Proposition 3.3 (Exact fan and quotient-filling morphism)**.** *Coning the Kuhn triangulation at height one gives a smooth toric fourfold Y_3 with height map locally equal to*

$$t = z_0 z_1 z_2 z_3.$$

The integral map

$$\widehat{P} : \mathbb{Z}^3 \oplus \mathbb{Z} \longrightarrow A_F \oplus \mathbb{Z}, \qquad \widehat{P}(x, r) = (Px, r),$$

is a map from this fan to the candidate's periodic A_2 fan and hence induces a toric morphism $F : Y_3 \to Y_2$ over the t-line.

On the dense torus define

$$\Psi^{(3)}_\lambda(x, t) = \big(\exp(2\pi i C_3(t)\lambda) t^\lambda x, t\big), \qquad \lambda \in \mathbb{Z}^3, \tag{19}$$

$$\Psi^{(2)}_\mu(y, t) = \big(\exp(2\pi i C_2(t)\mu) t^{B_0\mu} y, t\big), \qquad \mu \in \mathbb{Z}^2. \tag{20}$$

Then

$$F\Psi^{(3)}_\lambda = \Psi^{(2)}_{Q\lambda} F. \tag{21}$$

For a sufficiently small open cusp disc $\Delta_\epsilon = \{t \in \mathbb{C} : |t| < \epsilon\}$, set

$$Y_{r,\epsilon} = t^{-1}(\Delta_\epsilon), \qquad N_3^C = Y_{3,\epsilon}/\{\Psi^{(3)}_\lambda : \lambda \in \mathbb{Z}^3\}, \qquad N_0 = Y_{2,\epsilon}/\{\Psi^{(2)}_\mu : \mu \in \mathbb{Z}^2\}. \tag{22}$$

After shrinking Δ_ϵ, both actions are free and properly discontinuous, and F descends to a holomorphic filling morphism

$$\overline{F} : N_3^C \longrightarrow N_0 \tag{23}$$

whose map on smooth-fibre H_1 is Φ and whose monodromy compatibility is (13)*. Under $H_1(T^6) = L_3 \oplus L_3$ and $H_1(T^4) = A_F \oplus B_F$, the angular generators bound the toric coordinate discs while the deck generators survive. Hence the inclusion-induced specialization maps are exactly $c_3(a, b) = b$ and $c_F(a, b) = b$, and they form the commutative square*

$$\begin{array}{ccc} H_1(T^6; \mathbb{Z}) & \xrightarrow{\Phi} & H_1(T^4; \mathbb{Z}) \\ {\scriptstyle c_3}\downarrow & & \downarrow{\scriptstyle c_F} \\ \mathbb{Z}^3 & \xrightarrow{Q} & \mathbb{Z}^2. \end{array} \tag{24}$$

Proof. The four vertices of (18) map to

$$Pm, \quad Pm + n_{\pi_1}, \quad Pm - n_{\pi_3}, \quad Pm.$$

In target coordinates $(n_1, n_2, n_3) = (e_1, e_2 - e_1, -e_2)$. For the six permutations the three distinct images are the six A_2 triangles incident to Pm:

$$\begin{array}{c|c|c|c} (1,2,3) & \{v, v+e_1, v+e_2\} & (2,1,3) & \{v, v-e_1+e_2, v+e_2\} \\ (1,3,2) & \{v, v+e_1, v+e_1-e_2\} & (3,1,2) & \{v, v-e_2, v+e_1-e_2\} \\ (2,3,1) & \{v, v-e_1+e_2, v-e_1\} & (3,2,1) & \{v, v-e_2, v-e_1\}. \end{array}$$

Thus every source cone maps into a target cone. The toric fan criterion gives F, and preservation of the height coordinate gives $t \circ F = t$. The maximal Kuhn cones are unimodular, giving the displayed local monomial equation and smoothness of Y_3.

On the dense torus, $PC_3 = C_2Q$ and $P = B_0Q$ give

$$F\Psi_\lambda^{(3)}(x,t) = (\exp(2\pi i C_2 Q\lambda)t^{B_0Q\lambda}F(x), t),$$

which proves (21); the fan map extends it over every toric orbit.

For the analytic quotient, put $y = -\log|t|/(2\pi)$ and

$$\operatorname{trop}_t(x) = \frac{1}{\log|t|}(\log|x_1|, \log|x_2|, \log|x_3|).$$

The exact transformation law is

$$\operatorname{trop}_t(\Psi_\lambda^{(3)}x) = \operatorname{trop}_t(x) + A(t)\lambda, \qquad A(t) = I_3 + \frac{\operatorname{Im} C_3(t)}{y}.$$

Boundedness of C_3 permits the cusp to be shrunk until

$$\tfrac{1}{2}\|\lambda\| \le \|A(t)\lambda\| \le \tfrac{3}{2}\|\lambda\|.$$

For the target use

$$\operatorname{trop}_t^{(2)}(y) = \frac{1}{\log|t|}(\log|y_1|, \log|y_2|).$$

Its exact transformation law is

$$\operatorname{trop}_t^{(2)}(\Psi_\mu^{(2)}y) = \operatorname{trop}_t^{(2)}(y) + A_2(t)\mu, \qquad A_2(t) = B_0 + \frac{\operatorname{Im} C_2(t)}{y}.$$

Since B_0 is orthogonal, after one further shrinking the same bounds hold with $A_2(t), \mu$ in place of $A(t), \lambda$. Therefore a compact set can meet only finitely many of its translates for either action. On a boundary orbit the bounded cell label is translated by λ, so the same conclusion holds there; the target argument uses the invertible lattice map B_0 in exactly the same way. A fixed point in a smooth fibre would give $\operatorname{Im} Z_3\lambda = 0$, hence $\lambda = 0$; equality of a boundary cell with its translate likewise forces $\lambda = 0$. The target equalities give $\operatorname{Im} Z_2\mu = 0$ in the smooth fibre and $B_0\mu = 0$ on the boundary, hence $\mu = 0$. Rounding the coordinates of $A(t)^{-1}\operatorname{trop}_t(x)$ gives a uniformly bounded set of representatives. The closures of these representatives lie in the finite union of closed toric polydiscs belonging to the six Kuhn tetrahedra; that finite union is compact. This proves proper discontinuity. The same construction with $A_2(t)$ supplies target representatives. Over every smaller closed disc $\overline{\Delta}_{\epsilon'}$, $0 < \epsilon' < \epsilon$, the displayed finite unions are compact, proving properness there. The equivariant F therefore descends, and (24) is exactly the second identity in (13). □

3.2 Finite peripheral data and the precise globalization gap

In the manuscript lattice order the candidate matrices are

$$A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -6 & -1 & -1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -6 & 1 & 0 & 0 \\ 3 & 0 & 1 & 1 \end{pmatrix}. \tag{25}$$

The matrices A_1, A_2 are written in the manuscript order $(\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta}) = B_F \oplus A_F$, whereas M_F in (9) is written in the angular–deck order $A_F \oplus B_F$. With the explicit block swap

$$J = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \qquad M_F^{\mathrm{man}} = J M_F J = \begin{pmatrix} I_2 & 0 \\ B_0 & I_2 \end{pmatrix}, \tag{26}$$

one has $A_1^3 = A_2^4 = I$ and $A_1 A_2 M_F^{\mathrm{man}} = I$. Define

$$C_1 = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}, \quad C_2' = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad C_0 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}. \tag{27}$$

For uniform notation below set $D_1 = C_1$ and $D_2 = C_2'$. Then $(C_1)^3 = (C_2')^4 = I$, $C_1 C_2' = C_0^{-1}$, and $(C_0 - I)^2 = 0$ with rank one.

The kernel block in this presentation is not an unrelated direct summand of the rank-three cusp lattice. Put

$$\begin{gathered} r = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}, \qquad A = \begin{pmatrix} P \\ r \end{pmatrix} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} Q \\ r \end{pmatrix} = \begin{pmatrix} 0 & -1 & 1 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \\ D = B_0 \oplus [1] = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad U_6 = \mathrm{diag}(A, B) \in SL(6, \mathbb{Z}). \end{gathered} \tag{28}$$

Lemma 3.4 (The identical cusp lattice in quotient–kernel coordinates)**.** *Let*

$$\rho(Pa, ra; Qb, rb) = (Pa, Qb; ra, rb), \qquad \rho_{\mathrm{man}} = (J \oplus I_2)\rho.$$

Then

$$U_6 M_3 U_6^{-1} = \begin{pmatrix} I_3 & D \\ 0 & I_3 \end{pmatrix}, \tag{29}$$

$$\rho U_6 M_3 U_6^{-1} \rho^{-1} = M_F \oplus C_0, \tag{30}$$

$$\rho_{\mathrm{man}} U_6 M_3 U_6^{-1} \rho_{\mathrm{man}}^{-1} = M_F^{\mathrm{man}} \oplus C_0. \tag{31}$$

Projection onto the candidate block is exactly $\mathrm{pr}_{1,2}\, \rho U_6(a, b) = (Pa, Qb) = \Phi(a, b)$ *in angular–deck coordinates and* $J\Phi(a, b)$ *in manuscript coordinates.*

Proof. Both A and B have determinant one, and direct multiplication gives $A = DB$, so $AB^{-1} = D$. Conjugating $M_3 = \left(\begin{smallmatrix} I_3 & I_3 \\ 0 & I_3 \end{smallmatrix}\right)$ by U_6 proves (29). The first two rows of A, B are exactly P, Q, while $r(1, 1, 1)^{\mathsf{T}} = 1$; regrouping the first two and last coordinates therefore gives (30). Applying the already displayed block swap J proves (31) and both projection formulas. □

Proposition 3.5 (Exact finite-order affine lift)**.** *On $\widetilde{\Lambda} = \Lambda \oplus \mathbb{Z}^2$, identified with the original $\mathbb{Z}^3 \oplus \mathbb{Z}^3$ by lemma 3.4, set*

$$\widetilde{A}_1 = A_1 \oplus C_1, \qquad \widetilde{A}_2 = A_2 \oplus C_2', \qquad \widetilde{M}_0 = M_F^{\mathrm{man}} \oplus C_0.$$

Then

$$\widetilde{A}_1^3 = I, \qquad \widetilde{A}_2^4 = I, \qquad \widetilde{A}_1 \widetilde{A}_2 \widetilde{M}_0 = I.$$

The affine transformations

$$\widetilde{g}_j(x, y) = (A_j x + v_j/m_j, D_j y), \qquad (m_1, m_2) = (3, 4), \tag{32}$$

of $T^4 \times T^2$ have exact orders m_j, act freely, and project to the candidate finite-fibre actions.

Proof. All three matrix identities follow blockwise from the identities just displayed and the candidate identities. The vector $(v_j, 0)$ is fixed by $\widetilde{A}_j$. The m_j-th power of (32) is translation by the integral vector $(v_j, 0)$ and hence is the identity on the torus. If a smaller positive power fixed a point, its projection to T^4 would fix a point under the corresponding smaller power of the candidate affine transformation. The logarithmic-transform action in the candidate is free, so this is impossible. Projection to the first block is equivariant by construction. □

Remark 3.6 (Exact endpoint)**.** Theorems 3.1–3.2 and Propositions 3.3–3.5 prove the regular cusp, the entire toric cusp filling, and compatible finite *integral affine* data. They do not construct one holomorphic rank-three period family on the full $(3, 4, \infty)$ cover whose two finite analytic fillings glue to the cusp family. Even if that analytic gluing is constructed, proposition 2.3 proves that the unchanged three-fibre pattern has Euler characteristic six and hence is not S^8. This is a proved morphism with a proved endpoint, not an assertion that the rank-three and rank-two objects are unrelated.

4 The forced S^9, S^{10}, and rank-loss S^8

Put

$$\mathscr{V} = \mathrm{Mat}_{3\times 3}(\mathbb{C}), \qquad \mathcal{P}_3 = \{H \in \mathrm{Herm}_3(\mathbb{C}) : H \succeq 0\}, \qquad \mathcal{K}_3 = \{H \in \mathcal{P}_3 : \operatorname{tr} H = 1\}.$$

The superscript + denotes one-point compactification.

Theorem 4.1 (Complete $U(3)$ and $SU(3)$ orbit invariants)**.** *The maps*

$$\Gamma_U(A) = A^* A, \tag{33}$$

$$\Gamma_{SU}(A) = (A^* A, \det A) \tag{34}$$

induce homeomorphisms

$$\mathscr{V}/U(3) \xrightarrow{\sim} \mathcal{P}_3, \tag{35}$$

$$\mathscr{V}/SU(3) \xrightarrow{\sim} \mathcal{Y}_3 := \{(H, z) \in \mathcal{P}_3 \times \mathbb{C} : |z|^2 = \det H\}. \tag{36}$$

For a rank-r matrix A, the two orbit fibres and stabilizers are

$$U(3) \cdot A \cong U(3)/U(3-r), \qquad SU(3) \cdot A \cong SU(3)/SU(3-r),$$

with $SU(1)$ interpreted as the trivial group.

Proof. Both maps are invariant under the indicated left actions, and $|\det A|^2 = \det(A^*A)$. Every $H \succeq 0$ is the Gram matrix of $H^{1/2}$. If $A^*A = B^*B$, the rule $Ax \mapsto Bx$ is well-defined because the two matrices have the same kernel, and is isometric because

$$\langle Bx, By\rangle = x^*B^*By = x^*A^*Ay = \langle Ax, Ay\rangle.$$

Extending it orthogonally gives $U \in U(3)$ with $UA = B$. Thus (33) separates $U(3)$-orbits.

Given $(H, z) \in \mathcal{Y}_3$, if H is singular then $z = 0$ and $H^{1/2}$ realizes it. If H is positive definite, put

$$\omega = z/\sqrt{\det H}, \qquad A = \operatorname{diag}(\omega, 1, 1)H^{1/2}.$$

Then $(A^*A, \det A) = (H, z)$. If two matrices have both invariants equal, the unitary U above has determinant one when they are invertible. In rank at most two, multiply U on the left by a unitary that fixes $\operatorname{im} B$ pointwise and corrects the determinant on its nonzero orthogonal complement. Hence (34) separates $SU(3)$-orbits.

Both invariant maps are proper because $\operatorname{tr}(A^*A) = \|A\|_F^2$. The induced continuous bijections to the displayed closed Hausdorff sets are therefore homeomorphisms. A left stabilizer fixes $\operatorname{im} A$ pointwise and is arbitrary unitary, respectively special unitary, on $(\operatorname{im} A)^\perp$, proving the fibre statements. □

Let

$$E = \{X \in \operatorname{Herm}_3(\mathbb{C}) : \operatorname{tr} X = 0\}, \qquad \dim_{\mathbb{R}} E = 8, \qquad c = I_3/3.$$

Equip E with the Frobenius inner product $\langle X, Y\rangle_F = \operatorname{tr}(XY)$ and its norm; $B(E)$ and $S(E)$ below denote its closed unit ball and unit sphere. For $v \in S(E)$ define

$$R_*(v) = -\frac{1}{3\lambda_{\min}(v)}. \tag{37}$$

Since a nonzero traceless Hermitian matrix has a negative eigenvalue, $R_*(v)$ is finite and positive.

Lemma 4.2 (Explicit eight-ball chart)**.** *The maps*

$$\chi(c) = 0, \qquad \chi(c + sv) = \frac{s}{R_*(v)}v, \quad 0 < s \le R_*(v), \tag{38}$$

$$\chi^{-1}(rv) = c + rR_*(v)v, \quad 0 < r \le 1 \tag{39}$$

are inverse homeomorphisms $\mathcal{K}_3 \cong B(E) \cong B^8$. They send the rank-deficient locus onto $S(E)$.

Proof. The eigenvalues of $c + sv$ are $1/3 + s\lambda_j(v)$. Hence this matrix is positive semidefinite exactly for $0 \le s \le R_*(v)$ and is singular exactly at the upper endpoint. Every $H \neq c$ in $\mathcal{K}_3$ lies on one unique such ray. Continuity of the least eigenvalue makes $v \mapsto -\lambda_{\min}(v)$ a positive continuous function on the compact space $S(E)$, hence bounded above and bounded away from zero. Thus $R_*(v)$ is continuous and uniformly bounded above and away from zero. The formulas (38)–(39) are consequently continuous, including at c and at the origin, and prove all assertions. □

Theorem 4.3 (The forced sphere ladder and explicit inverses)**.** *Let $S_F^{17} = \{A \in \mathscr{V} : \|A\|_F = 1\}$. Then*

$$S_F^{17}/SU(3) \cong \mathcal{Y}_{3,1} := \{(H, z) \in \mathcal{Y}_3 : \operatorname{tr} H = 1\} \cong S^9. \tag{40}$$

An explicit second homeomorphism is

$$\Omega(H, z) = \begin{cases} \left(\chi(H), \sqrt{1 - \|\chi(H)\|^2}\, z/|z|\right), & H \succ 0, \\ (\chi(H), 0), & H \text{ singular}, \end{cases} \tag{41}$$

from $\mathcal{Y}_{3,1}$ *to* $S(E \oplus \mathbb{C})$.

Furthermore the weighted radial map

$$\Phi_{SU}(H,z) = \begin{cases} 0, & (H,z) = (0,0), \\ s\,\Omega(H/s^2, z/s^3), & s = \sqrt{\operatorname{tr} H} > 0 \end{cases} \tag{42}$$

is a homeomorphism $\mathcal{Y}_3 \cong E \oplus \mathbb{C} \cong \mathbb{R}^{10}$. *It extends to one-point compactifications and gives a homeomorphism of pairs*

$$\Big(\mathscr{V}^+/SU(3), \{\det A = 0\}^+/SU(3)\Big) \cong \Big(S^{10}, S^8_{\mathrm{lin}}\Big). \tag{43}$$

The complement is

$$S^{10} \setminus S^8 \cong E \times (\mathbb{C} \setminus \{0\}) \simeq S^1, \tag{44}$$

and $\arg\det A$ *is the angular deformation-retraction coordinate.*

Proof. The first identification in (40) follows from theorem 4.1. If $H \succ 0$, then $z \neq 0$; if H is singular, then $z = 0$ and lemma 4.2 gives $\|\chi(H)\| = 1$. Thus (41) lands on the unit sphere and is continuous even at rank loss because the coefficient of $z/|z|$ tends to zero. Given $(x,w) \in S(E \oplus \mathbb{C})$, put $H = \chi^{-1}(x)$. If $\|x\| < 1$, then $w \neq 0$ and the unique inverse determinant is

$$z = \sqrt{\det H}\, w/|w|. \tag{45}$$

If $\|x\| = 1$, then $w = 0$, H is singular, and $z = 0$. This proves the S^9 homeomorphism with its inverse.

For $(H,z) \neq 0$, the normalized pair $(H/s^2, z/s^3)$ belongs to $\mathcal{Y}_{3,1}$ because $|z/s^3|^2 = \det(H/s^2)$. Conversely, for $v \neq 0$ in $E \oplus \mathbb{C}$, set

$$s = \|v\|, \qquad (h,\zeta) = \Omega^{-1}(v/\|v\|), \qquad H = s^2 h, \qquad z = s^3\zeta.$$

This is the inverse of (42). The norm identity $\|\Phi_{SU}(H,z)\| = s$ proves continuity of both maps at zero and properness, so the map extends to compactifications.

Rank loss is exactly $z = 0$. Under Ω, the normalized rank-loss set maps to $S(E \oplus 0)$, and radialization maps the entire affine rank-loss set to $E \oplus 0$. Its one-point compactification is the standard linear $S^8 \subset S^{10}$. Removing it gives $E \times (\mathbb{C} \setminus 0)$; projection of the second factor to its phase is a deformation retraction. By construction this phase is $z/|z| = \det A/|\det A|$. □

Put

$$\Sigma_3 = \{H \in \mathrm{Herm}_3(\mathbb{C}) : H \succeq 0,\ \det H = 0\}.$$

Proposition 4.4 (The $U(3)$ ball and why its boundary is the S^8)**.** *The map*

$$E \times \mathbb{R}_{\geq 0} \longrightarrow \mathcal{P}_3, \qquad (K,t) \longmapsto K + (t - \lambda_{\min}(K))I_3 \tag{46}$$

is a homeomorphism with inverse

$$H \longmapsto \left(H - \frac{\operatorname{tr} H}{3} I_3, \lambda_{\min}(H)\right).$$

Consequently

$$\mathscr{V}^+/U(3) \cong \mathcal{P}_3^+ \cong B^9, \qquad \partial B^9 = \Sigma_3^+ \cong S^8. \tag{47}$$

By contrast, the quotient of the source unit sphere is

$$S^{17}_F/U(3) \cong \mathcal{K}_3 \cong B^8, \tag{48}$$

whose boundary is the trace-one rank-loss S^7. *Thus "take the source sphere quotient" and "take the boundary of the compactified full quotient" are different operations.*

Proof. The least eigenvalue of the image of (46) is t. Conversely, subtracting one third of the trace from H produces a traceless matrix K whose least eigenvalue is $\lambda_{\min}(H) - \operatorname{tr}(H)/3$; substitution recovers H. Eigenvalue continuity proves that the formulas are inverse homeomorphisms. The one-point compactification of a closed half-space is a closed ball, and its boundary is the one-point compactification of the boundary hyperplane. Now use theorem 4.1. Equation (48) follows from $\|A\|_F^2 = \operatorname{tr}(A^*A)$ and lemma 4.2. □

Corollary 4.5 (Residual determinant circle and its fibres)**.** *The residual group $U(3)/SU(3) \cong S^1$ acts on the pair in* (43) *by*

$$e^{i\theta} \cdot (x, w) = (x, e^{i\theta} w).$$

Its fixed set is precisely the rank-loss S^8, and the orbit map has

$$S^{10}/S^1 \cong B^9, \qquad q^{-1}(q(p)) \cong \begin{cases} S^1, & p \notin S^8, \\ \{p\}, & p \in S^8. \end{cases} \tag{49}$$

On the normalized section, $S^9/S^1 \cong B^8$ and the fixed set is the rank-deficient $S^7 = \partial B^8$.

Proof. Left multiplication by $U \in U(3)$ fixes $H = A^*A$ and sends $z = \det A$ to $(\det U)z$. Every phase is a determinant. Equations (41) and (42) therefore turn the residual action into scalar rotation of the complex coordinate. Its orbit space on $E \oplus \mathbb{C}$ is $E \times \mathbb{R}_{\geq 0}$, whose one-point compactification is a closed 9-ball; the zero complex coordinate is its boundary. The stabilizer is trivial for $w \neq 0$ and all of S^1 for $w = 0$, giving the fibres. On the unit sphere, replacing w by $|w|$ gives the closed unit ball in E and the last statement. □

The S^8 in (43) can also be defined without first forming an orbit quotient.

Proposition 4.6 (The exact residual-circle degeneration and join)**.** *Under* (40)*, forgetting determinant phase is exactly*

$$q_1 : S(E \oplus \mathbb{C}) = S^9 \longrightarrow B(E) = B^8, \qquad q_1(x, w) = x. \tag{50}$$

Its fibre is S^1 for $\|x\| < 1$ and a singleton for $\|x\| = 1$. More precisely,

$$S^9 = S(E \oplus \mathbb{C}) \cong S(E) * S(\mathbb{C}) \cong S^7 * S^1, \tag{51}$$

and q_1 collapses the S^1 join factor at the S^7 endpoint. On the full compactification the analogous map is

$$q_+ : S^{10} \longrightarrow B^9, \tag{52}$$

with circle fibre off the rank-loss S^8 and point fibre on that S^8. The integral class of the circle on either complement is generated by $\arg\det A$.

Proof. For fixed $H \succ 0$ of trace one, the equation $|z|^2 = \det H > 0$ leaves exactly its phase, an S^1; for singular H it forces $z = 0$. Formula (41) converts H into $x = \chi(H)$ and the determinant phase into the phase of w, so the orbit map is literally (50). Every unit pair has the unique form

$$(x, w) = (\cos\alpha\, u, \sin\alpha\, v), \qquad u \in S(E),\ v \in S(\mathbb{C}),\ 0 \leq \alpha \leq \pi/2,$$

away from the endpoints, with the usual endpoint identifications. This is the defining quotient of the join and proves (51). The full statement is corollary 4.5 and proposition 4.4. The generator claim follows from the loop

$$A_\theta = 3^{-1/2} \operatorname{diag}(e^{i\theta}, 1, 1), \qquad 0 \leq \theta \leq 2\pi,$$

whose Gram matrix is $I_3/3$ and whose determinant phase winds once. □

Proposition 4.7 (Spectral-floor chart and the marked fibres)**.** *The maps*

$$\mathcal{F} : E \longrightarrow \Sigma_3, \qquad \mathcal{F}(K) = K - \lambda_{\min}(K) I_3, \qquad \mathcal{F}^{-1}(H) = H - \frac{\operatorname{tr} H}{3} I_3 \tag{53}$$

are inverse homeomorphisms. Hence $\Sigma_3^+ \cong S^8$*, and it is exactly the* S^8 *of* (43)*.*

The fixed marked compression

$$\mathscr{O} : \Sigma_3 \longrightarrow \operatorname{Herm}_2(\mathbb{C})_{\succeq 0}, \qquad \mathscr{O}(H) = PHP^* \tag{54}$$

is onto. For a fixed $G \succeq 0$*, put*

$$X_P^{-1} H X_P^{-*} = \begin{pmatrix} G & a \\ a^* & \lambda \end{pmatrix}.$$

Its complete fibre in Σ_3 *is*

$$\mathscr{O}^{-1}(G) = \left\{ X_P \begin{pmatrix} G & a \\ a^* & a^* G^\dagger a + \tau \end{pmatrix} X_P^* : a \in \operatorname{Ran} G,\ \tau \geq 0,\ \left(\operatorname{rank} G < 2 \ \text{or}\ \tau = 0\right) \right\}. \tag{55}$$

Consequently the fibres are homeomorphic to

$$\begin{cases} \mathbb{C}^2, & G \succ 0, \\ \mathbb{C} \times \mathbb{R}_{\geq 0}, & \operatorname{rank} G = 1, \\ \mathbb{R}_{\geq 0}, & G = 0. \end{cases} \tag{56}$$

The positive section is $G \mapsto RGR^*$*. The affine map* (54) *is not proper, so these formulas do not assert an unproved extension through the one-point compactification point.*

Proof. For $K \in E$, the matrix $\mathcal{F}(K)$ is positive semidefinite and has least eigenvalue zero. If $H \in \Sigma_3$, then $K = H - (\operatorname{tr} H/3) I_3$ is traceless and has least eigenvalue $-\operatorname{tr} H/3$; therefore the two formulas in (53) are inverse. Continuity of the least eigenvalue proves the homeomorphism. Its one-point compactification is $E^+ \cong S^8$. The proof of theorem 4.3 identifies the same rank-loss affine space with $E \oplus 0$, so the two S^8 descriptions agree.

Equations (4)–(5) show that the upper-left block of $X_P^{-1} H X_P^{-*}$ is $PHP^* = G$. Congruence by the invertible X_P reduces positivity to positivity of the displayed block matrix. The exact generalized Schur-complement criterion is

$$G \succeq 0, \qquad a \in \operatorname{Ran} G, \qquad \tau := \lambda - a^* G^\dagger a \geq 0.$$

Indeed, multiplying on both sides by the invertible triangular matrices with off-diagonal entry $-G^\dagger a$ reduces it to $\operatorname{diag}(G, \tau)$; if a had a component on $\ker G$, testing the quadratic form on a large signed multiple of a kernel vector would violate positivity. The same congruence gives

$$\operatorname{rank} H = \operatorname{rank} G + \mathbf{1}_{\tau > 0}.$$

Thus H is singular precisely when $\operatorname{rank} G < 2$ or $\tau = 0$, proving (55). The three fibre types follow by using $\dim_{\mathbb{C}} \operatorname{Ran} G = 2, 1, 0$. Setting $a = 0$ and $\tau = 0$ gives $H = RGR^*$ and proves surjectivity and the section. Finally, for fixed $G \succ 0$ the parameter $a \in \mathbb{C}^2$ is unbounded, so the inverse image of the compact singleton $\{G\}$ is not compact; $\mathscr{O}$ is not proper. □

4.1 A global marked observation on the S^9 quotient

Although compression on the affine rank-loss S^8 is nonproper, the trace-one S^9 has a global compact observation with exactly computable fibres. Define

$$\mathscr{O}_9 : \mathcal{Y}_{3,1} \longrightarrow \mathcal{B}_P := \{G \succeq 0 : \operatorname{tr}(C^{-1}G) \le 1\}, \qquad \mathscr{O}_9(H, z) = PHP^*. \tag{57}$$

The set $\mathcal{B}_P$ is a compact convex body with nonempty interior in $\mathrm{Herm}_2(\mathbb{C})$, hence is homeomorphic to B^4 by radial projection from an interior point.

Theorem 4.8 (Every fibre of the S^9 marked observation)**.** *The map* (57) *is a continuous surjection. For $G \in \mathcal{B}_P$ set*

$$L(G) = \frac{1 - \operatorname{tr}(C^{-1}G)}{3}. \tag{58}$$

Its fibre is exactly the set of pairs

$$a \in \operatorname{Ran} G, \qquad a^*G^\dagger a \le L(G), \qquad z \in \mathbb{C}, \qquad |z|^2 = \det G\big(L(G) - a^*G^\dagger a\big), \tag{59}$$

where (H, z) is recovered uniquely from

$$H = X_P \begin{pmatrix} G & a \\ a^* & L(G) \end{pmatrix} X_P^*. \tag{60}$$

The fibre is S^5 when $G \succ 0$ and $L(G) > 0$; it is a point when $G \succ 0$ and $L(G) = 0$; it is a closed two-disc when $\operatorname{rank} G = 1$ *and $L(G) > 0$, degenerating to a point when $L(G) = 0$; and it is a point at $G = 0$. These are orbit-space fibres, not torus fibres.*

Proof. Since $\operatorname{tr} H = \operatorname{tr}(X_P^* X_P K)$ for $K = X_P^{-1} H X_P^{-*}$, (5) gives

$$1 = \operatorname{tr} H = \operatorname{tr}(C^{-1}G) + 3\lambda.$$

Thus $\lambda = L(G)$. The Schur-complement argument in proposition 4.7 gives exactly the first two conditions in (59). Since $\det X_P = 1$,

$$\det H = \det G\big(L(G) - a^*G^\dagger a\big)$$

when G is invertible, and the same identity follows polynomially, or from rank, when G is singular. The defining equation $|z|^2 = \det H$ of $\mathcal{Y}_{3,1}$ gives the last condition. Conversely these conditions construct a unique point by (60); taking $a = 0$ proves surjectivity for every $G \in \mathcal{B}_P$.

If $G \succ 0$, put $b = G^{-1/2}a$ and $w = z/\sqrt{\det G}$. Equation (59) becomes $\|b\|^2 + |w|^2 = L(G)$ in $\mathbb{C}^2 \oplus \mathbb{C}$. It is S^5 for $L(G) > 0$ and a point for $L(G) = 0$. If $\operatorname{rank} G = 1$, then $z = 0$ and the allowed a form the closed Hermitian norm disc in the one-dimensional complex space $\operatorname{Ran} G$. If $G = 0$, then $a = z = 0$ and $H = uu^*/3$ is the unique point. □

4.2 The exact period-to-S^9 and period-to-S^8 maps

For t in the punctured cusp and $G_6 \in \mathrm{Sym}_6^+(\mathbb{R})$, retain both arguments and put

$$G_4 = \Phi G_6 \Phi^{\mathsf{T}}, \qquad H_3(t; G_6) = \Omega_3(t) G_6 \Omega_3(t)^*, \qquad H_2(t; G_6) = \Omega_2(t) G_4 \Omega_2(t)^*. \tag{61}$$

Proposition 4.9 (Commuting marked Gram square)**.** *The matrices $G_4, H_3(t;G_6)$, and $H_2(t;G_6)$ are positive definite, and*

$$PH_3(t;G_6)P^* = H_2(t;G_6). \tag{62}$$

Equivalently, for every fixed t the square

$$\begin{array}{ccc} \mathrm{Sym}_6^+(\mathbb{R}) & \xrightarrow{G\mapsto \Omega_3(t)G\Omega_3(t)^*} & \mathrm{Herm}_3(\mathbb{C})_{\succ 0} \\ {\scriptstyle G\mapsto \Phi G\Phi^{\mathsf{T}}}\big\downarrow & & \big\downarrow{\scriptstyle H\mapsto PHP^*} \\ \mathrm{Sym}_4^+(\mathbb{R}) & \xrightarrow{G\mapsto \Omega_2(t)G\Omega_2(t)^*} & \mathrm{Herm}_2(\mathbb{C})_{\succ 0} \end{array} \tag{63}$$

commutes. Normalize without suppressing either argument:

$$h_3(t;G_6) = \frac{H_3(t;G_6)}{\operatorname{tr} H_3(t;G_6)}, \qquad z_3(t;G_6) = +\sqrt{\det h_3(t;G_6)}.$$

Then the explicit point

$$\jmath(t;G_6) = \big(h_3(t;G_6), z_3(t;G_6)\big) \in \mathcal{Y}_{3,1} \cong S^9, \qquad \mathscr{O}_9\big(\jmath(t;G_6)\big) = \frac{H_2(t;G_6)}{\operatorname{tr} H_3(t;G_6)}. \tag{64}$$

The positive square root is a specified phase-zero section; it does not erase the existence of the other determinant phases in S^9.

The spectral-floor map gives a second exact, generally noncompact-chart map

$$\begin{aligned} \sigma(t;G_6) &= H_3(t;G_6) - \lambda_{\min}\big(H_3(t;G_6)\big) I_3 \in \Sigma_3 \hookrightarrow \Sigma_3^+ \cong S^8, \\ \mathscr{O}\big(\sigma(t;G_6)\big) &= H_2(t;G_6) - \lambda_{\min}\big(H_3(t;G_6)\big) C. \end{aligned} \tag{65}$$

Proof. Surjectivity of Φ makes Φ^{T} injective; hence $G_4 = \Phi G_6 \Phi^{\mathsf{T}}$ is positive definite. The identity blocks in $\Omega_r(t)$ make $\Omega_r(t)^*$ injective, so both $H_r(t;G_6)$ are positive definite. Using $P\Omega_3(t) = \Omega_2(t)\Phi$ from (15),

$$PH_3(t;G_6)P^* = P\Omega_3(t)G_6\Omega_3(t)^*P^* = \Omega_2(t)\Phi G_6\Phi^{\mathsf{T}}\Omega_2(t)^* = H_2(t;G_6).$$

This proves the square and (64). Subtracting the least eigenvalue makes a positive matrix singular without altering its eigenspaces; applying P and $PI_3P^* = C$ proves (65). □

There is one useful cusp endpoint for a fixed, explicitly chosen metric. Take $G_6 = I_6$ in the real period frame, choose the standard logarithmic strip $0 \le x < 1$ for $s = x + iy$, and abbreviate $H_{3,I}(t) = H_3(t;I_6)$. The exact formula is

$$\begin{aligned} H_{3,I}(t) &= I_3 + Z_3(t)Z_3(t)^* \\ &= (1+|s|^2)I_3 + sC_3(t)^* + \overline{s}C_3(t) + C_3(t)C_3(t)^*. \end{aligned} \tag{66}$$

Put $h_{3,I}(t) = H_{3,I}(t)/\operatorname{tr} H_{3,I}(t)$, $z_{3,I}(t) = +\sqrt{\det h_{3,I}(t)}$, and $\jmath_I(t) = \jmath(t;I_6)$.

Corollary 4.10 (A proved cusp limit in the S^9 carrier)**.** *For* (66)*, boundedness of $C_3(t)$ gives*

$$h_{3,I}(t) \longrightarrow I_3/3, \qquad z_{3,I}(t) \longrightarrow 1/(3\sqrt{3}), \qquad \mathscr{O}_9\big(\jmath_I(t)\big) \longrightarrow C/3 \quad (t \to 0). \tag{67}$$

Thus the period metric has an exact continuous marked limit in the interior of the S^9 orbit carrier. This is a map of period–metric data, not a compactification of every point of the torus family and not a fourth singular filling.

Proof. Equation (66) and boundedness of C_3 give $y^{-2}H_{3,I} \to I_3$ as $y \to \infty$: indeed $0 \le x < 1$ gives $|s|^2/y^2 \to 1$, while the two terms linear in s are $O(y)$ and the remaining correction is $O(1)$. Taking traces and determinants yields $h_{3,I} \to I_3/3$ and the positive square-root limit for $z_{3,I}$. Compressing by P and using (64) gives the last limit. □

Theorem 4.11 (Which Gram carrier is strongest)**.** *Within the specified left-action invariant theory, the information order is strict:*

$$\mathcal{Y}_3 \xrightarrow{\textit{forget } \arg z} \mathcal{P}_3, \qquad \mathcal{Y}_{3,1} \xrightarrow{\textit{forget } \arg z} \mathcal{K}_3.$$

The maps have circle fibres on full-rank matrices and point fibres at rank loss. Hence:

(i) $S^9 = S^{17}_F/SU(3)$ *is the strongest* normalized single-sphere *carrier: it retains the complete Gram matrix, determinant phase, its integral winding, the exact circle degeneration* (50)*, and the global marked observation* (57)*.*

(ii) (S^{10}, S^8) *is the strongest* unnormalized compact pair*: it additionally retains the radial scale, and its* S^8 *is exactly the fixed rank-loss locus.*

(iii) *The standalone* $S^8 = \partial(\mathscr{V}^+/U(3))$ *is the strongest rank-loss sphere, but it has forgotten determinant phase and does not carry the global compact extension of the affine compression* (54)*.*

None of this ordering supplies a complex tangent structure, a T^6 *regular fibre, the base* $\mathbb{P}^1$*, the three special values, or their integral specialization maps.*

Proof. The orbit-separation theorem shows that (H, z) is the complete $SU(3)$ invariant and H the complete $U(3)$ invariant. For $H \succ 0$, changing the phase of z changes the $SU(3)$ orbit but not the $U(3)$ orbit; at rank loss $z = 0$, so no phase remains. This proves strict information loss and the fibre claim. Propositions 4.4 and 4.6, Theorem 4.8, and Equation (43) prove (i)–(iii). The absent structures cannot be inferred from the quotient map: its base has dimensions eight or nine, its generic fibre is S^1, its singular values form the full boundary S^7 or S^8, and its complement has fundamental group $\mathbb{Z}$. In contrast, since $f|_{X^\circ} : X^\circ \to B^\circ$ is a T^4-bundle and $\pi_2(B^\circ) = 0$, its homotopy exact sequence gives

$$1 \longrightarrow \mathbb{Z}^4 \longrightarrow \pi_1(X^\circ) \longrightarrow \pi_1(B^\circ) = F_2 \longrightarrow 1,$$

which surjects onto the nonabelian free group F_2. Thus even the regular complements cannot be homotopy equivalent. The tangent obstructions in theorems 2.1 and 2.2 finish the claimed scope. □

5 The 24-dimensional octonionic flag

We now construct the exceptional Jordan object instead of selecting it by dimension. Let

$$\mathfrak{h}_3(\mathbb{O}) = \left\{ \begin{pmatrix} x_1 & p & q \\ \overline{p} & x_2 & r \\ \overline{q} & \overline{r} & x_3 \end{pmatrix} : x_1, x_2, x_3 \in \mathbb{R},\ p, q, r \in \mathbb{O} \right\}, \qquad a \circ b = \frac{1}{2}(ab + ba).$$

Define

$$\mathbb{OP}^2 = \{a \in \mathfrak{h}_3(\mathbb{O}) : a^2 = a,\ \operatorname{tr} a = 1\}$$

and, without replacing the incidence equation by a mnemonic,

$$\mathrm{Fl}(\mathbb{O}) = \{(a,b) \in \mathbb{OP}^2 \times \mathbb{OP}^2 : \mathrm{Re}\,\mathrm{tr}(ab) = 0\}. \tag{68}$$

Let

$$F_4 = \mathrm{Aut}_{\mathbb{R}}(\mathfrak{h}_3(\mathbb{O}), \circ), \qquad d_1 = \mathrm{diag}(1,0,0), \qquad d_2 = \mathrm{diag}(0,1,0), \qquad x_{\mathbb{O},0} = (d_1, d_2).$$

Here F_4 denotes the compact real form. These definitions and all claims in the next theorem are recorded in the primary calculation of Mare and Willems [10, Definition 2.2, p. 488; Proposition 2.4, p. 489, and Proposition 2.7, p. 491, and the root-space discussion on pp. 491–492].

Theorem 5.1 (The exact triality tangent module)**.** *The diagonal F_4-action on* $\mathrm{Fl}(\mathbb{O})$ *is transitive, the stabilizer of $x_{\mathbb{O},0}$ is* $\mathrm{Spin}(8)$*, and hence*

$$\mathrm{Fl}(\mathbb{O}) \cong F_4/\,\mathrm{Spin}(8).$$

Fix pairwise distinct real numbers $\lambda_1, \lambda_2, \lambda_3$ with $\lambda_1 + \lambda_2 + \lambda_3 = 0$, and set $d_3 = I_3 - d_1 - d_2$ and $d_\lambda = \mathrm{diag}(\lambda_1, \lambda_2, \lambda_3)$. The exact comparison with the diagonal-orbit model is the equivariant diffeomorphism

$$\begin{aligned} \Xi_\lambda : \mathrm{Fl}(\mathbb{O}) &\longrightarrow F_4 d_\lambda \subset \mathfrak{h}_3^0(\mathbb{O}), \\ (a,b) &\longmapsto \lambda_1 a + \lambda_2 b + \lambda_3 (I_3 - a - b). \end{aligned} \tag{69}$$

For $z \in F_4 d_\lambda$, its inverse is $(p_1(z), p_2(z))$, where

$$p_i(z) = \frac{(z - \lambda_j I_3) \circ (z - \lambda_k I_3)}{(\lambda_i - \lambda_j)(\lambda_i - \lambda_k)}, \qquad \{i,j,k\} = \{1,2,3\}. \tag{70}$$

Define the three coordinate subspaces of $\mathfrak{h}_3^0(\mathbb{O})$ by

$$\begin{aligned} \mathfrak{h}_{\gamma_1} &= \left\{ \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & r \\ 0 & \overline{r} & 0 \end{pmatrix} : r \in \mathbb{O} \right\}, \\ \mathfrak{h}_{\gamma_2} &= \left\{ \begin{pmatrix} 0 & 0 & q \\ 0 & 0 & 0 \\ \overline{q} & 0 & 0 \end{pmatrix} : q \in \mathbb{O} \right\}, \\ \mathfrak{h}_{\gamma_3} &= \left\{ \begin{pmatrix} 0 & p & 0 \\ \overline{p} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} : p \in \mathbb{O} \right\}. \end{aligned} \tag{71}$$

As real $\mathrm{Spin}(8)$*-modules,*

$$T_{x_{\mathbb{O},0}}\,\mathrm{Fl}(\mathbb{O}) \xrightarrow[\cong]{d\Xi_\lambda} T_{d_\lambda}(F_4 d_\lambda) = \mathfrak{h}_{\gamma_1} \oplus \mathfrak{h}_{\gamma_2} \oplus \mathfrak{h}_{\gamma_3} \cong V_8 \oplus S_8^+ \oplus S_8^-. \tag{72}$$

In particular,

$$\dim_{\mathbb{R}} \mathrm{Fl}(\mathbb{O}) = 8 + 8 + 8 = 24.$$

Proof. Mare and Willems prove that the stabilizer of (d_1, d_2) consists exactly of the automorphisms fixing the diagonal Jordan subalgebra pointwise, which is $\mathrm{Spin}(8)$, and that the action is transitive. Their restricted-root decomposition of $\mathfrak{h}_3^0(\mathbb{O})$ has the three spaces in (71); the tangent space to the orbit is their direct sum. To verify the model comparison, write $(a,b) = (hd_1, hd_2)$ with $h \in F_4$.

Since $hI_3 = I_3$, $I_3 - a - b = hd_3$, so $\Xi_\lambda(a,b) = hd_\lambda$. The three orthogonal diagonal idempotents satisfy the polynomial identities $p_i(d_\lambda) = d_i$. Jordan automorphisms preserve these identities, whence $p_i(hd_\lambda) = hd_i$. This proves both inverse compositions in (69)– (70). All formulas are polynomial on the ambient real vector spaces with nonzero constant denominators, so the maps are smooth. Their differentials are inverse, which gives the displayed tangent identification at the correctly typed basepoints. Under the stabilizer, the three summands are respectively the vector representation and the two real half-spin representations [10, pp. 491–492, especially the discussion around Equation (2.10)]. Each is the displayed copy of the eight-dimensional real vector space $\mathbb{O}$, which proves the dimension calculation without suppressing any summand. □

For $k = 1, 2, 3$, let

$$\mathcal{E}_k = F_4 \times_{\mathrm{Spin}(8)} \mathfrak{h}_{\gamma_k}.$$

These are rank-eight real subbundles, and

$$T\,\mathrm{Fl}(\mathbb{O}) = \mathcal{E}_1 \oplus \mathcal{E}_2 \oplus \mathcal{E}_3. \tag{73}$$

They are associated bundles; the notation does not assert that any $\mathcal{E}_k$ is trivial.

Theorem 5.2 (Integral ring, cells, and Euler class)**.** *There are orientations and a basis $\beta_1, \beta_2 \in H^8(\mathrm{Fl}(\mathbb{O}); \mathbb{Z})$ for which*

$$\begin{aligned} e_1 &:= e(\mathcal{E}_1) = 2\beta_1 - \beta_2, \\ e_2 &:= e(\mathcal{E}_2) = -\beta_1 + 2\beta_2, \\ e_3 &:= e(\mathcal{E}_3) = e_1 + e_2 = \beta_1 + \beta_2, \end{aligned} \tag{74}$$

and

$$H^*(\mathrm{Fl}(\mathbb{O}); \mathbb{Z}) \cong \frac{\mathbb{Z}[\beta_1, \beta_2]}{(\beta_1^2 - \beta_1\beta_2 + \beta_2^2,\ \beta_1\beta_2(\beta_1 - \beta_2))}, \qquad |\beta_1| = |\beta_2| = 8. \tag{75}$$

Writing

$$u_{\mathbb{O}} = \beta_1^2\beta_2 = \beta_1\beta_2^2 \in H^{24}(\mathrm{Fl}(\mathbb{O}); \mathbb{Z})$$

with the induced orientation, $u_{\mathbb{O}}$ is a primitive top generator and

$$e\big(T\,\mathrm{Fl}(\mathbb{O})\big) = e_1e_2e_3 = 6u_{\mathbb{O}}, \qquad \langle e(T\,\mathrm{Fl}(\mathbb{O})), [\mathrm{Fl}(\mathbb{O})]\rangle = 6. \tag{76}$$

The integral cohomology is free and has Poincaré polynomial

$$1 + 2t^8 + 2t^{16} + t^{24}. \tag{77}$$

More precisely, on $\mathfrak{d}^0 = \{\mathrm{diag}(x_1, x_2, x_3) : x_1 + x_2 + x_3 = 0\}$, put

$$\gamma_1(x) = x_3 - x_2, \qquad \gamma_2(x) = x_1 - x_3, \qquad \gamma_3(x) = x_1 - x_2 = \gamma_1(x) + \gamma_2(x).$$

In the symmetric-pair normalization the actual restricted roots are $\{\pm\gamma_1/2, \pm\gamma_2/2, \pm\gamma_3/2\}$, of type A_2. The doubled labels γ_i name the same root spaces and reflection hyperplanes; this factor of two is the one stated in the source's root-normalization footnote [10, footnote 3, p. 491]. There is a cell decomposition

$$\mathrm{Fl}(\mathbb{O}) = \bigsqcup_{\pi \in S_3} C_\pi, \qquad C_\pi \cong \mathbb{R}^{8\ell(\pi)}. \tag{78}$$

For the stabilizer action,

$$\mathrm{Fl}(\mathbb{O})^{\mathrm{Spin}(8)} = \{\pi x_{\mathbb{O},0} : \pi \in S_3\}. \tag{79}$$

Thus both the cell calculation and the Euler-class calculation give $\chi(\mathrm{Fl}(\mathbb{O})) = 6$.

Proof. Mare and Willems' integral theorem gives a free ring generated by β_1, β_2, with relations

$$S_i(\beta_1, \beta_2 - \beta_1, -\beta_2) = 0, \qquad i = 2, 3,$$

and gives (74) [10, Theorem 1.1, p. 484, and Equation (3.2), p. 496]. Expanding without changing the generators,

$$S_2 = -\beta_1^2 + \beta_1\beta_2 - \beta_2^2, \qquad S_3 = \beta_1\beta_2(\beta_1 - \beta_2),$$

which is exactly (75). The second relation gives $\beta_1^2\beta_2 = \beta_1\beta_2^2$. Multiplying the first relation by β_1 and by β_2 then gives $\beta_1^3 = \beta_2^3 = 0$. Consequently

$$(2\beta_1 - \beta_2)(-\beta_1 + 2\beta_2)(\beta_1 + \beta_2) = 3\beta_1^2\beta_2 + 3\beta_1\beta_2^2 = 6u_{\mathbb{O}}.$$

The primary calculation states that one sixth of this Euler product is a basis of the top integral cohomology [10, Equation (3.3) and the following paragraph, p. 497]; hence $u_{\mathbb{O}}$ is primitive and (76) follows.

The same source constructs the S_3-indexed cells, with one eight-dimensional real root space for each inversion of π, and identifies the fixed-point set [10, Equations (2.11)–(2.12) and Proposition 2.10, pp. 493–494]. Their real dimensions are

$$0, \quad 8, 8, \quad 16, 16, \quad 24.$$

All cells are even-dimensional. The cellular chain groups therefore have zero differentials, yielding (77) and a second calculation of the Euler characteristic as the number of cells, namely six. Equivalently, the cellwise complex coordinates have dimensions $0, 4, 4, 8, 8, 12$, since $C_\pi \cong \mathbb{C}^{4\ell(\pi)}$ as real cells. This cell notation supplies no global complex tangent structure. □

Proposition 5.3 (The exact relation between the two Weyl data)**.** *The ordinary Coxeter number of the ambient root system F_4 is*

$$h(F_4) = 12.$$

The restricted root system that governs (78)*, however, is A_2; its Weyl group is S_3 and its Coxeter number is* 3*. For a common maximal torus of* Spin(8) *and F_4, there is a split exact sequence*

$$1 \longrightarrow W(D_4) \longrightarrow W(F_4) \longrightarrow S_3 \longrightarrow 1, \qquad [W(F_4) : W(D_4)] = 1152/192 = 6.$$

The quotient permutes the three triality representations. It gives an exact Weyl-group relation behind the number six; the scalar $h(F_4) = 12$ alone does not determine the cells or Euler characteristic.

Proof. For a finite root system the invariant degrees satisfy $d_i = e_i + 1$, and the largest degree is the Coxeter number. The authoritative F_4 table gives

$$(d_1, d_2, d_3, d_4) = (2, 6, 8, 12), \qquad (e_1, e_2, e_3, e_4) = (1, 5, 7, 11),$$

so $h(F_4) = 12$ [9, Section 10.1, p. 213, and the table on p. 216]. By contrast, the roots $\gamma_1, \gamma_2, \gamma_1 + \gamma_2$ in the proof of theorem 5.2 form A_2. Its two simple reflections act as the adjacent transpositions of three coordinates, and their product is a three-cycle; hence its Coxeter element has order 3. Mare and Willems explicitly identify this restricted Weyl group with S_3 [10, pp. 492–493].

For completeness the extension can be checked in root coordinates. Take an orthonormal basis $\varepsilon_1, \ldots, \varepsilon_4$. The long roots of F_4 are $\{\pm\varepsilon_i \pm \varepsilon_j : i < j\}$, the D_4 roots. Partition the short roots into

$$\mathcal{V} = \{\pm\varepsilon_i\}, \qquad \mathcal{S}_\pm = \left\{ \tfrac{1}{2} \sum_{i=1}^{4} \delta_i\varepsilon_i : \delta_i \in \{1, -1\},\ \prod_i \delta_i = \pm 1 \right\}.$$

These are exactly the three connected components of the graph joining orthogonal short roots: distinct components have inner products $\pm 1/2$, and within each component every non-antipodal distinct pair is orthogonal. Thus $W(F_4)$ permutes the three sets. An element in the kernel preserves $\mathcal{V}$, so is a signed permutation matrix; preserving both half-spin sets forces an even number of sign changes. These are exactly $W(D_4)$, of order $2^3 \cdot 4! = 192$. The reflections in ε_1 and $h = (\varepsilon_1+\varepsilon_2+\varepsilon_3+\varepsilon_4)/2$ induce two distinct transpositions: the first exchanges $\mathcal{S}_+, \mathcal{S}_-$, and the second exchanges $\mathcal{V}, \mathcal{S}_-$. Their product has order three, since $\langle \varepsilon_1, h\rangle = 1/2$. They generate a copy of S_3 mapping isomorphically onto the permutation quotient. This proves surjectivity and splitting, and gives $|W(F_4)| = 6 \cdot 192 = 1152$. The three displayed short-root sets are the weights of the three triality modules, so this is the asserted permutation of those modules. □

5.1 The exact six-chamber comparison

Let $\mathscr{K}_3^{\max}/\mathbb{Z}^3$ be the translation classes of the maximal Kuhn tetrahedra $\sigma_{m,\pi}$ in (18). Translation by $-m$ gives the exact identification

$$\mathscr{K}_3^{\max}/\mathbb{Z}^3 = \{[\sigma_{0,\pi}] : \pi \in S_3\}.$$

The zero-dimensional torus strata of W_3 are indexed by this set; call the stratum corresponding to $[\sigma_{0,\pi}]$ by z_π.

Theorem 5.4 (The exact marked map of finite indexing sets)**.** *After fixing the coordinate order (e_1, e_2, e_3) and the base flag $x_{\mathbb{O},0}$, the formula*

$$\Theta : W_3^{(0)} \longrightarrow \mathrm{Fl}(\mathbb{O})^{\mathrm{Spin}(8)}, \qquad \Theta(z_\pi) = \pi x_{\mathbb{O},0} \tag{80}$$

is an S_3-equivariant bijection. Here $W_3^{(0)} = \{z_\pi : \pi \in S_3\}$ is a finite set of torus zero-strata, and S_3 acts on the source by permuting coordinate directions and on the target as the restricted Weyl group. It identifies the six summands contributing to the two Euler characteristics, but it does not extend to an asserted identification of the spaces.

Proof. Every translation class of a maximal Kuhn tetrahedron has one and only one representative $\sigma_{0,\pi}$, so the source of Θ has exactly the six displayed elements. Equation (79) gives exactly the six displayed target elements. Thus Θ is bijective. For $\tau \in S_3$, permutation of coordinate directions gives

$$\tau \cdot [\sigma_{0,\pi}] = [\sigma_{0,\tau\pi}],$$

while the restricted Weyl action gives $\tau \cdot (\pi x_{\mathbb{O},0}) = (\tau\pi) x_{\mathbb{O},0}$. Hence Θ is equivariant.

For W_3, every positive-dimensional algebraic-torus stratum has Euler characteristic zero, so its six zero-strata produce $\chi(W_3) = 6$. For the flag, the six cells in (78) are all even-dimensional, so each contributes $+1$. The bijection matches the index sets; the two stratifications and their positive-dimensional pieces are not identified by Θ. More formally, let $I_{\mathrm{cell}} = \{\mathrm{label}(C_\pi) : \pi \in S_3\}$ be a set of six labels and put $\tau\,\mathrm{label}(C_\pi) = \mathrm{label}(C_{\tau\pi})$. Then $z_\pi \mapsto \pi x_{\mathbb{O},0} \mapsto \mathrm{label}(C_\pi)$ is a chain of equivariant bijections of finite sets. The last action is defined on labels only: cells of different dimensions occur, so no action carrying each cell homeomorphically to the cell with its translated label is asserted. □

Proposition 5.5 (A genuine embedded lower flag, and the degree boundary)**.** *Choose $\iota \in \mathbb{O}$ with $\iota^2 = -1$ and put $\mathbb{C}_\iota = \mathrm{span}_{\mathbb{R}}\{1, \iota\}$. Entrywise inclusion in $\mathfrak{h}_3(\mathbb{O})$ gives a topological embedding*

$$\jmath_\iota : \mathrm{Fl}_3(\mathbb{C}_\iota) \hookrightarrow \mathrm{Fl}(\mathbb{O}) \tag{81}$$

which sends the six coordinate flags to the six points in (79).

There is also an ungraded-ring isomorphism

$$\rho : H^*(\mathrm{Fl}_3(\mathbb{C});\mathbb{Z}) \xrightarrow{\cong} H^*(\mathrm{Fl}(\mathbb{O});\mathbb{Z}), \qquad x_i \longmapsto \beta_i, \quad i = 1, 2, \tag{82}$$

where the source has the presentation in (75) *with* $|x_i| = 2$*, while* $|\beta_i| = 8$*. No continuous map in either direction can induce this degree regrading.*

Proof. The subalgebra $\mathbb{C}_\iota$ is associative and isomorphic to $\mathbb{C}$. Rank-one Hermitian projectors in $\mathfrak{h}_3(\mathbb{C}_\iota)$ remain idempotent trace-one elements after entrywise inclusion, and orthogonality is the unchanged equation $\operatorname{Re}\operatorname{tr}(ab) = 0$. The resulting map is injective. Its domain is compact and its target Hausdorff, so it is a topological embedding. The diagonal projectors are unchanged, which proves the fixed-point assertion.

The standard complete-flag presentation is

$$H^*(\mathrm{Fl}_3(\mathbb{C});\mathbb{Z}) \cong \frac{\mathbb{Z}[x_1, x_2]}{(x_1^2 - x_1x_2 + x_2^2,\ x_1x_2(x_1 - x_2))}, \qquad |x_i| = 2;$$

Mare and Willems derive the octonionic presentation precisely by comparison with this A_2 flag ring [10, Theorem 1.1, p. 484, and the proof on pp. 496–497]. Thus ρ is an isomorphism after forgetting degrees. If $f : \mathrm{Fl}(\mathbb{O}) \to \mathrm{Fl}_3(\mathbb{C})$ were to induce ρ, then $f^*(x_i) \in H^2(\mathrm{Fl}(\mathbb{O});\mathbb{Z}) = 0$. If a map in the other direction were to induce ρ^{-1}, then the image of each β_i would lie in $H^8(\mathrm{Fl}_3(\mathbb{C});\mathbb{Z}) = 0$. Either conclusion contradicts the nonzero generators in (82). □

5.2 The octonionic determinant-boundary sphere

For $\lambda \in \mathbb{R}$ and $q_1, q_2, q_3 \in \mathbb{O}$, retain the ordered coordinates

$$Q_{\mathbb{O}}(\lambda; q_1, q_2, q_3) = \begin{pmatrix} \lambda & q_3 & \overline{q}_2 \\ \overline{q}_3 & \lambda & q_1 \\ q_2 & \overline{q}_1 & \lambda \end{pmatrix} \in \mathfrak{h}_3(\mathbb{O}). \tag{83}$$

The equal-diagonal slice and its positive cone are

$$\mathcal{T}(\mathbb{O}) = \{Q_{\mathbb{O}}(\lambda; q_1, q_2, q_3)\} \cong \mathbb{R} \oplus \mathbb{O}^3, \qquad \mathcal{C}(\mathbb{O}) = \mathcal{T}(\mathbb{O}) \cap \mathfrak{h}_3(\mathbb{O})_{\succeq 0}. \tag{84}$$

Thus $\dim_{\mathbb{R}} \mathcal{T}(\mathbb{O}) = 1 + 3 \cdot 8 = 25$. The Jordan determinant on this slice is the exact cubic

$$\det_{\mathbb{O}} Q_{\mathbb{O}} = \lambda^3 - \lambda(|q_1|^2 + |q_2|^2 + |q_3|^2) + 2\operatorname{Re}((q_1q_2)q_3). \tag{85}$$

The real part is unchanged if $q_1(q_2q_3)$ is used, because the octonionic associator has zero real part.

Theorem 5.6 (The exact octonionic S^{24} determinant carrier)**.** *The relative boundary of* $\mathcal{C}(\mathbb{O})$ *inside* $\mathcal{T}(\mathbb{O})$ *is*

$$\partial\mathcal{C}(\mathbb{O}) = \{Q \in \mathcal{C}(\mathbb{O}) : \det_{\mathbb{O}} Q = 0\}.$$

There are homeomorphisms of pairs

$$(\mathcal{C}(\mathbb{O})^+, (\partial\mathcal{C}(\mathbb{O}))^+) \cong (B^{25}, S^{24}). \tag{86}$$

In particular

$$S^{24}_{\det,\mathbb{O}} := (\partial\mathcal{C}(\mathbb{O}))^+ \cong S^{24}$$

is an exact determinant-boundary sphere. It is not the 24*-dimensional flag* $F_4/\operatorname{Spin}(8)$.

Proof. The nonnegative cone of the Euclidean Jordan algebra $\mathfrak{h}_3(\mathbb{O})$ is closed and convex. Its intersection

$$K_{\mathbb{O}} = \{(q_1, q_2, q_3) \in \mathbb{O}^3 : Q_{\mathbb{O}}(1; q_1, q_2, q_3) \succeq 0\}$$

with the affine hyperplane $\lambda = 1$ is therefore closed and convex. Every positive semidefinite 2×2 principal corner gives $|q_i| \leq 1$, so $K_{\mathbb{O}}$ is bounded and hence compact. The origin of $\mathbb{O}^3$ corresponds to I_3, which is positive definite, so it is an interior point of $K_{\mathbb{O}}$.

For $v \in S(\mathbb{O}^3)$, let

$$R_{\mathbb{O}}(v) = \max\{r \geq 0 : rv \in K_{\mathbb{O}}\}.$$

Convexity, compactness, and the interior-point property make $R_{\mathbb{O}} : S(\mathbb{O}^3) \to \mathbb{R}_{>0}$ continuous. Indeed, choose $0 < \epsilon < M$ with $\epsilon B^{24} \subset K_{\mathbb{O}} \subset MB^{24}$. The Minkowski functional

$$p_K(x) = \inf\{t > 0 : x \in tK_{\mathbb{O}}\}$$

is positive for $x \neq 0$, positively homogeneous, and subadditive: if $x \in aK_{\mathbb{O}}$ and $y \in bK_{\mathbb{O}}$, convexity gives $x + y \in (a + b)K_{\mathbb{O}}$. Taking infima proves subadditivity. As $p_K(w) \leq \|w\|/\epsilon$, subadditivity in both directions gives $|p_K(x) - p_K(y)| \leq \|x - y\|/\epsilon$. Hence $R_{\mathbb{O}}(v) = 1/p_K(v)$ is continuous. Radial rescaling

$$0 \longmapsto 0, \qquad rv \longmapsto \frac{r}{R_{\mathbb{O}}(v)} v$$

is consequently a homeomorphism $K_{\mathbb{O}} \cong B^{24}$ carrying $\partial K_{\mathbb{O}}$ onto S^{23}. Denote it by F_K; its inverse sends sv, $0 < s \leq 1$, to $sR_{\mathbb{O}}(v)v$, and sends zero to zero. Both maps are continuous at zero because the radial function is bounded above and below by positive constants.

A nonzero element of $\mathcal{C}(\mathbb{O})$ has $\lambda > 0$: if $\lambda = 0$, the same principal-corner inequalities force $q_1 = q_2 = q_3 = 0$. Positive scaling therefore reconstructs the whole cone as

$$\mathcal{C}(\mathbb{O}) = \{0\} \cup \{\lambda Q_{\mathbb{O}}(1; q_1, q_2, q_3) : \lambda > 0,\ (q_1, q_2, q_3) \in K_{\mathbb{O}}\}.$$

Here are explicit maps through the Lorentz cone, with $q = (q_1, q_2, q_3)$:

$$\begin{aligned}
\mathcal{C}(\mathbb{O}) &\longrightarrow L_{25} := \{(s, x) \in \mathbb{R} \oplus \mathbb{O}^3 : s \geq \|x\|\},\\
Q_{\mathbb{O}}(\lambda; q) &\longmapsto (\lambda, \lambda F_K(q/\lambda)) \quad (\lambda > 0),\\
0 &\longmapsto (0, 0),\\
L_{25} &\longrightarrow [0, \infty) \times \mathbb{O}^3,\\
(s, x) &\longmapsto (s - \|x\|, x).
\end{aligned}$$

The inverses are $(s, x) \mapsto Q_{\mathbb{O}}(s; sF_K^{-1}(x/s))$ for $s > 0$, with zero sent to zero, and $(t, x) \mapsto (t + \|x\|, x)$, respectively. The boundedness of F_K and F_K^{-1} proves continuity at the cone vertices. The boundary maps to $s = \|x\|$, then to $t = 0$, proving an explicit pair homeomorphism with a closed half-space and its boundary.

To identify this boundary with determinant zero in the relative topology, use the spectral decomposition in the Euclidean Jordan algebra. A positive definite element has three positive eigenvalues and is an interior point even in the full algebra, hence in the slice. Conversely, if a positive semidefinite Q has a zero eigenvalue, then $Q - \delta I_3 \in \mathcal{T}(\mathbb{O})$, for every $\delta > 0$, has a negative eigenvalue. Such points lie outside $\mathcal{C}(\mathbb{O})$ and converge to Q; thus Q is not a relative interior point. Since the cone is closed, these are exactly its relative boundary points. The determinant is the product of the three nonnegative eigenvalues, so its zero set in the cone is exactly that boundary.

Fixing an orthonormal identification $\mathbb{O}^3 \cong \mathbb{R}^{24}$, the map

$$(t,x) \longmapsto \left(\frac{2x}{D}, \frac{t^2 + \|x\|^2 - 1}{D} \right) \in B^{25}, \qquad D = 1 + t^2 + \|x\|^2,$$

extends at infinity to $(0,1) \in \partial B^{25}$. It is inverse stereographic projection to the closed hemisphere in S^{25} with first coordinate nonnegative, followed by deletion of that first coordinate. The hemisphere projection is bijective, with inverse adjoining $\sqrt{1 - \|y\|^2}$; hence the displayed map is a homeomorphism after one-point compactification. At $t = 0$ it is

$$x \longmapsto \left(\frac{2x}{1 + \|x\|^2}, \frac{\|x\|^2 - 1}{1 + \|x\|^2} \right) \in S^{24},$$

with infinity sent to $(0,1)$. This proves (86), including its specified boundary. □

5.3 The ordered Albert triality tensor and its global descent

5.3.1 Original coordinates and octonion identities

Let $\mathbb{O}$ be the real octonion division algebra, with its original conjugation, norm $N(a) = a\bar{a} = \bar{a}a$, and real inner product $h(a,b) = \mathrm{Re}(a\bar{b})$. Put $J = \mathfrak{h}_3(\mathbb{O})$, with its Albert Jordan product $X \circ Y = (XY + YX)/2$. Matrix multiplication here involves only products of two octonions in each entry; no product of three octonionic matrices is left unparenthesized. Set $e_i = \mathrm{diag}(\delta_{1i}, \delta_{2i}, \delta_{3i})$, $I = e_1 + e_2 + e_3$, and

$$E_1(a) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & a \\ 0 & \bar{a} & 0 \end{pmatrix}, \quad E_2(b) = \begin{pmatrix} 0 & 0 & \bar{b} \\ 0 & 0 & 0 \\ b & 0 & 0 \end{pmatrix}, \quad E_3(c) = \begin{pmatrix} 0 & c & 0 \\ \bar{c} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The specified equal-diagonal object remains exactly

$$Q_{\mathbb{O}}(\lambda; q_1, q_2, q_3) = \lambda I + E_1(q_1) + E_2(q_2) + E_3(q_3) = \begin{pmatrix} \lambda & q_3 & \bar{q}_2 \\ \bar{q}_3 & \lambda & q_1 \\ q_2 & \bar{q}_1 & \lambda \end{pmatrix}. \tag{87}$$

Let V_i be three labelled copies of $\mathbb{O}$ and $J_i = E_i(V_i)$. Labelling does not identify their group actions. The original trace metric on J is

$$\begin{aligned} G(X,Y) &= \mathrm{tr}(X \circ Y), \\ G\left(\sum_i \xi_i e_i + \sum_i E_i(a_i), \sum_i \eta_i e_i + \sum_i E_i(b_i) \right) &= \sum_i \xi_i \eta_i + 2 \sum_i h(a_i, b_i). \end{aligned} \tag{88}$$

In particular $E_i^* G = 2h$, not h.

We use alternativity, $N(ab) = N(a)N(b)$, and conjugation reversing products. The parenthesis changes used below follow as follows. Because $\bar{a} = 2\,\mathrm{Re}(a) - a$, alternativity gives

$$\bar{a}(ab) = N(a)b, \quad a(\bar{a}b) = N(a)b, \quad (ba)\bar{a} = N(a)b, \quad (b\bar{a})a = N(a)b. \tag{89}$$

Polarizing norm composition gives $h(ab, ac) = N(a)h(b,c)$ and $h(ba, ca) = N(a)h(b,c)$. For $a \neq 0$, these equations and (89) imply $L_a^* = L_{\bar{a}}$ and $R_a^* = R_{\bar{a}}$; for $a = 0$ the identities are immediate. Consequently

$$\begin{aligned} h(ab, c) &= h(b, \bar{a}c) = h(a, c\bar{b}), \\ \mathrm{Re}((ab)c) &= \mathrm{Re}(a(bc)) = \mathrm{Re}((bc)a). \end{aligned} \tag{90}$$

For the first real-part equality, both expressions equal $h(c, \bar{b}\bar{a})$ by the adjoint identities. For the second, use $\mathrm{Re}(xy) = h(x, \bar{y}) = h(y, \bar{x}) = \mathrm{Re}(yx)$. These are identities of real parts; they do not equate the octonions $(ab)c$ and $a(bc)$.

5.3.2 The complete Peirce product table

A cyclic triple means exactly $(i, j, k) = (1, 2, 3), (2, 3, 1), (3, 1, 2)$.

Proposition 5.7. *The decomposition $J = \bigoplus_i \mathbb{R}e_i \oplus \bigoplus_i J_i$ has the complete product table*

$$\begin{aligned} e_i \circ e_j &= \delta_{ij} e_i, \\ e_i \circ E_i(a) &= 0, & e_j \circ E_i(a) &= e_k \circ E_i(a) = \tfrac{1}{2} E_i(a), \\ E_i(a) \circ E_i(b) &= h(a, b)(e_j + e_k), \\ E_i(a) \circ E_j(b) &= \tfrac{1}{2} E_k(\overline{ab}). \end{aligned} \tag{91}$$

Reversing the Jordan arguments uses commutativity of $\circ$, without reversing the octonion order in the cyclic formula.

Proof. The first three formulas follow entry by entry. For example the two nonzero diagonal entries of $E_1(a) \circ E_1(b)$ are $(a\bar{b} + b\bar{a})/2$ and $(\bar{a}b + \bar{b}a)/2$, both $h(a, b)$. The mixed products have precisely the following nonzero entries:

product	first entry	second entry
$E_1(a) \circ E_2(b)$	$(1, 2) = \bar{b}\bar{a}/2$	$(2, 1) = ab/2$
$E_2(b) \circ E_3(c)$	$(2, 3) = \bar{c}\bar{b}/2$	$(3, 2) = bc/2$
$E_3(c) \circ E_1(a)$	$(3, 1) = \bar{a}\bar{c}/2$	$(1, 3) = ca/2$

These are respectively $E_3(\overline{ab})/2$, $E_1(\overline{bc})/2$, and $E_2(\overline{ca})/2$, by the original definitions of E_1, E_2, E_3. □

5.3.3 Three typed products, adjoints, and inverses

Define the trilinear form and all three bilinear maps by

$$\begin{aligned} \tau : V_1 \times V_2 \times V_3 &\longrightarrow \mathbb{R}, & \tau(a, b, c) &= \mathrm{Re}((ab)c), \\ \mu_{12} : V_1 \times V_2 &\longrightarrow V_3, & \mu_{12}(a, b) &= \overline{a\bar{b}} = \bar{b}\bar{a}, \\ \mu_{23} : V_2 \times V_3 &\longrightarrow V_1, & \mu_{23}(b, c) &= \overline{b\bar{c}} = \bar{c}\bar{b}, \\ \mu_{31} : V_3 \times V_1 &\longrightarrow V_2, & \mu_{31}(c, a) &= \overline{c\bar{a}} = \bar{a}\bar{c}. \end{aligned} \tag{92}$$

Every V_i carries h, with its label retained. Equation (90) and conjugation give

$$\tau(a, b, c) = h(\mu_{12}(a, b), c) = h(\mu_{23}(b, c), a) = h(\mu_{31}(c, a), b). \tag{93}$$

For example $h(\overline{a\bar{b}}, c) = \mathrm{Re}(\overline{a\bar{b}}\bar{c}) = \mathrm{Re}(\overline{c(ab)}) = \mathrm{Re}((ab)c)$. Thus these are exactly the three Riesz-curried maps for h. In the original matrix metric,

$$\tau(a, b, c) = G(E_1(a) \circ E_2(b), E_3(c)) = \mathrm{tr}((E_1(a) \circ E_2(b)) \circ E_3(c)). \tag{94}$$

Proposition 5.8. *For cyclic (i,j,k), $a \in V_i$, and $b \in V_j$,*

$$N(\mu_{ij}(a,b)) = N(a)N(b), \tag{95}$$
$$\mu_{ki}(\mu_{ij}(a,b),a) = N(a)b, \tag{96}$$
$$\mu_{jk}(b,\mu_{ij}(a,b)) = N(b)a. \tag{97}$$

For fixed $a \neq 0$, the real-linear isomorphism $M_a : V_j \to V_k$, $M_a(b) = \mu_{ij}(a,b)$, has

$$M_a^*(c) = \mu_{ki}(c,a), \quad M_a^* M_a = N(a)\,\mathrm{id}_{V_j}, \quad M_a M_a^* = N(a)\,\mathrm{id}_{V_k}, \quad M_a^{-1}(c) = \frac{\mu_{ki}(c,a)}{N(a)}.$$

For fixed $b \neq 0$, the inverse of $a \mapsto \mu_{ij}(a,b)$ is $c \mapsto \mu_{jk}(b,c)/N(b)$, and its numerator is also the adjoint.

Proof. Norm composition proves the first formula. The inverse numerators are, with their parentheses preserved,

$$\overline{\overline{ab}\,a} = \bar{a}(ab) = N(a)b, \qquad \overline{b\,\overline{ab}} = (ab)\bar{b} = N(b)a.$$

Their final equalities are (89). Relabelling by the three stated cyclic triples proves all six inverse identities. The adjoint statements are (93). Both adjoint-composition identities follow from the two inverse identities, including with the other argument nonzero. Division by $N(a)$ or $N(b)$ gives the inverses. No unit-length hypothesis has been imposed. □

Retaining G gives the following different numerical identities. For $X = E_i(a) \in J_i, Y = E_j(b) \in J_j$ with $i \neq j$, put $B_{ij}(X,Y) = 2X \circ Y = E_k(\mu_{ij}(a,b))$, using cyclic argument order. Then

$$G(B_{ij}(X,Y), B_{ij}(X,Y)) = \tfrac{1}{2}G(X,X)G(Y,Y), \qquad G(X \circ Y, X \circ Y) = \tfrac{1}{8}G(X,X)G(Y,Y). \tag{98}$$

Indeed the left sides are $2N(a)N(b)$ and $N(a)N(b)/2$, whereas $G(X,X)G(Y,Y) = 4N(a)N(b)$. The coefficient-one composition identity uses the explicitly stated h, not the trace metric G.

5.3.4 The determinant and the factor six in polarization

Proposition 5.9. *For arbitrary real ξ_i and the original octonionic coordinates q_i, let $X = \sum_i \xi_i e_i + \sum_i E_i(q_i)$. Its Jordan determinant is*

$$d(X) = \xi_1\xi_2\xi_3 - \sum_{i=1}^{3} \xi_i N(q_i) + 2\,\mathrm{Re}((q_1q_2)q_3). \tag{99}$$

In particular,

$$d(Q_{\mathbb{O}}(\lambda; q_1, q_2, q_3)) = \lambda^3 - \lambda\big(N(q_1) + N(q_2) + N(q_3)\big) + 2\,\mathrm{Re}((q_1q_2)q_3).$$

Proof. The Albert cubic norm is given by the Jordan trace formula

$$d(X) = \tfrac{1}{3}\,\mathrm{tr}(X \circ (X \circ X)) - \tfrac{1}{2}\,\mathrm{tr}(X)\,\mathrm{tr}(X \circ X) + \tfrac{1}{6}\,\mathrm{tr}(X)^3.$$

The product table computes the diagonal entries and three coordinates of X^2:

$$(X^2)_{ii} = \xi_i^2 + N(q_j) + N(q_k), \qquad (X^2)_i = (\xi_j + \xi_k)q_i + \overline{q_j q_k}, \quad (i,j,k) \text{ cyclic.}$$

With $s = \xi_1 + \xi_2 + \xi_3$, it follows that

$$\begin{aligned} \operatorname{tr}(X) &= s, \\ \operatorname{tr}(X^2) &= \sum_i \xi_i^2 + 2\sum_i N(q_i), \\ \operatorname{tr}(X \circ X^2) &= \sum_i \xi_i^3 + 3\sum_i (\xi_j + \xi_k) N(q_i) + 6\,\mathrm{Re}((q_1 q_2) q_3). \end{aligned}$$

Each scalar product of q_i with $\overline{q_j q_k}$ equals $\tau(q_1, q_2, q_3)$ by (93); the trace metric supplies the factor 2. Substitution into the cubic gives coefficient $(\xi_j + \xi_k) - s = -\xi_i$ for $N(q_i)$ and coefficient $6/3 = 2$ for τ. Direct expansion gives the scalar part

$$\tfrac{1}{3}\sum_i \xi_i^3 - \tfrac{1}{2} s \sum_i \xi_i^2 + \tfrac{1}{6} s^3 = \xi_1 \xi_2 \xi_3.$$

This proves (99). Setting all three $\xi_i = \lambda$ retains exactly the original equal-diagonal slice. □

Define the symmetric polarization $\mathcal{D}$ by $\mathcal{D}(X, X, X) = d(X)$, explicitly

$$\begin{aligned} 6\mathcal{D}(X, Y, Z) = {} & d(X + Y + Z) - d(X + Y) - d(X + Z) - d(Y + Z) \\ & + d(X) + d(Y) + d(Z). \end{aligned} \tag{100}$$

Expansion of homogeneous cubic monomials proves symmetry and trilinearity: inclusion–exclusion removes every term missing an argument, and degree three forces each remaining argument to occur once. On the diagonal the multiplier is $27 - 3 \cdot 8 + 3 = 6$. The same expansion proves

$$\frac{\partial^3}{\partial r \partial s \partial t} d(rX + sY + tZ)\bigg|_{r=s=t=0} = 6\mathcal{D}(X, Y, Z).$$

For the three distinct sectors one obtains

$$\begin{aligned} \partial_r \partial_s \partial_t d(rE_1(a) + sE_2(b) + tE_3(c))|_0 &= 2\tau(a, b, c), \\ \mathcal{D}(E_1(a), E_2(b), E_3(c)) &= \tfrac{1}{3}\tau(a, b, c). \end{aligned} \tag{101}$$

Indeed that determinant is $2rst\tau(a, b, c)$. Relative to the original G, the three Riesz representatives for fixed $E_1(a), E_2(b)$ are

scalar form on $J_1 \times J_2 \times J_3$	representative in J_3
τ	$E_3(\overline{ab})/2 = E_1(a) \circ E_2(b)$
$\partial_r \partial_s \partial_t d$	$E_3(\overline{ab}) = 2E_1(a) \circ E_2(b)$
$\mathcal{D}$	$E_3(\overline{ab})/6 = (E_1(a) \circ E_2(b))/3$

Thus the determinant coefficient 2 and polarization factor 6 are separate, fully specified operations.

5.3.5 The exact stabilizer and its three actions

Let $\kappa(a) = \bar{a}$ and define

$$\begin{aligned} H_\tau = \{ & (g_1, g_2, g_3) \in O(V_1, h) \times O(V_2, h) \times O(V_3, h) : \\ & \tau(g_1 a, g_2 b, g_3 c) = \tau(a, b, c) \text{ for every } a, b, c\}. \end{aligned} \tag{102}$$

By (93), orthogonality and nondegeneracy of h, this is equivalent to each of the equations

$$\begin{aligned} \mu_{12}(g_1a, g_2b) &= g_3\mu_{12}(a,b), \\ \mu_{23}(g_2b, g_3c) &= g_1\mu_{23}(b,c), \\ \mu_{31}(g_3c, g_1a) &= g_2\mu_{31}(c,a). \end{aligned} \tag{103}$$

With every conjugation retained, the first equation reads

$$(g_1a)(g_2b) = \kappa g_3\kappa(ab). \tag{104}$$

Consequently $(g_1, g_2, g_3) \mapsto (g_1, g_2, \kappa g_3 \kappa)$ is a group isomorphism onto the ordinary product-related triples $\mathcal{R} = \{(A,B,C) \in O(8)^3 : (Ax)(By) = C(xy) \text{ for all } x, y\}$.

Theorem 5.10. *Every element of H_τ has $g_i \in SO(8)$, and H_τ is isomorphic to the compact simply connected* Spin(8)*. The map*

$$\Phi : H_\tau \longrightarrow \{g \in \mathrm{Aut}(J,\circ) : g(e_i) = e_i \ (i = 1,2,3)\},$$

$$\Phi(g)\left(\sum_i \xi_i e_i + \sum_i E_i(a_i)\right) = \sum_i \xi_i e_i + \sum_i E_i(g_i a_i)$$

is an exact group isomorphism. Thus (103) *is* Spin(8) *equivariance for the actions on the original Albert coordinates.*

Proof. For unit $u, v \in \mathbb{O}$, reflection in $u^\perp$ is

$$D_u(x) = x - 2h(x,u)u = -u\bar{x}u.$$

To verify the second expression, multiply $x\bar{u} + u\bar{x} = 2h(x,u)$ on the right by u and use $(x\bar{u})u = x$. Define

$$A_{u,v}(x) = v(\bar{u}x), \quad B_{u,v}(y) = (y\bar{u})v, \quad C_{u,v}(z) = v(\bar{u}z\bar{u})v = D_vD_u(z).$$

Repeated outside letters are parenthesized by flexibility. The Moufang identity $(vx)(yv) = v(xy)v$, first for v and then for $\bar{u}$, proves

$$(v(\bar{u}x))((y\bar{u})v) = v(\bar{u}(xy)\bar{u})v.$$

This is a product-related triple. All its coordinates lie in $SO(8)$: multiplication by a unit octonion is orthogonal, and its determinant is continuously $+1$ on the connected unit sphere because it is $+1$ at 1. Every element of $SO(8)$ is a product of an even number of reflections. To obtain such a decomposition, reflect the image of the first vector of an orthonormal basis onto that vector, then repeat inside its orthogonal complement. After finitely many steps the map is the identity; the determinant records the parity of the reflections. Projection of $\mathcal{R} \cap SO(8)^3$ to C is therefore surjective.

Its kernel contains exactly two elements. If $(Ax)(By) = xy$, put $p = A1$, $q = B1$. Then $pq = 1$, $By = \bar{p}y$, and $Ax = xp$. The equation becomes $(xp)(\bar{p}y) = xy$. Replacing y by py gives $(xp)y = x(py)$ for every x, y. Hence p lies in the middle nucleus of $\mathbb{O}$, which is $\mathbb{R}$. The latter fact can be checked without an implicit associativity assumption. In the Cayley–Dickson presentation $\mathbb{O} = \mathbb{H} \oplus \mathbb{H}\ell$ with

$$(a + b\ell)(c + d\ell) = (ac - \bar{d}b) + (da + b\bar{c})\ell,$$

write $p = a + b\ell$ and use quaternion units $\mathsf{ij} = \mathsf{k}$. The ℓ coefficient of $(\mathsf{i}p)\mathsf{j} - \mathsf{i}(p\mathsf{j})$ is $-2b\mathsf{k}$, so $b = 0$. The ℓ coefficient of $(\ell a)x - \ell(ax)$ is $\bar{a}\bar{x} - \bar{x}\bar{a}$. Its vanishing for $x = \mathsf{i}, \mathsf{j}$ forces a real. Since $|p| = 1$, the kernel is $\{(I, I, I), (-I, -I, I)\}$.

There are no orientation-reversing related triples. First, the cyclic invariance of τ gives surjectivity onto $SO(8)$ in each of its three coordinates, and hence the same holds for the three projections of $\mathcal{R} \cap SO(8)^3$. If a product-related triple had $\det A = -1$, compose it with an orientation-preserving product-related triple whose first coordinate is κA^{-1}. The result has first coordinate κ, so satisfies $\bar{x}(By) = C(xy)$. Setting $x = 1$ yields $B = C$. Setting $y = 1$, and writing $p = B1$, yields $Bx = \bar{x}p$. Hence

$$\bar{x}(\bar{y}p) = \overline{xy}\, p \qquad \text{for all } x, y.$$

Set $y = p$ and multiply on the right by $\bar{p}$. The inverse property gives $\bar{x}\bar{p} = \bar{p}\bar{x}$ for all x. The center of $\mathbb{O}$ is $\mathbb{R}$: in the displayed Cayley–Dickson presentation, commuting $a + b\ell$ with i forces $b = 0$, and commuting a with i, j forces a real. Thus $p = \pm 1$. The preceding equation would imply $\bar{x}\bar{y} = \overline{xy} = \bar{y}\bar{x}$ for all x, y, contradicting $\mathsf{ij} = -\mathsf{ji}$. Thus $\det A = 1$. Cyclic permutation of H_τ excludes negative determinants in the other two coordinates too.

The group is connected. The reflection-pair triples are continuously parametrized by $S^7 \times S^7$, contain the identity at $u = v = 1$, and contain $(-I, -I, I)$ at $u = 1, v = -1$. Their third coordinates generate $SO(8)$, so every related triple is a product of these triples and a kernel element. Every such element therefore lies in the identity component. Projection to C is a two-sheeted covering: it is a surjective homomorphism of compact Lie groups with finite kernel, hence a local diffeomorphism with those two-element fibres. Here $\pi_1(SO(8)) = \mathbb{Z}/2$. One proof starts with the unit quaternion conjugation action on imaginary quaternions, which identifies $SO(3)$ with $S^3/\{\pm 1\}$, and uses the fibrations $SO(n-1) \to SO(n) \to S^{n-1}$ for $n \geq 4$; their bases have zero π_1 and π_2, so the fundamental group remains $\mathbb{Z}/2$. A connected two-sheeted cover of this group is simply connected. It is Spin(8), the simply connected cover of $SO(8)$.

The map Φ preserves every product in (91), by orthogonality and (103); bilinearity proves preservation of all Jordan products. Its inverse is $\Phi(g^{-1})$. Conversely, a Jordan automorphism fixing all e_i preserves

$$J_i = \{X \in J : e_i \circ X = 0, \quad e_j \circ X = e_k \circ X = X/2\}.$$

Its restrictions define linear maps g_i in the original E_i coordinates. Applying the automorphism to the same-sector Peirce product proves $h(g_i a, g_i b) = h(a, b)$; applying it to the mixed product proves (104). Thus the restrictions belong to H_τ. The direct-sum decomposition of J proves injectivity and surjectivity of Φ with no further kernel. □

Proposition 5.11. *Without the metrics, the stabilizer of the same typed tensor in $GL(V_1) \times GL(V_2) \times GL(V_3)$ is exactly*

$$\{(s_1 g_1, s_2 g_2, s_3 g_3) : s_i > 0, \quad s_1 s_2 s_3 = 1, \quad (g_1, g_2, g_3) \in H_\tau\}.$$

Thus the metric-preserving stabilizer is Spin(8)*; the unrestricted typed linear stabilizer has two additional independent positive scale parameters.*

Proof. Tensor invariance for (A, B, C) and the h adjoint imply $\mu_{12}(Aa, Bb) = C^{-T}\mu_{12}(a, b)$. For nonzero a, Proposition 5.8 gives $|\det M_a| = N(a)^4$ on dimension eight. Taking determinants yields $N(Aa)^4 = |\det B \det C|^{-1} N(a)^4$. Positivity therefore gives $A = s_1 g_1$, where $s_1 > 0$ and g_1 is orthogonal. The cyclic equations give $B = s_2 g_2$, $C = s_3 g_3$ likewise. For unit a, b, the two sides of the adjoint equation have norms $s_1 s_2$ and s_3^{-1}, so $s_1 s_2 s_3 = 1$. Dividing by those explicit scalars proves $(g_1, g_2, g_3) \in H_\tau$. The converse follows by trilinearity. □

The three actions can also be read directly as the vector and two real half-spin actions. On $S = V_2 \oplus V_3$ define

$$c(a)(b, c_0) = (-\mu_{31}(c_0, a), \mu_{12}(a, b)), \qquad a \in V_1. \tag{105}$$

The adjoint and inverse identities prove $c(a)^* = -c(a)$ and $c(a)^2 = -N(a)I$. Polarization gives

$$c(a)c(a') + c(a')c(a) = -2h(a, a')I.$$

This is the real Clifford action with negative norm convention. Even products preserve each of the two summands. Equation (103) gives the exact intertwining map

$$\operatorname{diag}(g_2, g_3)c(a) = c(g_1 a)\operatorname{diag}(g_2, g_3).$$

Conversely, an even product of unit Clifford operators covers the corresponding product of reflections on V_1, is orthogonal on S, and satisfies this equation. Its two diagonal blocks, together with its vector action, therefore form a triple in H_τ. This map from the spin double cover is onto: it covers $SO(V_1)$ and includes both elements over the identity, because the Clifford scalar -1 acts as $-I$ on both summands. The two groups are connected double covers and the map is an isomorphism. This identifies V_1 as the vector representation and V_2, V_3 as the two half-spin modules in the displayed Clifford realization, without conjugating any q_i. Their names S^+, S^- can be fixed by the parity summands; when the names are instead defined by an oriented volume element, that orientation must be specified as additional notation.

There is no nonzero Spin(8) equivariant linear map $V_i \to V_j$ for $i \neq j$. The four triples

$$(I, I, I), \quad (I, -I, -I), \quad (-I, I, -I), \quad (-I, -I, I)$$

belong to H_τ by the product equation. For any two different sectors, one of these acts as $+I$ on the source and $-I$ on the target. An equivariant linear map then equals its own negative and vanishes. Their exact positive relation is the three nonzero equivariant bilinear maps (92); fixing any nonzero argument gives the intersector isomorphisms and inverses already proved in Proposition 5.8.

5.3.6 An explicit tangent comparison and global bundle descent

Put $F_4 = \operatorname{Aut}(J, \circ)$ and use Theorem 5.10 to identify H_τ with the ordered frame stabilizer. Then $M = F_4/H_\tau$ parametrizes the orbit of the ordered frame (e_1, e_2, e_3).

Every Jordan automorphism preserves tr and G. Indeed the Peirce table gives $\operatorname{Tr}_{\operatorname{End}(J)} L_{e_i} = 1 + 16/2 = 9$ and $\operatorname{Tr}_{\operatorname{End}(J)} L_{E_i(a)} = 0$. The latter operator has zero diagonal blocks in the Peirce decomposition, as each product in its table changes summand. Thus $\operatorname{Tr} L_X = 9 \operatorname{tr} X$. Conjugating L_X by a Jordan automorphism gives L_{gX}, proving $\operatorname{tr}(gX) = \operatorname{tr} X$. Product preservation now gives $G(gX, gY) = G(X, Y)$.

Write $L_X(Y) = X \circ Y$. In any Jordan algebra $[L_X, L_Y]$ is a derivation, with the following algebraic proof. The coefficient of three distinct parameters in the Jordan identity gives

$$\begin{aligned} x \circ ((z \circ w) \circ y) + z \circ ((w \circ x) \circ y) + w \circ ((x \circ z) \circ y) = (z \circ w) \circ (y \circ x) + (w \circ x) \circ (y \circ z) \\ + (x \circ z) \circ (y \circ w). \end{aligned}$$

Subtracting this identity with x, y interchanged cancels the right sides and gives exactly

$$[L_x, L_y](z \circ w) = ([L_x, L_y]z) \circ w + z \circ ([L_x, L_y]w).$$

For cyclic (i, j, k) define

$$D_i(a) = 4[L_{E_i(a)}, L_{e_j}]. \tag{106}$$

Its complete values follow from the Peirce product table:

$$\begin{aligned} D_i(a)e_i &= 0, & D_i(a)e_j &= E_i(a), & D_i(a)e_k &= -E_i(a), \\ D_i(a)E_i(b) &= 2h(a,b)(e_k - e_j), & & & & \\ D_i(a)E_j(b) &= -E_k(\overline{ab}), & D_i(a)E_k(c) &= E_j(\overline{ca}). & & \end{aligned} \tag{107}$$

For example on e_j the commutator before multiplication by 4 is $E_i(a)/2 - E_i(a)/4 = E_i(a)/4$, and on $E_j(b)$ it is $0 - E_k(\overline{ab})/4$. The same product table supplies the other entries. Exponentiating a derivation gives Jordan automorphisms, since the derivation rule and the exponential power series give $e^{tD}(X \circ Y) = (e^{tD}X) \circ (e^{tD}Y)$.

Let $\mathfrak{f}_4$ be the derivation algebra and $\mathfrak{h}$ its subalgebra fixing each e_i. Put $\mathfrak{m}_i = \{D_i(a) : a \in V_i\}$ and $\mathfrak{m} = \bigoplus_i \mathfrak{m}_i$. Then

$$\mathfrak{f}_4 = \mathfrak{h} \oplus \mathfrak{m}, \qquad \mathfrak{m}_i \cong V_i, \qquad \mathrm{Ad}(h)D_i(a) = D_i(h_i a). \tag{108}$$

To check that no tangent directions are missing, differentiate $e_i^2 = e_i$ to get $De_i = 2e_i \circ De_i$. Thus De_i has only the components $J_j \oplus J_k$. Differentiating $I^2 = I$ gives $DI = 0$, so the two possible occurrences of any J_i component in the triple (De_1, De_2, De_3) are opposite. Equation (107) therefore supplies unique a_i such that $D - \sum_i D_i(a_i)$ fixes all three e_i. That equation also proves the sum direct. Conjugation of the commutator defining D_i by an automorphism fixing e_j proves the displayed equivariance.

Retain arbitrary distinct original spectral parameters $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$, with $\lambda_1 + \lambda_2 + \lambda_3 = 0$, and set

$$d_\lambda = \sum_i \lambda_i e_i, \qquad \Delta_1 = \lambda_2 - \lambda_3, \quad \Delta_2 = \lambda_3 - \lambda_1, \quad \Delta_3 = \lambda_1 - \lambda_2.$$

Every Δ_i is nonzero and

$$D_i(a)d_\lambda = \Delta_i E_i(a). \tag{109}$$

The equivariant map $\Xi_\lambda : M \to F_4 d_\lambda$, $[g] \mapsto g d_\lambda$, is a diffeomorphism. Its inverse recovers the ordered frame from the Jordan polynomials

$$p_i(z) = \frac{(z - \lambda_j I) \circ (z - \lambda_k I)}{(\lambda_i - \lambda_j)(\lambda_i - \lambda_k)}.$$

For $z = g d_\lambda$ these equal $g e_i$: evaluate on d_λ first, then apply the Jordan automorphism g. In particular the stabilizer of d_λ is exactly H_τ, and both maps are smooth. Equation (109) is the differential of this comparison with all scale factors retained.

Use the associated-bundle convention $(g, v) \sim (gh, h^{-1}v)$ for $F_4 \times_{H_\tau} V_i$, and denote this rank-eight real vector bundle by $\mathcal{V}_i$. Define

$$\begin{aligned} K_i &: \mathcal{V}_i \longrightarrow TM, & K_i[g,a] &= \left.\frac{d}{dt}\right|_0 g \exp(tD_i(a))H_\tau, \\ P_i &: \mathcal{V}_i \longrightarrow M \times J, & P_i[g,a] &= (gH_\tau, gE_i(a)). \end{aligned} \tag{110}$$

Both maps are well defined by the proved equivariance. Equation (108) shows that $\bigoplus_i K_i : \bigoplus_i \mathcal{V}_i \to TM$ is an isomorphism. The image of P_i above the frame $(p_1, p_2, p_3) = (ge_1, ge_2, ge_3)$ is precisely the intrinsic Peirce space

$$\{X \in J : p_i \circ X = 0, \quad p_j \circ X = p_k \circ X = X/2\}.$$

Suppressing the recorded base-point component of P_i when viewing it as a tangent vector to the orbit, the exact comparison is

$$d\Xi_\lambda \circ K_i = \Delta_i P_i. \tag{111}$$

Thus the original Albert slots and frame tangent slots are related by an explicit map, rather than an unrecorded change of coordinates.

The tensor and products descend by the formulas

$$\begin{aligned} \boldsymbol{\tau}([g,a],[g,b],[g,c]) &= \tau(a,b,c), \\ \boldsymbol{\mu}_{ij}([g,a],[g,b]) &= [g,\mu_{ij}(a,b)] \in \mathcal{V}_k. \end{aligned} \tag{112}$$

Representatives over one point can always be chosen with common g. Replacing it by gh replaces the arguments by $h_i^{-1}a, h_j^{-1}b, h_k^{-1}c$. Tensor invariance and (103) give exactly the same scalar and the same associated class. This proves well-defined global descent, beyond invariance at a single fibre. The associated metrics induced by h carry all norm, adjoint, and inverse formulas of Proposition 5.8 to identities of bundle maps.

Transport by K_i gives the three subbundles $\mathcal{T}_i = K_i(\mathcal{V}_i)$, the direct sum $TM = \mathcal{T}_1 \oplus \mathcal{T}_2 \oplus \mathcal{T}_3$, and global bilinear maps $\mathcal{T}_i \otimes \mathcal{T}_j \to \mathcal{T}_k$. Define a second, orbit-coordinate realization explicitly by $S_i = K_i/\Delta_i$. Then $d\Xi_\lambda S_i = P_i$. Transport of $\boldsymbol{\mu}$ by S_i is exactly the original coordinate map $E_i(a), E_j(b) \mapsto E_k(\overline{ab})$. These transports obey the following exact comparison. For $u_i \in J_i, u_j \in J_j$ at the reference orbit point, the K-transported product has orbit-coordinate value

$$\frac{\Delta_k}{\Delta_i \Delta_j}\, B_{ij}(u_i, u_j), \tag{113}$$

whereas the S-transported product has value $B_{ij}(u_i, u_j)$. Indeed the K-coordinate arguments are $E_i^{-1}u_i/\Delta_i$ and $E_j^{-1}u_j/\Delta_j$; bilinearity and the output multiplier Δ_k give the displayed coefficient. At every other point the same identity is transported by $g \in F_4$. The K-transported scalar in orbit coordinates is

$$\frac{\tau(E_1^{-1}u_1, E_2^{-1}u_2, E_3^{-1}u_3)}{\Delta_1\Delta_2\Delta_3},$$

and the S-transported scalar is its numerator. For the pullback $G_\lambda = \Xi_\lambda^* G$ of the original trace metric on the orbit, trace invariance and (88) give

$$G_\lambda(K_i[g,a], K_i[g,b]) = 2\Delta_i^2 h(a,b), \qquad G_\lambda(S_i[g,a], S_i[g,b]) = 2h(a,b).$$

No Δ_i is set to 1 and no coordinate is conjugated to absorb a sign or a metric factor.

There is a specified connection for which these descended tensors are parallel. On $F_4 \to F_4/H_\tau$ take

$$\omega = \mathrm{pr}_{\mathfrak{h}}(g^{-1}dg) \quad \text{for the decomposition (108).}$$

Because the projection commutes with $\mathrm{Ad}(H_\tau)$, this form is right equivariant and sends each fundamental vertical vector to its generating element of $\mathfrak{h}$. These are exactly the principal-connection conditions. Upstairs, the descended tensor and products are represented by the constant H_τ-invariant tensors τ, μ_{ij}. Their covariant derivatives are the ordinary derivatives of those constant coefficients plus the infinitesimal $\mathfrak{h}$ action. Both terms vanish. They are therefore parallel for this explicitly defined homogeneous connection and for the connections transported to TM by K or S. No assertion about Levi–Civita parallelism follows from this argument.

5.3.7 Source conventions and scope

The matrix convention, related-triple equation, and Jordan frame action agree entry for entry with Ichiro Yokota, *Exceptional Lie groups*, arXiv:0902.0431, Theorem 1.14.2, Lemmas 1.14.3–1.14.4, Theorem 1.16.2, and Theorem 2.7.1 [23]. The present proof supplies the coordinate calculations, norm and inverse maps, determinant coefficients, and global bundle maps in that convention. Its algebraic starting objects are the real alternative composition algebra $\mathbb{O}$ and its Albert Jordan algebra J; those defining structures are retained.

The proved result is an ordered three-sector tangent structure on $F_4/\operatorname{Spin}(8)$ with a normed triality tensor and three exact bilinear morphisms. It does not assert toroidal filling, an integrable complex structure, or physical dynamics. The obstruction to equivariant unary identifications and the positive bilinear relations between sectors were both proved. The dependence on the diagonal-orbit parameters is the exact morphism (111), with its product and metric consequences; it is not a missing comparison.

5.4 A coordinate-preserving triality map onto the determinant boundary

Retain the ordered matrix in (83) and write

$$E(q) = Q_{\mathbb{O}}(0; q_1, q_2, q_3), \qquad s(q) = |q_1|^2 + |q_2|^2 + |q_3|^2, \qquad \tau(q) = \operatorname{Re}((q_1 q_2) q_3).$$

Here the eigenvalues are the three real Jordan eigenvalues, with their multiplicities. They are the roots of $z^3 - s(q)z - 2\tau(q)$. Define

$$a(q) = -\lambda_{\min}(E(q)).$$

Thus $a(q)$ is exactly the largest real root of $\lambda^3 - s(q)\lambda + 2\tau(q) = 0$. This prescription keeps the complete cubic, including its mixed term and its factor two.

Theorem 5.12 (The spectral graph and its exact inverse)**.** *The map*

$$\mathcal{B} : \mathbb{O}^3 \longrightarrow \partial\mathcal{C}(\mathbb{O}), \qquad q \longmapsto Q_{\mathbb{O}}(a(q); q) \tag{114}$$

is a proper homeomorphism. Its inverse is the coordinate projection

$$\mathcal{P}\left(\begin{pmatrix} \lambda & q_3 & \overline{q}_2 \\ \overline{q}_3 & \lambda & q_1 \\ q_2 & \overline{q}_1 & \lambda \end{pmatrix}\right) = (q_1, q_2, q_3).$$

It extends to a based homeomorphism

$$\mathcal{B}^+ : (\mathbb{O}^3)^+ \xrightarrow{\cong} S^{24}_{\det,\mathbb{O}}.$$

Let $K = \operatorname{Spin}(8) \subset F_4$ be the pointwise stabilizer of the ordered diagonal Jordan frame. On $\mathbb{O}^3$, use the three representations defined by $kE_i(q_i) = E_i(\rho_i(k)q_i)$, with E_i the three ordered off-diagonal inclusions. Then $\mathcal{B}$ and $\mathcal{P}$ are K-equivariant. The norm and the cubic interaction are recovered on the boundary without discarding any coordinate:

$$2\tau(q) = a(q)s(q) - a(q)^3. \tag{115}$$

Proof. Use the Euclidean Jordan trace inner product $\langle u, v\rangle_J = \operatorname{tr}(u \circ v)$. For $q, p \in \mathbb{O}^3$, direct multiplication gives

$$\|E(q) - E(p)\|_J^2 = 2\sum_{i=1}^{3} |q_i - p_i|^2.$$

Let $\mathcal{I}$ be the compact set of primitive idempotents of trace one. The spectral theorem for the Euclidean Albert algebra gives

$$\lambda_{\min}(u) = \min_{e\in\mathcal{I}} \langle u, e\rangle_J.$$

For completeness, in a spectral Jordan frame write $u = \sum_i \mu_i e_i$. For every $e \in \mathcal{I}$, the numbers $\langle e_i, e\rangle_J$ are nonnegative and have sum one: positivity uses self-duality of the cone of squares, and the sum is $\langle I_3, e\rangle_J = 1$. Thus $\langle u, e\rangle_J \geq \min_i \mu_i$, with equality at an idempotent for a least eigenvalue. This proves the displayed formula. Every such e has $\|e\|_J^2 = \operatorname{tr}(e^2) = 1$. Cauchy–Schwarz and the minimum formula now give

$$|a(q) - a(p)| \leq \|E(q) - E(p)\|_J = \sqrt{2\sum_{i=1}^{3} |q_i - p_i|^2}. \tag{116}$$

In particular a is continuous.

Since $E(q)$ has trace zero, its least eigenvalue is nonpositive. If $q \neq 0$, all eigenvalues cannot be zero, since the spectral theorem would then give $E(q) = 0$. Their zero sum consequently forces a strictly negative least eigenvalue. Hence $a(0) = 0$ and $a(q) > 0$ for $q \neq 0$. Adding λI_3 adds λ to each of the three Jordan eigenvalues. Therefore

$$Q_{\mathbb{O}}(\lambda; q) \succeq 0 \quad \Longleftrightarrow \quad \lambda \geq a(q),$$

and within this positive cone the determinant vanishes exactly at $\lambda = a(q)$. This is the relative boundary established in theorem 5.6. It proves both inverse compositions in (114). Continuity of the graph and coordinate projection proves the homeomorphism. Moreover

$$\|Q_{\mathbb{O}}(a(q); q)\|_J^2 = 3a(q)^2 + 2s(q) \geq 2s(q),$$

so unbounded q leaves every compact subset of the boundary. This proves properness directly and the extension at infinity.

A member of K fixes I_3, preserves the Jordan product and trace, and sends each $E_i(q_i)$ to $E_i(\rho_i(k)q_i)$. It preserves the characteristic polynomial and the ordered eigenvalues. Thus $a(\rho_1(k)q_1, \rho_2(k)q_2, \rho_3(k)q_3) = a(q)$, which proves both equivariance identities. Substitution of $\lambda = a(q)$ into the unchanged determinant formula gives (115). □

The exact tangent map from the ordered flag. For pairwise different real $\lambda_1, \lambda_2, \lambda_3$ with zero sum, retain Ξ_λ of (69). At $x_0 = (d_1, d_2)$, the inverse of its differential, written in the original off-diagonal coordinates, is

$$\begin{aligned} A(q) &= E_2\left(\frac{q_2}{\lambda_1 - \lambda_3}\right) + E_3\left(\frac{q_3}{\lambda_1 - \lambda_2}\right), \\ B(q) &= E_1\left(\frac{q_1}{\lambda_2 - \lambda_3}\right) - E_3\left(\frac{q_3}{\lambda_1 - \lambda_2}\right), \\ (d\Xi_\lambda)_{x_0}(A(q), B(q)) &= E(q). \end{aligned} \tag{117}$$

Indeed, differentiate the three idempotents in a Jordan frame. The idempotence equations and the identity that their sum is I_3 force A to have only sectors two and three, B to have only

sectors one and three, and their sector-three parts to be negatives. To check that these are actual tangent vectors, the differential of the already proved diffeomorphism Ξ_λ is an isomorphism onto $E_1(\mathbb{O}) \oplus E_2(\mathbb{O}) \oplus E_3(\mathbb{O})$. Its formula is $(\lambda_1 - \lambda_3)A + (\lambda_2 - \lambda_3)B$. Solving its three sector equations gives precisely (117), so no extra tangent solutions or denominators are suppressed. Composing this tangent coordinate isomorphism with $\mathcal{B}$ gives an explicit equivariant homeomorphism from the complete tangent space at x_0 onto $\partial\mathcal{C}(\mathbb{O})$.

5.5 An explicit equivariant flag cell and its degree-one collapse

Write $J = \mathfrak{h}_3(\mathbb{O})$, $J_0 = \ker \operatorname{tr}$, $e = I_3$, and $d_i = \operatorname{diag}(\delta_{i1}, \delta_{i2}, \delta_{i3})$. The inner product on J is

$$\langle x, y \rangle_J = \operatorname{tr}(x \circ y), \qquad L_x(y) = x \circ y.$$

All operator products and brackets below belong to $\operatorname{End}_{\mathbb{R}}(J)$; thus $[A, B] = AB - BA$ is a commutator of real linear operators. An adjoint A^* in these formulas is the adjoint for $\langle\ ,\ \rangle_J$.

We preserve the ordered coordinates of the equal-diagonal slice by putting

$$\begin{aligned} E_1(r) &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & r \\ 0 & \bar{r} & 0 \end{pmatrix}, \qquad E_2(s) = \begin{pmatrix} 0 & 0 & \bar{s} \\ 0 & 0 & 0 \\ s & 0 & 0 \end{pmatrix}, \\ E_3(t) &= \begin{pmatrix} 0 & t & 0 \\ \bar{t} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad W = \mathbb{O} \oplus \mathbb{O} \oplus \mathbb{O}, \\ E(q) &= E_1(q_1) + E_2(q_2) + E_3(q_3) = Q_{\mathbb{O}}(0; q_1, q_2, q_3). \end{aligned} \tag{118}$$

In particular the second coordinate is the lower $(3,1)$ entry, and conjugating it in the upper $(1,3)$ entry remains part of every formula. Let $K = \operatorname{Aut}(J) = F_4$ and $M = \{m \in K : md_i = d_i\ (i = 1,2,3)\} = \operatorname{Spin}(8)$. The action of M on W is defined, without choosing three unrelated orthogonal actions, by

$$E_i((m \cdot q)_i) = mE_i(q_i), \qquad i = 1, 2, 3. \tag{119}$$

The three summands are the vector and the two real half-spin modules in this specified coordinate convention.

We use the real symmetric pair

$$G = E_{6(-26)}, \qquad \mathfrak{g} = \operatorname{Der}(J) \oplus L_{J_0}, \qquad \mathfrak{k} = \operatorname{Der}(J), \qquad \mathfrak{s} = L_{J_0}.$$

Here G is the connected reduced structure group in its real linear representation on J. Its Cartan involution on the Lie algebra is $\theta(A) = -A^*$; hence $\theta(D + L_x) = D - L_x$ for $D \in \operatorname{Der}(J)$ and $x \in J_0$. The symmetric-pair realization and restricted-root spaces are as in [10, Section 2.6, pp. 491–493]. The real-flag theorem we use is precisely [10, Appendix B, Theorem B.2, Proposition B.3 and Corollary B.4, p. 519]: for $P = MAN$ the space $G/P = K/M$ has cells Nw, and $X \mapsto \exp(X)wP$ is a diffeomorphism from the sum of the positive root spaces made negative by w^{-1} onto that cell. Its Cartan comparison is $\Theta(X) = X - \theta X$. We now determine that comparison, the exponential, its flag coordinates, its inverse, and the collapse with every coefficient specified.

Lemma 5.13 (The Cartan inverse with its exact coefficient)**.** *For $h = \operatorname{diag}(h_1, h_2, h_3) \in J_0$, set*

$$\rho_1(h) = \frac{h_3 - h_2}{2}, \qquad \rho_2(h) = \frac{h_1 - h_3}{2}, \qquad \rho_3(h) = \frac{h_1 - h_2}{2} = \rho_1(h) + \rho_2(h).$$

Fix any $h_* = \mathrm{diag}(h_{*,1}, h_{*,2}, h_{*,3}) \in J_0$ *with* $h_{*,1} > h_{*,3} > h_{*,2}$. *For* $u \in E_i(\mathbb{O})$ *define*

$$\nu_i(u) = \frac{1}{2}\left(L_u + \frac{[L_{h_*}, L_u]}{\rho_i(h_*)}\right). \tag{120}$$

Then $\nu_i(u)$ *is independent of the permitted choice of* h_*, *belongs to* $\mathfrak{g}_{\rho_i}$, *and satisfies*

$$[L_h, \nu_i(u)] = \rho_i(h)\nu_i(u), \qquad \nu_i(u) - \theta\nu_i(u) = L_u, \qquad \nu_i(u)e = \frac{1}{2}u. \tag{121}$$

Thus ν_i *is exactly the inverse of* Θ *on the* i*th summand; the factor* $1/2$ *in* (120) *is necessary. The isomorphism*

$$Z : W \longrightarrow \mathfrak{n} = \mathfrak{g}_{\rho_1} \oplus \mathfrak{g}_{\rho_2} \oplus \mathfrak{g}_{\rho_3}, \qquad Z(q) = \sum_{i=1}^{3} \nu_i(E_i(q_i))$$

is M*-equivariant.*

Proof. We first record the Jordan identity needed to ensure that the operators are in the correct Lie algebra. For any $a, b \in J$, $D_{a,b} = [L_a, L_b]$ is a derivation, and therefore

$$[D_{a,b}, L_c] = L_{D_{a,b}c}. \tag{122}$$

Here is the polarization proving this assertion. The coefficient of the product of three distinct real parameters in $(x^2 \circ y) \circ x = x^2 \circ (y \circ x)$ gives

$$\begin{aligned} &d \circ (b \circ (a \circ c)) + a \circ (b \circ (c \circ d)) + c \circ (b \circ (d \circ a)) \\ &\quad = (a \circ c) \circ (b \circ d) + (c \circ d) \circ (b \circ a) + (d \circ a) \circ (b \circ c). \end{aligned}$$

Subtract the same identity with a and b exchanged. The three terms on the right cancel, by commutativity of the Jordan product. The result, rearranged, is

$$D_{a,b}(c \circ d) = (D_{a,b}c) \circ d + c \circ (D_{a,b}d),$$

which proves the assertion and (122).

Let u have its two off-diagonal entries in positions (j,k) and (k,j), and write $D_h = [L_h, L_u]$ for real diagonal h. If h' is another real diagonal element, direct use of the binary Jordan product gives

$$\begin{aligned} D_h h' &= \left(\frac{(h_j + h_k)(h'_j + h'_k)}{4} - \frac{h_j h'_j + h_k h'_k}{2}\right) u \\ &= -\frac{(h_j - h_k)(h'_j - h'_k)}{4} u = -\rho_i(h)\rho_i(h')u. \end{aligned} \tag{123}$$

For completeness, the proportionality of D_h in h can also be checked on every $z \in J$ with no associativity assumption on three octonions. Expanding the two Jordan products, the fact that every entry of h is real gives the restricted identity

$$4[L_h, L_u]z = (hu - uh)z - z(hu - uh).$$

In this displayed identity only, the products are binary matrix products; the expansion is valid because every reassociation it uses contains a real entry from h. Since u has only the indicated two off-diagonal entries, $hu - uh$ depends linearly on the single real scalar $h_j - h_k$. Consequently

$$[L_h, L_u] = \frac{\rho_i(h)}{\rho_i(h_*)}[L_{h_*}, L_u]. \tag{124}$$

This assertion includes $\rho_i(h) = 0$. We use no corresponding identity with arbitrary octonionic h, and no octonionic matrix conjugation is used to define a group action.

Put $D_* = [L_{h_*}, L_u]$. Equations (122) and (123) give

$$[L_h, D_*] = -L_{D_* h} = \rho_i(h_*)\rho_i(h)L_u.$$

Combining this with (124) proves $[L_h, \nu_i(u)] = \rho_i(h)\nu_i(u)$ and also proves independence of h_*. Since $\theta L_u = -L_u$ and $\theta D_* = D_*$, subtracting $\theta\nu_i(u)$ from (120) gives L_u exactly. Every derivation kills the unit, so $\nu_i(u)e = u/2$. The map $u \mapsto L_u$ is injective because $L_u e = u$. The positive restricted-root spaces have the three stated eight-dimensional Cartan images; thus these injective maps exhaust them. Finally, an $m \in M$ fixes h_* and satisfies $mL_u m^{-1} = L_{mu}$ and $m[L_{h_*}, L_u]m^{-1} = [L_{h_*}, L_{mu}]$. These formulas prove the asserted equivariance. □

Lemma 5.14 (Finite exponential and the coefficient in the nilpotent law)**.** *For every $q \in W$,*

$$Z(q)^5 = 0, \qquad \exp Z(q) = I + Z(q) + \frac{Z(q)^2}{2} + \frac{Z(q)^3}{6} + \frac{Z(q)^4}{24}. \tag{125}$$

The same formula holds with $Z(q)$ replaced by $-Z(q)^*$. Writing $\nu_i(r)$ temporarily for $\nu_i(E_i(r))$, the brackets are

$$[\nu_1(r), \nu_2(s)] = \nu_3\left(-\frac{1}{2}\overline{rs}\right), \qquad [\nu_i(r), \nu_i(s)] = 0, \qquad [\nu_3(t), \nu_j(r)] = 0. \tag{126}$$

The last equality holds for $j = 1, 2, 3$. In exponential coordinates the group law on N is

$$\begin{aligned}(q_1, q_2, q_3) \star (q_1', q_2', q_3') = \Big(&q_1 + q_1',\ q_2 + q_2',\\ &q_3 + q_3' - \frac{1}{4}\overline{q_1 q_2'} + \frac{1}{4}\overline{q_1' q_2}\Big).\end{aligned} \tag{127}$$

Proof. The auxiliary operator $H_0 = L_{\mathrm{diag}(2,-2,0)}$ has the following eigenspaces in its representation on J:

eigenvalue	-2	-1	0	1	2
eigenspace	$\mathbb{R}d_2$	$E_1(\mathbb{O})$	$\mathbb{R}d_3 \oplus E_3(\mathbb{O})$	$E_2(\mathbb{O})$	$\mathbb{R}d_1$.

This table follows from $L_h d_j = h_j d_j$ and $L_h u = (h_j + h_k)u/2$ for an off-diagonal (j, k) element. The actual positive roots evaluated on $\mathrm{diag}(2, -2, 0)$ are $1, 1, 2$. Therefore every term of $Z(q)$ strictly raises these weights by at least one. Five successive applications give zero. Adjoints strictly lower the weights, proving the analogous assertion for $-Z(q)^*$. This proves both finite exponential formulas.

Brackets add restricted roots. There is no root $2\rho_i$, $\rho_1 + \rho_3$, or $\rho_2 + \rho_3$, so the brackets claimed to vanish in (126) do vanish. To compute the remaining coefficient, put $u = E_1(r)$, $v = E_2(s)$. Binary matrix multiplication gives, in the retained coordinate order,

$$u \circ v = \frac{1}{2}E_3(\overline{rs}) =: w.$$

The weight of u, v, and w under L_h is respectively $-h_1/2$, $-h_2/2$, and $-h_3/2$. Hence

$$[L_h, L_u]v = -\rho_1(h)w, \qquad [L_h, L_v]u = \rho_2(h)w,$$

and (120) gives $\nu_1(r)v = 0$ and $\nu_2(s)u = w$. Applying the commutator to e,

$$[\nu_1(r), \nu_2(s)]e = \frac{1}{2}(\nu_1(r)v - \nu_2(s)u) = -\frac{1}{4}E_3(\overline{rs}).$$

On the third root space the map $A \mapsto 2Ae$ is inverse to ν_3. This proves the first formula of (126). All iterated brackets of length three vanish. The exact Baker–Campbell–Hausdorff formula is therefore $\log(\exp X \exp Y) = X + Y + [X, Y]/2$; substituting the just-computed bracket gives (127), including both signs and the coefficient 1/4. □

Theorem 5.15 (A finite operator formula for the entire open cell)**.** *For $q \in W$ define*

$$\begin{aligned} A(q) &= \left(I + Z + \frac{Z^2}{2} + \frac{Z^3}{6} + \frac{Z^4}{24}\right) d_2, \\ B(q) &= \left(I - Z^* + \frac{(Z^*)^2}{2} - \frac{(Z^*)^3}{6} + \frac{(Z^*)^4}{24}\right) d_1, \qquad Z = Z(q), \\ \Psi(q) &= \left(\frac{A(q)}{\operatorname{tr} A(q)}, \frac{B(q)}{\operatorname{tr} B(q)}\right). \end{aligned} \tag{128}$$

Both traces in these formulas are strictly positive. The map Ψ is an M-equivariant diffeomorphism

$$W \xrightarrow{\Psi} U = \{(a, b) \in \mathrm{Fl}(\mathbb{O}) : a_{22} > 0, b_{11} > 0\} = C_{w_0}, \qquad w_0 = (12). \tag{129}$$

The inverse $q = \Psi^{-1}(a, b)$ is the exact formula

$$\boxed{q_1 = \frac{2a_{23}}{a_{22}}, \qquad q_2 = -\frac{2b_{31}}{b_{11}}, \qquad q_3 = \frac{2a_{12}}{a_{22}} - \frac{1}{4}\overline{q_1 q_2}.} \tag{130}$$

Every denominator in (130) *is a positive real scalar. In particular $\Psi(0) = (d_2, d_1)$.*

Proof. Let $\widehat{w}_0$ be the Jordan automorphism that interchanges the first and second real diagonal positions, obtained by conjugation with the real permutation matrix for (12). For this real matrix the conjugation is a well-defined Jordan automorphism. Its action on the retained positive roots is

$$w_0\rho_1 = -\rho_2, \qquad w_0\rho_2 = -\rho_1, \qquad w_0\rho_3 = -\rho_3.$$

It is consequently the longest Weyl element, and the real-flag cell theorem gives a diffeomorphism

$$W \longrightarrow N\widehat{w}_0 P/P, \qquad q \longmapsto \exp Z(q)\widehat{w}_0 P. \tag{131}$$

Here are the exact Iwasawa factors that identify the flag pair in (131). Write $g = kan$ uniquely with $k \in K$, $a = \exp L_h \in A$, $n \in N$. Each positive root operator kills d_1. This follows either by applying (120) to d_1, or from the H_0 weight table, whose highest weight is $\mathbb{R}d_1$. Each operator $-\nu_i^*$ kills d_2 by the lowest-weight statement. Thus

$$nd_1 = d_1, \qquad (n^*)^{-1}d_2 = d_2,$$

and consequently

$$gd_1 = e^{h_1}kd_1, \qquad (g^*)^{-1}d_2 = e^{-h_2}kd_2. \tag{132}$$

Since Jordan automorphisms preserve the trace, the traces on the right are exactly $e^{h_1} > 0$ and $e^{-h_2} > 0$. It follows that the flag corresponding to gP is

$$\left(\frac{gd_1}{\operatorname{tr}(gd_1)}, \frac{(g^*)^{-1}d_2}{\operatorname{tr}((g^*)^{-1}d_2)} \right) = (kd_1, kd_2).$$

For $g = \exp Z(q)\widehat{w}_0$, its two numerators are $\exp Z(q)d_2$ and $\exp(-Z(q)^*)d_1$, respectively. The finite exponential identities prove (128), all its positivity assertions, and the identification with (131).

We next verify the image description, so that no membership criterion is left implicit. For any $n \in N$, the H_0 weight table shows that the coefficient of d_2 in nd_j is 1 if $j = 2$ and 0 otherwise. Indeed n strictly raises weights apart from its identity term. Similarly the coefficient of d_1 in $(n^*)^{-1}d_j$ is 1 if $j = 1$ and 0 otherwise. A flag in $N\widehat{w}P/P$ is represented by the positive-trace multiples of $nd_{w(1)}$ and $(n^*)^{-1}d_{w(2)}$. It satisfies $a_{22} > 0$ precisely when $w(1) = 2$, and $b_{11} > 0$ precisely when $w(2) = 1$. Both conditions hold precisely for $w = w_0 = (12)$. The six-cell decomposition covers the flag, proving (129).

For the inverse we record the required coefficients of the finite exponentials. The same weights show

$$\begin{gathered} A(q)_{22} = 1, \qquad B(q)_{11} = 1, \qquad A(q)_{23} = \frac{1}{2}q_1, \qquad B(q)_{31} = -\frac{1}{2}q_2, \\ A(q)_{12} = \frac{1}{2}q_3 + \frac{1}{8}\overline{q_1 q_2}. \end{gathered} \tag{133}$$

To see also the last coefficient directly, the relevant first-order terms are

$$Zd_2 = \frac{1}{2}E_1(q_1) + \frac{1}{2}E_3(q_3), \qquad -Z^*d_1 = -\frac{1}{2}E_2(q_2) - \frac{1}{2}E_3(q_3).$$

An E_3 contribution to Z^2d_2 can arise only from $\nu_2(q_2)\nu_1(q_1)d_2$ or $\nu_1(q_1)\nu_2(q_2)d_2$; the latter is zero, and the former is $\frac{1}{4}E_3(\overline{q_1 q_2})$ by the calculation in Lemma 5.14. The other weight-two contribution $\nu_1(q_1)^2 d_2$ lies in $\mathbb{R}d_3$, and $\nu_2(q_2)^2 d_2 = 0$. The coefficient $1/2$ in the exponential gives $1/8$. Terms of degree three or four raise weight too far to have an E_3 component. The other four identities in (133) follow from the unique lowest, highest, and adjacent weights.

For $(a, b) = \Psi(q)$, the first two identities imply

$$A(q) = \frac{a}{a_{22}}, \qquad B(q) = \frac{b}{b_{11}}.$$

The remaining identities solve for q_1, q_2, q_3 in precisely the order (130). Since (131) is onto U, this proves both inverse compositions everywhere on U, not only a formal local inverse. The displayed formulas show their smoothness.

Finally, $Z(mq) = mZ(q)m^{-1}$ and m is orthogonal for the trace inner product. It fixes d_1, d_2 and preserves the trace. Each expression in (128) therefore commutes with the M action. This proves equivariance of Ψ and, by its proved bijectivity, of its inverse. □

Proposition 5.16 (The orientation sign, including the conjugated coordinate)**.** *Fix an orientation of $\mathbb{O}$, and give W the ordered product orientation (q_1, q_2, q_3). Give the flag the orientation specified in the integral Euler-class calculation: orient its first root summand by the chosen orientation of $E_1(\mathbb{O})$, its third by $n_2E_1(\mathbb{O})$, and its second by $n_1^{-1}n_2E_1(\mathbb{O})$, where $n_1 = (23)$ and $n_2 = (13)$ are real-permutation lifts. The total orientation is the ordered sum of root summands $(1, 2, 3)$, transported by K. Then $\Psi : W \to U$ is orientation preserving.*

Proof. These are the orientation transport conventions of [10, Section 3, pp. 495–496]. They are independent of the lifts within their cosets because M is connected and each of its orthogonal representations has positive real determinant. We compute the sign explicitly. For this computation only, call the product orientation of the three upper entries (r, q, p) in positions $(23), (13), (12)$ the upper-entry orientation. Octonionic conjugation fixes the real line and acts by -1 on the seven imaginary directions, so its real determinant is $(-1)^7 = -1$. The lift $n_2 = (13)$ takes an upper (23) entry r to the upper (12) entry $\bar{r}$. The lift $n_1^{-1} = (23)$ then takes that upper (12) entry to the upper (13) entry with the same value. The source orientations of summands $1, 2, 3$ thus have signs $+1, -1, -1$ relative to the upper-entry coordinates. Their product is $+1$, so the flag tangent orientation at d_λ agrees with the upper-entry orientation. Here d_λ has the original pairwise distinct parameters λ_i and the trace-zero relation; no ordering or rescaling of these parameters is needed.

The lift $\widehat{w}_0 = (12)$ interchanges the upper-entry (23) and (13) blocks and conjugates the (12) block. Interchanging two eight-dimensional blocks has determinant $(-1)^{8\cdot 8} = +1$, and conjugation has determinant -1. Thus the transported flag orientation at $\widehat{w}_0 d_\lambda$ is the negative of its upper-entry orientation. This transport preserves the flag orientation because it is the action of an element of connected K.

In the retained q coordinates, E conjugates exactly the second upper-entry block; its determinant relative to the upper-entry orientation is therefore -1. Differentiate (128) and the original orbit comparison $\Xi_\lambda(a, b) = \lambda_1 a + \lambda_2 b + \lambda_3(e - a - b)$ at $q = 0$. The traces have zero first derivative, and the result is

$$\begin{aligned} d(\Xi_\lambda \Psi)_0(q) &= \frac{\lambda_1 - \lambda_3}{2} E_1(q_1) + \frac{\lambda_3 - \lambda_2}{2} E_2(q_2) + \frac{\lambda_1 - \lambda_2}{2} E_3(q_3) \\ &= -\sum_{i=1}^{3} \rho_i(\widehat{w}_0 d_\lambda) E_i(q_i). \end{aligned} \tag{134}$$

Its three real scalar factors are nonzero; each acts on eight real coordinates, so their determinant contribution is positive. The total differential has sign -1 relative to upper-entry coordinates, exactly the sign of the transported flag orientation at this point. Therefore its sign as an oriented map is $+1$. The domain is connected and the map is a diffeomorphism, so this sign is $+1$ everywhere. □

Theorem 5.17 (The exact equivariant quotient onto the representation sphere)**.** *Let $F = \mathrm{Fl}(\mathbb{O})$ and put*

$$A = F \setminus U = \{(a, b) \in F : a_{22} = 0 \text{ or } b_{11} = 0\}.$$

It is the union of the five cells other than C_{w_0}, and is a closed 16-dimensional CW subcomplex. Define

$$c : F \longrightarrow W^+, \qquad c(a, b) = \begin{cases} \Psi^{-1}(a, b), & (a, b) \in U, \\ \infty, & (a, b) \in A. \end{cases} \tag{135}$$

Then c is continuous, surjective, and M-equivariant and induces a based M-equivariant homeomorphism

$$F/A \xrightarrow{\cong} W^+. \tag{136}$$

With the orientations in Proposition 5.16, c has degree $+1$. If $\omega_W \in \widetilde{H}^{24}(W^+; \mathbb{Z})$ is its positive generator, then

$$c_*[F] = [W^+], \qquad c^* \omega_W = u_{\mathbb{O}}, \qquad u_{\mathbb{O}} = \beta_1^2 \beta_2 = \beta_1 \beta_2^2. \tag{137}$$

On integral homology c_* is the identity in degree zero, an isomorphism in degree 24*, and the zero map on the two copies of* $\mathbb{Z}$ *in each of degrees* 8 *and* 16*. On fixed points its exact formula is*

$$c(w_0 x_{\mathbb{O},0}) = 0, \qquad c(w x_{\mathbb{O},0}) = \infty \quad (w \neq w_0).$$

These are the six points of F^M *mapping onto the two points* $(W^+)^M = \{0, \infty\}$*.*

Proof. All diagonal entries of primitive positive idempotents are nonnegative. The open-cell locus of Theorem 5.15 consequently gives the stated closed-set formula for A. The restricted roots have multiplicity eight. The six cell dimensions are $0, 8, 8, 16, 16, 24$, with the unique top cell C_{w_0}. The CW property of the real-flag decomposition implies that their union below degree 24 is the closed 16-skeleton, which proves the subcomplex assertion.

Continuity of c on U is already proved. Let $K_0 \subset W$ be compact. Its image $\Psi(K_0)$ is compact in the Hausdorff compact manifold F, and hence closed in F. Thus

$$c^{-1}(W^+ \setminus K_0) = F \setminus \Psi(K_0)$$

is open and contains A. Complements of compact subsets of W form a neighborhood basis of ∞. This proves continuity at every point of A, including all its lower-dimensional strata. It also proves the exact escape statement: a net in U approaching A eventually leaves $\Psi(K_0)$ for every compact $K_0 \subset W$. Surjectivity and the description of the fibers are immediate from the two inverse compositions of the chart: every finite point has one preimage, and $c^{-1}(\infty) = A$. Therefore the induced map $F/A \to W^+$ is a continuous bijection from a compact space to a Hausdorff space and hence a homeomorphism. The open cell is M-invariant by the proved chart equivariance; its complement is invariant too. The formula for c is therefore M-equivariant, with M fixing ∞.

Since A has dimension 16, its integral homology in degrees 23 and 24 is zero. The exact sequence of the pair gives an isomorphism

$$H_{24}(F; \mathbb{Z}) \xrightarrow{\cong} H_{24}(F, A; \mathbb{Z}).$$

Because A is a subcomplex, quotienting identifies this relative group with $\widetilde{H}_{24}(F/A; \mathbb{Z})$. At any point of U, the local map is the inverse of the orientation-preserving chart. It has local degree $+1$. There is exactly one preimage of that finite regular value. This fixes the sign of the preceding homology isomorphism and proves $c_*[F] = [W^+]$. Pairing with cohomology proves $c^* \omega_W = u_{\mathbb{O}}$ for the already fixed positive top generator of the flag. The remaining homology assertions follow because W^+ is a 24-sphere and F is connected with the stated six-cell homology. Each of the three nontrivial irreducible eight-dimensional M-modules has zero invariant subspace, so $W^M = \{0\}$. The point $w_0 x_{\mathbb{O},0}$ is $\Psi(0)$ and belongs to the open cell; the other five Weyl points lie in A. Substitution in the defining formula for c proves the fixed-point assertions. □

Corollary 5.18 (Composition with the retained spectral boundary graph)**.** *For* $q \in W$*, let*

$$\alpha(q) = -\lambda_{\min}(E(q)), \qquad \Gamma(q) = Q_{\mathbb{O}}(\alpha(q); q_1, q_2, q_3).$$

The exact boundary graph has inverse the off-diagonal coordinate projection $Q_{\mathbb{O}}(\lambda; q) \mapsto q$*. It extends to an* M*-equivariant based homeomorphism*

$$\Gamma^+ : W^+ \longrightarrow (\partial \mathcal{C}(\mathbb{O}))^+, \qquad \Gamma^+(\infty) = \infty.$$

Consequently the fully specified map

$$C_{\det} = \Gamma^+ c : F \longrightarrow S^{24}_{\det,\mathbb{O}} \tag{138}$$

is continuous, M-equivariant, and has degree $+1$ when the boundary sphere carries the orientation transported from its retained ordered coordinates by Γ^+. Its exact fibers are

$$C_{\det}^{-1}(\Gamma(q)) = \{\Psi(q)\} \quad (q \in W), \qquad C_{\det}^{-1}(\infty) = A.$$

Proof. The eigenvalues of $\lambda e + E(q)$ are $\lambda + \lambda_j(E(q))$, with their multiplicities retained. It is positive semidefinite precisely when $\lambda \geq -\lambda_{\min}(E(q))$, and lies on the relative boundary precisely when equality holds. This proves both compositions with the stated coordinate projection. The minimum eigenvalue is continuous, so Γ is continuous, and the projection is continuous. A homeomorphism is proper, because the inverse maps compact sets to compact sets; therefore it extends to the indicated one-point compactifications. An element of M fixes e, preserves eigenvalues as a Jordan automorphism, and sends $E(q)$ to $E(mq)$. Thus $\alpha(mq) = \alpha(q)$ and $\Gamma(mq) = m\Gamma(q)$. This proves equivariance also at infinity. The asserted properties, degree, and fibers of $C_{\det}$ now follow from Theorem 5.17 by composition with the explicitly oriented homeomorphism Γ^+. □

Corollary 5.19 (The exact limit on a reverse sphere map)**.** *Every continuous map $f : S^{24} \to F$ has degree zero. In particular neither c nor $C_{\det}$ admits a continuous section or a homotopy section. These assertions coexist with the degree-one equivariant maps just constructed.*

Proof. The cohomology groups $H^8(S^{24}; \mathbb{Z})$ vanish, so $f^*\beta_1 = f^*\beta_2 = 0$. Hence

$$f^* u_{\mathbb{O}} = f^*(\beta_1^2 \beta_2) = (f^*\beta_1)^2 f^*\beta_2 = 0.$$

This is exactly the statement that the degree is zero. A homotopy section of c would satisfy $f^* c^* \omega_W = \omega_W$, contradicting the equality just proved and (137). The same argument applies after the homeomorphism Γ^+. □

Proposition 5.20 (Exact maps among the three marked 24-dimensional constructions)**.** *The following constructions are related by the explicit spectral graph, flag-cell collapse, and smash-coordinate maps proved here.*

(i) *The octonionic flag is the homogeneous 24-manifold $F_4/\operatorname{Spin}(8)$, with $H^8 \cong H^{16} \cong \mathbb{Z}^2$ and $\chi = 6$.*

(ii) *After a based diffeomorphism $X \cong S^6$, the fourth smash power $X^{\wedge 4}$ is a based sphere $S^{24}_{\wedge}$; it has $\widetilde{H}^k = 0$ for $k \neq 24$ and $\chi = 2$, and its quotient maps depend on the chosen based candidate and cusp collapses.*

(iii) *The octonionic equal-diagonal cone has the determinant-boundary compactification $S^{24}_{\det,\mathbb{O}} = (\partial \mathcal{C}(\mathbb{O}))^+$. It is produced by the explicit cubic (85), not by a smash product.*

The first package is not homotopy equivalent to either sphere. The two sphere carriers in (ii) and (iii) are related by the based homeomorphism displayed below. The flag maps to the determinant sphere by the explicit degree-one map of Corollary 5.18; its entire infinite fibre is the stated 16-skeleton. The obstruction to a reverse homotopy section is Corollary 5.19.

The lower complex determinant orbit pair (S^{10}, S^8) from theorem 4.3 is a lower rung: its S^8 is the compactified complex rank-loss locus and its complement has determinant-phase fundamental group $\mathbb{Z}$. It is not a third 24-dimensional object.

Proof. Item (i) is theorems 5.1 and 5.2. The based smash identity $(S^6)^{\wedge 4} \cong S^{24}$ proves the sphere and cohomology assertions in (ii). Item (iii) is theorem 5.6. The nonzero degree-eight and degree-sixteen cohomology of the flag precludes a homotopy equivalence with either sphere; the Euler

characteristics give an independent obstruction. For an explicit choice-dependent comparison, let $\eta : (X, x_{\mathrm{base}}) \to (\mathbb{R}^6)^+$ be a based homeomorphism, write the four six-coordinate vectors consecutively as $(x_0, \dots, x_{23})$, and fix the explicit linear isometry $L : (\mathbb{R}^6)^4 \to \mathbb{O}^3$ given by $q_j = \sum_{a=0}^{7} x_{8(j-1)+a} u_a$ in the fixed ordered orthonormal octonion basis. Let $b = \mathcal{P}$ be the off-diagonal coordinate inverse of the spectral graph in Theorem 5.12. The smash compactification map

$$\kappa : ((\mathbb{R}^6)^+)^{\wedge 4} \longrightarrow ((\mathbb{R}^6)^4)^+$$

sends a tuple of finite vectors to their ordered tuple and sends the collapsed wedge to infinity. It is a continuous bijection from a compact space to a Hausdorff space, hence a homeomorphism. Thus

$$(b^+)^{-1} \circ L^+ \circ \kappa \circ \eta^{\wedge 4} : X^{\wedge 4} \xrightarrow{\cong} (\partial \mathcal{C}(\mathbb{O}))^+$$

is a based homeomorphism with inverse obtained by reversing these four maps. It depends explicitly on η and L; neither marked construction specifies them. The final lower-rung assertions are (43) and (44). □

6 What the mod–12 and mod–24 arithmetic actually is

6.1 The integral peripheral representation

The matrices C_1, C_2', C_0 in (27) are not merely of orders three, four, and infinite. Since C_2' and C_0 generate $SL_2(\mathbb{Z})$, their abelianization detects one exact modulus.

Theorem 6.1 (Peripheral modulus and the mod–24 boundary)**.** *Let*

$$\mathrm{ab} : SL_2(\mathbb{Z}) \twoheadrightarrow SL_2(\mathbb{Z})_{\mathrm{ab}}$$

be the abelianization map. There is a unique isomorphism

$$\overline{\nu} : SL_2(\mathbb{Z})_{\mathrm{ab}} \xrightarrow{\cong} \mathbb{Z}/12$$

for which the marked character

$$\chi = \overline{\nu} \circ \mathrm{ab} : SL_2(\mathbb{Z}) \longrightarrow \mathbb{Z}/12 \tag{139}$$

sends C_0 to -1. Its values on the three peripheral matrices are

$$\chi(C_1) = 4, \qquad \chi(C_2') = -3, \qquad \chi(C_0) = -1 \pmod{12}. \tag{140}$$

The residues 4 and $-3 = 9$ are complementary idempotents in $\mathbb{Z}/12$; additively they have exact orders 3 and 4.

Let

$$j : \mathbb{Z}/12 \hookrightarrow \mathbb{Z}/24, \qquad j(a) = 2a.$$

Every group homomorphism $\psi : SL_2(\mathbb{Z}) \to \mathbb{Z}/24$ has image in the even subgroup and factors uniquely through j. In particular

$$\chi_{24} = j\chi, \qquad (\chi_{24}(C_1), \chi_{24}(C_2'), \chi_{24}(C_0)) = (8, -6, -2) \pmod{24} \tag{141}$$

is the unique mod–24 re-encoding with

$$\exp\!\left(\frac{2\pi i\, \chi_{24}(M)}{24}\right) = \exp\!\left(\frac{2\pi i\, \chi(M)}{12}\right) \qquad (M \in SL_2(\mathbb{Z})).$$

Its image has order 12, not 24. If $r : \mathbb{Z}/24 \to \mathbb{Z}/12$ is ordinary reduction, then $r \circ j = [2]$, and there is no group homomorphism $\widetilde{\chi} : SL_2(\mathbb{Z}) \to \mathbb{Z}/24$ with $r \circ \widetilde{\chi} = \chi$.

Proof. Put

$$S_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad T = C_0.$$

Then $C_2' = S_0^3$, $C_1 = T^{-1}S_0$, and the standard integral modular presentation is

$$SL_2(\mathbb{Z}) = \langle S_0, T \mid S_0^4 = 1,\ S_0^2 = (S_0T)^3 \rangle.$$

Here is why these relations suffice. With $R = S_0T$, the presented group is $G = \langle S_0, R : S_0^4 = 1, S_0^2 = R^3 \rangle$. The element $c = S_0^2 = R^3$ is central and has order at most two; $G/\langle c \rangle = C_2 * C_3$. The modular normal-form theorem identifies $PSL_2(\mathbb{Z})$ with this same free product on the images of S_0, R [3, Theorem C.1, pp. 18–19]. The matrices S_0, T generate $SL_2(\mathbb{Z})$ [3, Theorem 1.1, pp. 1–2]; hence the map from G onto $SL_2(\mathbb{Z})$ is surjective and its kernel lies in $\langle c \rangle$. Since c maps to $-I_2 \neq I_2$, the kernel is trivial. This proves the presentation (equivalently $C_4 *_{C_2} C_6$) with the stated matrix generators. After abelianization, with additive generators s, t, its two relations are

$$4s = 0, \qquad 2s = 3(s + t),$$

or $4s = 0$ and $s + 3t = 0$. The relation matrix $\left(\begin{smallmatrix} 4 & 0 \\ 1 & 3 \end{smallmatrix}\right)$ has Smith normal form $\mathrm{diag}(1, 12)$, so the abelianization is cyclic of exact order 12, generated by t. Choosing $t \mapsto -1$ gives

$$s \mapsto 3, \qquad C_2' = S_0^3 \mapsto 9 = -3, \qquad C_1 = T^{-1}S_0 \mapsto 1 + 3 = 4,$$

which proves (140). Direct calculation in $\mathbb{Z}/12$ gives

$$4^2 = 4, \qquad 9^2 = 9, \qquad 4 + 9 = 1, \qquad 4 \cdot 9 = 0.$$

Every group homomorphism from $SL_2(\mathbb{Z})$ to an abelian group factors through its abelianization. An element $x \in \mathbb{Z}/24$ satisfies $12x = 0$ exactly when x is even, proving the unique factorization through j and excluding a surjective character. Also $rj(a) = 2a \pmod{12}$. Finally a reduction-compatible lift would send C_0 to either 11 or 23 modulo 24, both of which have order 24; an image of the order-12 abelianized generator cannot have that order. This proves the final assertion. □

Proposition 6.2 (The primitive cusp quotient singles out mod–3, not mod–12 or mod–24)**.** *For $N \geq 1$, let*

$$d_N : \mathbb{Z}^3 \longrightarrow \mathbb{Z}/N, \qquad d_N(a_1, a_2, a_3) = a_1 + a_2 + a_3 \pmod{N}.$$

There is a homomorphism $\overline{d}_N : \mathbb{Z}^2 \to \mathbb{Z}/N$ satisfying $d_N = \overline{d}_N P$ if and only if $N \mid 3$. For $N = 3$, it is explicitly

$$\overline{d}_3(y_1, y_2) = y_1 - y_2 \pmod{3}.$$

Similarly, d_N factors through Q if and only if $N \mid 3$; for $N = 3$ the exact formula is

$$d_3 = (1, 1)Q.$$

Thus neither primitive lattice quotient in theorem 3.1 produces a diagonal-sum character modulo 12 *or modulo* 24*.*

Proof. Both $\ker P$ and $\ker Q$ are generated by $k = (1, 1, 1)^{\mathsf{T}}$. A homomorphism out of $\mathbb{Z}^3$ factors through either quotient precisely when it kills that kernel. Since $d_N(k) = 3 \pmod{N}$, this occurs precisely when $N \mid 3$. At $N = 3$, direct multiplication with the matrices in (10) gives

$$(1, -1)P = (1, -2, 1) \equiv (1, 1, 1) \pmod{3}, \qquad (1, 1)Q = (1, -2, 1) \equiv (1, 1, 1) \pmod{3},$$

which proves both formulas and the claimed boundary. □

Proposition 6.3 (Coefficient reduction and determinant winding do not select a modulus)**.** *For every $N \geq 1$, coefficient reduction gives a graded-ring isomorphism*

$$H^*(\mathrm{Fl}(\mathbb{O}); \mathbb{Z}/N) \cong H^*(\mathrm{Fl}(\mathbb{O}); \mathbb{Z}) \otimes \mathbb{Z}/N,$$

and its reduced Euler class

$$e_N = 6u_{\mathbb{O}} \bmod N \in H^{24}(\mathrm{Fl}(\mathbb{O}); \mathbb{Z}/N)$$

has additive order $N/\gcd(N, 6)$. In particular e_{12} has order two and e_{24} has order four. Ordinary coefficient reduction is defined for every N; this operation alone makes no choice of $N = 12$ or $N = 24$.

Likewise, in the lower complex orbit pair (S^{10}, S^8), the determinant phase identifies the fundamental group of the rank-full complement in (44) *with $\mathbb{Z}$. Reducing its winding number modulo N gives a surjection to $\mathbb{Z}/N$ for every N, so this plain winding-number reduction makes no choice of* 12 *or* 24*. This phase statement is not being transferred to the real octonionic determinant in* (85)*.*

Proof. All integral cohomology groups in (77) are free. The universal coefficient sequence therefore has no Tor term, and multiplication is preserved by coefficient change. Equation (76) gives the stated reduced Euler class. Since the reduction of the primitive $u_{\mathbb{O}}$ generates $H^{24}(\mathrm{Fl}(\mathbb{O}); \mathbb{Z}/N) \cong \mathbb{Z}/N$, the least positive integer m annihilating e_N is the least m with $N \mid 6m$, namely $N/\gcd(N, 6)$. For the determinant complement, (44) is a deformation retraction onto the phase circle, and $\arg\det$ is that retraction coordinate. Its winding is an isomorphism on π_1, whose reduction modulo any N is surjective. □

6.2 The distinct stable $\mathbb{Z}/12 \hookrightarrow \mathbb{Z}/24$

The injection $j(a) = 2a$ is realized, after compatible generator choices, by the universal sixfold suspension homomorphism below. The cusp quotient supplies one particular element in its domain; it does not supply the injection itself. Neither construction comes from the octonionic flag or from a 24-dimensional smash source. We state all data needed to type these assertions.

Let v_0 be the suspension vertex to which the central cusp fibre is collapsed, write $W = q_2^{-1}(v_0)$, and choose explicitly $x_{\mathrm{base}} \in W$. The exact cusp collapse in the canonical audit then gives a based quotient

$$q_2 : (X, x_{\mathrm{base}}) \longrightarrow (\mathcal{A}, v_0), \qquad \mathcal{A} = \mathrm{Susp}(S^1 \times \overline{T}^2).$$

Let

$$K^{(1)} = (S^1 \times \{0\}) \cup (\{0\} \times S^1) \subset \overline{T}^2,$$

and let $c : \overline{T}^2 \to \overline{T}^2/K^{(1)} \cong S^2$ be the degree-one cell quotient, with $* = c(K^{(1)})$. Write

$$\Sigma_r(S^2, *) = \frac{[0, \epsilon] \times S^2}{(\{0, \epsilon\} \times S^2) \cup ([0, \epsilon] \times \{*\})}$$

for reduced suspension, the entire denominator being collapsed to one basepoint. The formula

$$\widehat{p}_2([r, e^{i\phi}, u]) = [r, c(u)] \in \Sigma_r(S^2, *) \cong S^3$$

defines a based map: it sends both suspension vertices of $\mathcal{A}$ and the entire cylinder over $K^{(1)}$ to that one basepoint. Thus

$$g = \widehat{p}_2 q_2 : (X, x_{\mathrm{base}}) \longrightarrow (S^3, *). \tag{142}$$

After the fixed based identification $X \cong S^6$, this determines $[g] \in \pi_6(S^3)$. The construction and its endpoint continuity are part of the cited candidate audit [2]. The exact research-file locators are the reduced-suspension construction in

and the one-copy map in

cusp_s12_carrier_homotopy.tex.

Theorem 6.4 (The stable even-subgroup morphism and its exact scope)**.** *Sixfold reduced suspension is an injection*

$$E^6 : \pi_6(S^3) \cong \mathbb{Z}/12 \hookrightarrow \pi_{12}(S^9) \cong \mathbb{Z}/24 \tag{143}$$

whose image is the even subgroup. With compatible generators α and ν_9,

$$E^6(a\alpha) = 2a\nu_9.$$

Consequently there is a unique $a \in \mathbb{Z}/12$ for which the particular map in (142) *satisfies*

$$[E^6 g] = 2a\nu_9. \tag{144}$$

The coefficient a is not determined by point-set surjectivity or by the ordinary homology map.

For every pair of based maps $x, y : S^6 \to S^3$,

$$[x \wedge y] = 0 \in \pi_{12}(S^6). \tag{145}$$

In particular,

$$g^{\wedge 4} : S^{24} \longrightarrow S^{12}$$

is based null-homotopic. Thus the $\mathbb{Z}/24$ in (143) *is the ambient target group of the sixfold-suspended one-copy map $S^{12} \to S^9$. The separate 24-dimensional fourth-smash map is null-homotopic.*

Proof. The relevant initial suspension chain is

$$\pi_6(S^3) \xrightarrow{E} \pi_7(S^4) \xrightarrow{E} \pi_8(S^5).$$

Hatcher computes these groups as $\mathbb{Z}/12$, $\mathbb{Z} \oplus \mathbb{Z}/12$, and $\mathbb{Z}/24$, respectively. The Hopf fibration identifies the torsion summand of the middle group with the suspended first group. In the quotient defining the second suspension, twice the Hopf generator of the free summand is identified with a generator of this torsion summand [5, Theorem 5.41(b), p. 573, and pp. 577–578]. Choose compatible signs and generators $\alpha \in \pi_6(S^3)$ and $\nu_5 = E\nu_4 \in \pi_8(S^5)$. The displayed quotient relation gives $E^2\alpha = 2\nu_5$. Multiplication by two from $\mathbb{Z}/12$ to $\mathbb{Z}/24$ is injective: $24 \mid 2a$ implies $12 \mid a$. For $n \geq 5$, Freudenthal gives an isomorphism

$$E : \pi_{n+3}(S^n) \longrightarrow \pi_{n+4}(S^{n+1}), \qquad n + 3 \leq 2n - 2.$$

Applying it for $n = 5, 6, 7, 8$ and putting $\nu_9 = E^4\nu_5$ gives $E^6\alpha = 2\nu_9$ and $\pi_{12}(S^9) \cong \pi_3^{\mathrm{S}} \cong \mathbb{Z}/24$. The stable group is independently tabulated in [8, Table 1, p. 113]. This proves (143) and the unique formula (144).

For (145), the ordered external smash has the exact factorization

$$[x \wedge y] = (E^3[x]) \circ (E^6[y])$$

by the reduced-join formula [11, Proposition 3.1]; its ordered-coordinate sign is $(-1)^{3\cdot 6} = +1$. The integral low-dimensional table gives $\pi_{12}(S^6) \cong \mathbb{Z}/2$ [6, opening table, p. 339]. The class $E^6[y]$ is even in $\pi_{12}(S^9) \cong \mathbb{Z}/24$, and postcomposition by $E^3[x]$ is a group homomorphism into the exponent-two group $\pi_{12}(S^6)$. Its value on an even element is therefore zero. Hence $g^{\wedge 2}$ is null-homotopic, and smashing its based null-homotopy with $g^{\wedge 2}$ proves the assertion for $g^{\wedge 4}$. □

Proposition 6.5 (The generator-free bridge between the two $\mathbb{Z}/12$-objects)**.** *The marked one-copy class defines a homomorphism*

$$\Theta_g : \mathbb{Z}/12 \longrightarrow \pi_6(S^3), \qquad \Theta_g([n]) = n[g]. \tag{146}$$

Therefore

$$\Theta_g \circ \chi : SL_2(\mathbb{Z}) \longrightarrow \pi_6(S^3)$$

is a well-defined homomorphism from the marked peripheral data to the one-copy homotopy group. If α and ν_9 are compatible generators as in theorem 6.4 and $[g] = a\alpha$, then

$$E^6\Theta_g\chi(M) = 2a\,\chi(M)\nu_9 \in \pi_{12}(S^9) \cong \mathbb{Z}/24. \tag{147}$$

For an integer lift $\widetilde{a}$ put $d = \gcd(\widetilde{a}, 12)$. Then

$$\ker\Theta_g = \langle [12/d] \rangle \subset \mathbb{Z}/12, \qquad \operatorname{im}\Theta_g = \langle [g] \rangle = \langle d\alpha \rangle, \qquad |\operatorname{im}\Theta_g| = 12/d.$$

The first group has order d. Since χ is surjective and E^6 injective, both composites have image order $12/d$, and $\ker(E^6\Theta_g\chi) = \chi^{-1}(\langle [12/d] \rangle)$. This bridge is canonical once the marked character χ, the based map g, and the based identification $S^6 \cong X$ are fixed. It does not prove $d = 1$, equivalently that $[g]$ is a generator or that Θ_g is an isomorphism, and it does not define an equivariant map of degenerations.

Proof. The source of Θ_g is well-defined because $\pi_6(S^3) \cong \mathbb{Z}/12$, so $12[g] = 0$. It is a homomorphism by the abelian group law on $\pi_6(S^3)$. Composition with the already defined χ is therefore a well-defined homomorphism. Writing $[g] = a\alpha$ and applying the homomorphism E^6 gives

$$E^6\Theta_g\chi(M) = \chi(M)E^6(a\alpha) = 2a\,\chi(M)\nu_9,$$

which is (147) for every $M \in SL_2(\mathbb{Z})$. The congruence $n\widetilde{a} \equiv 0 \pmod{12}$ is equivalent to $12/d \mid n$, giving the kernel. Bézout's identity gives $\langle a\alpha \rangle = \langle d\alpha \rangle$, of order $12/d$. Surjectivity of χ preserves that image, and injectivity of E^6 preserves its order and the composite kernel. This group calculation by itself does not compute a. The explicit framed calculation in theorem 6.6 below does: $a = 0$, so $d = 12$, $\ker\Theta_g = \mathbb{Z}/12$, and every image displayed here is zero. □

6.3 The original framed cusp coefficient is zero

Let X be the smooth compact source of the original construction, $f : X \to B$ its map, and t the specified cusp coordinate. Put $W = f^{-1}(0)$, choose the original $x_0 \in W$, and retain the ordered quotient lattice basis induced by $(\widehat{\gamma}, \widehat{u})$, with kernel generated by $(\widehat{w}, \widehat{\delta})$. Thus the quotient torus is $\overline{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$. The original periods are

$$Z(s) = sB_0 + C(t), \qquad e^{2\pi i s} = t, \qquad B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad C(t) = \begin{pmatrix} 6\mu(t) & h(t) \\ b(t) - h(t) & \mu(t) \end{pmatrix}. \tag{148}$$

Every entry of C is holomorphic at $t = 0$. The dense torus of the original toric cusp has coordinates (x_1, x_2, t), with $x = \exp(2\pi i\zeta)$. The residual lattice action is exactly

$$\Psi_\lambda(x, t) = (t^{B_0\lambda}\exp(2\pi i C(t)\lambda)x, t), \qquad \lambda \in \mathbb{Z}^2. \tag{149}$$

It extends freely and properly discontinuously across the original unimodular toric filling. In particular the quotient projection is a local biholomorphism at every toric fixed point. These are the reconstructed data and proved local properties of the original source [2, §4, source pp. 24–29].

Let

$$K^{(1)} = (\mathbb{R}/\mathbb{Z} \times \{0\}) \cup (\{0\} \times \mathbb{R}/\mathbb{Z}), \qquad c : \overline{T}^2 \longrightarrow \overline{T}^2/K^{(1)} \cong S^2$$

be the specified degree-one cell quotient. Its restriction to the open two-cell is taken with the smooth coordinates used in the original regular-fibre description. Both radial endpoints and the complete basepoint cylinder are collapsed in

$$\Sigma_r(S^2, *) = \frac{[0,\epsilon] \times S^2}{(\{0,\epsilon\} \times S^2) \cup ([0,\epsilon] \times \{*\})} \cong S^3.$$

The original based map is

$$g(x) = [|t|, c(\overline{\pi}_f(x))] \quad (0 < |t| < \epsilon), \qquad g(W) = g(X \setminus \operatorname{int} N_0) = *. \tag{150}$$

Theorem 6.6 (The original cusp map is null-homotopic)**.** *The based map $g : (X, x_0) \to (S^3, *)$ in* (150) *is based null-homotopic. More precisely, a regular inverse image with its actual ordered normal framing induced by (r, c_1, c_2) is framed nullbordant inside one original cusp chart times $[0,1]$, disjoint from $\{x_0\} \times [0,1]$.*

Proof. The proof constructs the complete inverse image, compares its actual framing with an explicitly extendible framing, and gives the extension.

1. The exact complete inverse image. Retain the lift

$$u^0 = \begin{pmatrix} -1/4 \\ 1/4 \end{pmatrix}, \qquad \overline{u}^0 = (3/4, 1/4) \pmod{\mathbb{Z}^2}, \qquad B_0 u^0 = \begin{pmatrix} 1/4 \\ 1/4 \end{pmatrix}.$$

Thus $\overline{u}^0 \notin K^{(1)}$, and $B_0 u^0$ lies in the interior of the original triangle $\operatorname{conv}(0, e_1, e_2)$. Use its original unimodular chart

$$z_0 = t/(x_1 x_2), \qquad z_1 = x_1, \qquad z_2 = x_2, \qquad t = z_0 z_1 z_2. \tag{151}$$

Set

$$\ell = \log|t| < 0, \qquad R(t) = -2\pi \operatorname{Im} C(t), \qquad A(t) = \ell B_0 + R(t).$$

For a representative $\zeta = Z(s)u + k$, where $u, k \in \mathbb{R}^2$, $\operatorname{Im} s = -\ell/(2\pi)$ gives the exact identity

$$\begin{pmatrix} \log|x_1| \\ \log|x_2| \end{pmatrix} = A(t)u. \tag{152}$$

The quotient value is $u \bmod \mathbb{Z}^2$. Since B_0 is orthogonal, for all $v \in \mathbb{R}^2$,

$$\|A(t)v\| \geq (|\ell| - \|R(t)\|)\|v\|. \tag{153}$$

The boundedness of R therefore makes $A(t)$ invertible on a sufficiently small punctured disc.

For any point of the fibre quotient over $\overline{u}^0$, subtracting the appropriate original period, equivalently applying $\Psi_{-\lambda}$, gives a representative with the exact lift $u = u^0$. This representative is unique: another residual translate has logarithmic moduli differing by $A(t)\lambda$, which vanishes only for $\lambda = 0$. Conversely every dense-torus point satisfying $\log|x| = A(t)u^0$ has this quotient value, since the remaining real parts of ζ are exactly the kernel coordinates k. For each fixed t, varying $k \bmod \mathbb{Z}^2$ gives all independent phases of x_1, x_2.

The following change of coordinates uses the full original period units:

$$h_j^0(t) = \exp(-2\pi i (C(t)u^0)_j), \qquad j = 1, 2, \tag{154}$$

$$\mathcal{F}(z_0, z_1, z_2) = (w_0, w_1, w_2) = \left(\frac{z_0}{h_1^0(t) h_2^0(t)},\, h_1^0(t) z_1,\, h_2^0(t) z_2 \right). \tag{155}$$

The functions h_j^0 are nowhere-zero holomorphic units, including at $t = 0$. The product $w_0 w_1 w_2$ is exactly t. The inverse of $\mathcal{F}$ is

$$(w_0, w_1, w_2) \longmapsto \left(w_0 h_1^0(t) h_2^0(t),\, \frac{w_1}{h_1^0(t)},\, \frac{w_2}{h_2^0(t)} \right), \qquad t = w_0 w_1 w_2.$$

Thus this is a biholomorphism of the original chart restricted to the cusp disc, not a replacement of $C(t)$.

For $r_0 \in (0, \epsilon)$, the entire inverse image of $y_0 = [r_0, c(\overline{u}^0)]$ becomes exactly

$$T_{r_0} = \{|w_0| = \rho_0,\ |w_1| = \rho_1,\ |w_2| = \rho_2\}, \qquad (\rho_0, \rho_1, \rho_2) = (r_0^{1/2}, r_0^{1/4}, r_0^{1/4}). \tag{156}$$

Indeed, $\log|h^0(t)| = 2\pi \operatorname{Im} C(t)u^0 = -R(t)u^0$, so (152) gives $\log|w_j| = \ell(B_0 u^0)_j$ for $j = 1, 2$; the product identity gives the exponent $1/2$ for w_0. Conversely these moduli recover $u = u^0$ by invertibility of $A(t)$. The angular map is explicitly

$$(\arg t, \arg w_1, \arg w_2) \longleftrightarrow (\arg w_0, \arg w_1, \arg w_2), \qquad \arg w_0 = \arg t - \arg w_1 - \arg w_2 \pmod{2\pi},$$

with inverse $\arg t = \arg w_0 + \arg w_1 + \arg w_2$. Its integral matrix has determinant one. This includes the complete angular mapping torus and all kernel phases, with their original signs.

Let o be the origin of (151). The original free properly discontinuous action supplies a neighborhood V of o on which the quotient projection is injective. To see the local injectivity directly, first choose a relatively compact neighborhood; proper discontinuity leaves only finitely many translates that can intersect it. None of those nonidentity translates fixes o. Shrinking finitely many times separates the neighborhood from all of them. The quotient map is a local biholomorphism, so its restriction to V is a chart of the original source. Choose a polydisc about 0 whose closure lies in $\mathcal{F}(V)$. For sufficiently small r_0, both T_{r_0} and

$$\{|w_0| \le \rho_0,\ |w_1| = \rho_1,\ |w_2| = \rho_2\} \tag{157}$$

lie in this polydisc. This proves that the whole inverse image embeds in one chart and supplies space for the bounding manifold below.

2. The actual normal framing, including its angle dependence. Put $L_j = \log|w_j|$, and use the unique real lift of the quotient coordinate near u^0. From the exact change of coordinates,

$$\begin{pmatrix} L_1 \\ L_2 \end{pmatrix} = \ell B_0 u^0 + A(t)(u - u^0), \qquad L_0 = \ell - L_1 - L_2. \tag{158}$$

Differentiating on $u = u^0$ annihilates the term $(dA(t))(u - u^0)$. Hence no derivative of $C(t)$ has been omitted: the exact differential there is

$$\begin{pmatrix} dL_0 \\ dL_1 \\ dL_2 \end{pmatrix} = J(t) \begin{pmatrix} d\ell \\ du_1 \\ du_2 \end{pmatrix}, \quad J(t) = \begin{pmatrix} 1/2 & -A_{11} - A_{21} & -A_{12} - A_{22} \\ 1/4 & A_{11} & A_{12} \\ 1/4 & A_{21} & A_{22} \end{pmatrix}. \tag{159}$$

Adding the second and third rows to the first gives $(1,0,0)$, and expansion in that row proves $\det J(t) = \det A(t)$.

Let $D = Dc_{\overline{u}^0}$ be the actual two-by-two derivative in the ordered open-cell target coordinates (c_1, c_2). It is a fixed invertible matrix; all its entries and signs are retained. Let

$$H = \begin{pmatrix} r_0 & 0 & 0 \\ 0 & D_{11} & D_{12} \\ 0 & D_{21} & D_{22} \end{pmatrix}.$$

Let n_j be the real radial vector w_j in the j-th complex coordinate, with other components zero. Then $dL_i(n_j) = \delta_{ij}$. Writing $N = (n_0, n_1, n_2)$ as the matrix with these vector columns, the actual dg-dual normal framing is, in the normal quotient bundle,

$$\mathfrak{n}(t) = NJ(t)H^{-1}. \tag{160}$$

Indeed, $dg(N) = HJ(t)^{-1}$, so multiplication by $J(t)H^{-1}$ gives the identity. In the Euclidean metric of this chart the radial vectors are perpendicular to the product torus; a different source metric changes their representatives by the canonical projection to its normal bundle and preserves this normal-quotient framing.

Put $\ell_0 = \log r_0$. Decrease r_0 further, if necessary, so that $|\ell_0| > 2\sup_{|t|\le r_0} \|R(t)\|$. For $0 \le \sigma \le 1$, define only a homotopy of frames by

$$A_\sigma(t) = \ell_0 B_0 + \sigma R(t),$$

and let $J_\sigma(t)$ be (159) with A_σ in place of A. For every $v \in \mathbb{R}^2$,

$$\|A_\sigma(t)v\| \ge (|\ell_0| - \|R(t)\|)\|v\| > 0 \quad (v \ne 0).$$

Thus all matrices are invertible. At $\sigma = 0$,

$$J_0 = \begin{pmatrix} 1/2 & \ell_0 & -\ell_0 \\ 1/4 & 0 & \ell_0 \\ 1/4 & -\ell_0 & 0 \end{pmatrix}, \qquad \det J_0 = \ell_0^2 > 0. \tag{161}$$

Continuity also proves $\det J_\sigma(t) > 0$ for every σ, t. The homotopy $NJ_\sigma(t)H^{-1}$, run from $\sigma = 1$ to $\sigma = 0$, compares the actual frame with the exact constant-matrix frame NP, where

$$P = J_0 H^{-1}. \tag{162}$$

The original embedded torus and original functions $C(t)$ have not changed. A framing homotopy is a framed cylinder: choose a smooth parameter function constant near its endpoints and use the displayed frames on $T_{r_0} \times [0,1]$.

3. A collared framed nullbordism with the exact constant matrix. Use a new time coordinate $v \in [0,1]$. Define

$$q(v) = \begin{cases} 0, & 0 \le v \le 1/4, \\ \rho_0^2 \exp(4 - 1/(v - 1/4)), & 1/4 < v \le 1. \end{cases} \tag{163}$$

This function is smooth: all right derivatives of $\exp(-1/(v-1/4))$ at $1/4$ vanish. It is strictly increasing for $v > 1/4$, satisfies $q(1/2) = \rho_0^2$, and $q'(1/2) = 16\rho_0^2 > 0$. In the chart times $[0,1]$, put

$$\mathcal{W} = \{|w_0|^2 + q(v) = \rho_0^2, \quad |w_1|^2 = \rho_1^2, \quad |w_2|^2 = \rho_2^2\}. \tag{164}$$

No point has $v > 1/2$. The three defining differentials are independent: the latter two have nonzero gradients in their separate complex factors; the first has nonzero w_0-gradient whenever $w_0 \neq 0$, and at $w_0 = 0$ it has the nonzero v-derivative $q'(1/2)$. Thus $\mathcal{W}$ is a smooth compact four-manifold. Its only boundary is $T_{r_0} \times \{0\}$, and on $0 \leq v \leq 1/4$ it is the exact product collar $T_{r_0} \times [0, 1/4]$. The north locus $w_0 = 0, v = 1/2$ is an interior cap: the implicit function theorem expresses v smoothly in terms of $(\operatorname{Re} w_0, \operatorname{Im} w_0)$ there. The first factor is a disk obtained by rotating its radial profile with this smooth cap and one boundary circle. Therefore $\mathcal{W} \cong D^2 \times T^2$. Its spatial coordinates lie in (157), so the complete manifold lies in the injective original chart.

In the Euclidean chart product metric define

$$V_0 = (w_0, 0, 0, q'(v)/2), \qquad V_1 = (0, w_1, 0, 0), \qquad V_2 = (0, 0, w_2, 0).$$

These are one half of the gradients of the three defining equations in (164). They are mutually orthogonal and never zero. For example, if $F_0 = |w_0|^2 + q(v) - \rho_0^2$, then

$$dF_0(V_0) = 2|w_0|^2 + \frac{1}{2}q'(v)^2 > 0$$

at every point of $\mathcal{W}$. On the entire collar $q' = 0$, so $(V_0, V_1, V_2) = (N, 0)$ exactly. Consequently

$$(V_0, V_1, V_2)P \tag{165}$$

is a framing of the normal three-plane bundle of $\mathcal{W}$, and its restriction to the collar is exactly the framing NP from (162), with every original target-coordinate sign retained. Concatenate the framing cylinder of the preceding paragraph with this nullbordism, rescaling the two interval lengths. Their framings agree exactly on the common product collar, so the concatenation is a smooth framed nullbordism of the actual inverse image.

4. The based map represented by this framed inverse image. Choose the injective chart to exclude x_0 if its center is not x_0. If its center is x_0, the constructed nullbordism still excludes x_0, since $|w_1| = \rho_1 > 0$ and $|w_2| = \rho_2 > 0$ everywhere on it. Compactness therefore gives tubular neighborhoods for the framed cylinder and cap disjoint from $\{x_0\} \times [0, 1]$.

The original continuous quotient map is smooth and a submersion near its entire inverse image of y_0. First make it constant in a neighborhood of x_0, by a based homotopy that changes nothing near that inverse image. Explicitly, take a target chart $\kappa : O \to \mathbb{R}^3$ with $\kappa(*) = 0$, whose image is a ball, and $y_0 \notin O$. By continuity choose a neighborhood U of x_0 with $g(U) \subset O$, and a smooth bump function β supported in U and equal to one near x_0. For $x \in U$, replace $g(x)$ during this homotopy by $\kappa^{-1}((1 - a\beta(x))\kappa(g(x)))$, $0 \leq a \leq 1$, and leave g fixed off U. Convexity of the target chart image makes the formula well-defined, the vanishing of β near the boundary of its support proves continuity of the pasted formula, and $\kappa(g(x_0)) = 0$ makes the homotopy based. It never introduces an inverse image of y_0. Its endpoint is smooth near x_0 and agrees with g near the entire old inverse image.

Denote the endpoint of the preceding homotopy by g_b. Choose one closed tubular neighborhood U_1 of the inverse image, contained in the smooth submersion region and with the entire inverse image contained in $\operatorname{int} U_1$. The compact set $X \setminus \operatorname{int} U_1$ has image under g_b missing y_0. In a fixed target metric put

$$d_0 = \min_{x \in X \setminus \operatorname{int} U_1} d(g_b(x), y_0) > 0.$$

Relative smooth approximation, fixed on the whole closed set U_1 and on a neighborhood of x_0, gives a smooth based representative g_{sm} whose error on $X \setminus \operatorname{int} U_1$ is less than $d_0/2$. It therefore has the

same inverse image and differential on U_1, and $d(g_{\mathrm{sm}}(x), y_0) > d_0/2$ on its complement. Thus there is no uncontrolled intervening region and no additional inverse image. Make the approximation also closer than a fixed target injectivity radius; its short geodesics give the based homotopy.

The relative Pontryagin construction now applies to the explicit framed nullbordism. It produces a based homotopy to a map missing y_0; stereographic contraction of $S^3 \setminus \{y_0\} \cong \mathbb{R}^3$ to the image of $*$ is based and contracts that last map. For completeness, the dependence on the framed inverse image in this argument is precisely the local product-neighborhood construction and the interpolation lemma of [1, §7, Theorem B and Lemma 4, pp. 43, 48–50]. Their relative versions fix the source basepoint because all tubular modifications are away from it and stereographic interpolation between equal values keeps that value fixed. A framed cylinder realizes a framing homotopy; the empty Pontryagin manifold realizes a map missing the chosen regular value. This proves the claimed based null-homotopy of the exact map g. □

Corollary 6.7. *Under the original fixed based identification $X \cong S^6$, the one-copy coefficient is*

$$[g] = 0 \in \pi_6(S^3) \cong \mathbb{Z}/12, \qquad a = 0 \pmod{12}, \qquad [E^6 g] = 0 \in \pi_{12}(S^9) \cong \mathbb{Z}/24.$$

The previously defined homomorphism $\Theta_g : \mathbb{Z}/12 \to \pi_6(S^3)$, $\Theta_g([n]) = n[g]$, and its composite with the marked peripheral character are zero.

Proof. The theorem states that the actual based map is null-homotopic. Precomposition with the fixed based identification preserves a based null-homotopy. Suspension preserves it as well. Thus the unique coefficient of $[g]$ in any generator of $\mathbb{Z}/12$ is zero. For every $[n]$, $\Theta_g([n]) = n \cdot 0 = 0$, and every composite with this homomorphism is zero. □

Exact scope. The proof computes this sphere-valued postcomposition. It does not compute the homotopy class of the earlier conical-target map $q_2 : X \to \mathrm{Susp}(S^1 \times \overline{T}^2)$, and it does not assert a null-homotopy for that different target. Point-set surjectivity of g and all of its stated metric restrictions remain properties of the original map: a homotopy-class computation does not alter the defining formula (150).

6.4 The retained-angle cusp quotient and its nonzero top-cell class

The exact object. The notation refers to the specified threefold X, its fixed based smooth identification with S^6, and its cusp filling from [12, §§3.7,4.1–4.5]. The native quotient and its compact extension are the maps in [13]; their full formulas are recalled below. This calculation uses the original period matrix and native period lattices. It computes a new postcomposition of that particular quotient; it does not use the homotopy class of the angle-forgetting map $g : X \to S^3$.

6.4.1 Original coordinates and the specified sphere-valued map

The ordered quotient lattice is

$$\overline{\Lambda} = \mathbb{Z}\langle \overline{\widehat{\gamma}}, \overline{\widehat{u}} \rangle, \qquad K_\Lambda = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle, \qquad \overline{T}^2 = \mathbb{R}^2/\mathbb{Z}^2.$$

In these ordered bases the original period data are

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad Z(s) = sB_0 + C(t), \qquad t = e^{2\pi i s}, \qquad C(t) = \begin{pmatrix} 6\mu(t) & h_{\mathrm{per}}(t) \\ b(t) - h_{\mathrm{per}}(t) & \mu(t) \end{pmatrix}. \tag{166}$$

Here $b = \beta + \tau$, $s = \tau - h_{\mathrm{per}}(t)$, and every entry of C is holomorphic at $t = 0$, exactly as in the original source. The subscript in h_{per} distinguishes that period function from the sphere-valued map below. Write

$$R(t) = -2\pi \operatorname{Im} C(t), \qquad \ell = \log|t|, \qquad A(t) = \ell B_0 + R(t). \tag{167}$$

Choose a closed cusp disc $|t| \le \epsilon$ inside the original coordinate disc and small enough for the quotient construction. On a slightly larger closed disc put $M = \sup \|R(t)\| < \infty$. We will choose a regular value with $0 < r_0 < \epsilon$ and $|\log r_0| > M$. This chooses a point of the already specified map; it does not change that map or the cutoff radius ϵ.

The original covering model on its dense torus is

$$x_j = e^{2\pi i \zeta_j}, \qquad \Psi_n(x,t) = \big(e^{2\pi i C(t) n} t^{B_0 n} x, t\big), \qquad n \in \mathbb{Z}^2. \tag{168}$$

For a branch of s, writing $\zeta = Z(s)u + k$ with $u, k \in \mathbb{R}^2$ gives

$$\log|x| = A(t)u. \tag{169}$$

Indeed, $\operatorname{Im} s = -\ell/(2\pi)$, so $-2\pi \operatorname{Im}(Z(s)u + k) = \ell B_0 u + R(t)u$. Since B_0 is orthogonal,

$$\big|(\ell B_0 + vR(t))\xi\big| \ge (|\ell| - vM)|\xi| > 0 \quad (0 \le v \le 1,\ \xi \ne 0) \tag{170}$$

whenever $|\ell| > M$. Thus $A(t)$ is invertible there. Equation (168) changes $A(t)^{-1}\log|x|$ by n, so the native quotient is exactly

$$\overline{\pi}_f([x,t]) = [A(t)^{-1}\log|x|] \in \mathbb{R}^2/\mathbb{Z}^2. \tag{171}$$

This also follows from the flat marking in [13].

Let

$$T^3 = S^1_\phi \times \overline{T}^2, \qquad S^1_\phi = \mathbb{R}/(2\pi\mathbb{Z}),$$

with coordinate basepoints $\phi = 0$, $u_1 = 0$, $u_2 = 0$. Its coordinate two-skeleton is the following specified closed subset:

$$K^{(2)} = (\{0\} \times \overline{T}^2) \cup (S^1_\phi \times \{u_1 = 0\}) \cup (S^1_\phi \times \{u_2 = 0\}). \tag{172}$$

The product CW structure has one open three-cell, with coordinates $(\phi, u_1, u_2) \in (0, 2\pi) \times (0,1)^2$. Consequently

$$c_3 : T^3 \longrightarrow T^3/K^{(2)} \cong S^3$$

is the degree-one quotient, where the orientation of S^3 is fixed by the displayed coordinate order. One explicit identification of its open cell with $\mathbb{R}^3$ is

$$(\phi, u_1, u_2) \longmapsto \big(-\cot(\phi/2), -\cot(\pi u_1), -\cot(\pi u_2)\big).$$

Each derivative in its own variable is strictly positive, and approach to any boundary face leaves every compact subset of $\mathbb{R}^3$. Extending by its point at infinity gives the stated based sphere identification and verifies its continuity on the collapsed skeleton.

The exact conical target and map are

$$\mathcal{A} = \frac{[0,\epsilon] \times T^3}{\{0\} \times T^3 \sim v_0, \quad \{\epsilon\} \times T^3 \sim v_1}, \tag{173}$$

$$q_2(x) = [r, e^{i\phi}, \overline{\pi}_f(x)] \qquad (t(f(x)) = re^{i\phi},\ 0 < r < \epsilon),$$

$$q_2(W) = v_0, \qquad q_2\big(X \setminus \operatorname{int} N_0\big) = v_1. \tag{174}$$

Fix the same source basepoint $x_{\mathrm{base}} \in W$. Define

$$p_3 : \mathcal{A} \longrightarrow \Sigma_r(S^3, *), \qquad p_3([r,z]) = [r, c_3(z)], \qquad H = p_3 q_2 : (X, x_{\mathrm{base}}) \longrightarrow (S^4, *). \tag{175}$$

The reduced suspension here collapses both endpoint copies of S^3 and the entire interval over $*$ to one point. Thus p_3 sends both v_0 and v_1 to $*$ and is well-defined and based. It is continuous by the defining quotient topology. An explicit identification of its open four-cell with $\mathbb{R}^4$ is

$$(r, \phi, u_1, u_2) \longmapsto (-\cot(\pi r/\epsilon), -\cot(\phi/2), -\cot(\pi u_1), -\cot(\pi u_2)). \tag{176}$$

This fixes the target orientation. Replacing r locally by $\ell = \log r$ preserves this orientation, since $dr/d\ell = r > 0$. The ordered coordinates in (176) specify the sphere identification used throughout this calculation.

Theorem 6.8. *For the exact map* (175)*,*

$$[H] = \eta_4 \circ \eta_5 \neq 0 \quad \textit{in} \quad \pi_6(S^4) \cong \mathbb{Z}/2, \tag{177}$$

where $\eta_n : S^{n+1} \to S^n$ is the $(n-2)$-fold reduced suspension of the complex Hopf map. In particular q_2 is not based null-homotopic. Its stabilized top-cell postcomposition is the element $\eta^2 \in \pi_2^{\mathrm{S}}$.

6.4.2 The entire regular inverse image lies in one original chart

Choose the actual lift

$$u_0 = (-1/4, 1/4)^{\mathsf{T}}, \qquad p_0 = B_0 u_0 = (1/4, 1/4)^{\mathsf{T}}, \qquad \phi_0 = \pi. \tag{178}$$

Modulo $\mathbb{Z}^2$, u_0 is $(3/4, 1/4)$ in the open cell. Put $t_0 = r_0 e^{i\phi_0}$ and $y_0 = [r_0, c_3(e^{i\phi_0}, [u_0])]$. The numbers r_0 and t_0 will remain nonzero throughout the framing calculation.

The original triangle $\{0, e_1, e_2\}$ has toric chart

$$\mathcal{U}_\sigma \cong \mathbb{C}^3, \qquad (z_0, z_1, z_2) = (t/(x_1 x_2), x_1, x_2), \qquad t = z_0 z_1 z_2. \tag{179}$$

These characters are dual to $(0,0,1), (1,0,1), (0,1,1)$, respectively, as direct pairing verifies. This is the chart of [12, Lemma 4.2(iii)], with its original coordinates.

For $j = 1, 2$, introduce the nowhere vanishing holomorphic functions

$$h_j(t) = \exp\bigl(-2\pi i (C(t) u_0)_j\bigr). \tag{180}$$

The following is an exact coordinate map, with the exact inverse displayed alongside it:

$$F(z) = (w_0, w_1, w_2) = \bigl(z_0/(h_1(t)h_2(t)),\ h_1(t) z_1,\ h_2(t) z_2\bigr), \tag{181}$$

$$F^{-1}(w) = \bigl(w_0 h_1(t) h_2(t),\ w_1/h_1(t),\ w_2/h_2(t)\bigr), \quad t = w_0 w_1 w_2. \tag{182}$$

Both maps are holomorphic wherever $|t|$ is in the period disc, both preserve t, and substitution proves that both composites are the identity. In particular the maps extend across the origin and across every coordinate divisor. No period matrix is replaced by a different matrix in this coordinate change.

Lemma 6.9. *For sufficiently small $r_0 > 0$, the whole inverse image $M = H^{-1}(y_0)$ is represented injectively inside one contractible coordinate neighborhood in X. In its w-coordinates it is exactly the embedded two-torus*

$$w(a,b) = \bigl(\rho_0 e^{i(\phi_0 - a - b)}, \rho_1 e^{ia}, \rho_2 e^{ib}\bigr), \qquad \rho_0 = r_0^{1/2}, \quad \rho_1 = \rho_2 = r_0^{1/4}, \quad (a,b) \in \bigl(\mathbb{R}/(2\pi\mathbb{Z})\bigr)^2. \tag{183}$$

Proof. The point y_0 lies off the collapsed subset of the target, so (175) and (174) imply

$$M = \{x : t(f(x)) = t_0,\ \overline{\pi}_f(x) = [u_0]\}.$$

Any lift of such a point to $(\mathbb{C}^*)^2 \times \{t_0\}$ has $A(t_0)^{-1} \log|x| = u_0 + n$ for one $n \in \mathbb{Z}^2$. Applying the actual transformation Ψ_{-n} from (168) produces a lift with $\log|x| = A(t_0)u_0$. If two such lifts represent the same point, Ψ_n relates them and changes their lifted quotient coordinate by n; hence $n = 0$, and the lifts are identical. This proves both completeness and injectivity of this chosen representative.

Since $\log|h_j(t)| = 2\pi \operatorname{Im}(C(t)u_0)_j = -(R(t)u_0)_j$, its w-coordinates satisfy

$$(\log|w_1|, \log|w_2|)^{\mathsf{T}} = A(t_0)u_0 - R(t_0)u_0 = \ell_0 p_0, \qquad \ell_0 = \log r_0.$$

Thus $|w_1| = |w_2| = r_0^{1/4}$, and the unchanged product $w_0 w_1 w_2 = t_0$ forces $|w_0| = r_0^{1/2}$ and the displayed phase. Conversely every pair (a, b) in (183) gives one such lift. The original kernel coordinates k_j had period one in $x_j = e^{2\pi i \zeta_j}$; the phase coordinates a, b have period 2π, with no cover or change of lattice.

The covering $Y_\epsilon \to N_0$ is injective on a neighborhood of the origin in (179), because the original action is free and properly discontinuous. Choose a ball V centered at that origin with closure inside such a neighborhood and the period disc. Choose a ball V_w about the origin with closure contained in $F(V)$. Every coordinate in (183) tends uniformly to zero as $r_0 \downarrow 0$. Therefore for sufficiently small r_0 the entire torus lies in V_w. The inverse image of V_w under F, followed by the injective covering projection, is the required contractible chart in X. □

6.4.3 The full differential and its ordered normal framing

We calculate in a neighborhood of the compact torus (183). The quotient coordinates admit a unique local lift near u_0. Set

$$v = (\log|w_1|, \log|w_2|)^{\mathsf{T}}, \quad \ell = \log|w_0 w_1 w_2|, \quad \phi = \arg(w_0 w_1 w_2)$$

using the branch of ϕ near $\phi_0 = \pi$. Equations (169) and (181) give the exact identity

$$v = \ell B_0 u + R(t)(u - u_0), \qquad u = u_0 + A(t)^{-1}(v - \ell p_0). \tag{184}$$

Differentiate this identity before restricting to the inverse image:

$$du = -A(t)^{-1}(dA(t))A(t)^{-1}(v - \ell p_0) + A(t)^{-1}(dv - p_0\, d\ell). \tag{185}$$

On M the vector $v - \ell p_0$ is exactly zero. Therefore

$$du|_M = A_0^{-1}(dv - p_0\, d\ell), \qquad A_0 = A(t_0). \tag{186}$$

This explicitly accounts for all derivatives of the original bounded period terms; none are omitted or estimated away.

Identify $T_w\mathbb{C}^3$ with $\mathbb{C}^3$ in the given coordinate chart, and define real tangent vectors

$$L_j = (0, \dots, w_j, \dots, 0), \qquad Q_j = (0, \dots, iw_j, \dots, 0) \qquad (0 \le j \le 2). \tag{187}$$

Thus $d\log|w_i|(L_j) = \delta_{ij}$, $d\arg w_i(Q_j) = \delta_{ij}$, and the crossed evaluations vanish. The ordered tangent vectors of the parametrization (183) are

$$T_a = -Q_0 + Q_1, \qquad T_b = -Q_0 + Q_2. \tag{188}$$

They are independent because their w_1 and w_2 components, respectively, are nonzero. For the target order (ℓ, ϕ, u_1, u_2), define the following four lifts of its coordinate vectors:

$$N_\ell = \tfrac{1}{2}L_0 + \tfrac{1}{4}L_1 + \tfrac{1}{4}L_2, \qquad N_\phi = Q_0, \tag{189}$$

$$N_i = -\big((A_0)_{1i} + (A_0)_{2i}\big)L_0 + (A_0)_{1i}L_1 + (A_0)_{2i}L_2 \qquad (i = 1, 2). \tag{190}$$

Let $P = (\ell, \phi, u_1, u_2)$. Evaluation using (186) gives the entire differential table

	N_ℓ	N_ϕ	N_1	N_2	T_a	T_b
$d\ell$	1	0	0	0	0	0
$d\phi$	0	1	0	0	0	0
du_1	0	0	1	0	0	0
du_2	0	0	0	1	0	0

. (191)

In particular P, and therefore H in its open target cell, is a submersion along M. The classes of $(N_\ell, N_\phi, N_1, N_2)$ in $TX|_M/TM$ give the specified normal four-framing. Orthogonal representatives are not required for this quotient-bundle definition; projecting the displayed lifts along TM to any chosen orthogonal complement preserves the quotient frame.

If the normal frame is specified using the literal four real coordinates in (176), their differential at the chosen value is

$$dH_{\text{cell}} = K\, dP, \qquad K = \operatorname{diag}\left(\frac{\pi r_0}{\epsilon}\csc^2\frac{\pi r_0}{\epsilon}, \frac{1}{2}, 2\pi, 2\pi\right).$$

The first factor includes $dr = r_0\, d\ell$, the second uses $\phi_0 = \pi$, and the final two use the actual representatives $(3/4, 1/4)$ of u_0. Thus, writing $\mathcal{N} = (N_\ell, N_\phi, N_1, N_2)$, the literal dual frame is $\mathcal{N}K^{-1}$. All four entries of K are positive constants on M. The explicitly positive diagonal matrices $(1-v)K^{-1} + vI_4$, $0 \le v \le 1$, give a homotopy from that literal frame to $\mathcal{N}$. This retains the original target coordinates and establishes that either coordinate description gives the same framed bordism class.

For completeness, the original period matrix admits the following explicit homotopy of this normal framing. Put

$$A_{0,v} = \ell_0 B_0 + vR(t_0), \qquad 0 \le v \le 1, \tag{192}$$

and replace A_0 by $A_{0,v}$ only in (190). Equation (170) proves invertibility for every v. Their images under the actual (du_1, du_2) are the columns of $A_0^{-1}A_{0,v}$, so these are normal frames throughout the homotopy. This is a homotopy of the computed frame on the same embedded torus, not a replacement of the original degeneration by a different one.

Lemma 6.10 (The exact orientation sign)**.** *Use the complex real-coordinate orientation, whose ordered coordinates are*

$$(\operatorname{Re} w_0, \operatorname{Im} w_0, \operatorname{Re} w_1, \operatorname{Im} w_1, \operatorname{Re} w_2, \operatorname{Im} w_2).$$

Then one has

$$\det(N_\ell, N_\phi, N_1, N_2, T_a, T_b) = -\det(A_0)|w_0 w_1 w_2|^2 < 0. \tag{193}$$

Hence the tangent orientation induced by the ordered normal frame is the orientation of $(-T_a, T_b)$.

Proof. The ordered frame $(L_0, Q_0, L_1, Q_1, L_2, Q_2)$ has determinant $|w_0|^2|w_1|^2|w_2|^2 > 0$ in real coordinates. Regroup its rows as $(L_0, L_1, L_2, Q_0, Q_1, Q_2)$; this permutation has three inversions and sign -1. Regroup the columns of the frame in (193) as $(N_\ell, N_1, N_2, N_\phi, T_a, T_b)$; this has two inversions and sign $+1$. The angular block is $\left(\begin{smallmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$, of determinant one. The radial block

has first column $(1/2, 1/4, 1/4)^{\mathsf{T}}$ and the other columns given by (190). Adding its last two rows to its first row gives first row $(1, 0, 0)$, with remaining lower-right block A_0. Its determinant is $\det A_0$. This proves the equality. Equation (192) joins A_0 to $\ell_0 B_0$ through invertible matrices, so $\operatorname{sign}\det A_0 = \operatorname{sign}\det(\ell_0 B_0) = +1$. The inequality and the tangent orientation follow. □

6.4.4 An explicit contraction of the full frame

Define a complex-linear matrix, also regarded as a real-linear matrix,

$$D(a, b) = \operatorname{diag}(e^{-i(a+b)}, e^{ia}, e^{ib}) \in \mathrm{SU}(3). \tag{194}$$

By (183), every L_j and Q_j at (a, b) is $D(a, b)$ times its value at $(0, 0)$. The coefficients in (189), (190), and (188) are constant on M, since $t = t_0$. It follows that the positively oriented full frame is exactly

$$(N_\ell, N_\phi, N_1, N_2, -T_a, T_b)_{(a,b)} = D(a, b)G_0, \tag{195}$$

where $G_0 \in \mathrm{GL}^+(6, \mathbb{R})$ is its single value at $(0, 0)$. Thus neither an unspecified phase nor a position-dependent real-linear matrix remains.

Here is a periodic contraction of D, with all parameters specified. For $v \in [0, 1]$ put

$$s_v = \sin(\pi v/2), \qquad c_v = \cos(\pi v/2), \qquad Q_v(\theta) = \begin{pmatrix} s_v e^{-i\theta} & -c_v \\ c_v & s_v e^{i\theta} \end{pmatrix}, \qquad K_v(\theta) = Q_v(\theta) Q_v(0)^*.$$

The two columns of Q_v have norm one and zero Hermitian inner product, and $\det Q_v = s_v^2 + c_v^2 = 1$. Thus $Q_v, K_v \in \mathrm{SU}(2)$. Direct substitution gives

$$K_0(\theta) = I, \quad K_1(\theta) = \operatorname{diag}(e^{-i\theta}, e^{i\theta}), \quad K_v(0) = I.$$

Let $\iota_{01}, \iota_{02} : \mathrm{SU}(2) \to \mathrm{SU}(3)$ act in the ordered coordinate planes $(0, 1)$ and $(0, 2)$, respectively, and as the identity in the unused coordinate. Then

$$D_v(a, b) = \iota_{01}(K_v(a))\, \iota_{02}(K_v(b)) \tag{196}$$

is a well-defined continuous map on $[0, 1] \times (\mathbb{R}/(2\pi\mathbb{Z}))^2$, with $D_0 = I$ and $D_1 = D$. This proves the needed null-homotopy directly, including both phase seams. In particular $D_v(a, b)G_0$ contracts the comparison matrix in (195) to one constant positive matrix.

Proposition 6.11. *The stable tangential framing induced by the actual normal four-frame of $H^{-1}(y_0)$ agrees, up to homotopy and the orientation sign* (193)*, with the Lie framing (∂_a, ∂_b) on the native two-torus.*

Proof. In the coordinate chart, the stable tangent trivialization is the isomorphism

$$\mathbb{R}^4 \oplus TM \longrightarrow T\mathbb{C}^3|_M, \qquad (v, V) \longmapsto v_1 N_\ell + v_2 N_\phi + v_3 N_1 + v_4 N_2 + V.$$

Using the intrinsic frame $(-\partial_a, \partial_b)$ on TM, its matrix is precisely (195). The homotopy (196) therefore makes this comparison constant. A constant positive matrix changes no homotopy class of stable framing: Gram–Schmidt deforms it to an element of $SO(6)$, and an orthogonal matrix of determinant one can be reduced to the identity by finitely many oriented plane rotations, each joined to the identity by varying its angle. This proves the asserted comparison within the chart.

It remains to compare the chart with the ambient sphere framing. Pull back the usual stable tangent trivialization $TS^6 \oplus \mathbb{R} \cong \mathbb{R}^7$ under the fixed based diffeomorphism $X \cong S^6$. On the

contractible w-ball containing M, it differs from the coordinate trivialization plus one real line by a continuous map into $\mathrm{GL}(7,\mathbb{R})$. Contracting the ball to its center contracts this comparison map to a constant. Thus its restriction to M contributes no varying stable framing class. If the chosen sphere orientation disagrees with the complex orientation, the remaining constant has negative determinant and reverses the resulting framed bordism class. Both that sign and the explicit sign (193) disappear in the order-two class computed below. This argument compares actual ambient trivializations on a contractible chart; it does not suppose that $\mathbb{C}^3$ globally trivializes TX. □

6.4.5 Nonzero framed bordism and the exact unstable class

We give the nonvanishing test for the resulting torus explicitly. A stable framing gives a spin structure by lifting its oriented frame to the trivial Spin-bundle. On the Lie-framed circle this is the disconnected double cover, whereas the spin structure induced from the boundary of a framed disc is the connected double cover: as the boundary tangent turns once, its 2π-rotation lifts from 1 to -1 in $\mathrm{Spin}(2)$. We call the former nonbounding and the latter bounding.

For a closed oriented spin surface S, the spin quadratic refinement

$$q_S : H_1(S;\mathbb{F}_2) \longrightarrow \mathbb{F}_2$$

takes a class represented by an embedded circle to zero or one according to whether the induced spin structure, using its normal line in the surface, is bounding or nonbounding. It satisfies

$$q_S(x+y) = q_S(x) + q_S(y) + (x \cdot y). \tag{197}$$

This is the usual spin quadratic form: resolving transverse intersections adds one tangent turn per intersection, and the 2π-spin-lift calculation above gives the last term. The well-defined correspondence between spin structures and such quadratic refinements is given in [18, Theorem 3.2 and Remark 3.3]; its original surface proof is [17].

For a symplectic basis $(e_i, f_i)_{i=1}^{g}$, put

$$\mathrm{Arf}(S) = \sum_{i=1}^{g} q_S(e_i) q_S(f_i) \in \mathbb{F}_2. \tag{198}$$

The basis independence can be seen without selecting a preferred basis: factor the finite sum over the symplectic planes to obtain

$$\sum_{x \in H_1(S;\mathbb{F}_2)} (-1)^{q_S(x)} = \prod_{i=1}^{g} \sum_{a,b \in \mathbb{F}_2} (-1)^{a q_S(e_i) + b q_S(f_i) + ab} = 2^g (-1)^{\sum_i q_S(e_i) q_S(f_i)}.$$

The left side has no choice of basis, proving the assertion.

Here is also the bordism-invariance argument needed for the present obstruction. A spin bordism has a handle decomposition. An index zero or three handle creates or removes a spin sphere, whose first homology and Arf invariant are zero. An index one handle either connects two components, in which case the first-homology quadratic forms add orthogonally, or increases the genus of one component. In the latter case the new symplectic pair contains the meridian of the handle. That meridian bounds the cocore disc with its induced spin structure, so its quadratic value is zero; the pair contributes zero to (198), regardless of the value on its dual. Existing cycles can be represented outside the attaching discs, so their quadratic values are preserved. An index two handle is the reverse of an index one handle. Therefore each handle preserves the total Arf invariant of the

boundary, and a spin boundary has Arf invariant zero. This also proves that a stably framed boundary has Arf invariant zero, because forgetting the framing gives a spin bordism. The full spin bordism statement is recorded in [18, Theorem 3.6].

For the intrinsic Lie framing on $M = (\mathbb{R}/(2\pi\mathbb{Z}))^2$, take

$$e = [a \mapsto (a, 0)], \qquad f = [b \mapsto (0, b)].$$

Both the tangent and normal vectors in the surface are constant in the Lie frame along either circle. Thus both circles have the nonbounding spin structure, and

$$e \cdot f = 1, \qquad q_M(e) = q_M(f) = 1, \qquad q_M(xe + yf) = x + xy + y, \qquad \operatorname{Arf}(M) = 1. \tag{199}$$

Reversing the first Lie tangent vector changes the orientation but does not change these mod-two values. In particular the specifically framed inverse image computed above is not a framed boundary.

To identify the element, let $\eta \in \pi_1^{\mathrm{S}}$ denote the stable class of the complex Hopf map. Its Pontryagin–Thom inverse image is the circle with its Lie framing. This can be verified directly on the Hopf map $S^3 \to \mathbb{CP}^1$: near the fibre $(z_1, z_2) = (e^{ia}, 0)$, the target coordinate is z_2/z_1, so the two horizontal normal vectors are $(0, e^{ia})$ and $(0, ie^{ia})$. Add the tangent vector $(ie^{ia}, 0)$ and the outward radial vector $(e^{ia}, 0)$. In $\mathbb{C}^2$ the resulting full frame makes one complete rotation in each of its two real coordinate planes. This loop is contractible explicitly: for the fixed real-linear involution $J(z_1, z_2) = (z_1, \overline{z}_2)$,

$$J \operatorname{diag}(e^{ia}, e^{ia}) J = \operatorname{diag}(e^{ia}, e^{-ia}) = K_1(-a),$$

and $JK_v(-a)J$ contracts the original loop to the identity using the periodic homotopy already constructed. Thus the induced stable tangent frame on the circle is the Lie frame. This is the basic framed-circle identification in [16].

The product of two normally framed inverse images represents the product in stable homotopy: its tubular collapse is the smash of the two tubular collapse maps, as follows immediately from the product tubular neighborhood $\nu_1 \times \nu_2$ and its ordered normal coordinates. Consequently the product of the two Lie circles represents η^2. Equation (199) proves $\eta^2 \neq 0$ by an explicit bordism invariant. The stable group $\pi_2^{\mathrm{S}} = \mathbb{Z}/2$ is Pontryagin's computation [15, 16]; the stable table in [19, pp. 384–385] gives the same group. Thus the sign from (193) satisfies $-\eta^2 = \eta^2$.

Proof of the theorem. The globally continuous quotient H is smooth on a neighborhood of its compact inverse image M, by (184) and (176). First make it constant near the source basepoint, before choosing any approximation bound. Choose a target chart $\kappa : O \to \mathbb{R}^4$ about $*$ with $\kappa(*) = 0$, convex ball image, and $y_0 \notin O$. Choose a source neighborhood U of x_{base}, disjoint from M, with $H(U) \subset O$. Let β be a smooth function supported in U, equal to one near x_{base}, with $0 \leq \beta \leq 1$. On U use the based homotopy

$$H_s(x) = \kappa^{-1}\big((1 - s\beta(x))\kappa(H(x))\big), \qquad 0 \leq s \leq 1,$$

and leave H unchanged off U. Its endpoint H_b is smooth and constant near the basepoint; it agrees with H near M and has no new inverse image of y_0.

Choose one closed tubular neighborhood U_1 of M, with $M \subset \operatorname{int} U_1$, contained in the unchanged smooth region and disjoint from the basepoint neighborhood. Compactness gives the positive bound

$$d_0 = \min_{x \in X \setminus \operatorname{int} U_1} d\big(H_b(x), y_0\big) > 0.$$

Relative smooth approximation, fixing the same closed set U_1 and a neighborhood of the basepoint, produces a smooth based map H_{sm} whose error from H_b is less than $d_0/2$ on its complement. Choose the approximation closer also than a fixed target injectivity radius; the short target geodesics give a based homotopy. Thus the exact inverse image and its differential are unchanged on U_1, and the complement acquires no new inverse image. The relative Pontryagin construction applies as in [14, §7, Theorem B and Lemma 4, pp. 43, 48–50].

Pontryagin–Thom therefore identifies the stabilization of $[H]$ with the bordism class of M carrying the four normal vectors (189)–(190). The comparison (195) and its explicit contraction identify that class with the Lie torus, with the harmless order-two orientation sign already computed. Hence

$$E^{\infty}[H] = \eta^2 \neq 0.$$

Freudenthal suspension is an isomorphism

$$E : \pi_{n+2}(S^n) \longrightarrow \pi_{n+3}(S^{n+1}) \qquad (n \geq 4),$$

because $n + 2 \leq 2n - 2$. Therefore stabilization from $\pi_6(S^4)$ is an isomorphism onto π_2^{S}. The composite $\eta_4 \circ \eta_5 : S^6 \to S^4$ stabilizes to η^2, so injectivity gives the exact equality (177). Finally, a based null-homotopy of q_2 would postcompose with p_3 to a based null-homotopy of H, contradicting (177). □

Exact scope of the comparison with the angle-forgetting map. The map here retains the angular coordinate before collapsing the coordinate two-skeleton. Its regular inverse image is the specific normally four-framed T^2 above. The angle-forgetting postcomposition has a normally three-framed T^3 inverse image. The calculation above is independent of its class and supplies an actual obstruction to null-homotopy of the original q_2, even when that other postcomposition is null-homotopic. The exact morphism carrying this obstruction is

$$(p_3)_* : [X, \mathcal{A}]_* \longrightarrow [X, S^4]_*, \qquad [q_2] \longmapsto [H] = \eta_4\eta_5.$$

No conclusion about an arbitrary earlier quotient follows merely from the diffeomorphism type of one inverse image; the specified four-framing is what proves this value.

Corollary 6.12 (The exact order-two quotient detected by the original cusp map)**.** *Let $d : (S^6, *) \to (X, x_{\mathrm{base}})$ be the fixed based diffeomorphism, and put*

$$\alpha = [q_2 \circ d] \in \pi_6(\mathcal{A}, v_0).$$

The original top-cell postcomposition induces a surjective homomorphism

$$\varepsilon = (p_3)_* : \pi_6(\mathcal{A}, v_0) \longrightarrow \pi_6(S^4, *) \cong \mathbb{Z}/2, \qquad \varepsilon(\alpha) = \eta_4 \circ \eta_5.$$

It gives the exact quotient isomorphism

$$\pi_6(\mathcal{A}, v_0)/\ker\varepsilon \xrightarrow{[\beta]\mapsto\varepsilon(\beta)} \mathbb{Z}/2.$$

Every odd multiple of α is nonzero. Its order is either infinite or an even positive integer; this calculation does not determine that order or assert a splitting of the displayed quotient.

Proof. Postcomposition distributes over the pinch sum on the source sphere, so $(p_3)_*$ is a group homomorphism. Theorem 6.8 gives $(p_3)_*(\alpha) = [p_3 q_2 d] = \eta_4\eta_5 \neq 0$. This is the nonzero element of the cyclic group of order two, proving surjectivity. The first isomorphism theorem gives the stated quotient with exactly the displayed map. For every integer n,

$$\varepsilon(n\alpha) = (n \bmod 2)\eta_4\eta_5.$$

An odd multiple therefore has nonzero image and cannot vanish. If $m\alpha = 0$, applying ε forces m to be even. These arguments prove all the assertions without inferring the exact order of α. □

6.5 The retained cusp class as a three-dimensional bundle inside the triality carrier

Retain the exact based map $H = p_3 q_2 : X \to S^4$, the fixed based diffeomorphism $d : S^6 \to X$, the original cutoff ϵ, the ordered native coordinates (r, ϕ, u_1, u_2), and every period appearing in (166). Theorem 6.8 has proved $[Hd] = \eta_4\eta_5 \neq 0$ in $\pi_6(S^4) \cong \mathbb{Z}/2$. We construct an exact bundle-valued image of this specific map and then its action on the original three Peirce sectors. A quaternion subalgebra and its ordered basis are explicit choices in this construction.

The literal Hopf pullback and its transition. Fix $\mathbb{H} = \mathbb{R}\langle 1, i, j, k\rangle$ with $i^2 = j^2 = k^2 = ijk = -1$. Identify the ordered open four-cell of the target with $\mathbb{H}$ by

$$z = -\cot(\pi r/\epsilon) - \cot(\phi/2)i - \cot(\pi u_1)j - \cot(\pi u_2)k. \tag{200}$$

This is exactly the ordered coordinate map (176), followed by the real linear map carrying the four standard basis vectors to $(1, i, j, k)$. Each of the four coordinate derivatives in the original open cell is positive. The based homeomorphism

$$\chi : S^4 = \mathbb{H} \cup \{\infty\} \longrightarrow \mathbb{HP}^1, \qquad \chi(z) = [1 : z], \qquad \chi(*) = [0 : 1] \tag{201}$$

uses right quaternionic lines. The orientation on this presentation of $\mathbb{HP}^1$ is the one induced by the ordered affine basis $(1, i, j, k)$. Thus no orientation convention on another presentation of $\mathbb{HP}^1$ is silently substituted.

Let

$$S^7 = \{(v_0, v_1) \in \mathbb{H}^2 : |v_0|^2 + |v_1|^2 = 1\}, \qquad \pi(v_0, v_1) = [v_0 : v_1].$$

The right action is $(v_0, v_1)u = (v_0 u, v_1 u)$ for $u \in \mathrm{Sp}(1) = \{u \in \mathbb{H} : |u| = 1\}$. It is free: at least one v_j is nonzero, and $v_j u = v_j$ implies $u = 1$. Two unit vectors on the same right line differ by a unique unit quaternion. Hence π is the quaternionic Hopf principal bundle. Its local sections in the affine coordinates $z = v_1 v_0^{-1}$ and $w = v_0 v_1^{-1}$ are

$$s_0(z) = \frac{(1, z)}{\sqrt{1 + |z|^2}}, \qquad s_\infty(w) = \frac{(w, 1)}{\sqrt{1 + |w|^2}}. \tag{202}$$

On $z \neq 0$, the coordinate change is $w = z^{-1}$ and

$$s_\infty(z^{-1}) = s_0(z)\, g(z), \qquad g(z) = \frac{\overline{z}}{|z|}. \tag{203}$$

Indeed, multiplying the right side gives $(\overline{z}/|z|, |z|)/\sqrt{1 + |z|^2}$, and the same expression results from the left side because $\sqrt{1 + |z^{-1}|^2} = \sqrt{1 + |z|^2}/|z|$. This calculation retains both the inverse and the conjugation.

Define

$$P_H = \{(x, v) \in X \times S^7 : \pi(v) = \chi(H(x))\} \longrightarrow X, \qquad (x, v)u = (x, vu). \tag{204}$$

The preimages under χH of the two affine charts cover X. Equations (202)–(203) are local sections and transitions for (204) after composition with χH. In particular, wherever both charts apply, the transition is the exact $g(z)$ from (203), with z given by the original four-coordinate expression (200). The angle ϕ has not been dropped.

Theorem 6.13. *The principal bundle P_H is nontrivial. Its clutching class is the nonzero element*

$$c(P_H) = \partial[\chi H d] \in \pi_5(\mathrm{Sp}(1)) \cong \mathbb{Z}/2, \tag{205}$$

where ∂ is the boundary of the specified right Hopf fibration. The associated real three-plane bundle

$$\mathcal{A}_H = P_H \times_{\mathrm{Ad}} \mathrm{Im}\,\mathbb{H}, \qquad \mathrm{Ad}_u(v) = uv\overline{u}, \tag{206}$$

is nontrivial, has nonzero order-two clutching class in $\pi_5(SO(3))$, and admits no nowhere-zero continuous section. Its oriented orthonormal frame bundle has P_H as its unique spin lift up to isomorphism over X. The maps from the retained homotopy group to these two clutching groups are isomorphisms:

$$\pi_6(S^4) \xrightarrow{\chi_*} \pi_6(\mathbb{HP}^1) \xrightarrow{\partial} \pi_5(\mathrm{Sp}(1)) \xrightarrow{\mathrm{Ad}_*} \pi_5(SO(3)). \tag{207}$$

Proof. The first map is an isomorphism because χ is a based homeomorphism. In the long exact sequence of the displayed Hopf bundle, the segment is

$$\pi_6(S^7) \longrightarrow \pi_6(\mathbb{HP}^1) \xrightarrow{\partial} \pi_5(S^3) \longrightarrow \pi_5(S^7).$$

Both outside groups vanish: cellular approximation homotopes a map from a sphere of dimension less than seven into the zero-cell of the CW structure on S^7. Exactness therefore makes ∂ an isomorphism. To fix its clutching convention, write the oriented source $S^6 = D^6_+ \cup D^6_-$, give the equator the boundary orientation of D^6_+, and choose local lifts $\sigma_\pm$ of the pulled-back bundle. On the equator define g_{+-} by $\sigma_- = \sigma_+ g_{+-}$. Choose the values of the two lifts equal at its fixed basepoint. We use the convention $\partial[\chi H d] = [g_{+-}]$. Changing disk lifts changes this map by null-homotopic factors and does not change its homotopy class. This is the defining boundary construction of the fibration exact sequence. The already proved nonzero class $[Hd]$ therefore gives the nonzero order-two class in (205).

One can also see nontriviality directly. A section of P_H gives a map $\widetilde{h} : S^6 \to S^7$ with $\pi\widetilde{h} = \chi H d$. Since $\pi_6(S^7) = 0$, this would make $\chi H d$ null-homotopic, contrary to Theorem 6.8. All spaces at issue are simply connected, so based and free null-homotopy agree here.

Conjugation by a unit quaternion preserves $\mathrm{Im}\,\mathbb{H}$ and its Euclidean norm. Its determinant is $+1$ by continuity in u and its value at $u = 1$. The kernel consists of the real unit quaternions $\{1, -1\}$: a quaternion commuting with i, j, k is real. For a unit imaginary quaternion n and $v \in \mathrm{Im}\,\mathbb{H}$, write $v = v_\parallel + v_\perp$ relative to the line $\mathbb{R}n$. Direct multiplication with $u = \cos\alpha + n\sin\alpha$ gives

$$uv\overline{u} = v_\parallel + \cos(2\alpha)v_\perp + \sin(2\alpha)(n \times v_\perp), \tag{208}$$

where $n \times v = (nv - vn)/2$. Every orientation-preserving orthogonal map of a real three-space fixes an axis and rotates its orthogonal plane: its real odd-degree characteristic polynomial has a real eigenvalue, norm preservation gives ± 1, and determinant $+1$ forces an eigenvalue $+1$. Thus (208) proves that $\mathrm{Ad} : S^3 \to SO(3)$ is onto with kernel $\{\pm 1\}$. It is the two-sheeted covering and induces

an isomorphism on π_5. Its value on (205) is the clutching class of (206), proving nontriviality and order two.

The bundle $P_H \to P_H/\{\pm 1\}$ is the spin double covering of the oriented orthonormal frames of $\mathcal{A}_H$. If that frame bundle had a section, its inverse image would be a double covering of the simply connected space $X \cong S^6$. This covering is trivial, so one of its two sections would give a section of P_H, again a contradiction. Any two lifts of an $SO(3)$ bundle differ by a principal $\mathbb{Z}/2$ bundle: locally the ratio between two lifts of each transition lies in the central kernel $\{\pm 1\}$, and the cocycle identity gives a $\mathbb{Z}/2$ cocycle. Every such bundle on the simply connected X is trivial. This proves the claimed uniqueness of the spin lift.

If $\mathcal{A}_H$ had a nowhere-zero section, divide that section by its positive length to obtain a unit section. Orthogonal complement gives $\mathcal{A}_H \cong \mathbb{R} \oplus \mathcal{B}$ for an oriented two-plane bundle $\mathcal{B}$. A two-plane bundle on S^6 has clutching map $S^5 \to SO(2) = S^1$. Every such map lifts to $\mathbb{R}$ and is null-homotopic, so $\mathcal{B}$ is trivial. This would trivialize $\mathcal{A}_H$, contradicting its nonzero clutching class. □

The fibrewise scalar product, orientation and cross product of $\operatorname{Im}\mathbb{H}$ descend to $\mathcal{A}_H$. For example,

$$(uv\overline{u})(uw\overline{u}) = u(vw)\overline{u}$$

by associativity in $\mathbb{H}$, so conjugation preserves $v \times w = (vw - wv)/2$. The induced fibrewise Lie bracket is exactly $[v, w] = 2(v \times w)$. No factor of two is discarded.

An explicit subgroup of the three-sector triality group. Fix the quaternion subalgebra above inside $\mathbb{O} = \mathbb{H} \oplus \mathbb{H}\ell$. Its ordered real basis is

$$(1, i, j, k, \ell, i\ell, j\ell, k\ell),$$

and its multiplication is

$$(a, b)(c, d) = (ac - \overline{d}b,\ da + b\overline{c}), \qquad K(a, b) = \overline{(a, b)} = (\overline{a}, -b). \tag{209}$$

This is the literal convention of Yokota, §1.10. In his cyclic related-triple presentation the group is

$$\mathrm{RT}(\mathbb{O}) = \{(T_1, T_2, T_3) \in SO(\mathbb{O})^3 : (T_1 x)(T_2 y) = K T_3 K(xy) \text{ for all } x, y \in \mathbb{O}\}. \tag{210}$$

Yokota, Theorem 1.16.2, identifies this group with Spin(8); Theorem 2.7.1 [23] gives its diagonal-fixing Jordan action. In particular the three projection representations in (210) are the vector and two half-spin sectors with that fixed marking.

For each $u \in \mathrm{Sp}(1)$ define

$$\begin{aligned} B_u(a, b) &= (ua\overline{u}, b), \\ A_u(a, b) &= (ua, b\overline{u}), \\ D_u(a, b) &= (a\overline{u}, b\overline{u}) = K A_u K(a, b). \end{aligned} \tag{211}$$

Theorem 6.14. *The formula*

$$\rho : \mathrm{Sp}(1) \longrightarrow \mathrm{Spin}(8) = \mathrm{RT}(\mathbb{O}), \qquad \rho(u) = (B_u, A_u, D_u) \tag{212}$$

is an injective continuous group homomorphism. It satisfies

$$(B_u x)(A_u y) = A_u(xy), \qquad \mathrm{Re}(((B_u x)(A_u y))(D_u z)) = \mathrm{Re}((xy)z) \tag{213}$$

for all octonions x, y, z and every unit quaternion u.

Proof. All three maps are real linear norm isometries by the quaternionic norm identity. Their determinants are $+1$: they vary continuously with u on the connected S^3 and equal the identity at $u = 1$. For $u, v \in \mathrm{Sp}(1)$, associativity in $\mathbb{H}$ and $\overline{uv} = \overline{v}\,\overline{u}$ give

$$B_u B_v = B_{uv}, \qquad A_u A_v = A_{uv}, \qquad D_u D_v = D_{uv}.$$

For example, the second coordinate of $A_u A_v(a, b)$ is $(b\overline{v})\overline{u} = b\overline{uv}$, retaining the order.

For $x = (a, b)$ and $y = (c, d)$, expansion of (209) gives

$$\begin{aligned}(B_u x)(A_u y) &= (ua\overline{u}, b)(uc, d\overline{u}) \\ &= \big((ua\overline{u})(uc) - \overline{d\overline{u}}\, b, (d\overline{u})(ua\overline{u}) + b\overline{uc}\big) \\ &= \big(u(ac - \overline{d}b), (da + b\overline{c})\overline{u}\big) = A_u(xy).\end{aligned}$$

This uses only quaternionic associativity in the displayed components. Since $A_u = K D_u K$, it proves membership in (210). If $\rho(u)$ is the identity, evaluate A_u at $(1, 0)$ to get $(u, 0) = (1, 0)$, so $u = 1$. Thus ρ is injective.

The other two cyclic pair identities, with every conjugation retained, are

$$\begin{aligned}(A_u x)(D_u y) &= (ua, b\overline{u})(c\overline{u}, d\overline{u}) \\ &= \big(u(ac - \overline{d}b)\overline{u},\ da + b\overline{c}\big) = B_u(xy) = K B_u K(xy), \\ (D_u x)(B_u y) &= (a\overline{u}, b\overline{u})(uc\overline{u}, d) \\ &= \big((ac - \overline{d}b)\overline{u}, (da + b\overline{c})\overline{u}\big) = D_u(xy) = K A_u K(xy).\end{aligned}$$

Put $\langle v, w\rangle = \mathrm{Re}(v\overline{w})$ on $\mathbb{O}$. Then $\mathrm{Re}(vz) = \langle v, Kz\rangle$. The already proved product relation and $K D_u = A_u K$ imply

$$\begin{aligned}\mathrm{Re}(((B_u x)(A_u y))(D_u z)) &= \langle A_u(xy), K D_u z\rangle \\ &= \langle A_u(xy), A_u K z\rangle = \langle xy, Kz\rangle = \mathrm{Re}((xy)z).\end{aligned}$$

No reassociation of a non-real octonionic triple product was used. □

The exact original Jordan coordinates and all three bundles. Write the original Hermitian matrix with upper entries (p, q, r) as

$$J(\lambda; p, q, r) = \begin{pmatrix} \lambda_1 & p & q \\ \overline{p} & \lambda_2 & r \\ \overline{q} & \overline{r} & \lambda_3 \end{pmatrix}, \qquad (q_1, q_2, q_3) = (r, \overline{q}, p). \tag{214}$$

Thus q_2 here is the cyclic Jordan coordinate, not the previously defined cusp quotient map; only the explicitly displayed ordered triple in this paragraph uses this coordinate notation. Under (212), the cyclic coordinates transform by

$$(q_1, q_2, q_3) \longmapsto (B_u q_1, A_u q_2, D_u q_3).$$

Since $K A_u K = D_u$, the action in the original upper entries is

$$(r, q, p) \longmapsto (B_u r, D_u q, D_u p), \qquad J(\lambda; p, q, r) \longmapsto J(\lambda; D_u p, D_u q, B_u r). \tag{215}$$

In particular this is an action on the three specific spaces $\mathfrak{h}_{\gamma_1}, \mathfrak{h}_{\gamma_2}, \mathfrak{h}_{\gamma_3}$ of (71), in their original order.

The determinant and trace form are

$$\det J = \lambda_1\lambda_2\lambda_3 - \lambda_1|r|^2 - \lambda_2|q|^2 - \lambda_3|p|^2 + 2\,\mathrm{Re}((r\overline{q})p), \tag{216}$$

$$\mathrm{tr}(J^2) = \lambda_1^2 + \lambda_2^2 + \lambda_3^2 + 2(|p|^2 + |q|^2 + |r|^2). \tag{217}$$

Theorem 6.14 and norm preservation prove that (215) preserves every term, with its displayed sign and coefficient. Its being a Jordan automorphism also follows directly from Yokota's diagonal-fixing action in Theorem 2.7.1, with exactly the identification (214). Equivalently, the Peirce multiplication rules $F_i(x) \circ F_i(y) = \langle x, y\rangle(E_{i+1} + E_{i+2})$ and $F_i(x) \circ F_{i+1}(y) = \frac{1}{2}F_{i+2}(\overline{xy})$, where indices are cyclic modulo three, are preserved by the norm identity and the three cyclic pair identities proved above. Here $F_1(x)$ has upper entry x at $(2,3)$, $F_2(x)$ has upper entry $\overline{x}$ at $(1,3)$, and $F_3(x)$ has upper entry x at $(1,2)$; all other entries are zero apart from their Hermitian conjugates. Direct matrix multiplication gives the displayed rules, with coefficient $1/2$ from the Jordan product. Finally $E_i \circ E_j = \delta_{ij}E_i$, $E_i \circ F_i(x) = 0$, and $E_i \circ F_j(x) = F_j(x)/2$ for $i \neq j$, and these rules are preserved because all E_i are fixed. The displayed identities specify the Jordan product on every pair by real bilinearity.

For a right principal bundle our associated-bundle convention is $(pu, v) \sim (p, \sigma(u)v)$. Let

$$\mathcal{L}_H = P_H \times_{u:a\mapsto ua} \mathbb{H}.$$

This is the tautological right quaternionic line pulled back by χH: the fibre map $[(x, v), a] \mapsto (x, va)$ is well-defined, right quaternionic linear and invertible onto the line $\chi H(x)$. Its unit sphere bundle is P_H, by $p \mapsto [p, 1]$. Consequently it is nontrivial as a quaternionic line. It is also nontrivial as a real four-plane bundle: if it had a nowhere-zero real section, dividing by length would give a section of its unit sphere bundle and hence of P_H.

The original ordered Peirce representation produces the real rank-24 bundle

$$\mathcal{W}_H = \mathcal{V}_{r,H} \oplus \mathcal{V}_{q,H} \oplus \mathcal{V}_{p,H}, \quad \mathcal{V}_{r,H} = P_H \times_B \mathbb{O}, \quad \mathcal{V}_{q,H} = \mathcal{V}_{p,H} = P_H \times_D \mathbb{O}. \tag{218}$$

The equality between the last two names specifies isomorphic representations with two separately retained labelled copies. Their explicit decompositions are

$$\begin{aligned} \mathcal{V}_{r,H} &\cong \mathbb{R} \oplus \mathcal{A}_H \oplus \mathbb{R}^4, && (\mathrm{Re}\, a, \mathrm{Im}\, a, b), \\ \mathcal{V}_{q,H} &\cong \mathcal{L}_H \oplus \mathcal{L}_H, && (a, b) \longmapsto (\overline{a}, \overline{b}), \\ \mathcal{V}_{p,H} &\cong \mathcal{L}_H \oplus \mathcal{L}_H, && (a, b) \longmapsto (\overline{a}, \overline{b}). \end{aligned} \tag{219}$$

For the last two maps, $\overline{a\overline{u}} = u\overline{a}$ proves equivariance, and similarly for b. Each quaternionic conjugation has real determinant -1 in $(1, i, j, k)$, so their ordered pair has determinant $+1$. The first line keeps the actual coordinate order $(1, \mathrm{Im}\,\mathbb{H}, \mathbb{H}\ell)$; moving the one-dimensional first factor across the imaginary three-space would contribute the sign $(-1)^3 = -1$, and no such reorder is being made. In cyclic coordinates the middle action is A; the equivariant map to two left standard representations is $(a, b) \mapsto (a, \overline{b})$, of determinant -1. This is why the original $\overline{q}$ is retained explicitly.

All fibrewise maps in (213) and (216) descend to the associated bundles: replacing a local frame p by pu changes inputs by precisely the three displayed actions, whose product and scalar identities have been proved. In particular

$$\mathbb{R}^3 \oplus \mathcal{W}_H = P_H \times_\rho J_3(\mathbb{O})$$

is a bundle of Jordan algebras with its three fixed diagonal idempotents, its full product, trace form and determinant. The three-sector structure on (218) includes the explicit nonzero invariant trilinear map $(r, q, p) \mapsto 2\,\mathrm{Re}((r\overline{q})p)$; it is nonzero because its value at $r = q = p = 1$ is exactly 2.

What extension to the full triality group forgets. Form the principal bundle

$$Q_H = P_H \times_\rho \mathrm{Spin}(8) \longrightarrow X, \quad (pu, s) \sim (p, \rho(u)s). \tag{220}$$

It contains the reduction P_H by $p \mapsto [p, 1]$. This is equivariant because $[pu, 1] = [p, \rho(u)]$.

Proposition 6.15. *The principal bundle Q_H is trivial. All three rank-eight bundles in* (218) *are therefore trivial as real vector bundles, while their specified subbundles $\mathcal{A}_H$ and $\mathcal{L}_H$ remain nontrivial. After any chosen trivialization of Q_H, its reduction determines a continuous map*

$$f_H : X \longrightarrow \mathrm{Spin}(8)/\rho(\mathrm{Sp}(1)) \tag{221}$$

whose pullback of the principal bundle $\mathrm{Spin}(8) \to \mathrm{Spin}(8)/\rho(\mathrm{Sp}(1))$ *is P_H. This map is non-null-homotopic.*

Proof. Let $g : S^5 \to \mathrm{Sp}(1)$ be a based clutching representative for P_H in the convention used in (205). Its homotopy class has order two. The transition D_g on an eight-dimensional sector is conjugate by $C(a, b) = (\overline{a}, \overline{b})$ to $\mathrm{diag}(L_g, L_g)$ on $\mathbb{H}^2$. For $0 \le t \le \pi/2$ put

$$R_t = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix} \quad \text{on } \mathbb{H}^2.$$

This is an orthogonal path commuting with quaternionic scalar right multiplication. Its conjugation carries $\mathrm{diag}(L_g, I)$ at $t = 0$ to $\mathrm{diag}(I, L_g)$ at $t = \pi/2$. The maps into $SO(8)$ represented by these two matrices are therefore based homotopic. Pointwise product of maps from a based sphere into a topological group represents addition in its homotopy group: concatenation in one loop coordinate and pointwise multiplication have the same unit and satisfy interchange, yielding the same operation. Consequently

$$[\mathrm{diag}(L_g, L_g)] = 2[\mathrm{diag}(L_g, I)] = \iota_*(2[g]) = 0 \quad \text{in } \pi_5(SO(8)),$$

where $\iota(u) = \mathrm{diag}(L_u, I)$. Conjugating back by C gives $[D_g] = 0$.

The third projection of (210) is a double covering $\mathrm{Spin}(8) \to SO(8)$: this is the triality covering in the third marking, with the same conjugation K as in (210). Coverings induce isomorphisms on π_n for $n \ge 2$ by unique lifting of based sphere maps and their homotopies. The projection sends $[\rho g]$ to $[D_g] = 0$, so $[\rho g] = 0$. A bundle obtained by gluing two trivial bundles over disks with a null-homotopic clutching map is trivial: extend that clutching map across a disk and change its local frame by the extension. This proves triviality of Q_H directly. Every associated vector bundle of a trivial principal bundle is trivial; applying the three projections gives the asserted rank-eight trivializations. The earlier proofs of nontriviality of $\mathcal{A}_H$ and $\mathcal{L}_H$ still apply to these actual subbundles, so forgetting their inclusions loses information.

Choose an isomorphism $\tau : Q_H \to X \times G$, where $G = \mathrm{Spin}(8)$, and put $K_0 = \rho(\mathrm{Sp}(1))$. For $p \in (P_H)_x$ write $\tau([p, 1]) = (x, a_p)$. Then $a_{pu} = a_p \rho(u)$, so $f_H(x) = a_p K_0$ is well-defined, independent of the choice of p, and continuous in local sections. The explicit principal bundle isomorphism to the pullback is

$$P_H \longrightarrow f_H^*(G \to G/K_0), \qquad p \longmapsto (x, a_p),$$

using the identification of $\mathrm{Sp}(1)$ with K_0 by the proved injective map ρ. The formula is fibrewise bijective and has inverse in each local trivialization. If f_H were null-homotopic, its pullback bundle would be isomorphic to the pullback along a constant map and hence trivial, contradicting Theorem 6.13. Equivalently, in the fibration exact sequence its clutching boundary is precisely $c(P_H) \neq 0$. □

There is also an explicit map into the space of oriented three-planes in the first eight-dimensional sector. If $T_1 : G \to SO(\mathbb{O})$ is the first projection of (210), define

$$\begin{aligned} \Gamma : G/K_0 &\longrightarrow \mathrm{Gr}_3^+(\mathbb{O}), \\ gK_0 &\longmapsto \mathrm{span}_{\mathbb{R}}^+(T_1(g)i, T_1(g)j, T_1(g)k). \end{aligned} \tag{222}$$

Here the superscript records the orientation of the displayed ordered basis. This formula is well-defined: multiplication of g on the right by $\rho(u)$ changes that ordered basis by the orientation-preserving map Ad_u of $\mathrm{Im}\,\mathbb{H}$, because $T_1\rho(u) = B_u$. Continuity follows from local sections of $G \to G/K_0$. Let $\mathcal{T}_3$ be the tautological oriented three-plane bundle on this Grassmannian. The explicit fibre map

$$G \times_{K_0} \mathrm{Im}\,\mathbb{H} \longrightarrow \Gamma^*\mathcal{T}_3, \qquad [g, v] \longmapsto (gK_0, T_1(g)v)$$

is well-defined since $T_1(g\rho(u))v = T_1(g)\,\mathrm{Ad}_u\, v$. It is a linear isometry on every fibre, hence a bundle isomorphism. Pulling back by (221) therefore gives

$$(\Gamma f_H)^*\mathcal{T}_3 \cong \mathcal{A}_H.$$

In particular $\Gamma f_H : X \to \mathrm{Gr}_3^+(\mathbb{O})$ is non-null-homotopic, since pullback along a null-homotopic map would make the tautological bundle trivial. This locates the three-plane in the actual first triality sector, with its orientation and its nonzero clutching class retained.

Thus the exact bridge is the sequence of explicit constructions

$$H \;\longmapsto\; P_H \;\longmapsto\; \mathcal{A}_H \;\longmapsto\; (\mathcal{W}_H, \rho\text{-reduction}, 2\,\mathrm{Re}((r\overline{q})p)).$$

The first two steps retain the entire nonzero order-two class by (207). In the last step the specified reduction, or equivalently the distinguished three-plane with its spin frame data, retains that class. The triple of unrestricted rank-eight bundles alone does not. The space G/K_0 parametrizes reductions acting on this very same rank-24 carrier; no claim that $\dim(G/K_0) = 24$ is made.

The precise half-angle sign law and its relation to the cusp. For the fixed imaginary unit i define the path

$$u(\theta) = \cos(\theta/2) + i\sin(\theta/2), \qquad 0 \le \theta \le 2\pi. \tag{223}$$

Equation (208) gives

$$\mathrm{Ad}_{u(\theta)}\, i = i, \quad \mathrm{Ad}_{u(\theta)}\, j = \cos\theta\, j + \sin\theta\, k, \quad \mathrm{Ad}_{u(\theta)}\, k = -\sin\theta\, j + \cos\theta\, k.$$

This is one full 2π rotation in the oriented (j, k) plane. Its lift starts at 1 and ends at -1. It is therefore the nontrivial loop in $SO(3)$: a null-homotopy of the loop would lift over a disk and force its lifted endpoints to agree. Substitution in (211) gives exactly

$$\rho(-1) = (I, -I, -I). \tag{224}$$

It is central in (210) because each component is scalar. Thus a full rotation in the three-dimensional adjoint sector acts by minus the identity on both associated spinor sectors while acting by the identity on the vector sector. This conclusion follows from actual related-triple maps and their product identity.

The parameter θ in (223) is a structure-group parameter; it has not been equated with the original cusp phase ϕ. The actual dependence on that cusp phase is already specified, with all four coordinates, by (200) and (203). No connection or holonomy is part of the definition of P_H. In

particular the order-two class proved here lies in $\pi_5(\mathrm{Sp}(1))$ through degree-six clutching, and is not being identified with a degree-one line-bundle class. Every real line bundle on $X \cong S^6$ is trivial: its unit sphere bundle is a double covering of a simply connected space, so it has a section. The exact relation established here is instead the faithful order-two detection (207), the nontrivial real three-plane (206), its explicit three-sector triality action, and the structural sign law (224).

6.6 A nonzero Hopf–cusp morphism to the stable mod–24 group

The retained-angle input used below is the complete Theorem 6.8 above.

6.6.1 The exact source maps and conventions

Retain the original marked source (X, x_0), its fixed based diffeomorphism $d : (S^6, *) \to (X, x_0)$, the cusp coordinate t, the fixed cutoff ϵ, and the native quotient $\overline{\pi}_f$ of the cusp fibres by $\mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle$. The quotient coordinates are ordered by $(\overline{\widehat{\gamma}}, \overline{\widehat{u}})$. Thus the fixed continuous based map is

$$q_2 : X \longrightarrow \mathcal{A} = \mathrm{Susp}(S^1 \times \overline{T}^2), \qquad q_2(x) = [|t|, t/|t|, \overline{\pi}_f(x)] \quad (0 < |t| < \epsilon).$$

It sends the entire central fibre W to the first suspension vertex and $X \setminus \operatorname{int} N_0$ to the second. The unreduced suspension $\mathcal{A}$ is based at its first vertex. In the original product torus set

$$K^{(2)} = (\{1\} \times \overline{T}^2) \cup (S^1 \times \{u_1 = 0\}) \cup (S^1 \times \{u_2 = 0\}), \qquad c_3 : T^3 \longrightarrow T^3/K^{(2)} \cong S^3.$$

The retained-angle sphere map and its sphere-source representative are

$$p_3([r, z]) = [r, c_3(z)], \qquad H = p_3 q_2 : X \longrightarrow \Sigma_r(S^3, *) \cong S^4, \qquad h = H \circ d : S^6 \longrightarrow S^4. \tag{225}$$

Both suspension vertices and the whole cylinder over the basepoint are identified in $\Sigma_r(S^3, *)$. The preceding complete retained-angle framing theorem proves

$$[h] = \eta_4 \circ \eta_5 \neq 0 \quad \text{in} \quad \pi_6(S^4) \cong \mathbb{Z}/2. \tag{226}$$

Its proof identifies the actual derivative-framed inverse torus with the Lie-framed torus, with Arf invariant one. We use this already proved class, retaining the particular map (225); see [20].

Fix the complex Hopf map

$$\eta_2 : (S^3, (1, 0)) \longrightarrow (\mathbb{CP}^1, [1 : 0]) \cong (S^2, *), \qquad (z_1, z_2) \longmapsto [z_1 : z_2].$$

Use ordered smash coordinates on spheres. In this note $E^k f = f \wedge \mathrm{id}_{S^k}$ denotes right reduced suspension; the usual ordered sphere homeomorphisms are understood. Put $\eta_n = E^{n-2}\eta_2$, and write η for the stable Hopf class. The order of each factor is retained below.

6.6.2 The Hopf–cusp homomorphism and its nonzero value

Theorem 6.16. *The assignment*

$$\mu_\eta : \pi_6(S^4) \longrightarrow \pi_9(S^6), \qquad \mu_\eta([x]) = [\eta_2 \wedge x] \tag{227}$$

is an injective group homomorphism.

Its literal representative has the complete domain and codomain

$$\eta_2 \wedge x : S^3 \wedge S^6 \cong S^9 \longrightarrow S^2 \wedge S^4 \cong S^6.$$

Choose any generator ν_6 of $\pi_9(S^6) \cong \mathbb{Z}/24$. Then the particular original cusp map supplies

$$\kappa_H = [\eta_2 \wedge h] = \mu_\eta([h]) = 12\nu_6 \neq 0. \tag{228}$$

It has exact order two. Its value is independent of orientation reversals in the sphere identifications and of the choice of the generator ν_6.

Proof. A based homotopy of x smashes with the fixed based Hopf map, so (227) is well-defined. The exact ordered point-set factorization is

$$\eta_2 \wedge x = (\eta_2 \wedge \mathrm{id}_{S^4}) \circ (\mathrm{id}_{S^3} \wedge x) : S^3 \wedge S^6 \longrightarrow S^3 \wedge S^4 \longrightarrow S^2 \wedge S^4. \tag{229}$$

The first applied map has source dimension nine and target dimension seven; the second has source dimension seven and target dimension six. The second map is $E^4\eta_2$. To compare the first with $E^3 x = x \wedge \mathrm{id}_{S^3}$, interchange its sphere factors at source and target. The interchange degrees are respectively $(-1)^{3\cdot 6} = +1$ and $(-1)^{3\cdot 4} = +1$. A based map of a sphere of degree one is based homotopic to the identity, so

$$\mu_\eta([x]) = (E^4\eta_2)_*(E^3[x]). \tag{230}$$

Suspension is a group homomorphism. Postcomposition is also a group homomorphism on homotopy groups, since it distributes over the source pinch map defining the sum. Thus (230) proves additivity, including zero and inverses, before passage to stable homotopy.

Freudenthal makes

$$E : \pi_{n+3}(S^n) \longrightarrow \pi_{n+4}(S^{n+1})$$

an isomorphism for $n \geq 5$, because $n + 3 < 2n - 1$. In particular stabilization $\pi_9(S^6) \to \pi_3^{\mathrm{S}}$ is an isomorphism. Stabilization from $\pi_6(S^4)$ is likewise an isomorphism onto π_2^{S}, because $n+2 < 2n-1$ for $n \geq 4$. The exact factorization (229) shows that its stable class is the stable composition product of η with $[x]$. Using the proved original class (226),

$$E^\infty \kappa_H = \eta\, E^\infty[h] = \eta\, \eta^2 = \eta^3 = 12\nu \neq 0 \quad \text{in} \quad \pi_3^{\mathrm{S}} \cong \mathbb{Z}/24. \tag{231}$$

The integral relation here is the one explicitly stated in [22, p. 385], not the coefficient 4 in the two-primary quotient. More precisely, the two-primary relation is $\eta^3 = 4\nu$ modulo eight. Bilinearity and $2\eta = 0$ give $2\eta^3 = 0$ integrally. In $\mathbb{Z}/24$ the only elements killed by two are 0 and 12ν, and only the second has nonzero image 4ν modulo eight. This recovers the exact integral coefficient and its nonvanishing.

The inverse of the stabilization isomorphism therefore gives (228). Since the source of μ_η is cyclic of order two, generated by $[h]$, its nonzero value proves injectivity and identifies its image as $\{0, 12\nu_6\}$.

An orientation reversal at a sphere source or target multiplies the corresponding stable homotopy class by -1; here $-12\nu_6 = 12\nu_6$. A sign change in the chosen Hopf representative also leaves the stable product unchanged since $-\eta = \eta$. Finally, replacing the generator by $u\nu_6$, with u a unit modulo twenty-four, does not change the element: every such u is odd, and $12u \equiv 12 \pmod{24}$. All orientation and generator assertions consequently hold for the actual order-two element. □

Corollary 6.17 (The same nonzero class in the twelve-dimensional receiver)**.** *Put $\nu_9 = E^3\nu_6$. The particular based map*

$$E^3(\eta_2 \wedge h) : S^{12} \longrightarrow S^9$$

has class $12\nu_9 \neq 0$ in $\pi_{12}(S^9) \cong \mathbb{Z}/24$.

Proof. Each of these three suspensions is an isomorphism in the preceding stable range. Apply their composite to (228); it preserves the exact order two. □

6.6.3 The actual marked peripheral character

Retain the original three elliptic-kernel matrices, with their original signs and ordering:

$$C_0 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad C_1 = \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix}, \qquad C_2' = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{232}$$

The existing peripheral theorem [21] defines the surjective marked abelianization character

$$\chi : SL_2(\mathbb{Z}) \longrightarrow \mathbb{Z}/12, \qquad (\chi(C_0), \chi(C_1), \chi(C_2')) = (-1, 4, -3). \tag{233}$$

For an explicit check of this normalization, let

$$S_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad T = C_0.$$

Then $C_2' = S_0^3$, $C_1 = T^{-1}S_0$, and direct matrix multiplication gives $C_1C_2' = C_0^{-1}$. In the modular presentation $S_0^4 = 1$, $S_0^2 = (S_0T)^3$, the abelianized generators satisfy $4s = 0$ and $s + 3t = 0$. Thus $s = -3t$ and $12t = 0$. The assignment $t \mapsto -1$, $s \mapsto 3$ satisfies both relations and maps onto $\mathbb{Z}/12$; consequently the order of the abelianized generator is exactly twelve. It gives, without a change of peripheral matrices, precisely $\chi(C_0) = -1$, $\chi(C_1) = 1 + 3 = 4$, and $\chi(C_2') = 3 \cdot 3 = -3 \pmod{12}$.

Theorem 6.18 (An exact nonzero peripheral homotopy morphism)**.** *Define*

$$\Theta_H : \mathbb{Z}/12 \longrightarrow \pi_6(S^4), \qquad \Theta_H([n]) = n[h], \qquad \Psi_H = \mu_\eta \circ \Theta_H \circ \chi : SL_2(\mathbb{Z}) \longrightarrow \pi_9(S^6). \tag{234}$$

These are group homomorphisms. The complete coefficient formula is

$$\Psi_H(M) = 12\,\widetilde{\chi(M)}\,\nu_6, \tag{235}$$

where $\widetilde{\chi(M)}$ is any integer lift of the marked residue. In the original matrix order of (232)*, its exact values are*

$$(\Psi_H(C_0), \Psi_H(C_1), \Psi_H(C_2')) = (12\nu_6, 0, 12\nu_6). \tag{236}$$

Its image and kernel are exactly

$$\operatorname{im}\Psi_H = \{0, 12\nu_6\}, \qquad \ker\Psi_H = \chi^{-1}(2\mathbb{Z}/12). \tag{237}$$

In particular its kernel has index two in $SL_2(\mathbb{Z})$.

Proof. The element $[h]$ has exact order two, so replacing n by $n + 12k$ leaves $n[h]$ unchanged. Addition in its abelian homotopy group proves that Θ_H is a group homomorphism. Moreover

$$\ker\Theta_H = 2\mathbb{Z}/12, \qquad \operatorname{im}\Theta_H = \pi_6(S^4).$$

The composite in (234) is consequently a group homomorphism, and (228) gives $\mu_\eta(n[h]) = 12n\nu_6$. Changing the integer lift by $12k$ changes its coefficient by $144k$, which is a multiple of twenty-four. This proves (235) with its independence of lifts. The three coefficients are respectively $-12 \equiv 12$, $48 \equiv 0$, and $-36 \equiv 12 \pmod{24}$, proving (236). Since $12n \equiv 0 \pmod{24}$ holds exactly when n is even, the asserted kernel is exact. Surjectivity of χ and nonvanishing at C_0 prove that the image is the full order-two subgroup, so the first isomorphism theorem gives the index assertion. □

Here is the complete parity factorization. Let $\mathrm{red}_2 : \mathbb{Z}/12 \to \mathbb{Z}/2$ be ordinary parity reduction, let $\iota_2([e]) = 12e \pmod{24}$, and let $c_{\nu_6} : \pi_9(S^6) \to \mathbb{Z}/24$ send ν_6 to one. Every arrow and its codomain in the diagram are fixed:

$$\begin{array}{ccc} SL_2(\mathbb{Z}) & \xrightarrow{\Psi_H} & \pi_9(S^6) \\ {\scriptstyle \mathrm{red}_2 \circ \chi}\downarrow & & \downarrow{\scriptstyle c_{\nu_6}} \\ \mathbb{Z}/2 & \xrightarrow{\iota_2} & \mathbb{Z}/24. \end{array} \tag{238}$$

Commutativity is exactly the coefficient calculation above; the bottom arrow is injective because $12e$ is zero modulo twenty-four exactly when e is zero modulo two.

6.6.4 Comparison with the earlier maps, with their exact types

Write $\psi_H = c_{\nu_6} \circ \Psi_H$. Ordinary reduction $r : \mathbb{Z}/24 \to \mathbb{Z}/12$ gives

$$r \circ \psi_H = 0, \qquad (r \circ \psi_H)(C_0) = 0 \neq -1 = \chi(C_0). \tag{239}$$

Thus this nonzero character is not a reduction-compatible lift of χ. Its natural commuting square is (238). The existing nonexistence theorem for a reduction-compatible lift remains valid: every homomorphism from $SL_2(\mathbb{Z})$ to an abelian group factors through its abelianization $\mathbb{Z}/12$, whereas a lift of $\chi(C_0) = -1$ would require C_0 to map to 11 or 23 modulo twenty-four, each of order twenty-four. The abelianized generator has order twelve and cannot have either image.

The universal even-subgroup injection is the separate map

$$j : \mathbb{Z}/12 \hookrightarrow \mathbb{Z}/24, \qquad j([a]) = [2a].$$

The new coefficient morphism factors through it by the exact identity

$$\psi_H = j \circ [6] \circ \chi, \qquad [6] : \mathbb{Z}/12 \longrightarrow \mathbb{Z}/12, \quad [a] \longmapsto [6a]. \tag{240}$$

Indeed $2(6a) = 12a \pmod{24}$. The map j is injective with image of order twelve; the composite (240) has image of order two because [6] retains precisely parity. The map j is realized by the already proved universal suspension $E^6 : \pi_6(S^3) \to \pi_{12}(S^9)$ after compatible generator choices. Equation (240) is a coefficient identity; the actual new geometric representative is $\eta_2 \wedge h$, followed, when desired, by three suspensions as above.

For the original angle-forgetting map $g = \widehat{p}_2 q_2 : X \to S^3$, the separate framed calculation has proved $[g \circ d] = 0$. Therefore the previous morphism

$$\Theta_g : \mathbb{Z}/12 \longrightarrow \pi_6(S^3), \qquad [n] \longmapsto n[g \circ d]$$

and $E^6 \Theta_g \chi$ are zero. In contrast, $E^3 \Psi_H(C_0) = 12\nu_9 \neq 0$ in this same stable receiver $\pi_{12}(S^9)$. These statements agree because the exact postcompositions from $\mathcal{A}$ are different: p_3 retains the angular coordinate in its top-cell map, while $\widehat{p}_2$ forgets it. The new homomorphism uses the already proved value of $p_3 q_2$ and the specified Hopf map, with all maps displayed in (225)–(234).

The constructed representative has source $S^3 \wedge X \cong S^9$; its threefold suspension has source S^{12}. The number twenty-four is the order of the receiving stable homotopy group. This calculation does not identify either source with a twenty-four-dimensional carrier, does not supply new periods or analytic overlap maps, and does not complete the global higher torus family. It supplies the exact nonzero homotopy and peripheral morphisms just proved.

Source checks. The three matrices and the marked values were checked directly against equations *elliptic-peripheral-matrices* and *peripheral-mod12-values* in the existing higher-rung draft [21]. The calculation above reproduces their signs using the same S_0, T. The retained-angle input is the complete derivative-framing theorem in [20]. For the stable facts, the author-hosted text [22] was read at Corollary 4.24, printed p. 360, and at printed pp. 384–385. Page 385 explicitly distinguishes the two-primary coefficient four from the integral coefficient twelve. Proposition 4.56 on that page, with its proof on p. 388, gives the stable composition product and its smash description. The fixed-factor proof (229)–(230) establishes the needed homomorphism here directly.

6.7 Boundary of the three tested numerical mechanisms

Theorem 6.19 (Boundary of the three tested numerical mechanisms)**.** *For the chamber indexing, peripheral abelianization, stable suspension, and the three additional mechanisms tested below, the verified relations are the following.*

(i) *The six Kuhn zero-strata and the six octonionic fixed flags have the marked S_3-equivariant finite-set bijection* (80)*. Their common labels also index the flag cells, and both Euler characteristics are* 6*.*

(ii) *The elliptic-kernel peripheral matrices produce the surjective character $\chi : SL_2(\mathbb{Z}) \to \mathbb{Z}/12$, with no reduction-compatible or surjective mod–*24 *lift.*

(iii) *The separate stable suspension map embeds $\mathbb{Z}/12$ as $2(\mathbb{Z}/24)$, and proposition 6.5 gives the exact composite determined by g. The actual coefficient is $a = 0$ by theorem 6.6; the composite is zero, and the associated S^{24} fourth-smash map is null-homotopic.*

(iv) *The ambient F_4 root system has $h = 12$, and $W(F_4)/W(D_4) \cong S_3$. The three specific tests give: diagonal-sum factorization through P or Q exactly for $N \mid 3$; ordinary cohomology reduction for every N, with e_N of order $N/\gcd(N, 6)$; and plain determinant-winding reduction onto $\mathbb{Z}/N$ for every N. None of these specified operations selects $N = 12$ or $N = 24$ from the dimension or Coxeter number alone.*

No arrow between these four lines is defined merely by equality of their integers.

Proof. Item (i) is theorem 5.4; item (ii) is theorem 6.1; item (iii) is theorem 6.4 and proposition 6.5; and the Coxeter assertion in (iv) is proposition 5.3. The three stated tests for an additional arithmetic arrow are evaluated by propositions 6.2 and 6.3: the primitive lattice diagonal-sum factorization allows only $N \mid 3$, cohomology reduces for every modulus, and determinant winding reduces onto every modulus. These computations prove the stated boundary for those mechanisms without asserting that arbitrary future structures cannot be added. □

7 The full Albert orbit quotient and the Cartan family

The fixed-spectrum comparison of theorem 5.1 and section 5.3 extends to the whole Albert algebra with all three eigenvalue variables retained. We give the invariant quotient, its smooth regular part, the exact normal framing, and the two focal strata. The diagonalization and flag geometry are established in [23, Proposition 2.8.2 and Theorems 2.7.4, 2.8.3] and [10, Definition 2.2 and Sections 2.3–2.6]; the calculations below specify their maps in the retained coordinates.

7.1 Original space, invariants, and direct diagonalization

Let $\mathbb{O}$ be the real octonion division algebra with its retained conjugation, norm N, and inner product $h(a,b) = \mathrm{Re}(a\bar{b})$. Set $J = \mathfrak{h}_3(\mathbb{O})$, with Jordan product $X \circ Y = (XY + YX)/2$, and retain the cyclic coordinates of (87):

$$X = \begin{pmatrix} \xi_1 & q_3 & \bar{q}_2 \\ \bar{q}_3 & \xi_2 & q_1 \\ q_2 & \bar{q}_1 & \xi_3 \end{pmatrix} = \sum_{i=1}^{3} \xi_i e_i + \sum_{i=1}^{3} E_i(q_i). \tag{OCB1}$$

Here e_i are the original diagonal idempotents, E_i are the original off-diagonal inclusions, and $I = e_1 + e_2 + e_3$. Matrix products in the Jordan product involve only products of two octonions. Powers of a single Jordan element are unambiguous by the Jordan identity; all subsequent polynomial formulas use these powers.

The positive trace metric and determinant are

$$\begin{aligned} G(X,Y) &= \mathrm{tr}(X \circ Y), \\ G(X,X) &= \xi_1^2 + \xi_2^2 + \xi_3^2 + 2\sum_i N(q_i), \\ d(X) &= \xi_1\xi_2\xi_3 - \sum_i \xi_i N(q_i) + 2\,\mathrm{Re}((q_1 q_2) q_3). \end{aligned} \tag{OCB2}$$

Thus $\dim_{\mathbb{R}} J = 27$, with every diagonal coordinate, metric factor, and mixed cubic retained. Define

$$t(X) = \mathrm{tr}\, X, \qquad u(X) = G(X,X), \qquad e_2(X) = \frac{t(X)^2 - u(X)}{2}, \qquad \Phi(X) = (t(X), u(X), d(X)). \tag{OCB3}$$

Let $F_4 = \mathrm{Aut}_{\mathbb{R}}(J, \circ)$. An automorphism fixes I, preserves trace and G, and therefore preserves the determinant by the trace identity

$$d(X) = \tfrac{1}{3}\,\mathrm{tr}(X^3) - \tfrac{1}{2} t(X) u(X) + \tfrac{1}{6} t(X)^3. \tag{OCB4}$$

For completeness, trace preservation follows without choosing a preferred matrix realization: the Peirce table gives $\mathrm{Tr}_{\mathrm{End}(J)}(L_X) = 9\,\mathrm{tr}\, X$; conjugation of L_X by an automorphism proves the assertion. Product preservation is a closed polynomial condition, so F_4 is closed in the orthogonal group of (J, G) and hence compact.

Proposition 7.1 (Diagonalization and the exact invariant fibres)**.** *Every $X \in J$ is F_4-conjugate to a real diagonal matrix. The fibres of Φ are exactly the F_4 orbits, including repeated-eigenvalue and scalar orbits. The map $\Phi : J \to \Phi(J)$ is proper.*

Proof. On the compact orbit of X, maximize $\xi_1^2 + \xi_2^2 + \xi_3^2$. If $q_i \neq 0$ at a maximizing point, put $a = q_i/|q_i|$, let (i,j,k) be cyclic, and use the already proved derivation

$$D_i(a) = 4[L_{E_i(a)}, L_{e_j}]. \tag{OCB5}$$

Its action satisfies $D_i(a)(e_j - e_k) = 2E_i(a)$ and $D_i(a)E_i(a) = -2(e_j - e_k)$. The remaining off-diagonal slots do not contribute diagonal entries under this action. Write the relevant component as $v(e_j - e_k) + wE_i(a)$, with $v = (\xi_j - \xi_k)/2$ and $w = |q_i| > 0$. Exponentiation rotates (v, w) through angle $2s$. Some rotation gives squared diagonal-difference coordinate $v^2 + w^2$, increasing the diagonal-square sum by $2w^2$. This contradicts maximality. Hence every off-diagonal coordinate vanishes at a maximum, proving diagonalization with the diagonal mean retained.

If $X = g\sum_i \lambda_i e_i$, then (OCB3)–(OCB4) show that the multiset of real eigenvalues is exactly the multiset of roots of

$$p_X(z) = z^3 - t(X)z^2 + e_2(X)z - d(X). \tag{OCB6}$$

It is independent of the diagonalizing automorphism. Every permutation of a real diagonal is realized by conjugation with a real orthogonal permutation matrix. Such conjugation preserves the Jordan product; a commuting sign adjustment can be used if determinant $+1$ is desired. Equal invariant triples thus give the same eigenvalue multiset and the same orbit. Conversely, the invariants are constant on orbits, as proved above. This includes repeated roots and scalar matrices. Finally, a compact set of invariant triples bounds $u = G(X, X)$, and its inverse image is closed and bounded in J, hence compact. □

This is the compact-orbit diagonalization argument of [23, Proposition 2.8.2, pp. 56–57], reproduced in the retained derivation coordinates. It also proves the polynomial identities used below by transporting their diagonal versions.

7.2 Exact projectors, smooth orbit inverse, and regular family

Fix labelled, pairwise distinct real numbers $\lambda_1, \lambda_2, \lambda_3$, with arbitrary sum. Put $d_\lambda = \sum_i \lambda_i e_i$ and $\mathcal{O}_\lambda = F_4 d_\lambda$. The ordered frame space is

$$\mathcal{F} = \{(p_1, p_2, p_3) : p_i^2 = p_i,\ \operatorname{tr} p_i = 1,\ p_i \circ p_j = 0\ (i \neq j),\ \textstyle\sum_i p_i = I\}.$$

The correspondence $(p_1, p_2, p_3) \mapsto (p_1, p_2)$ and its inverse $(p_1, p_2) \mapsto (p_1, p_2, I - p_1 - p_2)$ identify this space with the flag space of (68); in particular $\mathcal{F} = F_4/\operatorname{Spin}(8)$ by theorem 5.1. Define

$$\Xi_\lambda : \mathcal{F} \longrightarrow \mathcal{O}_\lambda, \qquad (p_i) \longmapsto \sum_i \lambda_i p_i, \tag{OCB7}$$

and, for $\{i, j, k\} = \{1, 2, 3\}$, define

$$P_i^\lambda(X) = \frac{(X - \lambda_j I) \circ (X - \lambda_k I)}{(\lambda_i - \lambda_j)(\lambda_i - \lambda_k)}. \tag{OCB8}$$

Proposition 7.2 (The ordered spectral diffeomorphism)**.** *The map Ξ_λ is an F_4-equivariant diffeomorphism, with inverse $(P_1^\lambda, P_2^\lambda, P_3^\lambda)$. In particular* $\dim \mathcal{O}_\lambda = 24$. *If $\mathcal{C} = \{\lambda_1 > \lambda_2 > \lambda_3\}$ and J_{reg} is the set of elements with distinct eigenvalues, then $\mathcal{F} \times \mathcal{C} \to J_{\mathrm{reg}}$, given by the same spectral sum, is a diffeomorphism.*

Proof. For $X = \sum_l \lambda_l p_l$, orthogonality and idempotency give

$$\begin{aligned}(X - \lambda_j I) \circ (X - \lambda_k I) &= \sum_l (\lambda_l - \lambda_j)(\lambda_l - \lambda_k) p_l \\ &= (\lambda_i - \lambda_j)(\lambda_i - \lambda_k) p_i.\end{aligned}$$

Thus (OCB8) recovers every ordered projector and proves both inverse compositions. It also proves that the stabilizer of d_λ is the pointwise diagonal-frame stabilizer Spin(8). The quotient orbit map is an injective immersion of the compact homogeneous space $F_4/\operatorname{Spin}(8)$ into the Hausdorff vector space J, hence an embedding. Its inverse is polynomial in X divided by nonzero fixed real constants, proving smoothness of the inverse and the asserted equivariant diffeomorphism.

The differential at fixed λ is explicitly

$$dP_i^{\lambda}|_X(H) = \frac{2X \circ H - (\lambda_j + \lambda_k)H}{(\lambda_i - \lambda_j)(\lambda_i - \lambda_k)}. \tag{OCB9}$$

At d_λ, the sector $E_i(\mathbb{O})$ joins eigenspaces j, k. On this sector $dP_j(H) = H/(\lambda_j - \lambda_k)$, $dP_k(H) = -H/(\lambda_j - \lambda_k)$, and $dP_i(H) = 0$. Substitution into $\sum_l \lambda_l dP_l(H)$ returns H, verifying the differential inverse with its signs.

The original derivations satisfy

$$\begin{gathered} D_i(a)d_\lambda = (\lambda_j - \lambda_k)E_i(a) = \Delta_i E_i(a), \\ (\Delta_1, \Delta_2, \Delta_3) = (\lambda_2 - \lambda_3, \lambda_3 - \lambda_1, \lambda_1 - \lambda_2), \qquad (i,j,k) \text{ cyclic.} \end{gathered} \tag{OCB10}$$

All three multipliers are nonzero, so the three eight-dimensional off-diagonal sectors constitute the tangent space. Their transported Peirce subbundles at every orbit point have metric $2\Delta_i^2 h$ in the homogeneous tangent coordinates of (111).

For the full map

$$\mathcal{F} \times \mathcal{C} \longrightarrow J_{\mathrm{reg}}, \qquad ((p_i), (\lambda_i)) \longmapsto \sum_i \lambda_i p_i, \tag{OCB11}$$

surjectivity follows from diagonalization and sorting. Injectivity follows from the unique ordered roots and (OCB8). At a simple root, $\partial p_X/\partial z = \prod_{j \neq i}(\lambda_i - \lambda_j) \neq 0$; the implicit function theorem therefore gives a smooth local root function. Ordering glues these functions globally. Equation (OCB8), with the smoothly varying roots, provides the smooth inverse of (OCB11). □

7.3 Retained mean, radius, and angular parameter

For every original X, put

$$\mu = \frac{t(X)}{3}, \qquad Y = X - \mu I, \qquad s = G(Y,Y) = u(X) - \frac{t(X)^2}{3}, \qquad d_0 = d(Y). \tag{OCB12}$$

Write $J_0 = \mathfrak{h}_3^0(\mathbb{O}) = \{Y \in J : \operatorname{tr} Y = 0\}$. The displayed change of variables has inverse $X = \mu I + Y$; μ and s retain their actual values. The eigenvalues $a_i = \lambda_i - \mu$ of Y satisfy $\sum_i a_i = 0$, so expansion gives

$$d(X) = \mu^3 - \frac{\mu s}{2} + d_0. \tag{OCB13}$$

For $s = r^2 > 0$, the ordered eigenvalues have the exact parametrization

$$\begin{aligned} \lambda_1 &= \mu + \sqrt{2/3}\, r \cos\theta, \\ \lambda_2 &= \mu + \sqrt{2/3}\, r \cos(\theta - 2\pi/3), \qquad 0 \le \theta \le \pi/3. \\ \lambda_3 &= \mu + \sqrt{2/3}\, r \cos(\theta + 2\pi/3), \end{aligned} \tag{OCB14}$$

The cosines sum to zero, their squared sum is $3/2$, and their product is $\frac{1}{4}\cos 3\theta$. Consequently

$$\begin{aligned} t(X) &= 3\mu, & u(X) &= 3\mu^2 + r^2, \\ d_0 &= \frac{r^3}{3\sqrt{6}}\cos 3\theta, & d(X) &= \mu^3 - \frac{\mu r^2}{2} + \frac{r^3}{3\sqrt{6}}\cos 3\theta. \end{aligned} \tag{OCB15}$$

For $0 < \theta < \pi/3$ the ordered inequalities are strict. The inverse is

$$\begin{aligned} \mu = t/3, \qquad r = \sqrt{u - t^2/3}, \\ \theta = \tfrac{1}{3}\arccos\left(\frac{3\sqrt{6}\,[d - \mu^3 + \mu r^2/2]}{r^3}\right). \end{aligned} \tag{OCB16}$$

To prove that every ordered triple is listed, put $a_3 = -a_1 - a_2$ and expand the discriminant:

$$\mathcal{D} = \prod_{i<j}(a_i - a_j)^2 = \frac{s^3}{2} - 27d_0^2 \geq 0. \tag{OCB17}$$

Thus the arccos argument lies in $[-1, 1]$. The triple in (OCB14) has the same characteristic polynomial and the correct ordering, hence equals the original triple. Equality in (OCB17) means repeated eigenvalues. If $r = 0$, positivity of G gives $Y = 0$ and $X = \mu I$; this scalar stratum requires no angle.

The full invariant image is therefore exactly

$$\left\{(t, u, d) : s = u - t^2/3 \geq 0, \quad 27(d - \mu^3 + \mu s/2)^2 \leq s^3/2, \quad \mu = t/3\right\}. \tag{OCB18}$$

Every triple in this set is attained by (OCB14), including the scalar case just described. This gives the whole three-dimensional orbit quotient; specified mean and radius select its corresponding fibre.

The full regular metric also records these parameters. Let h_i be the metric h on the ith associated bundle, expressed by the homogeneous tangent coordinate $K_i[g, a]$ of (110). The pullback of G by (OCB11)–(OCB14) is

$$3\,d\mu^2 + dr^2 + r^2 d\theta^2 + \sum_i 2\Delta_i(\mu, r, \theta)^2 h_i. \tag{OCB19}$$

Indeed the diagonal differential has coefficient vectors $\mathbf{1}$, a/r, and $\partial_\theta a$. Their pairwise inner products vanish, and their squared norms are $3, 1, r^2$. Every diagonal differential is orthogonal to every off-diagonal Peirce sector. The sector terms follow from (OCB10) and $E_i^* G = 2h$, with every eigenvalue difference and radial factor retained.

7.4 Exact codimension and global normal framing

The same 24-dimensional regular orbit has codimension three in J, two in the affine fixed-trace space $\mu I + J_0$, and one in the fixed-trace, fixed-radius sphere $\mu I + S_r^{25}$. The change of variables (OCB12) and the radius-level inclusion give the actual maps between these ambient descriptions.

For a fixed distinct triple, its normal space in J is

$$\nu_X \mathcal{O}_\lambda = \operatorname{span}_{\mathbb{R}}\{P_1^\lambda(X), P_2^\lambda(X), P_3^\lambda(X)\}. \tag{OCB20}$$

At a diagonal point this is the orthogonal complement of the three off-diagonal tangent sectors; transport by F_4 proves the equality everywhere. The projectors are pairwise G-orthogonal with squared norm one.

The differential of the invariant map has gradients

$$\nabla t = I, \qquad \nabla u = 2X, \qquad \nabla d = X^\# = X^2 - tX + e_2 I. \tag{OCB21}$$

To check the determinant gradient, diagonalize X. By (OCB2), its derivative on a diagonal variation is $\sum_i \lambda_j \lambda_k H_{ii}$, where $\{i, j, k\} = \{1, 2, 3\}$, and its derivative on an off-diagonal variation is zero. The

right side of (OCB21) is the diagonal vector of these coefficients; automorphism invariance proves the formula for every X, H. In the projector basis the coefficient determinant is

$$\det \begin{pmatrix} 1 & 1 & 1 \\ 2\lambda_1 & 2\lambda_2 & 2\lambda_3 \\ \lambda_2\lambda_3 & \lambda_1\lambda_3 & \lambda_1\lambda_2 \end{pmatrix} = -2(\lambda_1-\lambda_2)(\lambda_1-\lambda_3)(\lambda_2-\lambda_3). \tag{OCB22}$$

Hence Φ has rank three exactly on the regular stratum, where its tangent kernel is the orbit tangent space.

In the fixed-trace hyperplane define the two normal vectors

$$Y = X - \mu I, \qquad Z = Y^2 - \frac{s}{3} I. \tag{OCB23}$$

They have trace zero and are normal by (OCB20). The trace-zero eigenvalue identities give

$$\operatorname{tr} Y^3 = 3d_0, \qquad \operatorname{tr} Y^4 = s^2/2, \qquad \operatorname{Gram}(Y,Z) = \begin{pmatrix} s & 3d_0 \\ 3d_0 & s^2/6 \end{pmatrix}. \tag{OCB24}$$

Specifically, $\sum_i a_i^3 = 3a_1a_2a_3$ follows from $\sum_i a_i = 0$. Squaring $\sum_{i<j} a_ia_j = -s/2$ gives $\sum_{i<j} a_i^2a_j^2 = s^2/4$, since the remaining cross term is $2a_1a_2a_3\sum_i a_i = 0$. Expanding s^2 now gives $\sum_i a_i^4 = s^2/2$. These identities and $Z = Y^2 - sI/3$ prove every Gram entry. Therefore

$$\det \operatorname{Gram}(Y,Z) = s^3/6 - 9d_0^2 = \mathcal{D}/3 > 0 \tag{OCB25}$$

for distinct eigenvalues. Thus (Y, Z) is a smooth ordered global two-framing of the orbit normal bundle in $\mu I + J_0$.

The corresponding explicitly specified orthonormal frame is

$$n_1 = \frac{Y}{\sqrt{s}}, \qquad n_2 = \frac{Z - (3d_0/s)Y}{\sqrt{s^2/6 - 9d_0^2/s}}. \tag{OCB26}$$

Every square root here is the positive real root; its argument is positive by (OCB25). In J, prepend $n_0 = I/\sqrt{3}$ to obtain (n_0, n_1, n_2). In the radius sphere the normal direction is n_2. These formulas identify the three normal bundles explicitly.

These normal fields are parallel for the normal connection along the fixed orbit. Each is a linear combination of the P_i^λ with coefficients constant on that orbit. Along $g(s)X$, its ambient derivative is $g'(s)$ applied to the corresponding diagonal normal vector, and lies in the transported off-diagonal tangent sectors by (OCB10). Its normal component is zero. Thus the normal bundle is flat with the displayed global parallel orthonormal frame.

For any transported normal $\eta = \sum_i \nu_i P_i^\lambda$ with fixed real ν_i, use the convention $A_\eta H = -(D_H\eta)^T$. Then

$$A_\eta\big|_{E_i(\mathbb{O})} = -\frac{\nu_j - \nu_k}{\lambda_j - \lambda_k}\,\mathrm{id}, \qquad (i,j,k) \text{ cyclic}, \tag{OCB27}$$

with the sectors transported by g at other points. Indeed a path generated by $D_i(a)$ has tangent $(\lambda_j - \lambda_k)E_i(a)$, whereas the derivative of its normal field is $(\nu_j - \nu_k)E_i(a)$. This proves the formula and its minus sign. The shape operator for n_0 is zero, and the one for the radial normal n_1 is $-\,\mathrm{id}\,/r$.

7.5 Cartan equations and spherical principal curvatures

On the whole trace-zero space J_0, with its original trace metric, define the explicitly scaled cubic

$$C(Y) = 3\sqrt{6}\, d(Y). \tag{OCB28}$$

The inverse relation is $d(Y) = C(Y)/(3\sqrt{6})$, and (OCB13) recovers the full determinant with its mean and radius. Orthogonal projection of (OCB21) to J_0 gives $\nabla_{J_0} d = Y^2 - sI/3 = Z$. Hence (OCB24) gives

$$\|\nabla C\|_G^2 = 54\|Z\|_G^2 = 9s^2, \qquad \Delta_{J_0} C = 0. \tag{OCB29}$$

For the second equality, the full J Laplacian of (OCB2) is $-8\,\mathrm{tr}\, X$. Each eight-dimensional octonion slot has metric $2h$ and contributes $-8\xi_i$; each pure diagonal second derivative is zero, and the mixed cubic is separately linear in each real coordinate. Restricting to J_0 subtracts the second derivative in the unit scalar direction $I/\sqrt{3}$. The identity

$$d(Y + zI/\sqrt{3}) = d(Y) - \frac{sz}{2\sqrt{3}} + \frac{z^3}{3\sqrt{3}}$$

shows that this second derivative is zero at $z = 0$. The full Laplacian also vanishes at trace zero, proving (OCB29) with the original metric factor two.

On each specified radius sphere $S_r^{25} \subset J_0$, $r > 0$, set $f_r = C/r^3$. Then (OCB15) gives $f_r = \cos 3\theta$, and homogeneity yields

$$\|\nabla_{S_r} f_r\|^2 = \frac{9}{r^2}(1 - f_r^2), \qquad \Delta_{S_r} f_r = -\frac{81}{r^2} f_r. \tag{OCB30}$$

Indeed the radial component of ∇C is $(3C/r)n_1$. The polar Laplacian on the 26-dimensional Euclidean space, applied to the degree-three homogeneous harmonic function, is $r[3(3 + 24)f_1 + \Delta_{S_1} f_1]$. These calculations prove both constants in (OCB30). Its regular level hypersurfaces are therefore isoparametric: both the squared gradient and the Laplacian depend only on the function value. By (OCB15)–(OCB18), the regular levels are precisely the fixed-mean, fixed-radius regular $F_4/\operatorname{Spin}(8)$ orbits.

The principal curvatures admit a direct calculation. On the diagonal section let $N_\theta = (1/r)\partial_\theta Y$, a unit vector perpendicular to Y, and transport it by the projectors. The exact eigenvalue differences in (OCB14) are

$$\Delta_1 = \sqrt{2} r \sin\theta, \qquad \Delta_2 = -\sqrt{2} r \sin(\theta + \pi/3), \qquad \Delta_3 = -\sqrt{2} r \sin(\theta - \pi/3). \tag{OCB31}$$

Apply (OCB27) with $\nu_i = (\partial_\theta \lambda_i)/r$. For the declared normal N_θ, the three principal curvatures are

$$-\frac{1}{r}\cot\theta, \qquad -\frac{1}{r}\cot(\theta + \pi/3), \qquad -\frac{1}{r}\cot(\theta - \pi/3), \tag{OCB32}$$

each on its original eight-dimensional Peirce sector. They are distinct for $0 < \theta < \pi/3$. In the invariant frame (OCB26), one has $n_2 = -N_\theta$: differentiating $C = r^3 \cos 3\theta$ gives $\nabla_{S_r} C = -3r^2 \sin 3\theta\, N_\theta$, whereas n_2 is the positive-length version of the projected determinant gradient. The principal curvatures for n_2 are consequently the opposites of (OCB32). This proves the three-curvature, multiplicity-eight Cartan family with both normal orientations, mean, and radius specified.

7.6 Focal projective planes and their endpoint maps

At $\theta = 0$ one has $\lambda_2 = \lambda_3 = \mu - r/\sqrt{6}$ and $\lambda_1 = \mu + 2r/\sqrt{6}$; at $\theta = \pi/3$ one has $\lambda_1 = \lambda_2 = \mu + r/\sqrt{6}$ and $\lambda_3 = \mu - 2r/\sqrt{6}$. The two singular orbits in $\mu I + S_r^{25}$ are exactly

$$\begin{aligned} \mathcal{P}^{+}_{\mu,r} &= \{\mu I + \tfrac{r}{\sqrt{6}}(3p - I) : p^2 = p,\ \operatorname{tr} p = 1\}, \\ \mathcal{P}^{-}_{\mu,r} &= \{\mu I - \tfrac{r}{\sqrt{6}}(3p - I) : p^2 = p,\ \operatorname{tr} p = 1\}. \end{aligned} \tag{OCB33}$$

Their inverse maps are, respectively,

$$p = \tfrac{1}{3}\left(I + \tfrac{\sqrt{6}}{r}(X - \mu I)\right), \qquad p = \tfrac{1}{3}\left(I - \tfrac{\sqrt{6}}{r}(X - \mu I)\right). \tag{OCB34}$$

These affine maps are smooth, have the displayed two-sided inverses on the specified sets, and are F_4-equivariant. The primitive idempotent space is the sixteen-dimensional Cayley projective plane $\mathbb{OP}^2 = F_4/\operatorname{Spin}(9)$ [23, Theorems 2.7.4 and 2.8.3] [10, Section 2.3]. Hence the formulas give embedded projective-plane strata.

The endpoint maps of the regular family are the explicit projections

$$\begin{aligned} (p_1, p_2, p_3) &\longmapsto \mu I + \tfrac{r}{\sqrt{6}}(3p_1 - I), \\ (p_1, p_2, p_3) &\longmapsto \mu I - \tfrac{r}{\sqrt{6}}(3p_3 - I). \end{aligned} \tag{OCB35}$$

Each fibre is $\operatorname{Spin}(9)/\operatorname{Spin}(8) \cong S^8$: fixing one primitive idempotent leaves two ordered complementary idempotents, the octonionic projective line. These frame bundles are identified in [10, Sections 2.3–2.6]. They are also the focal maps. Indeed the diagonal path $Y(\theta)$ is a great circle because $\partial_\theta^2 Y = -Y$ and $|\partial_\theta Y| = r$. Transport by F_4 therefore gives the normal geodesics to the regular hypersurfaces. At each endpoint one of the tangent multipliers in (OCB31) vanishes with its eight-dimensional sector, giving exactly the collapse in (OCB35). Each focal stratum has codimension nine in the ambient radius sphere and a normal sphere of dimension eight.

For $r = 0$, the construction gives the scalar line $X = \mu I$, each point fixed by F_4. For $r > 0$ and endpoint angle, the full repeated-eigenvalue stratum retains the two parameters (μ, r) in addition to its projective-plane coordinate, and has dimension eighteen in J. These descriptions retain every scalar, radial, and mean stratum.

7.7 Exact map from the equal-diagonal triality slice

For the original matrix $Q_{\mathbb{O}}(\lambda; q) = \lambda I + E(q)$, with $E(q) = \sum_i E_i(q_i)$ and $\tau(q) = \operatorname{Re}((q_1 q_2) q_3)$, the maps above give

$$\mu = \lambda, \qquad Y = E(q), \qquad r^2 = 2\sum_i N(q_i), \qquad d_0 = 2\tau(q), \qquad C(Y) = 6\sqrt{6}\,\tau(q). \tag{OCB36}$$

Thus its invariant triple is

$$\begin{aligned} \Phi(Q_{\mathbb{O}}(\lambda; q)) = \big(&3\lambda,\ 3\lambda^2 + 2\sum_i N(q_i), \\ &\lambda^3 - \lambda \sum_i N(q_i) + 2\tau(q)\big). \end{aligned} \tag{OCB37}$$

This is the explicit invariant morphism from the original triality coordinates to the full orbit quotient. At every simple spectrum, its ordered-projector lift is (OCB8) applied to $Q_{\mathbb{O}}(\lambda; q)$. The positive boundary graph of theorem 5.12 chooses the original scalar $\lambda = -\lambda_{\min}(E(q))$. In the full orbit coordinates, this says that the least eigenvalue is zero and the other two are nonnegative. Equations (OCB36)– (OCB37) prove the relation to the whole Albert orbit family with all coefficients and this scalar choice retained.

Attribution and scope. Yokota's diagonalization is Proposition 2.8.2, pp. 56–57; his primitive-idempotent stabilizer and Cayley-plane identification are Theorems 2.7.4 and 2.8.3, pp. 54 and 57 [23]. Mare and Willems give the primitive idempotents, full real diagonal space, frame stabilizer, projective plane, S^8 frame projections, and the polar isotropy representation in Definition 2.2 and Sections 2.3–2.6, pp. 488–493 [10]. Their description identifies J_0 with the isotropy representation of $E_{6(-26)}/F_4$ and displays its diagonal two-plane and three eight-dimensional root spaces. The Cartan-family equations and focal maps used here are proved in (OCB29)–(OCB35). The orbit and flag geometry is established literature; these calculations assert neither complex integrability nor physical dynamics.

8 The exact spectral map of the common marked lattice

The literal common-lattice embedding of theorem 14.6 has a matrix-valued and frame-valued lift into the Albert orbit family of section 7. We calculate this map with the three octonionic slots separately labelled, the trace metric, the arbitrary scalar diagonal, and the original determinant coefficient two.

8.1 Domain and the original matrix

Use the ordered quaternion subalgebra

$$\mathbb{H} = \mathbb{R}\langle 1, i, j, k\rangle \subset \mathbb{O}, \qquad i^2 = j^2 = k^2 = ijk = -1.$$

For $a = (a_0, a_1, a_2, a_3)$, write

$$\begin{gathered} v = a_1 i + a_2 j + a_3 k, \qquad t = \sqrt{a_1^2 + a_2^2 + a_3^2}, \qquad a = a_0 + v, \\ Q = a_0^2 + t^2, \qquad P = a_0(a_0^2 - 3t^2). \end{gathered} \tag{NCS1}$$

Every square root in this section means the nonnegative real root. Here t denotes the length of the imaginary quaternion, while the trace invariant will always be written $\operatorname{tr} X$. The integral domain and its retained embedding are

$$D_4 = \{a \in \mathbb{Z}^4 : a_0 + a_1 + a_2 + a_3 \in 2\mathbb{Z}\}, \qquad j(a) = \sqrt{2}(a, a, a) \in \mathbb{O}^3. \tag{NCS2}$$

The literal intersection theorem theorem 14.6 identifies $j(D_4)$ with the common lattice of the two specified marked Niemeier/Leech embeddings and the specified Wilson embedding. Its original Niemeier coordinates are

$$x_{i,m} = (-1)^m a_{i-1} \quad (1 \le i \le 4,\ 0 \le m \le 5), \qquad d = 0. \tag{NCS3}$$

The six coordinates in each A_5 block sum to zero. Pairing with the retained Weyl vector gives $3\sum_i a_i$, which is divisible by six for $a \in D_4$; hence these vectors belong to the retained neighbor sublattice K. The full Wilson membership certificate and both intersection equalities are proved in theorem 14.6.

For arbitrary $\mu \in \mathbb{R}$, define

$$Y(a) = \begin{pmatrix} 0 & \sqrt{2}a & \sqrt{2}\bar{a} \\ \sqrt{2}\bar{a} & 0 & \sqrt{2}a \\ \sqrt{2}a & \sqrt{2}\bar{a} & 0 \end{pmatrix}, \qquad X(\mu, a) = \mu I + Y(a). \tag{NCS4}$$

This is the original cyclic convention $(q_1, q_2, q_3) = (\sqrt{2}a, \sqrt{2}a, \sqrt{2}a)$, including the conjugated upper-right entry. On $J = \mathfrak{h}_3(\mathbb{O})$, the product is $A \circ B = (AB + BA)/2$ and the metric is $G(A, B) = \mathrm{tr}(A \circ B)$. Consequently

$$\|j(a)\|^2 = 6Q, \qquad r^2 := G(Y, Y) = 12Q. \tag{NCS5}$$

This retains the factor two between the two metrics. The map $(\mu, a) \mapsto X(\mu, a)$ from $\mathbb{R} \times D_4$ to J is injective: its inverse on its image is $\mu = \mathrm{tr}\, X/3$ and $a = X_{12}/\sqrt{2}$.

8.2 Complete determinant and invariant map

The quaternion identity $v^2 = -t^2$ and associativity give

$$\mathrm{Re}(a^3) = a_0^3 - 3a_0 t^2 = P.$$

Thus the triality cubic and determinant of Y are exactly

$$\begin{aligned} \tau(j(a)) &= \mathrm{Re}((\sqrt{2}a\sqrt{2}a)\sqrt{2}a) = 2\sqrt{2}P, \\ \det Y &= 2\tau(j(a)) = 4\sqrt{2}P. \end{aligned} \tag{NCS6}$$

The complete invariant map is therefore

$$\boxed{(\mathrm{tr}\, X, \mathrm{tr}\, X^2, \det X) = (3\mu,\ 3\mu^2 + 12Q,\ \mu^3 - 6\mu Q + 4\sqrt{2}P).} \tag{NCS7}$$

Indeed each of the three squared off-diagonal norms is $2Q$. The negative terms in the original determinant formula thus sum to $-6\mu Q$, while the mixed term is precisely (NCS6). This proves the displayed invariant map with every diagonal variable and original lattice scale retained. The characteristic polynomial of Y is

$$p_Y(z) = z^3 - 6Qz - 4\sqrt{2}P. \tag{NCS8}$$

8.3 Exact eigenvalues, projectors, and directional inverse

The three labelled eigenvalues of Y are

$$\nu_0 = 2\sqrt{2}a_0, \qquad \nu_1 = -\sqrt{2}a_0 - \sqrt{6}t, \qquad \nu_2 = -\sqrt{2}a_0 + \sqrt{6}t. \tag{NCS9}$$

Their sum is zero, their squared sum is $12Q$, and their product is $4\sqrt{2}P$. Direct multiplication of $\prod_{j=0}^{2}(z - \nu_j)$ gives (NCS8), proving that these are the eigenvalues. The eigenvalues of X are $\mu + \nu_j$, with μ arbitrary.

For $t > 0$, let $n = v/t$, so $n^2 = -1$ and $v = tn$. The plane $\mathbb{C}_n = \mathbb{R}\langle 1, n\rangle$ is a commutative associative subalgebra. Set

$$\begin{aligned} T &= \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \qquad T^3 = I, \\ \omega_n &= -\frac{1}{2} + \frac{\sqrt{3}}{2}n, \qquad \omega_n^3 = 1, \qquad 1 + \omega_n + \omega_n^2 = 0. \end{aligned} \tag{NCS10}$$

Every entry of (NCS4) lies in $\mathbb{C}_n$, so associative matrix multiplication is legitimate throughout this calculation. Define

$$\Pi_j(n) = \frac{1}{3}\left(I + \omega_n^{-j}T + \omega_n^{-2j}T^2\right), \qquad j = 0, 1, 2. \tag{NCS11}$$

These matrices are Hermitian: $T^* = T^2$, $\bar{\omega}_n = \omega_n^{-1}$, and the two coefficients are conjugates. Each trace is one. To verify all projector identities, use

$$\sum_{k=0}^{2} \omega_n^{mk} = \begin{cases} 3, & 3 \mid m, \\ 0, & 3 \nmid m. \end{cases}$$

Expanding (NCS11) gives

$$\Pi_i \Pi_j = \delta_{ij} \Pi_j, \qquad \sum_{j=0}^{2} \Pi_j = I, \qquad T\Pi_j = \omega_n^j \Pi_j. \tag{NCS12}$$

Furthermore $Y = \sqrt{2}(aT + \bar{a}T^2)$, whence

$$\begin{aligned} Y\Pi_j &= \sqrt{2}(a\omega_n^j + \bar{a}\omega_n^{2j})\Pi_j = \nu_j \Pi_j, \\ X &= \sum_{j=0}^{2} (\mu + \nu_j)\Pi_j. \end{aligned} \tag{NCS13}$$

Multiplication in $\mathbb{C}_n$ gives exactly the three signs in (NCS9). Thus (NCS11) gives an actual map into the ordered Jordan-frame space $F_4/\operatorname{Spin}(8)$.

When the eigenvalues are pairwise distinct, these are also the intrinsic spectral projectors:

$$\Pi_j = \frac{(Y - \nu_k I) \circ (Y - \nu_l I)}{(\nu_j - \nu_k)(\nu_j - \nu_l)}, \qquad \{j,k,l\} = \{0,1,2\}. \tag{NCS14}$$

Substitution of (NCS13), together with (NCS12), proves this formula. Its denominators are retained; their exceptional loci are computed below.

The labelled frame retains the direction n. Specifically,

$$(\Pi_1)_{12} = -\frac{1}{6} - \frac{\sqrt{3}}{6} n, \qquad n = -2\sqrt{3}\, \operatorname{Im}(\Pi_1)_{12}. \tag{NCS15}$$

Together with $a_0 = \nu_0/(2\sqrt{2})$, $t = (\nu_2 - \nu_1)/(2\sqrt{6})$, and $v = tn$, this recovers the original quaternion from the labelled spectrum and frame. The scalar parameter is recovered from $\operatorname{tr} X/3$. Thus the matrix and frame lift has an exact inverse on its labelled image. The invariant triple (NCS7) alone loses n; the lift (NCS11) records it explicitly.

At $t = 0$, one has $Y = \sqrt{2}a_0(T + T^2)$. The rank-one projector $\Pi_0 = (I + T + T^2)/3$ remains fixed, and $I - \Pi_0$ is the rank-two projector of the repeated eigenvalue $-\sqrt{2}a_0$ when $a_0 \neq 0$. No direction n is assigned to a zero imaginary part. At $a = 0$, the matrix is the scalar μI.

8.4 Singular loci and the exact Cartan angle

The discriminant of the original polynomial (NCS8), with its full scalar factor, is

$$\boxed{\prod_{i<j} (\nu_i - \nu_j)^2 = 864(Q^3 - P^2) = 864 t^2 (3a_0^2 - t^2)^2.} \tag{NCS16}$$

Indeed expansion of $(a_0^2 + t^2)^3 - a_0^2(a_0^2 - 3t^2)^2$ gives $t^2(3a_0^2 - t^2)^2$. The first equality also follows by multiplying the three differences in (NCS9). The spectrum is simple exactly when $t > 0$ and $t^2 \neq 3a_0^2$.

For nonzero a, use the Cartan polynomial of (OCB28) in the original trace metric. Equations (NCS5)–(NCS6) give

$$C(Y) = 3\sqrt{6}\det Y = 24\sqrt{3}P, \qquad r^2 = 12Q,$$
$$\boxed{C(Y) = r^3\cos 3\theta, \qquad \theta = \frac{1}{3}\arccos\left(\frac{P}{Q^{3/2}}\right) \in [0, \pi/3].} \tag{NCS17}$$

The discriminant identity proves that the arccos argument belongs to $[-1, 1]$. This is an identity for C with the whole amplitude retained: Y still has radius $r = \sqrt{12Q}$, and its scalar-shifted determinant is still (NCS7).

To link this to the quaternionic argument, define $\varphi \in [0, \pi]$ by

$$a_0 = \sqrt{Q}\cos\varphi, \qquad t = \sqrt{Q}\sin\varphi.$$

When $t > 0$, the inverse is $a = \sqrt{Q}(\cos\varphi + n\sin\varphi)$, retaining Q, φ, n. The identity $4\cos^3\varphi - 3\cos\varphi = \cos 3\varphi$ gives

$$\cos 3\theta = \cos 3\varphi. \tag{NCS18}$$

The ordered Cartan chamber thus folds the quaternionic angular variable by this explicit triple-angle map. The labelled frame, (NCS9), and (NCS15) retain the inverse information. This quaternionic angle is defined by (NCS1); its definition makes no identification with a cusp angular coordinate.

For $t > 0$, the decreasing orders of the labelled spectrum are

$$\begin{array}{c|l} a_0 > t/\sqrt{3} & \nu_0 > \nu_2 > \nu_1 \\ -t/\sqrt{3} < a_0 < t/\sqrt{3} & \nu_2 > \nu_0 > \nu_1 \\ a_0 < -t/\sqrt{3} & \nu_2 > \nu_1 > \nu_0. \end{array} \tag{NCS19}$$

The differences $\nu_0 - \nu_2 = \sqrt{2}(3a_0 - \sqrt{3}t)$ and $\nu_0 - \nu_1 = \sqrt{2}(3a_0 + \sqrt{3}t)$ prove each row. At $a_0 = t/\sqrt{3}$, precisely $\nu_0 = \nu_2$; at $a_0 = -t/\sqrt{3}$, precisely $\nu_0 = \nu_1$. Thus these rows specify how to order the projectors before applying (OCB7). At the two double-eigenvalue loci, the formulas (NCS11) remain defined, and (NCS12)– (NCS13) give repeated-eigenspace projectors $\Pi_0 + \Pi_2$ and $\Pi_0 + \Pi_1$, respectively. Their complementary simple-eigenvalue projectors are Π_1 and Π_2. This follows directly by grouping the equal coefficients in (NCS13); no vanishing denominator in (NCS14) is used.

8.5 Actual integral witnesses in the orbit family

The image proof of theorem 15.1 supplies three original-coordinate lattice lines:

$$\begin{aligned} L_\varepsilon(h) &= (-3h - \varepsilon, h, h, h + \varepsilon), \qquad \varepsilon = \pm 1, \\ L_0(h) &= (-3h, h + 1, h - 1, h), \qquad h \in \mathbb{Z}. \end{aligned} \tag{NCS20}$$

Their coordinate sums are zero. Direct expansion gives

a	$P(a)$	$Q(a)$
$L_\varepsilon(h)$	$6h + 2\varepsilon$	$12h^2 + 8\varepsilon h + 2$
$L_0(h)$	$18h$	$12h^2 + 2.$

(NCS21)

The eigenvalues, projectors, full radius, and scalar-shifted determinant of each actual common-lattice vector are therefore given by (NCS7)– (NCS17), evaluated at (NCS21) and its original

imaginary vector v. The Cartan angles satisfy

$$\begin{aligned}\cos 3\theta_\varepsilon(h) &= \frac{6h+2\varepsilon}{(12h^2+8\varepsilon h+2)^{3/2}},\\ \cos 3\theta_0(h) &= \frac{18h}{(12h^2+2)^{3/2}}.\end{aligned} \tag{NCS22}$$

As $|h|\to\infty$, both ratios tend to zero: the numerators have degree one, while their positive denominators grow as $|h|^3$. Hence the angles tend to $\pi/6$, and simultaneously $r^2=12Q\to\infty$. The original matrices and their radii remain those in (NCS4)– (NCS5).

On the shared direction $a=h(-3,1,1,1)$ one has $P=0$ and $Q=12h^2$. For $h\neq 0$, the decreasing spectrum is exactly

$$(6\sqrt{2}|h|,\ 0,\ -6\sqrt{2}|h|),\qquad r=12|h|,\qquad \theta=\pi/6. \tag{NCS23}$$

The discriminant (NCS16) is strictly positive, so this zero-cubic direction lies on a regular Cartan orbit. The spectrum of $X(\mu,a)$ is obtained by adding μ to each displayed eigenvalue.

Sources and scope of the bridge. The literal-intersection theorem theorem 14.6 supplies the marked Niemeier/Leech and Wilson membership proofs for (NCS2)–(NCS3); the mixed cubic is (345). Wilson's construction is [31]. The full invariant quotient, retained mean and radius, ordered spectral diffeomorphism, and focal maps are proved in section 7, using the flag geometry of [10, Sections 2.3–2.6] and [23, Proposition 2.8.2 and Theorems 2.7.4, 2.8.3]. The complete image theorem theorem 15.1 proves

$$P(D_4)=18\mathbb{Z}\cup(6\mathbb{Z}+2)\cup(6\mathbb{Z}-2)$$

using exactly (NCS20). This section provides its matrix-valued and frame-valued lift with an inverse on the labelled image; it makes no assertion that every Cartan orbit meets this rank-four lattice. The construction is independent of the assertion that the original completed complex threefold is diffeomorphic to S^6. No interacting quantum Hamiltonian is specified here. The retained cusp bundle has its separate explicit triality-subgroup map; its source angular coordinate and the quaternionic angular coordinate above retain their respective definitions.

9 The modulo-840 lattice, triality, and exact shell maps

This section constructs the original modulo-840 lattice N with root system $A_5^4D_4$, its full integral isometry group, and the exact maps from denominator shells and finite prime samples to the retained residue coordinates. The original data and source labels are recorded in [28]. The shared continuation proves the index-six Leech neighbour Λ and the marked isometry $\mathcal{L}$ into the three octonion sectors. In the specified Cayley frame, put $\tau(q)=\mathrm{Re}((q_1q_2)q_3)$ and $F=\tau\circ\mathcal{L}$; the Albert determinant is $\lambda^3-\lambda(y,y)+2F(y)$. The value-generated groups have quotient $\mathcal{V}_F(\Lambda)/\mathcal{V}_F(N)\cong\mathbb{Z}/12$ (Theorem 11.3). The three residue-cycled neighbours meet in J, with determinant 81 and index 9 in each; the fixed fourth extension has exactly 72 roots of type A_2^{12} (Theorems 12.3 and 12.5). The common lattice and this fourth extension have the exact same value-generated group $\frac{1}{27}\mathbb{Z}\oplus\frac{\sqrt{2}}{216}\mathbb{Z}\oplus\frac{\sqrt{3}}{18}\mathbb{Z}$. The inclusions from $\mathcal{V}_F(N)$ and into $\mathcal{V}_F(\Lambda)$ have matrices $\mathrm{diag}(1,4,1)$ and $\mathrm{diag}(1,1,3)$ and quotients $\mathbb{Z}/4$, $\mathbb{Z}/3$ (Theorems 13.2 and 13.3). Wilson's construction [31] is realized with both metrics and the direction $c_{\mathrm{W}}=c^{-1}$ retained. Its fixed Gram matrix is $3G_{E_8}$, its complement is explicitly isometric to $A_2\otimes E_8$, and the index is 3^8, with quotient $\mathbb{F}_3^8$. Its literal intersections with $\mathcal{L}(N)$ and $\mathcal{L}(\Lambda)$ are the same rank-four lattice of Gram matrix $6G_{D_4}$, minimum squared norm 12

and value-generated subgroup $4\sqrt{2}\mathbb{Z}$ for τ ($8\sqrt{2}\mathbb{Z}$ for 2τ). This is not the actual image: $12\sqrt{2}$ is not attained by τ on that intersection. All coordinates, multiplication conventions, domain lattices and maps are specified in the complete proofs below.

9.1 The finite multiplicative carrier and its four fibres

For an odd prime q and a unit r modulo q, write $\left(\frac{r}{q}\right) = 1$ when r is a square modulo q, and -1 otherwise. Let

$$\begin{aligned} V_{840} &= \{r \in (\mathbb{Z}/840\mathbb{Z})^\times : r \equiv 1 \pmod{24}\}, \\ \beta(r) &= \left(\frac{1-\left(\frac{r}{5}\right)}{2}, \frac{1-\left(\frac{r}{7}\right)}{2}\right) \in \mathbb{F}_2^2, \\ S_{840} &= \ker\beta, \qquad F_\eta = \beta^{-1}(\eta) \quad (\eta \in \mathbb{F}_2^2). \end{aligned} \tag{241}$$

Multiplication is used in V_{840} and addition in $\mathbb{F}_2^2$.

Theorem 9.1 (The marked cyclic fibre)**.** *The map β is an onto group homomorphism, each F_η has six elements, and*

$$\begin{aligned} S_{840} &= \{1, 121, 169, 289, 361, 529\} = \langle 289\rangle, \\ (289^j)_{j=0}^5 &= (1, 289, 361, 169, 121, 529) \pmod{840}. \end{aligned} \tag{242}$$

Each fibre is a torsor under multiplication by S_{840}. In particular the marking

$$\kappa : S_{840} \xrightarrow{\cong} \mathbb{Z}/6\mathbb{Z}, \qquad r = 289^{\kappa(r)} \pmod{840}, \tag{243}$$

is an isomorphism of groups.

Proof. The Chinese remainder map modulo $24, 5, 7$, whose moduli are pairwise coprime, identifies V_{840} with $(\mathbb{Z}/5)^\times \times (\mathbb{Z}/7)^\times$. The powers of 2 modulo 5 are $1, 2, 4, 3$, and the powers of 3 modulo 7 are $1, 3, 2, 6, 4, 5$; these prove the respective cyclic orders four and six. In a cyclic group of either even order, the squares are exactly the even powers. Therefore each Legendre character is the homomorphism recording exponent parity and is onto. Their product is onto $\mathbb{F}_2^2$. Its kernel has $(4/2)(6/2) = 6$ elements and every fibre has that same cardinality.

Successive multiplication of the six displayed residues by 289 gives the next entry and then returns from 529 to 1. They are distinct. Moreover $289 \equiv 1 \pmod{24}$, $289 \equiv 4 \pmod 5$, and $289 \equiv 2 \pmod 7$, so it belongs to the simultaneous square kernel. Its six powers therefore exhaust that kernel. If $r, s \in F_\eta$, then $sr^{-1} \in S_{840}$ and $s = (sr^{-1})r$, uniquely. This proves the torsor assertion and all claims about κ. □

Fix the coordinate order

$$(\eta_1, \eta_2, \eta_3, \eta_4) = (00, 10, 01, 11)$$

and choose one origin $f_i \in F_{\eta_i}$ in each fibre, with $f_1 = 1$. The six ordered elements of its fibre are $f_i 289^j$, $0 \le j \le 5$. These origins are part of the coordinate identifications used below; no selection of them is inferred from an individual denominator certificate.

9.2 The root lattice and its exact discriminant form

For a finite set F, give the free abelian group $\mathbb{Z}[F] = \bigoplus_{r\in F} \mathbb{Z}e_r$ the metric $(e_r, e_s) = 1$ for $r = s$ and zero otherwise. Put

$$A(F) = \left\{ \sum_{r\in F} n_r e_r : \sum_{r\in F} n_r = 0 \right\}.$$

The ordered fibre coordinates identify each $A(F_{\eta_i})$ isometrically with

$$A = \{(n_0, \dots, n_5) \in \mathbb{Z}^6 : \textstyle\sum_j n_j = 0\}.$$

In a separate Euclidean space with orthonormal basis e_1, e_2, e_3, e_4, define

$$D = \{x \in \mathbb{Z}^4 : x_1 + x_2 + x_3 + x_4 \equiv 0 \pmod 2\}, \qquad R_{\mathrm{ES}} = \left(\bigoplus_{i=1}^{4} A(F_{\eta_i}) \right) \perp D. \tag{244}$$

All five summands are orthogonal with precisely their displayed metrics. The symbol R_{ES} denotes this lattice; the unadorned R later used for an integer shell modulus has a different declared domain.

For a full integral lattice L in its real span, let

$$L^\vee = \{x : (x, L) \subset \mathbb{Z}\}, \qquad D_L = L^\vee / L.$$

If L is even integral, its discriminant quadratic form is $q_L(x + L) = (x, x) \pmod{2\mathbb{Z}}$. This is well-defined because replacing x by $x + \ell$ changes its squared norm by $2(x, \ell) + (\ell, \ell) \in 2\mathbb{Z}$.

Theorem 9.2 (Root metric and marked discriminant)**.** *The lattice R_{ES} is even integral of rank 24 and has Gram matrix*

$$G_{R_{\mathrm{ES}}} = G_A^{\oplus 4} \oplus G_D,$$

where

$$G_A = \begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{pmatrix}, \qquad G_D = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & -1 \\ 0 & -1 & 2 & 0 \\ 0 & -1 & 0 & 2 \end{pmatrix}.$$

Their determinants are 6 and 4, respectively, and $\det R_{\mathrm{ES}} = 6^4 \cdot 4 = 5184$.

In each identified A summand use $\lambda = (5, -1, -1, -1, -1, -1)/6$ as the discriminant generator. For D, use the representatives

$$v = (1, 0, 0, 0), \qquad s = \tfrac{1}{2}(1, 1, 1, 1), \qquad c = \tfrac{1}{2}(-1, 1, 1, 1) \tag{245}$$

and the exact marking $10 \mapsto v + D$, $01 \mapsto s + D$, $11 \mapsto c + D$. Then

$$\begin{aligned} D_{R_{\mathrm{ES}}} &= (\mathbb{Z}/6\mathbb{Z})^4 \oplus \mathbb{F}_2^2, \\ q_{R_{\mathrm{ES}}}(x_1, x_2, x_3, x_4; y) &= \frac{5}{6} \sum_{i=1}^{4} x_i^2 + \mathbf{1}_{y\neq 0} \pmod{2\mathbb{Z}}. \end{aligned} \tag{246}$$

Proof. For a six-element ordered fibre the five differences $e_j - e_{j+1}$, $0 \le j \le 4$, form a basis of its augmentation lattice: the coefficients of a zero-sum vector in this basis are its first five successive partial sums. Dot products give G_A. The determinant d_n of the analogous tridiagonal n-by-n matrix satisfies $d_n = 2d_{n-1} - d_{n-2}$, $d_0 = 1, d_1 = 2$, by first-row expansion; induction gives $d_n = n + 1$. The

four vectors $e_1 - e_2, e_2 - e_3, e_3 - e_4, e_3 + e_4$ lie in D and their coordinate determinant has absolute value 2. Their integer span therefore has index 2 in $\mathbb{Z}^4$. The parity homomorphism $\mathbb{Z}^4 \to \mathbb{F}_2$, $x \mapsto \sum x_i \pmod 2$, is onto and has kernel D, so this span equals D. Their dot products give G_D, whose determinant is the square of the coordinate determinant, namely 4. Integer coordinates make all pairings integral. In either lattice the squared norm is congruent modulo two to the coordinate sum, hence is even. Rank and determinant now follow from the orthogonal direct sum.

The vector λ has sum zero and integral pairing with each augmentation root. Its norm is $5/6$, and $k\lambda \in A$ holds exactly when 6 divides k. Thus it has exact order six in D_A. In an integral basis of an integral lattice with Gram matrix G, the dual coordinates are $G^{-1}\mathbb{Z}^n$; therefore $|L^\vee/L| = |\det G|$. Since $|D_A| = 6$, this element generates the full discriminant group and gives its stated form.

A vector pairing integrally with all $e_i - e_j$ has common fractional parts in its four coordinates. Pairing also with $e_i + e_j$ forces this common fractional part to be either zero or one half. Conversely every entirely integral or entirely half-integral vector pairs integrally with every root of D. There are two integer classes according to sum parity, and two half-integer classes according to the parity of their difference from s. The four representatives $0, v, s, c$ are therefore all the classes, and their squared norms are $0, 1, 1, 1$. Their addition is as marked because $v + s - c = (2, 0, 0, 0) \in D$. The dual and quadratic form of an orthogonal sum are the direct sums of those of its summands, which proves (246). □

9.3 The index-72 glue and all glue-class minima

Define the binary and ternary codes

$$\begin{aligned} C_2 &= \operatorname{span}_{\mathbb{F}_2}\{(1111;00), (1100;10), (1010;01)\}, \\ C_3 &= \operatorname{span}_{\mathbb{F}_3}\{z_1 = (1,1,1,0),\ z_2 = (1,2,0,1)\}. \end{aligned} \tag{247}$$

The semicolon distinguishes the four binary A5 coordinates from the two D4 discriminant coordinates. Let

$$\begin{aligned} \mathcal{H} &= \{(3u + 2z; y) : (u; y) \in C_2, z \in C_3\} \subset D_{R_{\mathrm{ES}}}, \\ N &= \{x \in R_{\mathrm{ES}}^\vee : x + R_{\mathrm{ES}} \in \mathcal{H}\}. \end{aligned} \tag{248}$$

Here $3u + 2z$ is computed coordinatewise modulo six using the integer representatives of u and z.

Theorem 9.3 (Even unimodular completion with exactly 144 roots)**.** *The subgroup $\mathcal{H}$ is isotropic of order 72. Its inverse image N is a positive-definite even unimodular lattice of rank 24, with $[N : R_{\mathrm{ES}}] = 72$. Among the 71 nonzero glue classes, 46 have minimum squared norm 4 and 25 have minimum squared norm 6. The roots of N are exactly those of R_{ES}, with root system $A_5^4 D_4$: 30 roots in each A5 summand and 24 in D4.*

Proof. The binary generators are independent, hence give eight words. Listing their sums shows that the seven nonzero words comprise one word $(1111;00)$ and six words whose first part has weight two and whose second part is nonzero. A ternary word is

$$az_1 + bz_2 = (a + b, a + 2b, a, b).$$

Its last two coordinates prove injectivity of the parameter pair (a, b), so there are nine words. If $a = 0 \neq b$ the third coordinate alone vanishes; if $b = 0 \neq a$ the fourth alone vanishes; and if both are nonzero exactly one of $a + b, a + 2b$ vanishes. Thus every nonzero word has weight three, and each of the four coordinates is the unique vanishing coordinate for two words.

The map $(u_i, z_i) \mapsto 3u_i + 2z_i \pmod 6$ is bijective: reduction modulo two recovers u_i, and reduction modulo three recovers $2z_i$, from which multiplication by 2 recovers z_i. It is a homomorphism of additive groups, since changing binary or ternary representatives by their moduli changes $3u_i + 2z_i$ by a multiple of six. Its product with the retained binary y-coordinate therefore maps $C_2 \times C_3$ injectively onto the subgroup $\mathcal{H}$. It follows that $|\mathcal{H}| = 8 \cdot 9 = 72$. For $u_i \in \{0, 1\}$ and $z_i \in \{0, 1, 2\}$,

$$\frac{5}{6}(3u_i + 2z_i)^2 = \frac{15}{2}u_i + 10u_i z_i + \frac{10}{3}z_i^2 \equiv \frac{3}{2}u_i + \frac{4}{3}z_i^2 \pmod{2\mathbb{Z}}.$$

The cross-term is even; a nonzero z_i has square 1 or 4, whose contributions in the last expression differ by 4. The binary contribution including D4 is therefore zero, 6, or $3 + 1 = 4$, and the ternary contribution is zero or $(4/3)3 = 4$ modulo $2\mathbb{Z}$. This proves isotropy.

The inverse image is a lattice because it is a subgroup between the full lattices R_{ES} and $R_{\mathrm{ES}}^{\vee}$. Isotropy gives $(x, x) \in 2\mathbb{Z}$ for every $x \in N$, and

$$(x, y) = \frac{(x + y, x + y) - (x, x) - (y, y)}{2} \in \mathbb{Z} \qquad (x, y \in N).$$

Thus it is even integral. Its index is $|\mathcal{H}| = 72$. For a finite-index inclusion of equal-rank lattices, the covolume changes by that index, and the Gram determinant is the square of the covolume in orthonormal coordinates. Hence

$$\det N = \frac{6^4 \cdot 4}{72^2} = 1.$$

The dual-index formula proved above implies $N = N^{\vee}$. Its rank and positive definiteness are inherited from the unchanged real span and metric.

For $0 \le k \le 5$, every vector in the A5 class $k\lambda + A$ has the form

$$m - \frac{k}{6}(1, 1, 1, 1, 1, 1), \qquad m \in \mathbb{Z}^6, \quad \sum m_i = k.$$

Its squared norm is $\sum m_i^2 - k^2/6$. Since $m_i^2 \ge m_i$ for every integer m_i, the minimum is at least $k - k^2/6$. It is attained by choosing exactly k of the six entries equal to one and the others zero. Thus

$$\mu_A(k) = \frac{k(6 - k)}{6}. \tag{249}$$

Every nonzero integral D4 class has minimum one, attained by v; every half-integral vector has norm at least $4(1/2)^2 = 1$, attained by s, c in their respective classes. The four A5 minima and the D4 minimum add and are simultaneously attained because the metric is an orthogonal sum.

For a coordinate with $k_i = 3u_i + 2z_i \pmod 6$, its minimum contribution is 0 for $(u_i, z_i) = (0, 0)$, 3/2 for $(1, 0)$, 4/3 for $(0, z_i \ne 0)$, and 5/6 for $(1, z_i \ne 0)$. The complete nonzero case table is

$\mathrm{wt}(u)$	$\mathrm{wt}(z)$	$\lvert\mathrm{supp}(u) \cap \mathrm{supp}(z)\rvert$	classes	minimum squared norm
2	0	0	6	$2(3/2) + 1 = 4$
4	0	0	1	$4(3/2) = 6$
0	3	0	8	$3(4/3) = 4$
2	3	1	24	$(3/2) + 2(4/3) + (5/6) + 1 = 6$
2	3	2	24	$(4/3) + 2(5/6) + 1 = 4$
4	3	3	8	$(3/2) + 3(5/6) = 4.$

(250)

For each of the six weight-two binary words, four nonzero ternary words omit a coordinate inside its support, and four omit a coordinate outside. This proves the two counts $6 \cdot 4 = 24$; all other

counts follow from the enumerated binary and ternary words. Thus the total number at minimum 4 is $6+8+24+8=46$, and the total at minimum 6 is $1+24=25$. Their sum is 71, leaving exactly the zero glue class.

No nonzero glue class can contain a norm-two vector. In the zero class, an A5 norm-two vector has exactly one coordinate 1 and another -1, so is $e_i - e_j$, with $6 \cdot 5 = 30$ possibilities. A D4 norm-two vector has exactly two coordinates of absolute value one, so is $\pm e_i \pm e_j$, with $4\binom{4}{2} = 24$ possibilities. An orthogonal sum of two nonzero root-lattice vectors has norm at least 4 because each summand is positive even. These are therefore all $4 \cdot 30 + 24 = 144$ roots of N. □

A Niemeier lattice means a positive-definite even unimodular lattice of rank 24. The theorem constructs one with root system $A_5^4 D_4$ directly. Its root type appears with lambency 6 in the primary umbral table [24, Table 3, p. 21]. This table reference is not needed to prove the inverse-image construction or any minimum above.

For comparison with the earlier labeled glue presentation in the source [27, 28], put

$$h_1 = (1,3,1,1;00), \qquad h_2 = (1,4,2,3;11), \qquad h_3 = (3,0,3,0;01),$$

and let $\mathcal{H}_{\mathrm{old}}$ be their subgroup. The exact transport is

$$P(x_1,x_2,x_3,x_4;y) = (x_3,x_4,x_1,x_2;y), \qquad P(\mathcal{H}_{\mathrm{old}}) = \mathcal{H}. \tag{251}$$

Indeed, if $rh_1 + th_2 + wh_3 = 0$ with $r,t \in \mathbb{Z}/6$ and $w \in \mathbb{Z}/2$, the final two coordinates imply that t is even and $w = 0$. The first and third coordinates then give $r + t = 0$ and $r + 2t = 0$ modulo six, whence $t = r = 0$. Thus $|\mathcal{H}_{\mathrm{old}}| = 6 \cdot 6 \cdot 2 = 72$. Directly,

$$\begin{aligned} P(h_1) &= 3(1111;00) + 2(2,2,2,0;00), \\ P(h_2) &= 3(0110;11) + 2(1,0,2,2;00), \\ P(h_3) &= 3(1010;01). \end{aligned}$$

The displayed ternary parts are $2z_1$ and $2z_1 + 2z_2$. Consequently $P(\mathcal{H}_{\mathrm{old}}) \subset \mathcal{H}$, and the equal orders prove equality. This transporter is realized by the corresponding isometry permuting the full identified A5 summands.

9.4 The common D4 root lattice, shortest weights, and triality

Use orthonormal coordinates $\varepsilon_1, \dots, \varepsilon_4$ for the ordinary F4 root realization. Its long roots and the three parts of its short roots are

$$\begin{aligned} \Phi_D &= \{\pm\varepsilon_i \pm \varepsilon_j : 1 \le i < j \le 4\}, \\ \mathcal{V} &= \{\pm\varepsilon_i : 1 \le i \le 4\}, \\ \mathcal{S}_\pm &= \left\{ \frac{1}{2}\sum_{i=1}^{4} \delta_i \varepsilon_i : \delta_i \in \{1,-1\},\ \prod_i \delta_i = \pm 1 \right\}, \\ \Phi_F &= \Phi_D \cup \mathcal{V} \cup \mathcal{S}_+ \cup \mathcal{S}_-. \end{aligned} \tag{252}$$

For a nonzero vector a, reflection means the exact map $s_a(x) = x - 2(x,a)a/(a,a)$. Write W_D and W_F for the groups generated by the reflections in Φ_D and Φ_F, respectively.

Theorem 9.4 (Integral D4 and the marked triality quotient)**.** *The map*

$$\iota_D : D \longrightarrow \mathbb{Z}\langle \Phi_D \rangle, \qquad (x_1,x_2,x_3,x_4) \longmapsto \sum_{i=1}^{4} x_i \varepsilon_i \tag{253}$$

is an integral isometry onto the long-root lattice. It extends to duals and discriminant forms, sending the shortest-vector sets of the marked classes v, s, c respectively to $\mathcal{V}, \mathcal{S}_+, \mathcal{S}_-$.

The two matrices

$$R_D = \operatorname{diag}(-1,1,1,1), \qquad H_D = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \tag{254}$$

are the short-root reflections s_{ε_1} and s_c, where $c = (-1,1,1,1)/2$ is read in the ε coordinates. They induce $(s\ c)$ and $(v\ s)$ on the three nonzero discriminant classes. There is a split exact sequence

$$1 \longrightarrow W_D \longrightarrow W_F \longrightarrow S_3 \longrightarrow 1, \qquad W_D \cong \{\textit{even signed permutations of four coordinates}\}, \tag{255}$$

whose displayed section is $\langle R_D, H_D\rangle \cong S_3$. In particular $|W_D| = 192$ and $|W_F| = 1152$.

Proof. The four D4 basis vectors in Theorem 9.2 are among Φ_D, and every vector of Φ_D has even coordinate sum. Thus the displayed identity of coordinates has exactly the stated image, preserves the metric, and extends to duals by the integer-pairing definition. In a nonzero integral dual class, norm one forces one coordinate ± 1 and all others zero. In a half-integral class, norm one forces all four coordinates to be $\pm 1/2$. The difference from s has even coordinate sum exactly when the number of negative signs is even. This proves the three shortest-vector identifications with the stated marking.

The reflection identity for R_D is immediate, and $H_D = I - 2cc^T$ follows by multiplying the four coordinates of c. Both are orthogonal involutions. The first preserves the parity of an integer coordinate sum. For $x \in D$, the four coordinates of $H_D x$ are integers because each numerator is congruent to $\sum x_i$ modulo two, and their sum is $2x_1$. Thus both preserve D. Their effects on the representatives show that R_D fixes the class v and exchanges s, c, whereas $H_D v = s$, $H_D s = v$, and $H_D c = -c \equiv c \pmod{D}$. They therefore preserve all of Φ_F by the shortest-vector description. On the plane with ordered basis ε_1, c, their product has matrix

$$\begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix},$$

whose cube is the identity, while both reflections fix the orthogonal complement. Consequently $R_D^2 = H_D^2 = (R_D H_D)^3 = 1$. Every word in two such involutions reduces to $(R_D H_D)^k$ or $(R_D H_D)^k R_D$, $0 \le k \le 2$. Their discriminant permutations are the six elements of S_3, so these six words are distinct and give the stated section.

Reflection in $\varepsilon_i - \varepsilon_j$ swaps coordinates i, j. Reflection in $\varepsilon_i + \varepsilon_j$ swaps them and changes both signs. Hence the long-root reflections generate every permutation and every even sign change, and each long-root reflection is of this form. This proves the assertion about W_D and its order $2^3 \cdot 4! = 192$. Such a reflection acts trivially on $D^\vee/D$: for a root a of squared norm two, $s_a(x) - x = -(x,a)a \in D$ for all $x \in D^\vee$.

Let $K = \langle W_D, R_D, H_D\rangle$. Each of its generators preserves the long roots, so W_D is normal in K by $g s_a g^{-1} = s_{ga}$. The even signed permutations act transitively on each of $\mathcal{V}, \mathcal{S}_+, \mathcal{S}_-$, and the displayed section permutes these three sets transitively. Thus every short root is $g\varepsilon_1$ for some $g \in K$, and its reflection $g R_D g^{-1}$ belongs to K. All generators of W_F therefore belong to K; the reverse inclusion follows from their definitions, proving $W_F = K$. An element in the kernel of its permutation action on the three short sets preserves $\mathcal{V}$, so it is a signed permutation matrix. Preservation of $\mathcal{S}_+$ forces an even number of sign changes, so the kernel is W_D. This proves exactness, splitting, and the order $6 \cdot 192 = 1152$. □

The exact geometric datum and weight marking. Let $\mathbb{O}$ denote the real normed octonion algebra and let $J = \mathfrak{h}_3(\mathbb{O})$ be the real vector space of Hermitian three-by-three matrices, with Jordan product $a \circ b = (ab + ba)/2$. Define

$$\mathbb{OP}^2 = \{a \in J : a^2 = a,\ \operatorname{tr} a = 1\}, \qquad \mathrm{Fl}(\mathbb{O}) = \{(a, b) \in (\mathbb{OP}^2)^2 : \operatorname{Re}\operatorname{tr}(ab) = 0\}.$$

For the compact group $F_4 = \mathrm{Aut}_{\mathbb{R}}(J, \circ)$, Mare and Willems prove the homogeneous identification $\mathrm{Fl}(\mathbb{O}) = F_4/\operatorname{Spin}(8)$ and identify its tangent representation as $V_8 \oplus S_8^+ \oplus S_8^-$ [25, Definition 2.2, Proposition 2.4, Proposition 2.7, and pp. 491–492]. These are the prior geometric results used here. With their torus coordinate forms L_i, their equations (5.4)–(5.6) assign the vector, positive half-spin, and negative half-spin weights respectively the patterns $\pm L_i$, even half-sign sums, and odd half-sign sums [25, pp. 510–511]. The coordinate map $L_i \mapsto \varepsilon_i$ retains that sign marking and identifies these three weight sets with those of (252).

For completeness the half-spin weight patterns admit the following direct calculation. In the complex exterior algebra $\Lambda^*\mathbb{C}^4$, let a_i be exterior multiplication by the i-th basis vector and b_i contraction by its dual. On a basis wedge, insertion and deletion give

$$a_i a_j + a_j a_i = b_i b_j + b_j b_i = 0, \qquad a_i b_j + b_j a_i = \delta_{ij} I.$$

Thus $C_i = a_i + b_i$ and $D_i = \mathrm{i}(b_i - a_i)$ square to one and mutually anticommute for distinct Clifford generators. Writing $N_i = a_i b_i$, direct multiplication gives

$$\tfrac{1}{2} C_i D_i = \mathrm{i}(N_i - \tfrac{1}{2} I).$$

The four commuting infinitesimal torus operators therefore have weights $\sum_i (\mathbf{1}_{i \in I} - 1/2)\varepsilon_i$ on the basis wedge indexed by $I \subset \{1, 2, 3, 4\}$. The even exterior part has an even number of negative signs, and the odd part has an odd number. There are eight one-dimensional weight spaces in each part. For the vector representation the four real coordinate planes complexify into their two rotation eigenspaces, with weights $\varepsilon_i, -\varepsilon_i$. This proves the displayed weight formulas directly in the stated marking.

The map proved here is consequently the actual chain

$$D \xrightarrow{\iota_D} \mathbb{Z}\langle \Phi_D \rangle, \qquad \{v, s, c\} \longleftrightarrow \{\mathcal{V}, \mathcal{S}_+, \mathcal{S}_-\} = \{\mathrm{wt}(V_8), \mathrm{wt}(S_8^+), \mathrm{wt}(S_8^-)\}.$$

The same S_3 acts through the explicitly proved permutations of these three sets. The companion higher-rung treatment records the geometric datum and root-coordinate comparison at the respective labels

`h24:thm:octonionic-triality`
`h24:prop:two-Coxeter-systems`.

The exact source pointers and passages are included in [28]. The integral metric map in this section is the map on D4 and its dual. The definitions make no map from the whole rank-24 lattice N into the flag tangent space.

9.5 The full indicated glue stabilizer and its triality quotient

All vector spaces over finite fields in this subsection retain their displayed bases. Write

$$j : \mathbb{F}_3^2 \longrightarrow \mathbb{F}_3^4, \qquad j(a, b) = (a + b, a + 2b, a, b), \qquad C_3 = j(\mathbb{F}_3^2).$$

The four row forms defining this injection are

$$\ell_1 = (1,1), \quad \ell_2 = (1,2), \quad \ell_3 = (1,0), \quad \ell_4 = (0,1).$$

They represent all four one-dimensional subspaces of the dual of $\mathbb{F}_3^2$. Indeed each nonzero row has either first coordinate zero, in which case its line is represented by ℓ_4, or a unique scalar multiple with first coordinate one, namely one of $(1,0), (1,1), (1,2)$.

Let $E \subset \mathbb{F}_2^4$ be the even-weight subspace and define

$$\tau : E \longrightarrow \mathbb{F}_2^2, \qquad \tau(u_1,u_2,u_3,u_4) = (u_2+u_4, u_3+u_4).$$

The binary code in the index-72 glue is exactly

$$C_2 = \{(u, \tau(u)) : u \in E\}.$$

To verify this, the independent vectors $1111, 1100, 1010$ form a basis of the three-dimensional space E, and their images under τ are respectively $00, 10, 01$. These are precisely the three stated binary code generators. In particular τ is onto and

$$\ker \tau = \langle 1111 \rangle.$$

Explicitly, $\tau(u) = 0$ gives $u_2 = u_3 = u_4$, and the even-weight equation then gives $u_1 = u_4$.

For a permutation p of $\{1,2,3,4\}$, use the coordinate convention

$$(P_p u)_i = u_{p(i)}.$$

The permutation preserves E and $\langle 1111 \rangle$, so it induces a unique invertible map B_p satisfying

$$B_p \tau(u) = \tau(P_p u) \qquad (u \in E). \tag{256}$$

For an entirely explicit formula, the two columns of B_p are $\tau(P_p(1100))$ and $\tau(P_p(1010))$. The convention implies $P_p P_q = P_{q \circ p}$. Accordingly $B_p B_q = B_{q \circ p}$; no reversal is suppressed in what follows.

The subgroup of (248) can therefore be written

$$\mathcal{H} = \{(3u + 2z; \tau(u)) : u \in E,\ z \in C_3\} \subset (\mathbb{Z}/6)^4 \oplus \mathbb{F}_2^2.$$

Let G be its stabilizer among the transformations

$$g = (\varepsilon, p, B), \qquad (x;y) \longmapsto \big((\varepsilon_i x_{p(i)})_{i=1}^4 ; By\big),$$

where $\varepsilon_i \in \{1,-1\}$, $p \in S_4$, and $B \in GL_2(\mathbb{F}_2)$. Denote the signed permutation matrix on the first four coordinates by M_g.

Theorem 9.5 (An explicit isomorphism to $GL_2(\mathbb{F}_3)$)**.** *Restriction to the ternary code gives an isomorphism*

$$\Theta : G \xrightarrow{\cong} GL_2(\mathbb{F}_3), \qquad M_g j = j\Theta(g).$$

For every $A \in GL_2(\mathbb{F}_3)$, its inverse image is specified uniquely by

$$\ell_i A = \varepsilon_i \ell_{p(i)} \quad (1 \le i \le 4), \qquad B = B_p. \tag{257}$$

In particular $|G| = 48$.

Proof. The elements of $\mathcal{H}$ of order dividing two are precisely $(3u;\tau(u))$, and those of order dividing three are precisely $(2z;0)$. Every indicated ambient transformation preserves these primary parts, so an element of G satisfies $M_gC_3 = C_3$. The injection j therefore yields one and only one $\Theta(g) \in GL_2(\mathbb{F}_3)$. The binary part is preserved exactly when $B\tau(u) = \tau(P_pu)$, because all signs become one modulo two. Surjectivity of τ forces $B = B_p$.

If $\Theta(g) = I$, the equality $M_gj = j$ implies $\varepsilon_i\ell_{p(i)} = \ell_i$ for every i. The four row lines are distinct, so $p(i) = i$. Since each ℓ_i is nonzero and $1 \neq -1$ in $\mathbb{F}_3$, also $\varepsilon_i = 1$ for every i. It follows that $B_p = I$ and $g = 1$. Thus Θ is injective.

Conversely fix $A \in GL_2(\mathbb{F}_3)$. The rows ℓ_iA are nonzero and belong to four distinct row lines. Since the ℓ_i represent every row line, each ℓ_iA has a unique expression $\varepsilon_i\ell_{p(i)}$, where the nonzero scalar in $\mathbb{F}_3$ is represented by the exact integer sign $\varepsilon_i = 1$ or -1. Distinctness of the row lines makes p a permutation. The resulting signed permutation satisfies $M_gj = jA$, hence preserves C_3. Choosing $B = B_p$ preserves C_2 as proved above, hence preserves $\mathcal{H}$. This constructs an element of G mapping to A, proving surjectivity and the stated inverse formula.

Finally

$$M_{gh}j = M_gM_hj = M_gj\Theta(h) = j\Theta(g)\Theta(h),$$

so $\Theta(gh) = \Theta(g)\Theta(h)$. An invertible 2-by-2 matrix over $\mathbb{F}_3$ has eight choices for its first nonzero column and six choices for its second column outside the span of the first. Thus its group order is $8 \cdot 6 = 48$. □

The map to the permutation of the four component labels is more precisely $g \mapsto p^{-1}$, because a vector supported on component k is sent to component $p^{-1}(k)$. Its kernel under Θ is $\{I, -I\}$. In fact a matrix preserving all four row lines is diagonal by preservation of ℓ_3 and ℓ_4, and its diagonal entries are equal by preservation of ℓ_1. Conversely scalar matrices preserve all four lines. Therefore the component permutation image has order $48/2 = 24$, so is all S_4, with

$$G/\Theta^{-1}(\{\pm I\}) \xrightarrow{\overline{\Theta}} PGL_2(\mathbb{F}_3) \cong S_4.$$

The center of $GL_2(\mathbb{F}_3)$ is exactly $\{\pm I\}$: a central matrix commutes with $\mathrm{diag}(1,-1)$, forcing its off-diagonal entries to vanish, and commuting with $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ forces the diagonal entries to agree.

Theorem 9.6 (The exact triality quotient and its quaternion kernel)**.** *The projection $\pi : G \to GL_2(\mathbb{F}_2)$, $(\varepsilon, p, B) \mapsto B$, is onto, and under Θ its kernel is the quaternion subgroup generated by*

$$I_q = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}, \qquad J_q = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

There is a split exact sequence

$$1 \longrightarrow Q_8 \longrightarrow GL_2(\mathbb{F}_3) \xrightarrow{\pi\Theta^{-1}} GL_2(\mathbb{F}_2) \longrightarrow 1.$$

The splitting is the subgroup generated by

$$A_R = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \qquad A_H = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}.$$

It identifies $GL_2(\mathbb{F}_3)$ with $Q_8 \rtimes S_3$ for the conjugation action of this displayed subgroup on this displayed Q_8.

Proof. The nonzero elements of $E/\langle 1111\rangle$ are represented by the complementary pairs of two-element subsets

$$\{12 \mid 34\}, \qquad \{13 \mid 24\}, \qquad \{14 \mid 23\},$$

marked respectively by $10, 01, 11$ via τ. Thus B_p is exactly the permutation action on these three partitions, expressed in the displayed binary marking.

Using (257) gives

$$M(A_R)x = (x_2, x_1, x_3, -x_4), \quad B_R = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

$$M(A_H)x = (x_1, x_3, x_2, -x_4), \quad B_H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

These binary matrices act respectively as the transpositions (01 11) and (10 01), so they generate all permutations of the three nonzero binary vectors. There are $(4-1)(4-2) = 6$ invertible binary matrices; a matrix fixing all nonzero vectors fixes the basis. Hence this permutation action identifies $GL_2(\mathbb{F}_2)$ with S_3 and proves surjectivity of π. Its kernel therefore has order eight.

For I_q the row rule gives

$$p = (2, 1, 4, 3), \qquad \varepsilon = (-1, 1, 1, -1),$$

and for J_q it gives

$$p = (3, 4, 1, 2), \qquad \varepsilon = (-1, -1, 1, 1).$$

Both permutations fix each of the three displayed complementary partitions, so both matrices lie in the kernel. Direct matrix multiplication over $\mathbb{F}_3$ gives

$$I_q^2 = J_q^2 = -I, \qquad I_q J_q = -J_q I_q, \qquad I_q J_q = \begin{pmatrix} 1 & 2 \\ 2 & 2 \end{pmatrix}.$$

The eight matrices $I, -I, \pm I_q, \pm J_q, \pm I_q J_q$ are pairwise distinct by their displayed entries. The relations show this set is a group: powers reduce using $I_q^2 = J_q^2 = -I$, and factors can be reordered using $J_q I_q = -I_q J_q$. These relations identify it with the quaternion group Q_8. Since the kernel has order eight, this is the whole kernel.

The matrices A_R, A_H satisfy

$$A_R^2 = A_H^2 = I, \qquad A_R A_H = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad (A_R A_H)^3 = I.$$

Every word in the two involutions reduces to one of $(A_R A_H)^k$ or $(A_R A_H)^k A_R$, $k = 0, 1, 2$, by alternation and the cubic relation. Their images under π are the six distinct permutations generated by B_R, B_H. Hence the subgroup has exactly six elements and maps isomorphically onto $GL_2(\mathbb{F}_2)$. This gives the splitting. For any g, the unique splitting element with image $\pi(g)$ leaves g times its inverse in the kernel; the intersection of kernel and splitting is trivial. Multiplication therefore realizes the asserted semidirect product with the indicated conjugation action. □

The matrices in the kernel have an additional marked interpretation: their projective component permutations are the four permutations $1, (12)(34), (13)(24), (14)(23)$. Each fixes all three partitions, and these are exactly the kernel of the onto action $S_4 \to S_3$, whose kernel has order $24/6 = 4$. Thus the preceding exact sequence is the precise map from the order-48 ternary-code symmetry to the three D_4 discriminant classes.

Theorem 9.7 (A faithful action on the actual index-72 lattice)**.** *Fix the four identifications of the A_5 coordinate lattices used in the glue presentation, and the discriminant marking $10 \mapsto v, 01 \mapsto s, 11 \mapsto c$ of D_4. The group $G \cong GL_2(\mathbb{F}_3)$ has a faithful action by integral isometries of the index-72 overlattice. Write $\varphi_i : A \longrightarrow A(F_{\eta_i})$ for the restriction of the fixed free-coordinate isometry $e_j \mapsto e_{f_i 289^j}$. For $X_i \in A(F_{\eta_i})$, the i-th output component is exactly $\varepsilon_i \varphi_i \varphi_{p(i)}^{-1} X_{p(i)}$. In the identified A coordinates this is the action*

$$L_g(X_1, X_2, X_3, X_4; d) = \big(\varepsilon_1 X_{p(1)}, \ldots, \varepsilon_4 X_{p(4)}; \delta(B) d\big),$$

where $\delta : GL_2(\mathbb{F}_2) \to O(D_4)$ is the section given by

$$\delta(B_R) = R_D = \mathrm{diag}(-1, 1, 1, 1), \qquad \delta(B_H) = H_D = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

Under Θ, the two displayed full triality lifts are exactly A_R and A_H.

Proof. The four identified A_5 summands carry the same Euclidean metric, so the displayed signed permutation is a root-lattice isometry. The matrices R_D, H_D preserve D_4: R_D negates one integral coordinate and preserves parity of the sum; for $x \in D_4$, each coordinate of $H_D x$ is integral because its numerator has the same parity as $\sum x_i$, and the coordinate sum of $H_D x$ is $2x_1$. Both matrices are orthogonal involutions by multiplication. Furthermore $R_D = s_{e_1}$, $H_D = s_c$, with $c = (-1, 1, 1, 1)/2$, and $(e_1, c) = -1/2$. On the plane with basis e_1, c, their product has matrix $\left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$ and has cube the identity; both reflections fix the orthogonal complement pointwise. The discriminant actions are the two distinct transpositions B_R, B_H. The same word reduction as in the preceding proof therefore shows that $\langle R_D, H_D \rangle$ has exactly six elements and that its discriminant action is an isomorphism onto $GL_2(\mathbb{F}_2)$. Its inverse is the homomorphism δ.

Thus $g \mapsto L_g$ is a homomorphism of root-lattice isometries. It extends to the dual lattice because an isometry preserves the integer-pairing condition defining the dual. Its discriminant action is the specified (ε, p, B), which preserves $\mathcal{H}$. Consequently it preserves the inverse image

$$N = \{x \in R_{\mathrm{ES}}^{\vee} : x + R_{\mathrm{ES}} \in \mathcal{H}\}.$$

It is therefore an integral isometry of N. If L_g is the identity on N, it is the identity on the full real span, because N contains the full-rank lattice R_{ES}. It is therefore also the identity on $R_{\mathrm{ES}}^{\vee}/R_{\mathrm{ES}}$, in particular on the ternary code. Hence $\Theta(g) = I$, and injectivity of Θ gives $g = 1$. This proves faithfulness. The exact formulas for A_R, A_H were established above, so their lifts are the stated full triality transformations. □

This constructs an explicit finite subgroup of $O(N)$, including its action on all five root summands and on their glue, and its exact D_4-triality quotient. The group and its projective quotient agree with the $A_5^4 D_4$ row of [24, Table 3, p. 21]; the isomorphisms here have been proved by their actual code actions.

For clarity, the two full triality elements, denoted $\widehat{R} = L_{\Theta^{-1}(A_R)}$ and $\widehat{H} = L_{\Theta^{-1}(A_H)}$, act in the fixed coordinates by

$$\widehat{R}(X_1, X_2, X_3, X_4; d) = (X_2, X_1, X_3, -X_4; R_D d),$$
$$\widehat{H}(X_1, X_2, X_3, X_4; d) = (X_1, X_3, X_2, -X_4; H_D d).$$

Their squares and the cube of their product are the identity on the entire lattice, by the proved faithful group homomorphism. The product on the first three A5 summands is the indicated three-cycle; its two negations on the fourth summand cancel.

9.6 The exact reduced-shell divisor bijection

Let p be a positive prime with $p \equiv 1 \pmod{24}$, and let the positive integer shell R satisfy $R \equiv 3 \pmod{4}$. Retain

$$a = a_R(p) = \frac{p+R}{4}, \qquad N_{\mathrm{sh}} = pa, \qquad g = \gcd(R, N_{\mathrm{sh}}), \qquad R_0 = \frac{R}{g}, \quad N_0 = \frac{N_{\mathrm{sh}}}{g}. \tag{258}$$

The integer N_{sh} is the source's scalar $N = pa$; the subscript distinguishes it from the lattice N without changing either object. In particular a, R_0, N_0 are positive integers, $\gcd(R_0, N_0) = 1$, and the prescribed first denominator equation is

$$\frac{4}{p} - \frac{1}{a} = \frac{4a-p}{pa} = \frac{R}{N_{\mathrm{sh}}} = \frac{R_0}{N_0} = \frac{1}{b} + \frac{1}{c}. \tag{259}$$

Here b, c are positive integer denominators, and their order is initially retained. Define the finite set

$$\mathcal{D}_{R,p} = \{d \in \mathbb{Z}_{>0} : d \mid N_0^2,\ d \equiv -N_0 \pmod{R_0}\}, \qquad A_R(p) = |\mathcal{D}_{R,p}|.$$

Theorem 9.8 (Certificates, ordered pairs, and complementary pairs)**.** *There is an exact bijection*

$$\begin{aligned} \mathcal{D}_{R,p} &\longrightarrow \{(b,c) \in \mathbb{Z}_{>0}^2 : 4/p = 1/a + 1/b + 1/c\}, \\ d &\longmapsto \left(\frac{N_0 + d}{R_0}, \frac{N_0 + N_0^2/d}{R_0} \right), \end{aligned} \tag{260}$$

with inverse $d = R_0 b - N_0$. The involution $d \mapsto N_0^2/d$ corresponds exactly to $(b,c) \mapsto (c,b)$. Consequently the number of unordered pairs of remaining denominators is

$$U_R(p) = \frac{A_R(p) + \delta_{R,p}}{2}, \qquad \delta_{R,p} = \begin{cases} 1, & R_0 \mid 2, \\ 0, & R_0 \nmid 2. \end{cases} \tag{261}$$

The unique possible fixed certificate is $d = N_0$, giving $b = c = 2N_0/R_0$.

Proof. For a positive solution of (259), $R_0 b/N_0 = 1 + b/c > 1$, so $d = R_0 b - N_0 > 0$; likewise $e = R_0 c - N_0 > 0$. Multiplying the fraction equation by $N_0 bc$ gives $R_0 bc = N_0(b+c)$, and therefore

$$(R_0 b - N_0)(R_0 c - N_0) = N_0^2.$$

Thus $d \mid N_0^2$, $d \equiv -N_0 \pmod{R_0}$, and $e = N_0^2/d$.

Conversely, let $d \in \mathcal{D}_{R,p}$ and $e = N_0^2/d$. For $R_0 > 1$, the coprimality $\gcd(R_0, N_0) = 1$ makes $d \equiv -N_0$ a unit, and hence

$$e \equiv N_0^2(-N_0)^{-1} \equiv -N_0 \pmod{R_0}.$$

For $R_0 = 1$, this congruence is vacuous and holds automatically. Thus both expressions in (260) are positive integers. Substitution into $(R_0 b - N_0)(R_0 c - N_0) = N_0^2$ and expansion give the original fraction equation. Each construction recovers exactly $d = R_0 b - N_0$ and $e = R_0 c - N_0$, so they are inverse, proving the bijection and the involution statement.

For a positive divisor, $d = N_0^2/d$ holds exactly when $d = N_0$. This divisor is a certificate exactly when $N_0 \equiv -N_0 \pmod{R_0}$, that is, $R_0 \mid 2N_0$. Coprimality makes this equivalent to $R_0 \mid 2$. Every other orbit contains two certificates, so counting the fixed orbits and the two-element orbits gives $U_R(p) = \delta_{R,p} + (A_R(p) - \delta_{R,p})/2$, exactly (261). □

Factor the exact integer N_0 as $N_0 = \prod_{j=1}^{m} q_j^{e_j}$, with distinct positive primes q_j and positive exponents e_j, and define

$$\begin{aligned} B(N_0) &= \prod_{j=1}^{m} \{-e_j, -e_j+1, \dots, e_j\}, \\ \lambda_{R_0,N_0} : B(N_0) &\longrightarrow (\mathbb{Z}/R_0\mathbb{Z})^{\times}, \qquad \lambda_{R_0,N_0}(\beta) = \prod_{j=1}^{m} q_j^{\beta_j} \pmod{R_0}. \end{aligned} \tag{262}$$

The group for modulus one is taken to be the one-element group, so that the same notation also expresses the vacuous congruence in that case. For $R_0 > 1$, every q_j is a unit by coprimality, so all negative powers in this formula are defined.

Theorem 9.9 (The exact exponent-box map and the two raw channels)**.** *The map*

$$\beta \longmapsto d = \prod_{j=1}^{m} q_j^{e_j+\beta_j} \tag{263}$$

bijects $\lambda_{R_0,N_0}^{-1}(-1)$ with $\mathcal{D}_{R,p}$. It retains each valuation and both original integers R_0, N_0.

In addition, suppose $\gcd(R, pa) = 1$ *and* $p \nmid a$*. Write*

$$a = \prod_{j=1}^{s} q_j^{e_j}, \qquad B(a) = \prod_{j=1}^{s} \{-e_j, \dots, e_j\},$$

and define, for each unit t modulo R,

$$c_R(a;t) = \#\left\{ \beta \in B(a) : \prod_j q_j^{\beta_j} \equiv t \pmod{R} \right\}.$$

Then the number of raw divisor certificates, equivalently ordered denominator pairs, is

$$A_R(p) = c_R(a;-1) + 2c_R(a;-p^{-1}). \tag{264}$$

Its complementary-divisor quotient is the count in (261)*, with $R_0 = R$ under these additional coprimality hypotheses.*

Proof. A divisor of N_0^2 has one and only one exponent vector $\alpha_j \in \{0, \dots, 2e_j\}$. The exact change $\beta_j = \alpha_j - e_j$ gives the entire displayed box and $dN_0^{-1} \equiv \prod_j q_j^{\beta_j} \pmod{R_0}$. Multiplication of $d \equiv -N_0$ by the unit N_0^{-1} gives precisely the target -1; modulo one both statements are vacuous. This proves the first bijection and its inverse.

Under the additional hypotheses, $R_0 = R$, $N_0 = pa$, and the exponent of p in pa is exactly one. Every positive integer divisor d of p^2a^2 therefore has exactly one valuation-index representation

$$d = p^{t+1} \prod_j q_j^{e_j+\beta_j}, \qquad (t, \beta) \in \{-1, 0, 1\} \times B(a).$$

After division by pa, its unit-group residue is

$$d(pa)^{-1} \equiv p^t \prod_j q_j^{\beta_j} \pmod{R}.$$

Distinct valuation-index pairs may have the same unit-group residue; they remain distinct integer divisors in the count. The three conditions for this to equal -1 are

t	$\prod_j q_j^{\beta_j} \pmod R$
-1	$-p$
0	-1
1	$-p^{-1}$.

The negation $\beta \mapsto -\beta$ is a bijection of $B(a)$ and inverts the unit-group value, so it bijects the fibres at $-p$ and $-p^{-1}$. Summing the sizes of these three disjoint raw exponent classes proves (264). The complementary divisor sends (t, β) to $(-t, -\beta)$, so identifying it changes the raw count as proved in Theorem 9.8; it does not leave the factor two in (264) unchanged. □

This counting convention corrects the wording attached to the source's two-channel formula: the displayed sum is a raw count, whereas a count after identifying complementary divisors is (261). The exact source locator and formula are retained in [26, 28].

9.7 A fixed-shell count does not descend to the residue fibre

Theorem 9.10 (Two primes in the same residue with different shell counts)**.** *The two primes* 2521 *and* 3361 *both reduce to* $1 \in S_{840}$*, but for the same shell* $R = 3$ *their raw and unordered counts are*

$$A_3(2521) = 0, \qquad A_3(3361) = 6, \qquad U_3(2521) = 0, \quad U_3(3361) = 3. \tag{265}$$

One actual denominator certificate in the second case is

$$\frac{4}{3361} = \frac{1}{841} + \frac{1}{974690} + \frac{1}{28266010}. \tag{266}$$

Thus the raw count, its unordered quotient, and the indicator of completion of this fixed shell each fail to factor through prime reduction modulo 840.

Proof. The residue equations are the literal equalities $2521 = 3 \cdot 840 + 1$ and $3361 = 4 \cdot 840 + 1$. For an explicit finite primality verification, the primes at most 57, in increasing order, are

$$\mathcal{P} = (2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53).$$

The remainders against its first 15 entries for 2521, all 16 entries for 3361, and its first nine entries for 631, are respectively

$$(1, 1, 1, 1, 2, 12, 5, 13, 14, 27, 10, 5, 20, 27, 30),$$
$$(1, 1, 1, 1, 6, 7, 12, 17, 3, 26, 13, 31, 40, 7, 24, 22),$$
$$(1, 1, 1, 1, 4, 7, 2, 4, 10).$$

All are nonzero, and the respective integer square roots are 50, 57, and 25. The prime 29 has remainders $1, 2, 4$ against $2, 3, 5$, the primes at most its integer square root 5. Every composite positive integer greater than one has a prime divisor at most its square root: in a factorization into two factors greater than one, the smaller factor is at most the square root and has a prime divisor no larger than itself. These nonzero remainder lists therefore prove each stated primality.

For $R = 3$,

$$a_3(2521) = 631, \qquad a_3(3361) = 841 = 29^2.$$

Both primes are 1 modulo three; also $631 \equiv 1 \pmod 3$ and $841 \equiv 1 \pmod 3$. Thus $\gcd(3, pa) = 1$ in both cases. Both a values are smaller than their respective primes, so $p \nmid a$. Every additional hypothesis of Theorem 9.9 is now verified. The two targets -1 and $-p^{-1}$ coincide modulo three.

For $a = 631$, all exponents $-1, 0, 1$ give residue one, so the target -1 has count zero. For $a = 29^2$, among the five exponents $-2, -1, 0, 1, 2$, exactly $-1, 1$ give -1 modulo three because $29 \equiv -1 \pmod 3$. Both target counts are therefore 2 and the raw count is $2 + 2 \cdot 2 = 6$. Here $R_0 = 3 \nmid 2$, so both unordered counts are half the corresponding raw counts. This proves (265).

For the second prime, retain

$$N_{\mathrm{sh}} = 3361 \cdot 841 = 2826601, \qquad d = 3361 \cdot 29 = 97469, \qquad \frac{N_{\mathrm{sh}}^2}{d} = 3361 \cdot 29^3 = 81971429.$$

Then

$$\frac{N_{\mathrm{sh}} + d}{3} = 974690, \qquad \frac{N_{\mathrm{sh}} + N_{\mathrm{sh}}^2/d}{3} = 28266010.$$

The bijection of Theorem 9.8 now proves (266). The first prime has no completion with this prescribed first denominator by its zero certificate count. Two arguments with the same residue and unequal function values cannot be the pullback of a single residue function: if $f(p) = h(p \bmod 840)$, equality of the residues forces equality of the values. Applying this observation to the three displayed functions proves the claimed failures of factorization. □

9.8 An actual residue cocycle and a retained-sample Fourier map

For $r \in S_{840}$, multiplication by r permutes each fibre. Let $\rho(r)$ act by

$$\rho(r)e_s = e_{rs} \quad (s \in V_{840}), \qquad \rho(r)|_D = I, \qquad b(r) = e_r - e_1 \in A(S_{840}) \subset N. \tag{267}$$

The two free coordinates in $b(r)$ belong to the principal fibre.

Theorem 9.11 (The full-lattice residue action and its exact cocycle)**.** *The permutations $\rho(r)$ extend to a faithful homomorphism $\rho : S_{840} \to O(N)$. The map $b : S_{840} \to N$ satisfies*

$$b(rs) = b(r) + \rho(r)b(s) \qquad (r, s \in S_{840}), \qquad b(1) = 0, \qquad (b(r), b(r)) = 2 \quad (r \neq 1). \tag{268}$$

Thus reduction of any prime into S_{840}, followed by b, is an explicit lattice-valued map on that prime domain.

Proof. Multiplication permutes the orthonormal free coordinates, preserves their augmentation kernels, and obeys $\rho(r)\rho(s) = \rho(rs)$. On the i-th fibre its dual discriminant generator is

$$\lambda_i = e_{f_i} - \frac{1}{6} \sum_{t \in F_{\eta_i}} e_t.$$

Its image differs from it by $e_{rf_i} - e_{f_i} \in A(F_{\eta_i})$. Hence $\rho(r)$ acts trivially on the entire A5 discriminant group, since λ_i generates it. It fixes D and therefore the whole discriminant group, in particular $\mathcal{H}$. It follows from the inverse-image definition that it preserves N. If $\rho(r)$ were the identity, applying it to $e_1 - e_{289}$ would give $e_r - e_{289r} = e_1 - e_{289}$. Since $289 \neq 1$, the positive coordinate on the left is exactly r, so comparison of free-coordinate coefficients forces $r = 1$. This proves faithfulness.

Finally

$$b(r) + \rho(r)b(s) = (e_r - e_1) + (e_{rs} - e_r) = e_{rs} - e_1 = b(rs).$$

For $r \neq 1$, the two basis vectors in the difference are orthogonal unit vectors, giving squared norm two. The stated prime map is the composition $p \mapsto p \bmod 840 \mapsto b(p \bmod 840)$, whose domains have all been specified. □

Fix a positive shell $R \equiv 3 \pmod 4$, and let P be any finite set of positive primes whose reduction lies in S_{840}. For every $p \in P$, use its actual count $A_R(p)$ defined by the reduced divisor set, including the full data of (258). Define

$$\begin{aligned} w_{R,P}(r) &= \sum_{\substack{p \in P \\ p \equiv r \pmod{840}}} A_R(p), \\ \widehat{w}_{R,P}(k) &= \sum_{r \in S_{840}} w_{R,P}(r) \exp\left(-\frac{2\pi \mathrm{i} k \kappa(r)}{6}\right), \qquad k \in \mathbb{Z}/6\mathbb{Z}. \end{aligned} \tag{269}$$

This is the pushforward along prime reduction, followed by the finite Fourier transform with the already fixed generator 289.

Theorem 9.12 (Inversion and an integral augmentation-valued count map)**.** *The function $w_{R,P}$ is recovered by*

$$w_{R,P}(r) = \frac{1}{6} \sum_{k=0}^{5} \widehat{w}_{R,P}(k) \exp\left(\frac{2\pi \mathrm{i} k \kappa(r)}{6}\right). \tag{270}$$

Let $u = \sum_{r \in S_{840}} e_r$, $W_{R,P} = \sum_r w_{R,P}(r) e_r$, and $M_{R,P} = \sum_r w_{R,P}(r) = \widehat{w}_{R,P}(0)$. Then

$$Q_{R,P} = 6W_{R,P} - M_{R,P} u \in A(S_{840}) \subset N \tag{271}$$

is a well-defined integral lattice-valued count map. Put $\zeta_6 = e^{2\pi \mathrm{i}/6}$ and

$$f_k = \sum_{j=0}^{5} \zeta_6^{kj} e_{289^j} \qquad (0 \le k \le 5).$$

The five vectors $f_1, \dots, f_5$ form a basis of $A(S_{840}) \otimes \mathbb{C}$, with

$$\rho(289) f_k = \zeta_6^{-k} f_k, \qquad Q_{R,P} = \sum_{k=1}^{5} \widehat{w}_{R,P}(k) f_k. \tag{272}$$

Both the prime sample P and the shell R are retained in the domain of these maps.

Proof. For an integer m, the sum $\sum_{k=0}^{5} \zeta_6^{km}$ equals 6 if $6 \mid m$. Otherwise it is the geometric sum $(1 - \zeta_6^{6m})/(1 - \zeta_6^m) = 0$. Substituting (269) into the right side of (270) therefore gives

$$\sum_{s \in S_{840}} w_{R,P}(s) \frac{1}{6} \sum_{k=0}^{5} \zeta_6^{k(\kappa(r) - \kappa(s))} = w_{R,P}(r),$$

because κ is bijective. This proves inversion.

The coefficient sum of $6W_{R,P} - M_{R,P}u$ is $6M_{R,P} - 6M_{R,P} = 0$, and its coefficients are integers, so it belongs to the exact augmentation lattice. The same geometric sum shows that f_k has coefficient sum zero for $1 \le k \le 5$ and $f_0 = u$. Moreover the six f_k are independent: the coefficient Fourier transform of f_k is six in coordinate k and zero in all other coordinates, by that same identity. Since the augmentation space has dimension five, its five nonzero-frequency vectors form a basis. Replacing j by $j + 1$ in $\rho(289) f_k$ gives the multiplier ζ_6^{-k}, including its displayed sign.

Inversion in each free coordinate gives $W_{R,P} = (1/6) \sum_{k=0}^{5} \widehat{w}_{R,P}(k) f_k$. Multiplication by 6 and subtraction of $M_{R,P} u = \widehat{w}_{R,P}(0) f_0$ prove the second identity in (272). All sums were taken over the specified finite set P and its residue image. The transform therefore recovers these aggregate counts; it makes no identification between the separate counts of distinct primes with the same residue. Their possible inequality is proved in Theorem 9.10. □

9.9 The entire lattice automorphism group and its Weyl quotient

Cheng, Duncan and Harvey define the umbral group as the quotient of the Niemeier lattice automorphism group by its root-reflection subgroup; their Table 3 identifies this quotient as $GL_2(3)$ for $A_5^4D_4$ [24, Section 2.4 and Table 3]. The argument here proves that identification in the present residue, metric, discriminant and glue coordinates, and locates the arithmetic six-cycle in the full group. It does not claim the abstract group as new. The displayed complement is the particular lift already constructed here; no equality of its representation with a simple-root-preserving section is assumed.

Retain the exact lattice N, the root lattice R_{ES}, the four fibre identifications φ_i, the discriminant marking, the glue subgroup $\mathcal{H}$, and the lift $g \mapsto L_g$ constructed above. For a lattice K in a Euclidean space, $O(K)$ means the group of real linear isometries of its real span that map K onto itself. For the present root lattice define

$$W(R_{\mathrm{ES}}) = \langle s_\alpha : \alpha \in R_{\mathrm{ES}},\ (\alpha,\alpha) = 2\rangle, \qquad s_\alpha(x) = x - (x,\alpha)\alpha.$$

All actions below retain the previously fixed six-coordinate order in each A_5 summand and the marking $10 \mapsto v$, $01 \mapsto s$, $11 \mapsto c$ in $D^\vee/D$.

Theorem 9.13 (All isometries of the displayed A_5 lattice)**.** *Let $A = \{x \in \mathbb{Z}^6 : \sum_{j=0}^{5} x_j = 0\}$ with its displayed Euclidean metric, and for a permutation σ of $\{0,1,2,3,4,5\}$ let $Q_\sigma e_j = e_{\sigma(j)}$. Every element of $O(A)$ has exactly one expression $\epsilon Q_\sigma|_{\mathbb{R}A}$, where $\epsilon \in \{1,-1\}$. Thus*

$$O(A) = \langle Q_\sigma|_{\mathbb{R}A} : \sigma \in S_6\rangle \times \{I,-I\}, \qquad W(A) = \langle Q_\sigma|_{\mathbb{R}A} : \sigma \in S_6\rangle \cong S_6.$$

On the marked discriminant group $D_A = (\mathbb{Z}/6)\lambda$, $\lambda = e_0 - \frac{1}{6}\sum_{j=0}^{5} e_j$, this isometry acts by $k\lambda + A \mapsto \epsilon k\lambda + A$. The kernel of this discriminant action is exactly $W(A)$.

Proof. The norm-two vectors are precisely $\alpha_{ij} = e_i - e_j$ for $i \neq j$. For two distinct such roots,

$$(\alpha_{ij}, \alpha_{kl}) = \delta_{ik} - \delta_{il} - \delta_{jk} + \delta_{jl}.$$

Their inner product is one exactly when they have the same first index or the same second index. Indeed equality of first indices, with distinct second indices, makes the displayed expression one; equality of second indices, with distinct first indices, does the same. If both positive Kronecker terms vanish the expression is nonpositive, and if both are one then the roots are equal, the excluded case.

Put

$$C_i = \{\alpha_{ij} : j \neq i\}, \qquad -C_i = \{\alpha_{ji} : j \neq i\} \qquad (0 \le i \le 5).$$

These twelve sets are exactly the maximal sets of roots in which every two distinct members have inner product one. To prove this, a singleton can be enlarged by keeping its first index and choosing a third index, so a maximal such set contains a pair. If the pair is α_{ij}, α_{ik}, with $j \neq k$, a further root α_{lm} has inner product one with both only if

$$(l = i \text{ or } m = j) \quad \text{and} \quad (l = i \text{ or } m = k).$$

Since $j \neq k$, these conditions force $l = i$. Thus the whole set is contained in C_i, and maximality makes it equal to C_i. For a pair with common second index the same argument, after negating both roots and every possible further member, gives $-C_i$. Each C_i and $-C_i$ has five roots and is maximal by this argument.

Form a graph whose vertices are these twelve maximal sets, with an edge when the two sets have nonempty intersection. Its intersections are exactly

$$C_i \cap C_j = (-C_i) \cap (-C_j) = \varnothing \quad (i \neq j), \qquad C_i \cap (-C_j) = \begin{cases} \{\alpha_{ij}\}, & i \neq j, \\ \varnothing, & i = j. \end{cases}$$

Consequently it is the bipartite graph on the two displayed six-element families with every cross-edge except the six pairs $(C_i, -C_i)$. It is connected: two distinct vertices on the same side have a common neighbor indexed by any index different from both; vertices on opposite sides with different indices are adjacent; and C_i and $-C_i$ are joined by the path $C_i, -C_j, C_k, -C_i$ for distinct i, j, k. A connected bipartite graph has only two choices for its ordered bipartition, because the side of a fixed vertex determines the side of each other vertex along any connecting path. Hence an isometry of A, which permutes roots and preserves their inner products and set intersections, either preserves both displayed families or exchanges them.

In the first case write $C_i \mapsto C_{\sigma(i)}$ and $-C_i \mapsto -C_{\tau(i)}$. The unique opposite-family vertex not adjacent to C_i is $-C_i$. Preservation of that missing edge forces $\tau(i) = \sigma(i)$ for each i. The singleton intersection formula then gives $\alpha_{ij} \mapsto \alpha_{\sigma(i)\sigma(j)}$. In the case exchanging the families, composition with $-I$ reduces to the first case, so the root action is $\alpha_{ij} \mapsto -\alpha_{\sigma(i)\sigma(j)}$. The roots $e_j - e_5$, $0 \le j \le 4$, span $\mathbb{R}A$, so the root action determines the whole linear isometry. These cases give precisely the asserted expressions.

The permutation σ is determined by the permutation of the six vertices C_i, after the sign is determined by whether the two families are preserved or exchanged. Thus the expression is unique. Every Q_σ and $-I$ preserves A, and $-I$ commutes with every linear map; uniqueness proves the direct product. Reflection in $e_i - e_j$ interchanges free coordinates i, j and fixes the others, as follows on substituting this root in $x \mapsto x - (x, e_i - e_j)(e_i - e_j)$. Every permutation is a product of transpositions: each cycle $(i_1\ i_2\ \cdots\ i_m)$ is $(i_1\ i_m)(i_1\ i_{m-1}) \cdots (i_1\ i_2)$. These reflections therefore generate precisely the displayed S_6. Finally

$$Q_\sigma \lambda - \lambda = e_{\sigma(0)} - e_0 \in A.$$

Thus the discriminant action is the stated sign. Since $\lambda + A$ has exact order six, its negative is distinct from itself; the action is trivial exactly when $\epsilon = 1$. □

Theorem 9.14 (All isometries of the displayed D_4 lattice)**.** *For the exact lattice $D = \{x \in \mathbb{Z}^4 : \sum_i x_i \equiv 0 \pmod 2\}$, the action on its marked discriminant group gives the split exact sequence*

$$1 \longrightarrow W_D \longrightarrow O(D) \xrightarrow{d_D} GL_2(\mathbb{F}_2) \longrightarrow 1, \qquad O(D) = W_D \rtimes \delta(GL_2(\mathbb{F}_2)).$$

Here W_D consists of all signed coordinate permutations with an even number of minus signs, and δ is exactly the section generated by the earlier matrices R_D, H_D. In particular $O(D) = W_F$ in the root and weight coordinates of Theorem 9.4.

Proof. Every isometry of D preserves its dual: if $T \in O(D)$, $x \in D^\vee$, and $a \in D$, then $(Tx, a) = (x, T^{-1}a) \in \mathbb{Z}$. It therefore induces an additive automorphism of $D^\vee/D \cong \mathbb{F}_2^2$, giving the homomorphism d_D. The three nonzero classes have the exact norm-one representative sets

$$\begin{aligned} \mathcal{V} &= \{\pm e_i : 1 \le i \le 4\}, \\ \mathcal{S}_+ &= \{\tfrac{1}{2}\textstyle\sum_i \delta_i e_i : \prod_i \delta_i = 1\}, \\ \mathcal{S}_- &= \{\tfrac{1}{2}\textstyle\sum_i \delta_i e_i : \prod_i \delta_i = -1\}, \qquad \delta_i \in \{1, -1\}. \end{aligned}$$

in the respective marked classes v, s, c. For the integral nonzero class, norm one forces exactly one coordinate ± 1. A half-integral vector has norm at least $4(1/2)^2 = 1$, with equality exactly when every coordinate is $\pm 1/2$. The difference of such a vector from $s = (1,1,1,1)/2$ has one coordinate -1 per minus sign, so it belongs to D exactly when their number is even. This proves the three sets and their marking directly.

If $d_D(T) = I$, then T preserves $\mathcal{V}$. The vectors Te_i are orthonormal members of $\mathcal{V}$, so $Te_i = \delta_i e_{\sigma(i)}$ for a permutation σ of four indices and signs δ_i. Preservation of $\mathcal{S}_+$ forces $Ts \in \mathcal{S}_+$, hence $\prod_i \delta_i = 1$. Thus the kernel consists only of even signed permutations.

Conversely, reflection in $e_i - e_j$ interchanges coordinates i, j, and reflection in $e_i + e_j$ interchanges them and negates both. Their composition produces a sign change in exactly those two coordinates. Any even set of sign changes is a product of such disjoint pairs, and transpositions generate all permutations. The root reflections therefore generate every even signed permutation, and each root reflection is itself such a matrix. If $\alpha \in D$ has norm two and $x \in D^\vee$, then $s_\alpha x - x = -(x,\alpha)\alpha \in D$. Every root reflection acts trivially on the discriminant group. This proves $\ker d_D = W_D$ in both directions.

For completeness the section has the original exact matrices

$$R_D = \operatorname{diag}(-1,1,1,1), \qquad H_D = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

They preserve D, are orthogonal involutions, and act on (v, s, c) by $(s\ c)$ and $(v\ s)$, respectively, as calculated in Theorem 9.4. Their binary matrices in the retained marking are

$$B_R = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B_H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The same earlier coordinate calculation gives $R_D^2 = H_D^2 = (R_D H_D)^3 = I$; thus their subgroup has at most six elements, by reducing words to $(R_D H_D)^k$ or $(R_D H_D)^k R_D$, $k = 0, 1, 2$. The induced two transpositions generate the six permutations of the three nonzero binary vectors. There are exactly six invertible binary matrices, by the $3 \cdot 2$ choices of their independent columns, and their action on the three nonzero vectors is faithful. Hence the six words have distinct discriminant actions, and the restriction of d_D to $\langle R_D, H_D \rangle$ is an isomorphism onto $GL_2(\mathbb{F}_2)$. Its inverse is the specified δ. For every $T \in O(D)$, $T\delta(d_D(T))^{-1} \in \ker d_D = W_D$. The two factors intersect trivially because $d_D \delta = I$. This proves the asserted splitting and exhausts $O(D)$. The earlier theorem proved $W_F = \langle W_D, R_D, H_D \rangle$, so it also proves the last equality. □

Theorem 9.15 (The exact full automorphism quotient of N)**.** *Every isometry of N has a unique factorization*

$$T = wL_g, \qquad w \in W(R_{\mathrm{ES}}), \quad g \in G.$$

The Weyl group is normal, and the induced discriminant action gives the split exact sequence and the indicated isomorphisms

$$\begin{aligned} 1 \longrightarrow W(R_{\mathrm{ES}}) \longrightarrow O(N) &\xrightarrow{d_{\mathrm{ES}}} G \longrightarrow 1, \\ O(N) &= W(R_{\mathrm{ES}}) \rtimes L(G), \\ L(G) &\xrightarrow{\cong} O(N)/W(R_{\mathrm{ES}}) \cong G \xrightarrow{\Theta} GL_2(\mathbb{F}_3), \\ W(R_{\mathrm{ES}}) &\cong S_6^4 \times W_D, \qquad |O(N)| = 48 \cdot (6!)^4 \cdot (2^3 \cdot 4!). \end{aligned} \tag{273}$$

In identified coordinates, if $w = (Q_{\sigma_1}, Q_{\sigma_2}, Q_{\sigma_3}, Q_{\sigma_4}; w_D)$ *and* $g = (\varepsilon, p, B)$*, the exact semidirect-product action is*

$$L_g w L_g^{-1} = (Q_{\sigma_{p(1)}}, Q_{\sigma_{p(2)}}, Q_{\sigma_{p(3)}}, Q_{\sigma_{p(4)}}; \delta(B) w_D \delta(B)^{-1}). \tag{274}$$

In the original fibre coordinates the i*-th permutation factor on the right means* $\varphi_i Q_{\sigma_{p(i)}} \varphi_i^{-1}$*.*

Proof. Theorem 9.3 proves that the norm-two vectors of N are exactly the 144 roots of its five original orthogonal summands. Join two distinct roots by an edge when their inner product is nonzero. The five summands are exactly the connected components of this graph. There are no edges between different summands by orthogonality. Within an A_5 summand, the simple roots $e_0 - e_1, e_1 - e_2, e_2 - e_3, e_3 - e_4, e_4 - e_5$ have consecutive inner product -1, forming a connected graph, and span its real space. Every root not already among them has nonzero inner product with at least one of them; otherwise it is orthogonal to its entire real space and has norm zero, contrary to norm two. Thus its whole root graph is connected. In D, the four roots

$$e_1 - e_2, \quad e_2 - e_3, \quad e_3 - e_4, \quad e_3 + e_4$$

span $\mathbb{R}D$, and the second has inner product -1 with each of the other three. The same spanning argument proves connectedness of the D_4 root graph.

An element $T \in O(N)$ permutes the norm-two vectors and preserves these edges, hence permutes their connected components. Their cardinalities are $30, 30, 30, 30, 24$, so T preserves the D_4 component and permutes the four A_5 components. The roots generate each original lattice summand: the five and four displayed simple roots are their integral bases by Theorem 9.2. Consequently T preserves their integer span R_{ES}. The same component argument applies to every element of $O(R_{\mathrm{ES}})$, since its norm-two vectors are the same 144 roots. It follows that every such element has, in the fixed identified coordinates, the form

$$T(X_1, X_2, X_3, X_4; d) = \left(\varepsilon_i Q_{\sigma_i} X_{p(i)}\right)_{i=1}^{4} ; T_D d, \tag{275}$$

where $p \in S_4$, $\varepsilon_i \in \{1, -1\}$, $\sigma_i \in S_6$, and $T_D \in O(D)$. This follows by transporting each isometry from its input A_5 summand to its output summand using the fixed φ_i, then applying Theorem 9.13. All choices in (275) are unique. The convention is precisely the earlier convention: a vector supported on component k is sent to component $p^{-1}(k)$.

Let $\mathcal{A}_{\mathrm{ES}}$ be the group of transformations of the exact discriminant group $(\mathbb{Z}/6)^4 \oplus \mathbb{F}_2^2$ given by

$$(x; y) \longmapsto ((\varepsilon_i x_{p(i)})_{i=1}^4; By), \qquad \varepsilon_i \in \{1, -1\}, \quad p \in S_4, \quad B \in GL_2(\mathbb{F}_2).$$

Each tuple gives a distinct transformation: the image of the order-six generator in each individual summand determines its output summand and its sign, and the images of the two binary basis vectors determine B. The discriminant action of (275), calculated by the two preceding theorems, is exactly this tuple with $B = d_D(T_D)$. This defines a homomorphism

$$d_{\mathrm{ES}} : O(R_{\mathrm{ES}}) \longrightarrow \mathcal{A}_{\mathrm{ES}}.$$

It is a homomorphism because extending an isometry to the dual and then passing to the quotient commutes with composition. In particular the component permutation rule remains $P_p P_q = P_{q \circ p}$; there is no change of the earlier convention.

This homomorphism is onto. Given any tuple $a = (\varepsilon, p, B) \in \mathcal{A}_{\mathrm{ES}}$, define

$$\widetilde{L}_a(X_1, X_2, X_3, X_4; d) = \left(\varepsilon_i X_{p(i)}\right)_{i=1}^{4} ; \delta(B) d.$$

It is a root-lattice isometry and has discriminant action a. The map $a \mapsto \widetilde{L}_a$ is a homomorphism: if $a = (\varepsilon, p, B)$ and $a' = (\varepsilon', q, B')$, its composite has i-th output $\varepsilon_i \varepsilon'_{p(i)} X_{q(p(i))}$ and D-component $\delta(B)\delta(B')d = \delta(BB')d$, which are exactly the tuple product in the stated convention.

The kernel of d_{ES} consists exactly of the tuples in (275) with $p = 1$, every $\varepsilon_i = 1$, and $T_D \in W_D$. It is therefore the direct product $S_6^4 \times W_D$. Every root reflection acts on its own orthogonal summand and fixes the other four pointwise, and the preceding two theorems identify the groups generated inside those summands. Hence this direct product is exactly $W(R_{\mathrm{ES}})$.

Every root reflection also preserves N. Indeed $\alpha \in N$ and integrality of N give $s_\alpha x = x - (x, \alpha)\alpha \in N$ for every $x \in N$. The reflection is an involution, so inclusion is equality. Thus $W(R_{\mathrm{ES}}) \subset O(N)$. It is normal there, since for $T \in O(N)$

$$T s_\alpha T^{-1}(x) = x - (x, T\alpha)T\alpha = s_{T\alpha}(x),$$

and $T\alpha$ is again a norm-two root.

It remains to impose the actual glue, which completely determines the remaining isometries. For $T \in O(R_{\mathrm{ES}})$ and $x \in R_{\mathrm{ES}}^\vee$, the inverse-image definition gives

$$Tx \in N \quad \Longleftrightarrow \quad d_{\mathrm{ES}}(T)(x + R_{\mathrm{ES}}) \in \mathcal{H}.$$

Consequently $T(N) = N$ if and only if $d_{\mathrm{ES}}(T)(\mathcal{H}) = \mathcal{H}$. For the forward implication, represent each class of $\mathcal{H}$ by a vector of N and apply both T and T^{-1}. For the reverse implication, the displayed equivalence applies to T and T^{-1} because the action on $\mathcal{H}$ is bijective. The group preserving $\mathcal{H}$ inside precisely $\mathcal{A}_{\mathrm{ES}}$ is the group G of Theorem 9.5, proved there to be exactly $GL_2(\mathbb{F}_3)$ through the code map Θ. We have therefore proved the equality of subgroups

$$O(N) = d_{\mathrm{ES}}^{-1}(G) \quad \text{inside } O(R_{\mathrm{ES}}).$$

The section $\widetilde{L}$ restricts on G to the already specified lift L. This proves surjectivity of the restricted map $d_{\mathrm{ES}} : O(N) \to G$, with exactly the asserted kernel. For $T \in O(N)$, set $g = d_{\mathrm{ES}}(T)$. Then $w = TL_g^{-1}$ lies in the kernel and $T = wL_g$. If $wL_g = w'L_{g'}$, applying d_{ES} gives $g = g'$, then cancellation gives $w = w'$. In particular $L(G) \cap W(R_{\mathrm{ES}}) = \{I\}$. Normality and this unique factorization prove the semidirect product and the quotient isomorphism, including its surjectivity. The four S_6 factors have order $(6!)^4$; W_D has 2^3 sign choices and $4!$ permutation choices, and $|G| = 48$. Unique factorization proves the displayed order.

Finally put $Z = L_g^{-1}X$. From the i-th output equation $X_i = \varepsilon_i Z_{p(i)}$ follows $Z_{p(i)} = \varepsilon_i X_i$, with each vector read in the fixed identified A coordinates. Therefore the i-th output of $L_g w L_g^{-1} X$ is

$$\varepsilon_i Q_{\sigma_{p(i)}} Z_{p(i)} = \varepsilon_i Q_{\sigma_{p(i)}} \varepsilon_i X_i = Q_{\sigma_{p(i)}} X_i.$$

The D-component is $\delta(B) w_D \delta(B)^{-1} d$. It belongs to W_D by normality of $\ker d_D$. These are exactly the components of (274), proving the asserted action with all signs and component indices retained. □

Theorem 9.16 (The residue six-cycle in the full automorphism group)**.** *For $r \in S_{840} = \langle 289 \rangle$, let $\rho(r)e_s = e_{rs}$ on each of the four fibres and $\rho(r)|_D = I$, with the exact fibre action used in* (267)*. Then*

$$\rho(S_{840}) \subset W(R_{\mathrm{ES}}), \qquad d_{\mathrm{ES}}(\rho(r)) = 1, \qquad L_g \rho(r) L_g^{-1} = \rho(r) \quad (g \in G,\ r \in S_{840}). \tag{276}$$

In particular the map

$$S_{840} \times G \longrightarrow O(N), \qquad (r, g) \longmapsto \rho(r)L_g$$

is an injective homomorphism with image of order $6 \cdot 48 = 288$, isomorphic to $C_6 \times GL_2(\mathbb{F}_3)$.

Proof. Put $Ce_j = e_{j+1 \bmod 6}$ on the free six-coordinate space and restrict it to $\mathbb{R}A$. If $r = 289^m$, with $m \in \{0,1,2,3,4,5\}$, the exact fibre identification gives

$$\rho(r)e_{f_i289^j} = e_{f_i289^{j+m}}, \qquad \rho(r)|_{A(F_{\eta_i})} = \varphi_i C^m \varphi_i^{-1}.$$

Thus in all four identified summands $\rho(r)$ is the same coordinate permutation C^m, and it is the identity on D. Theorem 9.13 puts it in the corresponding product of Weyl groups, giving the first two assertions. Each sign $\varepsilon_i I$ commutes with C^m, and the same operator C^m occurs in every component. Directly the i-th outputs of $L_g\rho(r)$ and $\rho(r)L_g$ are respectively $\varepsilon_i C^m X_{p(i)}$ and $C^m \varepsilon_i X_{p(i)}$, which agree; the D-outputs also agree since $\rho(r)|_D = I$. This proves the conjugation identity without changing the chosen origins.

The six restrictions $C^m|_{\mathbb{R}A}$ are distinct: if $C^m(e_0 - e_1) = e_0 - e_1$, comparison of the unique coordinate with coefficient $+1$ gives $m = 0 \pmod 6$. Thus ρ is faithful. Commutation makes the displayed product map a homomorphism. If $\rho(r)L_g = I$, then $L_g = \rho(r)^{-1}$ lies in $L(G) \cap W(R_{\mathrm{ES}}) = \{I\}$. Faithfulness of both factors gives $r = 1$ and $g = 1$. This proves injectivity and the exact image order. □

10 The exact Leech neighbor of the retained index-72 lattice

Retain the lattice, metric, discriminant marking, and glue literally:

$$\begin{aligned}
A &= \{(x_0,\dots,x_5) \in \mathbb{Z}^6 : \textstyle\sum x_j = 0\},\\
D &= \{(d_1,\dots,d_4) \in \mathbb{Z}^4 : \textstyle\sum d_i \equiv 0 \pmod 2\}, \qquad R = A^{\perp 4} \perp D,\\
C_2 &= \langle b_0 = (1111;00), b_1 = (1100;10), b_2 = (1010;01)\rangle_{\mathbb{F}_2},\\
C_3 &= \langle z_1 = (1,1,1,0), z_2 = (1,2,0,1)\rangle_{\mathbb{F}_3},\\
H &= \{(3u + 2z; y) : (u;y) \in C_2,\ z \in C_3\},\\
N &= \{x \in R^\vee : x + R \in H\}.
\end{aligned}$$

The coordinates in each A remain the six ordered elements f_i289^j of its original residue fibre. The A discriminant generator is $\lambda = (5,-1,-1,-1,-1,-1)/6$. The D markings $10, 01, 11$ remain represented by $v_D = (1,0,0,0)$, $s_D = (1,1,1,1)/2$, $c_D = (-1,1,1,1)/2$. The preceding proof establishes that N is even unimodular, $[N : R] = 72$, $\det R = 5184$, and its roots are exactly those of $A_5^4 D_4$. Every root has squared norm two. No metric rescaling occurs anywhere below.

10.1 The Weyl vector and the exact neighbor

Use the original simple roots $a_j = e_j - e_{j+1}$, $0 \le j \le 4$, in each A component and

$$d_1 = e_1 - e_2, \quad d_2 = e_2 - e_3, \quad d_3 = e_3 - e_4, \quad d_4 = e_3 + e_4$$

in D. Their highest roots are $\theta_A = e_0 - e_5 = \sum_j a_j$ and $\theta_D = e_1 + e_2 = d_1 + 2d_2 + d_3 + d_4$. Both highest-root heights are five, so the common Coxeter number is six. Define

$$\rho_A = (5,3,1,-1,-3,-5)/2, \qquad \rho_D = (3,2,1,0), \qquad \rho = \rho_A^{\oplus 4} \perp \rho_D.$$

Each simple root pairs to one with ρ. Moreover $\rho_A - 3\lambda = (0,2,1,0,-1,-2) \in A$ and $\rho_D \in D$. Thus the exact glue class of ρ is $(3,3,3,3;00)$, the image of b_0, and $\rho \in N$. Directly

$$\rho_A^2 = 35/2, \quad \rho_D^2 = 14, \quad \rho^2 = 4(35/2) + 14 = 84 = 2 \cdot 6 \cdot 7.$$

Consequently $(\rho, N) \subset \mathbb{Z}$. Fix $\alpha = a_0$ in the first A component and put

$$\ell_\rho : N \to \mathbb{Z}/6, \quad x \mapsto (\rho, x) \bmod 6, \qquad K = \ker \ell_\rho, \quad R_0 = R \cap K, \quad v = \rho/6 - \alpha, \quad \Lambda = K + \mathbb{Z}v. \tag{NL1}$$

The pairing $(\rho, \alpha) = 1$ proves $[N : K] = [R : R_0] = 6$. For $x \in K$, the number $(v, x) = (\rho, x)/6 - (\alpha, x)$ is integral, and

$$v^2 = 84/36 - 2/6 + 2 = 4, \qquad 6v = \rho - 6\alpha \in K, \qquad (\rho, 6v) = 78.$$

Hence Λ is positive definite even integral of rank 24. If $mv \in N$, its pairing after adding $m\alpha$ with α is $m/6 \in \mathbb{Z}$, so $6 \mid m$. It follows that

$$N \cap \Lambda = K, \qquad [\Lambda : K] = 6, \qquad \det K = 36, \qquad \det \Lambda = 36/6^2 = 1. \tag{NL2}$$

Changing α to another root of ρ-pairing one changes v by an element of K, and so does not change Λ. We shall use the equivalent exact congruence description

$$\Lambda = \{x + k\rho/6 : x \in N,\ k \in \mathbb{Z},\ (\rho, x) + k \equiv 0 \pmod 6\}. \tag{NL3}$$

Indeed write $x + k\rho/6 = (x + k\alpha) + kv$; the first summand is in K exactly under this congruence. Changing k by six can be absorbed in x using $\rho \in N$, and preserves the congruence since $\rho^2 = 84$.

10.2 A complete rootlessness proof

Theorem 10.1 (The Leech neighbor in the retained coordinates)**.** *The lattice $\Lambda = K + \mathbb{Z}(\rho/6 - \alpha)$ is positive definite even unimodular of rank 24 and has minimum squared norm exactly four. It is the holy-construction Leech lattice of the retained $A_5^4 D_4$ glue, and $N \cap \Lambda = K$ with $[N : K] = [\Lambda : K] = 6$.*

Proof. The rank, parity, unimodularity and indices were proved in (NL1)–(NL2). The holy-presentation identification is proved in (NL8)–(NL11) below. We now prove the required minimum directly. For $k = 0$ in (NL3), a possible root is an original root of N. An A root pairs with ρ to one of $\pm 1, \ldots, \pm 5$. For D, all roots $\pm e_i \pm e_j$ have nonzero ρ_D-pairing of absolute value at most five. Thus none belongs to K. Negation and the reduction of k modulo six leave $k = 1, 2, 3$.

Define the exact shifted class minima by

$$m_A(k, t) = \min_{x \in t\lambda + A} |x + k\rho_A/6|^2, \qquad m_D(k, y) = \min_{x + D = y} |x + k\rho_D/6|^2.$$

Their values are

$k\backslash t$	0	1	2	3	4	5
1	35/72	35/72	35/72	35/72	35/72	35/72
2	11/18	4/9	11/18	4/9	11/18	4/9
3	3/8	17/24	17/24	3/8	17/24	17/24

$k\backslash y$	00	10	01	11
1	7/18	7/18	7/18	7/18
2	2/9	5/9	5/9	5/9
3	1/2	1/2	1/2	1/2

. (NL4)

We prove the table and its needed equality information. The shifted A coordinates are precisely

$$\left(n_j - \frac{t}{6} + \frac{k(5 - 2j)}{12}\right)_{j=0}^{5}, \qquad n_j \in \mathbb{Z}, \quad \sum n_j = t. \tag{NL5}$$

At $k = 1$, coordinatewise nearest representatives are the six numbers $\pm 1/12, \pm 3/12, \pm 5/12$ in an order depending on t. They sum to zero and their squares sum to 35/72. At $k = 2$ and even t, the

nearest coordinates consist of two copies of each sign of $1/6$ and one copy of each sign of $1/2$; the half-integer ties admit exactly the needed zero-sum choice and the squared sum is $11/18$. At odd t, there are two each of $0, 1/3, -1/3$, giving $4/9$. For $t = 1, 3, 5$ their actual ordered shifted vectors are

$$(-1, 1, 0, -1, 1, 0)/3, \quad (1, 0, -1, 1, 0, -1)/3, \quad (0, -1, 1, 0, -1, 1)/3. \tag{NL6}$$

Subtracting $\rho_A/3$ gives original squared norms $17/6, 3/2, 17/6$, respectively. At $k = 3, t = 0, 3$, the nearest coordinates alternate $\pm 1/4$, giving $3/8$ with zero sum. At each other t, the coordinatewise nearest representatives have three absolute values $1/12$ and three $5/12$, square sum $13/24$, and coordinate sum 1 or -1. Meeting the zero-sum constraint requires an integer change. Every nonzero coordinate change increases the squared sum by at least $(7/12)^2 - (5/12)^2 = 1/6$. Changing one suitable $5/12$ coordinate attains this increase and restores the zero sum. Thus the constrained minimum is $17/24$. At $t = 0, 3$, the original minimizing vectors are

$$(-1, -1, 0, 0, 1, 1), \qquad (-3, -1, -1, 1, 1, 3)/2, \tag{NL7}$$

of squared norms $4, 11/2$.

For D and $k = 1$ the shift is $(1/2, 1/3, 1/6, 0)$. The nearest integer coordinate distances are $(1/2, 1/3, 1/6, 0)$; the first-coordinate tie meets either integer parity. The nearest half-integer distances are $(0, 1/6, 1/3, 1/2)$; the last-coordinate tie meets either half-integer class. Both squared sums are $7/18$. At $k = 2$ the shift is $(1, 2/3, 1/3, 0)$. The unique nearest integer vector before shifting is $(-1, -1, 0, 0) \in D$, giving $2/9$. Changing parity costs at least $1/3$, attained by changing the second or third coordinate, giving $5/9$. The half-integer coordinate distances are $(1/2, 1/6, 1/6, 1/2)$, giving $5/9$; ties meet either marking. At $k = 3$ the shift is $(3/2, 1, 1/2, 0)$. Integer-class minimum distances are $(1/2, 0, 1/2, 0)$, and half-integer-class distances $(0, 1/2, 0, 1/2)$. Ties meet every marking and all square sums are $1/2$. In class 00, the two original minimizing vectors are

$$(-1, -1, 0, 0), \qquad (-2, -1, -1, 0),$$

of squared norms $2, 6$. In class 10, they are

$$(-1, -1, -1, 0), \qquad (-2, -1, 0, 0),$$

of squared norms $3, 5$. In either half-integer class they have the shapes $(-3/2, -1/2, -1/2, \pm 1/2)$ and $(-3/2, -3/2, -1/2, \pm 1/2)$. Exactly one last-coordinate sign in each shape meets each original marking; their squared norms are $3, 5$. This proves every entry and the relevant equality cases in (NL4).

For $k = 1$, every full shifted vector has squared norm at least $4(35/72) + 7/18 = 7/3 > 2$. For $k = 2$, the lower bound from (NL4) is two. Equality forces all t_i odd and $y = 00$. In the actual glue this forces $u = 1111$. If $z = 0$, all $t_i = 3$ and (NL6) gives $x^2 = 4(3/2) + 2 = 8$. If $z \neq 0$, its weight is three, so exactly one $t_i = 3$ and three $t_i \in \{1, 5\}$; then $x^2 = 3/2 + 3(17/6) + 2 = 12$. The norm-two equation is

$$2 = x^2 + \tfrac{2}{3}(\rho, x) + \tfrac{28}{3}, \qquad (\rho, x) = -11 - \tfrac{3}{2}x^2.$$

Its pairing is -23 or -29, always 1 modulo six. Condition (NL3) requires residue 4, so all nine equality candidates, one per ternary word, are excluded.

For $k = 3$, the lower bound two requires $z = 0$: every nonzero ternary word raises the minimum by $3(17/24 - 3/8) = 1$. If $u = 0$, the possible original squared norms are $18, 22$. If $u = 1111$, they are $24, 28$. If u has weight two, its D class is nonzero and they are $2(4) + 2(11/2) + 3 = 22$ or $2(4) + 2(11/2) + 5 = 24$. These are all sixteen equality candidates. But

$$2 = x^2 + (\rho, x) + 21, \qquad (\rho, x) = -19 - x^2,$$

and (NL3) would require $x^2 \equiv 2 \pmod 6$. None of $18, 22, 24, 28$ has that residue. Thus Λ has no roots. Together with (NL2), this proves that it is positive definite even unimodular of rank 24, with minimum squared norm exactly four, attained by v. □

10.3 The exact holy-construction presentation

For $0 \le t \le 5$ set

$$\omega_t = (\underbrace{1,\dots,1}_{t},\underbrace{0,\dots,0}_{6-t}) - \tfrac{t}{6}(1,\dots,1).$$

Then $\omega_t + A = t\lambda + A$ and $(\rho_A, \omega_t) = t(6-t)/2 = 3\omega_t^2$. For the D markings set

$$\omega_{00} = 0, \quad \omega_{10} = e_1, \quad \omega_{01} = (1,1,1,1)/2, \quad \omega_{11} = (1,1,1,-1)/2.$$

The last differs from the original c_D by $(1,0,0,-1) \in D$, so retains its exact marking. For $a = (t_1,t_2,t_3,t_4;y) \in H$ define

$$u_a = (\omega_{t_1},\omega_{t_2},\omega_{t_3},\omega_{t_4};\omega_y), \qquad g_a = u_a - \rho/6. \tag{NL8}$$

These give all 72 representatives of N/R. Each nonzero D representative has norm one and pairing three; hence

$$(\rho, u_a) = 3u_a^2 \in 6\mathbb{Z}, \quad u_a \in K, \quad g_a^2 = u_a^2 - (\rho,u_a)/3 + 84/36 = 7/3. \tag{NL9}$$

Let $\mathcal{F}$ be the 29 vectors consisting, in each original component, of the negative original simple roots and the positive highest root. Their pairings with ρ are -1 or 5. There are positive integral zero relations of coefficient sum six:

$$\theta_A + \sum_{j=0}^{4}(-a_j) = 0, \qquad \theta_D + (-d_1) + 2(-d_2) + (-d_3) + (-d_4) = 0. \tag{NL10}$$

Every component of g_a pairs to $1/6$ with the corresponding vertices except one tip, where the pairing is $-5/6$. For A this follows because ω_t has simple-root pairing one only at a_{t-1} when $t \ne 0$, and has $(\omega_t, \theta_A) = 1$ for $t \ne 0$. For D the three nonzero representatives have simple-root pairing one only at d_1, d_4, d_3, respectively; all have highest-root pairing one. The zero representative points to the highest root. These computations fix all orientations in the holy generator diagram.

The following coefficient presentations are exact equalities:

$$\begin{aligned} N &= \left\{ \sum_f a_f f + \sum_a b_a g_a : a_f, b_a \in \mathbb{Z},\ \sum_a b_a = 0 \right\}, \\ \Lambda &= \left\{ \sum_f a_f f + \sum_a b_a g_a : a_f, b_a \in \mathbb{Z},\ \sum_f a_f + \sum_a b_a = 0 \right\}. \end{aligned} \tag{NL11}$$

For the first, the f span R and $g_a - g_0 = u_a$ generate its given extension N; the zero coefficient sum cancels $\rho/6$. For the second, write $A = \sum a_f$, $B = \sum b_a$, $r = \sum a_f f$, and $x = r + \sum b_a u_a$. The vector is $x - B\rho/6$, and by (NL9) $(\rho, x) \equiv -A \equiv B$, exactly (NL3) with $k = -B$. Conversely, for any $x + k\rho/6$ in (NL3), write $x = r + \sum b_a u_a$. The coefficient of $u_0 = 0$ can be set so $B = -k$. Write r with the negative simple-root vertices. The congruence gives $A + B \equiv 0 \pmod 6$. Adding a suitable multiple of a zero relation (NL10) makes $A + B = 0$ exactly. Thus both directions of (NL11) are proved.

These are exactly the holy coefficient rules of Conway–Sloane [29, pp. 277–280]. Borcherds gives the uniform proof of the holy-construction theorem [30, Section 7, especially the coefficient rules (i)–(ii) and Lemma 7.3]. Thus (NL11) is an explicit realization of that Leech construction from the actual 72 marked glue words; all its metric, determinant, integrality and rootlessness assertions have also been independently proved here.

10.4 The indices, discriminant morphism, and cyclic order twelve

Reduction modulo R gives an isomorphism $K/R_0 \cong H$: its kernel is R_0, and (NL9) supplies representatives in K for every H class. Consequently

$$[K : R_0] = 72, \quad [N : R_0] = [\Lambda : R_0] = 432, \quad \det R_0 = 6^2 \cdot 5184 = 186624. \tag{NL12}$$

Thus the original and new inclusions, with every index, are

$$R_0 \subset R \subset N, \quad [R : R_0] = 6,\ [N : R] = 72; \qquad R_0 \subset K \subset \Lambda, \quad [K : R_0] = 72,\ [\Lambda : K] = 6.$$

In particular $R \cap \Lambda = R_0$, an exact common sublattice.

Theorem 10.2 (The exact order-twelve quotient generator)**.** *The quotient of Λ by the original height-zero root sublattice R_0 is $\Lambda/R_0 \cong \mathbb{Z}/12 \oplus (\mathbb{Z}/6)^2$. The class of $v = \rho/6 - \alpha$ has exact order twelve.*

Proof. Let $\bar{b}_i$ denote the image in H of the binary generator b_i, namely its first four coordinates multiplied by three and its last two coordinates retained. Since $6v = \rho - 6\alpha$ and $6\alpha \in R_0$,

$$6[v] = \bar{b}_0 \quad \text{in } \Lambda/R_0. \tag{NL13}$$

This is a nonzero order-two element. The classes $\bar{b}_0, \bar{b}_1, \bar{b}_2, 2z_1, 2z_2$ form the displayed independent binary and ternary generators of H. Eliminate $\bar{b}_0$ using (NL13). The resulting surjective presentation is

$$\Lambda/R_0 \cong \mathbb{Z}/12 \oplus (\mathbb{Z}/2)^2 \oplus (\mathbb{Z}/3)^2 \cong \mathbb{Z}/12 \oplus (\mathbb{Z}/6)^2. \tag{NL14}$$

There are no further relations: this presentation has order $12 \cdot 4 \cdot 9 = 432$, equal to (NL12). Thus $[v]$ has exact order twelve. This particular occurrence of twelve is proved from the Coxeter denominator six and the actual order-two Weyl-vector glue class. □

Set $M = N + \Lambda = N + \mathbb{Z}(\rho/6)$. Then

$$M = K^\vee, \qquad K^\vee/K = \langle [\alpha], [v] \rangle \cong (\mathbb{Z}/6)^2. \tag{NL15}$$

Indeed $M \subset K^\vee$, its index over N is six by pairing with α, and $[K^\vee : N] = [N : K] = 6$. The cyclic images of N/K and Λ/K have order six and trivial intersection, proving the stated generators. Their exact discriminant data are

$$q_K([\alpha]) = q_K([v]) = 0, \quad b_K([\alpha], [v]) = 1/6 \pmod{\mathbb{Z}}, \quad q_K(a[\alpha] + b[v]) = ab/3 \pmod{2\mathbb{Z}}. \tag{NL16}$$

These follow from $\alpha^2 = 2$, $v^2 = 4$ and $(\alpha, v) = 1/6 - 2 = -11/6$. The two original lattices are therefore the inverse images of the two indicated order-six isotropic subgroups in this specified discriminant form.

10.5 An exact Lorentzian quotient and symmetry transport

Take $\mathcal{I} = N \perp U$ in coordinates (x, m, n) with

$$\langle (x,m,n),(x',m',n')\rangle = (x,x') - mn' - m'n.$$

Thus U has Gram matrix $\left(\begin{smallmatrix} 0 & -1 \\ -1 & 0 \end{smallmatrix}\right)$. The vectors $z = (0,0,1)$ and $w = (\rho, 6, 7)$ are primitive isotropic: $w^2 = 84 - 2 \cdot 6 \cdot 7 = 0$, and six and seven are coprime. There are integral quotient isometries

$$\begin{aligned} z^\perp/\mathbb{Z}z &\longrightarrow N, \quad [(x,0,n)] \mapsto x, \\ w^\perp/\mathbb{Z}w &\longrightarrow \Lambda, \quad [(x,m,n)] \mapsto x - m\rho/6. \end{aligned} \tag{NL17}$$

For the second map, orthogonality means $(\rho, x) = 7m + 6n$, giving (NL3) with $k = -m$. Conversely take $m = -k$ and $n = ((\rho,x) + 7k)/6$ for a vector in (NL3). If the image vanishes, $x = m\rho/6$; pairing with α forces $m = 6j$, and then $n = 7j$, so the kernel is exactly $\mathbb{Z}w$. Finally

$$|x - m\rho/6|^2 = x^2 - \tfrac{m}{3}(\rho,x) + \tfrac{7}{3}m^2 = x^2 - 2mn.$$

Polarization gives the full pairing, proving the isometry.

For every $T \in O(N)$ the exact transport laws are

$$T(K_\rho) = K_{T\rho}, \qquad T(\Lambda_\rho) = \Lambda_{T\rho}, \qquad (x,m,n) \mapsto (Tx,m,n) : w_\rho \mapsto w_{T\rho}. \tag{NL18}$$

They follow by substitution in (NL3). In particular the stabilizer of ρ acts on the fixed Λ_ρ. General isometries act on this specified family of neighbors; no assertion that an arbitrary chosen automorphism complement fixes this Weyl vector is required.

10.6 The exact F_4 Coxeter action of order twelve

The order twelve in (NL14) is a lattice quotient invariant of this explicitly marked construction. There is also the ambient F_4 Coxeter action on the same original D_4 real root space. Use the precise F_4 simple roots

$$\beta_1 = e_2 - e_3, \quad \beta_2 = e_3 - e_4, \quad \beta_3 = e_4, \quad \beta_4 = (e_1 - e_2 - e_3 - e_4)/2, \qquad C = s_{\beta_1} s_{\beta_2} s_{\beta_3} s_{\beta_4},$$

where $s_\beta(x) = x - 2(x,\beta)\beta/\beta^2$ and the rightmost factor acts first. The first two roots have squared norm two, and the last two squared norm one. No rescaling of the D_4 roots has occurred.

Theorem 10.3 (Coxeter twelve and its marked triality quotient)**.** *In the retained four-coordinate metric,*

$$C = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \end{pmatrix}, \qquad C^3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{NL19}$$

It satisfies $C^4 - C^2 + I = 0$, $C^6 = -I$ and has exact order twelve. Its marked discriminant action is $v_D + D \mapsto c_D + D \mapsto s_D + D \mapsto v_D + D$. The restriction of the original triality quotient is the exact sequence

$$1 \longrightarrow \langle C^3\rangle \cong \mathbb{Z}/4 \longrightarrow \langle C\rangle \cong \mathbb{Z}/12 \longrightarrow \langle (v_D\, c_D\, s_D)\rangle \cong \mathbb{Z}/3 \longrightarrow 1, \tag{NL20}$$

with $\langle C^3\rangle \subset W(D_4)$.

Proof. The first reflection exchanges coordinates two and three; the second exchanges coordinates three and four; the third changes the sign of coordinate four; the last is $I - 2\beta_4\beta_4^T$. Their product gives the first matrix in (NL19). Multiplying it three times gives the second, whose square is $-I$. Multiplying the first matrix also gives $C^4 - C^2 + I = 0$, and hence $C^{12} = I$. Its first column gives $Cv_D = c_D + (1, -1, 0, 0)$, with the displayed correction in D. Directly $Cs_D = v_D$ and $Cc_D = (0, 0, -1, -1) + s_D$. The displayed integer correction lies in D and proves the stated three-cycle with the original v_D, s_D, c_D markings. The first two reflections preserve D as root reflections. The last two preserve D because a sign change preserves integer sum parity, and reflection in β_4 sends an integer even-sum vector to an integer vector of even sum. The inverse reflections have the same property, so these maps are isometries of D. Consequently their induced action on $D^\vee/D$ is defined and has the exact order-three image calculated above. The matrix C^3 is an even signed permutation of exact order four, and thus lies in the already identified group $W(D_4)$. If $C^m = I$, its discriminant image forces $3 \mid m$, and the exact order of C^3 then forces $4 \mid m/3$. Thus C has exact order twelve and the discriminant kernel on its cyclic subgroup is exactly $\langle C^3 \rangle$. This proves every map and group order in (NL20). □

In particular C^2 still has nontrivial order-three discriminant action, while C^3 is in the long-root Weyl group. The Coxeter denominator six in (NL1), the cyclic order twelve in (NL14), and the F_4 Coxeter twelve in (NL20) therefore have their stated, distinct domains and explicit relationships through the original D_4 marking.

10.7 An exact cyclic-triality isometry of the entire Niemeier space

Retain the ordered fibres $f_i 289^j$, $1 \le i \le 4$, $0 \le j \le 5$, the Euclidean augmentation coordinates of $A(F_{\eta_i})$, the ordered orthonormal coordinates of $D \subset \mathbb{R}^4$, and the 72-element glue $\mathcal{H}$ of (248). Thus the actual real span is

$$V_N = N \otimes_{\mathbb{Z}} \mathbb{R} = \left\{ (x_{i,j}; d_k) \in \mathbb{R}^{24} \oplus \mathbb{R}^4 : \sum_{j=0}^{5} x_{i,j} = 0 \ (1 \le i \le 4) \right\},$$

with squared norm $\sum_{i,j} x_{i,j}^2 + \sum_k d_k^2$. The four displayed linear constraints, rather than an implicit change of metric, give its dimension 24. Choose an ordered orthonormal octonion basis $(u_0, \dots, u_7)$ with $u_0 = 1$. Every octonion product below is the product of the chosen real octonion algebra, without a change of multiplication convention.

For $p \in \{0, 1\}$ and $j \in \{0, 1, 2\}$, set

$$m_{i,p} = \frac{1}{3} \sum_{j=0}^{2} x_{i,p+2j}, \qquad t_{i,p,j} = x_{i,p+2j} - m_{i,p}, \tag{277}$$

$$\alpha_i = \frac{\sum_{j=0}^{2} x_{i,2j} - \sum_{j=0}^{2} x_{i,1+2j}}{\sqrt{6}}, \qquad (b_0, \dots, b_7) = (\alpha_1, \alpha_2, \alpha_3, \alpha_4, d_1, d_2, d_3, d_4). \tag{278}$$

In particular $m_{i,0} = \alpha_i/\sqrt{6}$, $m_{i,1} = -\alpha_i/\sqrt{6}$, and $\sum_j t_{i,p,j} = 0$. The ordered map is

$$\begin{aligned} \mathcal{L} : V_N &\longrightarrow \mathbb{O}^3, \\ (x; d) &\longmapsto (q_1, q_2, q_3), \\ q_{j+1} &= \sum_{i=1}^{4} \sum_{p=0}^{1} \left(t_{i,p,j} + \frac{b_{2(i-1)+p}}{\sqrt{3}} \right) u_{2(i-1)+p}. \end{aligned} \tag{279}$$

Theorem 10.4 (Actual rank-24 map, inverse, pairing, and residue action)**.** *The map $\mathcal{L}$ is a real linear isometry onto $\mathbb{O}^3$ with its product norm. Write $q_{j+1} = \sum_{a=0}^{7} q_{j,a} u_a$, and define*

$$b_a = \frac{1}{\sqrt{3}} \sum_{j=0}^{2} q_{j,a}, \qquad t_{i,p,j} = q_{j,2(i-1)+p} - \frac{1}{3} \sum_{h=0}^{2} q_{h,2(i-1)+p}. \tag{280}$$

Its inverse is exactly

$$x_{i,p+2j} = t_{i,p,j} + \frac{(-1)^p b_{i-1}}{\sqrt{6}}, \qquad d_k = b_{k+3}. \tag{281}$$

If $r = \rho(289^2) = \rho(361)$, with the residue action of Theorem 9.16, and $c(q_1, q_2, q_3) = (q_3, q_1, q_2)$, then

$$\mathcal{L} r = c \mathcal{L}, \qquad r^3 = c^3 = I, \qquad rN = N. \tag{282}$$

Thus $\mathcal{L}|_N$ embeds the original even unimodular lattice, with every pairing and glue class retained, into the actual three octonion sectors and intertwines an actual order-three lattice isometry with cyclic triality.

Proof. The two means sum to zero because each six-coordinate fibre has total sum zero. The equations for $m_{i,p}$ follow from $\alpha_i = 6m_{i,0}/\sqrt{6}$. Orthogonality to constant triplets gives

$$\sum_{j=0}^{5} x_{i,j}^2 = \sum_{p=0}^{1} \sum_{j=0}^{2} t_{i,p,j}^2 + 3m_{i,0}^2 + 3m_{i,1}^2 = \sum_{p,j} t_{i,p,j}^2 + \alpha_i^2.$$

For each $a = 2(i-1) + p$, the same zero-sum identity gives

$$\sum_{j=0}^{2} \left(t_{i,p,j} + \frac{b_a}{\sqrt{3}} \right)^2 = \sum_{j=0}^{2} t_{i,p,j}^2 + b_a^2.$$

Summing over all eight values of a proves $\sum_{h=1}^{3} |q_h|^2 = \sum_{i,j} x_{i,j}^2 + \sum_k d_k^2$. Polarization proves equality of all pairings. Equations (280) recover the means and residuals of each output triplet. Equation (281) then recovers the original means, the original fibre coordinates, and d. Conversely those formulas applied to arbitrary q yield $\sum_j x_{i,j} = 0$ and return the prescribed q. This proves bijectivity without a dimension-only argument.

The residue action sends the coordinate vector $e_{f_i 289^h}$ to $e_{f_i 289^{h+2}}$, so the coefficient in output position $p+2j$ is the old coefficient in position $p+2(j-1)$, with j read modulo three. Both means, all α_i, and all d_k are fixed. Hence the three output octonions become (q_3, q_1, q_2), proving the intertwining formula. Each coordinate permutation acts trivially on $A^\vee/A$: for $\lambda = e_0 - \frac{1}{6}\sum_h e_h$, its difference from λ is an augmentation root. The D-coordinates are fixed. Thus r acts trivially on the full discriminant group, preserves $\mathcal{H}$, and preserves its exact inverse image N. Its order is three, as seen on $e_0 - e_1$ in any fibre. □

The exact octonion-coordinate glue test. This embedding has the following fully explicit image, denoted $\Lambda_{\mathbb{O}} = \mathcal{L}(N)$. Given q, first recover $(x; d)$ by (280)–(281). For each fibre require $x_{i,h} - x_{i,k} \in \mathbb{Z}$ for every h, k, and put

$$k_i = -6x_{i,0} \pmod 6 \in \mathbb{Z}/6\mathbb{Z}.$$

This is defined because $6x_{i,0} = \sum_h (x_{i,0} - x_{i,h}) \in \mathbb{Z}$. Require also that the four d_k are all integral or all half-integral. Let $y(d) \in \mathbb{F}_2^2$ be the unique class with

$$d + D = 0 + D,\ v + D,\ s + D,\ c + D \quad \text{marked respectively by } 00, 10, 01, 11.$$

Then the final and exact test is

$$(k_1, k_2, k_3, k_4; y(d)) = \big(3u + 2(a+b, a+2b, a, b); \tau(u)\big) \tag{283}$$

for $u \in \mathbb{F}_2^4$ of even weight, $(a, b) \in \mathbb{F}_3^2$, with the first four coordinates read modulo six, and $\tau(u) = (u_2+u_4, u_3+u_4)$. Indeed the integral-difference conditions are precisely membership in $A^\vee$, and the common fractional part equals $-k_i/6 \pmod{\mathbb{Z}}$, which is the fractional part of $k_i\lambda$. The half-integral test is precisely membership in $D^\vee$, and (283) is the original $\mathcal{H}$. This proves both directions of the image description. No new lattice or rescaled glue is substituted.

The determinant and the factor two in the Albert metric. The matrix convention, also specifying this construction independently of its host document, is

$$Q_{\mathbb{O}}(\lambda; q) = \begin{pmatrix} \lambda & q_3 & \overline{q}_2 \\ \overline{q}_3 & \lambda & q_1 \\ q_2 & \overline{q}_1 & \lambda \end{pmatrix} \in \mathfrak{h}_3(\mathbb{O}), \qquad X \circ Y = (XY + YX)/2.$$

Put

$$E(q) = Q_{\mathbb{O}}(0; q_1, q_2, q_3), \qquad \mathfrak{t}(q_1, q_2, q_3) = \operatorname{Re}((q_1q_2)q_3), \qquad \mathfrak{t}_N(X) = \mathfrak{t}(\mathcal{L}X).$$

The original ordered Albert determinant becomes the exact polynomial

$$\det_{\mathbb{O}}\big(\lambda I + E(\mathcal{L}X)\big) = \lambda^3 - \lambda(X, X) + 2\mathfrak{t}_N(X). \tag{284}$$

Moreover

$$\operatorname{tr}\big(E(\mathcal{L}X) \circ E(\mathcal{L}Y)\big) = 2(X, Y). \tag{285}$$

For completeness, direct binary multiplication gives $\operatorname{tr} E = 0$, $\operatorname{tr}(E \circ E) = 2\sum_i |q_i|^2$, and $\operatorname{tr}(E \circ (E \circ E)) = 6\mathfrak{t}(q)$. For the last equality, the three cyclic off-diagonal coordinates of $E \circ E$ are $\overline{q_2q_3}, \overline{q_3q_1}, \overline{q_1q_2}$; their trace pairings with the corresponding q_i each give $2\mathfrak{t}(q)$. The Jordan trace formula $\det X = \operatorname{tr}(X \circ (X \circ X))/3 - \operatorname{tr} X \operatorname{tr}(X \circ X)/2 + (\operatorname{tr} X)^3/6$ therefore gives the first identity after expanding $X = \lambda I + E$ and substituting the proved norm equality. For the second, each off-diagonal octonion occurs in both transposed matrix entries, giving exactly $2\sum_i \operatorname{Re}(q_i\overline{p_i})$. Real associativity and cyclicity of octonion real parts give $\mathfrak{t}(q_3, q_1, q_2) = \mathfrak{t}(q_1, q_2, q_3)$:

$$\operatorname{Re}((q_3q_1)q_2) = \operatorname{Re}(q_3(q_1q_2)) = \operatorname{Re}((q_1q_2)q_3).$$

Consequently $\mathfrak{t}_N(rX) = \mathfrak{t}_N(X)$, and the whole determinant in (284) is r-invariant with λ fixed. Equivalently, for the real permutation matrix $Pe_i = e_{i+1}$, indices modulo three,

$$P\big(\lambda I + E(q)\big)P^T = \lambda I + E(q_3, q_1, q_2).$$

This is also a Jordan automorphism: multiplication by a real permutation matrix merely reindexes the two-factor matrix products, so it preserves $(AB + BA)/2$. It therefore carries the positive cone and its determinant boundary to themselves. The cyclic bridge is simultaneously linear, isometric on $\mathbb{O}^3$, integral on the stated lattice image, and compatible with the actual Albert cubic and positive cone.

Dependence and the four-dimensional lattice summand. The map uses precisely the four specified origins f_i, the generator 289, the even/odd triplet order, the displayed order of b, the ordered D-coordinates, and the chosen orthonormal octonion basis. Changing one of these changes the displayed map; the construction makes no claim of independence from these data. Its restriction to the actual D-summand is

$$d \longmapsto \frac{1}{\sqrt{3}}(z(d), z(d), z(d)), \qquad z(d) = d_1u_4 + d_2u_5 + d_3u_6 + d_4u_7. \tag{286}$$

This is an isometry on D, and on $D^\vee$ in its real span. In particular its 24 shortest nonzero dual vectors become 24 specified points in this four-dimensional diagonal subspace. The separate Cartan-weight map $d \mapsto \sum_k d_k\varepsilon_k$ is related to it by the exact isometry $\sum_k d_k\varepsilon_k \mapsto (z(d), z(d), z(d))/\sqrt{3}$.

10.8 The Leech neighbour in the same three octonion coordinates

The preceding map also transports the entire neighbour construction, not only the real span of N. Put $N_{\mathbb{O}} = \mathcal{L}(N)$ and retain the actual ρ, α, K, Λ of Theorem 10.1. In the already fixed basis let

$$U = \sum_{a=0}^{7} u_a, \qquad V = \frac{u_0 + u_1 + u_2 + u_3}{\sqrt{2}} + \sqrt{3}u_4 + \frac{2u_5 + u_6}{\sqrt{3}}.$$

Substitution in the original isometry gives the explicit vectors

$$\begin{aligned} \varrho &= \mathcal{L}\rho = (V + 2U, V, V - 2U), \\ \delta &= \mathcal{L}\alpha = \left(\frac{(2+\sqrt{2})u_0 - 2u_1}{3}, \frac{(\sqrt{2}-1)u_0 + u_1}{3}, \frac{(\sqrt{2}-1)u_0 + u_1}{3} \right). \end{aligned} \tag{287}$$

Indeed each even triplet of ρ_A has mean $1/2$, each odd triplet has mean $-1/2$, and both residual triplets are $(2, 0, -2)$. The four mean-difference coordinates are $\sqrt{6}/2$, while the four D coordinates are $(3, 2, 1, 0)$. For α, the two residual triplets in the first fibre are $(2, -1, -1)/3$ and $(-2, 1, 1)/3$; its only nonzero mean-difference coordinate is $2/\sqrt{6}$. These facts prove every coefficient in (287). In particular $|\varrho|^2 = 84$, $|\delta|^2 = 2$, and $\langle \varrho, \delta \rangle = 1$ in the product octonion metric.

Define

$$\begin{aligned} K_{\mathbb{O}} &= \{q \in N_{\mathbb{O}} : \langle \varrho, q \rangle \equiv 0 \pmod 6\}, \\ \Lambda_{\mathrm{Leech},\mathbb{O}} &= K_{\mathbb{O}} + \mathbb{Z}(\varrho/6 - \delta). \end{aligned} \tag{288}$$

The isometry and its inverse prove, in both directions, $\mathcal{L}(K) = K_{\mathbb{O}}$ and $\mathcal{L}(\Lambda) = \Lambda_{\mathrm{Leech},\mathbb{O}}$. Thus (288) is a full coordinate description of the Leech lattice inside the very same three sectors. The glue test above, followed by its displayed height congruence, is an exact membership test for $K_{\mathbb{O}}$; adjoining the six cosets $k(\varrho/6 - \delta)$, $0 \le k < 6$, gives all of the Leech lattice. The common-sublattice indices, discriminant form, and order-twelve quotient are transported without alteration.

The cyclic action also acts on the marked family of neighbours. Retain $r = \rho(361)$ and its corresponding map $c(q_1, q_2, q_3) = (q_3, q_1, q_2)$. For any transported Weyl vector $\rho' = T\rho$, $T \in O(N)$, the neighbour is defined using $T\alpha$, so that $\langle \rho', T\alpha \rangle = 1$. Then

$$c\,\mathcal{L}(K_{\rho'}) = \mathcal{L}(K_{r\rho'}), \qquad c\,\mathcal{L}(\Lambda_{\rho'}) = \mathcal{L}(\Lambda_{r\rho'}).$$

This follows directly from $\mathcal{L}r = c\mathcal{L}$ and (NL18). The inverse maps use r^{-1} and c^{-1}. It specifies exactly which marked neighbour is the target; it does not silently assume that the chosen ρ is fixed by r. Finally the common cubic evaluates on every one of these lattice points as

$$\det Q_{\mathbb{O}}(\lambda; \mathcal{L}X) = \lambda^3 - \lambda(X, X) + 2\,\mathrm{Re}(((\mathcal{L}X)_1(\mathcal{L}X)_2)(\mathcal{L}X)_3).$$

This identity holds for all $X \in N_{\mathbb{R}}$, hence also on the explicitly embedded N, K and Λ.

10.9 Exactly which root isometries preserve this glue

Retain the coordinate convention $(P_p x)_i = x_{p(i)}$. An arbitrary root-lattice isometry has the original form

$$T(X_1, X_2, X_3, X_4; d) = \big((\varepsilon_i Q_{\sigma_i} X_{p(i)})_{i=1}^4;\ T_D d\big),$$

where $\varepsilon_i \in \{\pm 1\}$, $\sigma_i \in S_6$, $p \in S_4$, and $T_D \in O(D)$. If $B = d_D(T_D)$, the necessary and sufficient condition for $T(N) = N$ is

$$B\tau(u) = \tau(P_p u)\ (u \in E), \qquad \big(\varepsilon_i \ell_{p(i)}\big)_{i=1}^4 = \big(\ell_i A\big)_{i=1}^4 \quad \text{for some } A \in GL_2(\mathbb{F}_3), \tag{289}$$

where $\ell_1 = (1,1), \ell_2 = (1,2), \ell_3 = (1,0), \ell_4 = (0,1)$. The A, and hence $B = B_p$, are unique. Indeed the two-primary part of the glue is the graph of τ, and signs are trivial modulo two. Its three-primary part is the image of the injective map $j(a,b) = (a+b, a+2b, a, b)$, so preserving it is equivalent to $M_\varepsilon P_p j = jA$. These are exactly the two conditions in (289); the primary decomposition proves their sufficiency as well as necessity. Since the four ℓ_i represent the four distinct projective row lines over $\mathbb{F}_3$, each A gives exactly one signed permutation by this equation. This recovers the precise 48-element glue quotient, not merely its action on three labels.

The canonical full-automorphism theorem proves, with these same coordinates and section, that

$$O(N) = \big(S_6^4 \times W(D_4)\big) \rtimes L(G), \qquad G \cong GL_2(\mathbb{F}_3).$$

Conjugation by the actual map $\mathcal{L}$ now gives the entire orthogonal stabilizer of the embedded lattice, with explicit matrices obtained from (279) and its inverse:

$$O(\Lambda_{\mathbb{O}}) = \mathcal{L} O(N) \mathcal{L}^{-1}, \qquad \Gamma_G(g) = \mathcal{L} L_g \mathcal{L}^{-1}, \qquad \mathcal{L} L_g = \Gamma_G(g) \mathcal{L}. \tag{290}$$

Both inclusions in the first equality follow by conjugating an isometry preserving the specified lattice. This is a faithful action on the whole 24-dimensional space. Each $\Gamma_G(g)$ commutes with c, because the canonical equation $L_g \rho(289^2) = \rho(289^2) L_g$ conjugates to that equation.

For completeness, the previously displayed D_4-triality section and geometric frame permutations have a precise comparison even though that particular proposed linear conjugacy fails. On V_N, each of $\widehat{R}, \widehat{H}$ has trace $5 - 5 + 2 = 2$: the transposed A_5 pair contributes zero, the other two components contribute $5, -5$, and the D_4 reflection contributes 2. Their three-cycle has trace $5 + 1 = 6$. A geometric Albert-frame transposition acts on $\mathbb{O}^3$ by swapping two sectors with conjugation and by conjugating the fixed sector; its trace is $1 - 7 = -6$. Its three-cycle has trace zero. Therefore an invertible map cannot conjugate these two particular S_3 representations, since trace is invariant under conjugation. The complete real character decompositions are, respectively,

$$V_N|_{\langle \widehat{R}, \widehat{H} \rangle} \cong \mathbf{71} \oplus 5\,\mathrm{sgn} \oplus 6\,\mathrm{Std}, \qquad \mathbb{O}^3|_{S_3^{\mathrm{frame}}} \cong \mathbf{1} \oplus 7\,\mathrm{sgn} \oplus 8\,\mathrm{Std}\,.$$

These multiplicities follow by taking inner products with the characters $(1,1,1), (1,-1,1), (2,0,-1)$, on classes of sizes $1, 3, 2$. The exact whole-space relation for the original section is (290); the exact determinant-compatible cyclic relation is (282). In the latter relation the residue element has trivial discriminant action while its octonion action cycles the sectors. Those are two explicitly calculated actions of the same operator on two explicitly different constructions; no identification between their quotient homomorphisms is assumed.

Finally, the connected group Spin(8) acts on the ambient triality module, and its subgroup preserving $\Lambda_{\mathbb{O}}$ is exactly its intersection with (290). It is finite: the columns of a lattice isometry in a fixed lattice basis have prescribed bounded lengths, so each column has finitely many possible

lattice values. A connected group preserving a discrete full lattice pointwise as a set acts trivially, since each orbit of a lattice vector is connected and discrete and the lattice spans. The nontrivial connected triality action therefore does not preserve the whole lattice. The proved cyclic subgroup $\langle c\rangle$ does preserve both the lattice and its Albert cubic. Equations (289) and (290) specify the full glue stabilizer; they do not assert that every element of that finite orthogonal group also preserves the cubic.

11 The arithmetic of the retained Albert cubic on both lattices

11.1 Original data and the exact polynomial

Retain the ordered coordinates $X = (x_{i,j}; d_k)$, $1 \le i \le 4$, $0 \le j \le 5$, $1 \le k \le 4$, with $\sum_{j=0}^{5} x_{i,j} = 0$, and their original Euclidean metric. Let

$$A_5 = \{x \in \mathbb{Z}^6 : \sum_j x_j = 0\}, \qquad D_4 = \{d \in \mathbb{Z}^4 : \sum_k d_k \equiv 0 \pmod 2\}, \qquad R = A_5^{\perp 4} \perp D_4.$$

Write $\lambda = (5,-1,-1,-1,-1,-1)/6$, $v_D = (1,0,0,0)$, and $s_D = (1,1,1,1)/2$. The original discriminant glue is generated by

$$\begin{aligned} &(3,3,3,3;00), \quad (3,3,0,0;10), \quad (3,0,3,0;01),\\ &(2,2,2,0;00), \quad (2,4,0,2;00) \end{aligned}$$

in $(\mathbb{Z}/6)^4 \oplus \mathbb{F}_2^2$, with the same marking $10 = v_D + D_4$, $01 = s_D + D_4$. Its order is 72: the first three generators have independent two-primary reductions, and the last two independent three-primary reductions. The lattice N is its inverse image in $R^\vee$.

Retain

$$\rho_A = (5,3,1,-1,-3,-5)/2, \qquad \rho_D = (3,2,1,0), \qquad \rho = \rho_A^{\oplus 4} \perp \rho_D, \qquad \alpha = (e_0 - e_1, 0, 0, 0; 0),$$

and the exact neighbour

$$K = \{X \in N : (\rho, X) \equiv 0 \pmod 6\}, \qquad v = \rho/6 - \alpha, \qquad \Lambda = K + \mathbb{Z}v.$$

Here $\rho \in N$, $\rho^2 = 84$, and $(\rho, \alpha) = 1$, as follows immediately from these coordinates and the first glue generator. The name Λ always denotes this neighbour.

Fix an ordered orthonormal octonion basis $u_0 = 1, u_1, \dots, u_7$. For the first part of the calculation no multiplication table is imposed beyond the actual multiplication of these basis elements. Define, in the original coordinates,

$$s_i = \sum_{j=0}^{5} (-1)^j x_{i,j}, \tag{291}$$

$$A_{2(i-1)+p} = x_{i,p} - \frac{x_{i,p} + x_{i,p+2} + x_{i,p+4}}{3}, \qquad p = 0, 1, \tag{292}$$

$$B_{2(i-1)+p} = x_{i,p+2} - \frac{x_{i,p} + x_{i,p+2} + x_{i,p+4}}{3}, \qquad p = 0, 1. \tag{293}$$

Write $A = A_0 + \mathbf{a}$, $B = B_0 + \mathbf{b}$, where $\mathbf{a} = \sum_{h=1}^{7} A_h u_h$ and $\mathbf{b} = \sum_{h=1}^{7} B_h u_h$. The common mean octonion is

$$M = \frac{\sqrt{2}}{6} \sum_{a=0}^{3} s_{a+1} u_a + \frac{\sqrt{3}}{3} \sum_{k=1}^{4} d_k u_{k+3}. \tag{294}$$

The retained isometry therefore has precisely the ordered form

$$\mathcal{L}X = (A + M, B + M, -A - B + M).$$

This is an identity with its original three triplets: the residuals sum to zero, the first four common mean coefficients are $s_i/\sqrt{18} = \sqrt{2}s_i/6$, and the last four are $d_k/\sqrt{3} = \sqrt{3}d_k/3$. For completeness the original squared norm and the output norm are both

$$|A|^2 + |B|^2 + |A + B|^2 + \frac{1}{6}\sum_i s_i^2 + \sum_k d_k^2.$$

In the original fibres this follows by separating the means $s_i/6, -s_i/6$ from their zero-sum triplets; in the output it follows from $A + B + (-A - B) = 0$ and $3|M|^2 = \sum_i s_i^2/6 + \sum_k d_k^2$. The inverse recovers $M = (q_1 + q_2 + q_3)/3$, then $s_i = 3\sqrt{2}M_{i-1}$, $d_k = \sqrt{3}M_{k+3}$, and $A = q_1 - M, B = q_2 - M$. Adding $(-1)^p s_i/6$ to the p-th residual triplet in the i-th fibre recovers each original $x_{i,p+2j}$. Thus the map is onto and preserves the original pairing by polarization.

For imaginary octonions put

$$\mathbf{a} \times \mathbf{b} = \frac{\mathbf{ab} - \mathbf{ba}}{2}, \qquad \phi(\mathbf{a}, \mathbf{b}, \mathbf{c}) = \langle \mathbf{a} \times \mathbf{b}, \mathbf{c} \rangle.$$

Let $\phi_{ijk} = \phi(u_i, u_j, u_k)$. Thus ϕ is the actual alternating octonion three-form in the chosen frame, including all signs. Define

$$D = A_0^2 + A_0B_0 + B_0^2 - \langle \mathbf{a}, \mathbf{a} \rangle - \langle \mathbf{a}, \mathbf{b} \rangle - \langle \mathbf{b}, \mathbf{b} \rangle, \tag{295}$$

$$Z = (2A_0 + B_0)\mathbf{a} + (A_0 + 2B_0)\mathbf{b} - 3\mathbf{a} \times \mathbf{b} = \sum_{h=1}^{7} Z_h u_h. \tag{296}$$

Equivalently, the component in the final formula is the exact quadratic polynomial

$$Z_h = (2A_0 + B_0)A_h + (A_0 + 2B_0)B_h - 3\sum_{i,j=1}^{7} \phi_{ijh}A_iB_j.$$

Theorem 11.1 (Exact pullback with every radical and coefficient)**.** *The original cubic* $F(X) = \mathrm{Re}(((\mathcal{L}X)_1(\mathcal{L}X)_2)(\mathcal{L}X)_3)$ *has the complete polynomial expression*

$$F(X) = P_0(X) + \sqrt{2}P_2(X) + \sqrt{3}P_3(X), \tag{297}$$

where every variable is the displayed linear function of the original $x_{i,j}, d_k$*, and*

$$P_0 = -A_0B_0(A_0 + B_0) + B_0\langle \mathbf{a}, \mathbf{a} \rangle + A_0\langle \mathbf{b}, \mathbf{b} \rangle + 2(A_0 + B_0)\langle \mathbf{a}, \mathbf{b} \rangle, \tag{298}$$

$$P_2 = \frac{-s_1D + s_2Z_1 + s_3Z_2 + s_4Z_3}{6} + \frac{s_1(4s_1^2 - 3\sum_{i=1}^4 s_i^2)}{108} - \frac{s_1\sum_{k=1}^4 d_k^2}{6}, \tag{299}$$

$$P_3 = \frac{d_1Z_4 + d_2Z_5 + d_3Z_6 + d_4Z_7}{3}. \tag{300}$$

These identities hold throughout the original real vector space, in particular on N*,* K*, and* Λ*. They retain the ordered Albert determinant as*

$$\det Q_{\mathbb{O}}(\lambda; \mathcal{L}X) = \lambda^3 - \lambda(X, X) + 2F(X).$$

Proof. For $a = a_0 + \mathbf{a}$, $b = b_0 + \mathbf{b}$, $c = c_0 + \mathbf{c}$, binary octonion multiplication gives

$$\mathrm{Re}((ab)c) = a_0b_0c_0 - a_0\langle \mathbf{b}, \mathbf{c}\rangle - b_0\langle \mathbf{a}, \mathbf{c}\rangle - c_0\langle \mathbf{a}, \mathbf{b}\rangle - \phi(\mathbf{a}, \mathbf{b}, \mathbf{c}).$$

Call the first four, symmetric, terms $S(a, b, c)$. Substitute $(a, b, c) = (A+M, B+M, -A-B+M)$. Terms with exactly two copies of M in S sum to $S(M, M, A + B - A - B) = 0$. Terms with exactly one copy sum to $-S(M, A, A) - S(M, A, B) - S(M, B, B)$. The term with no copy is $S(A, B, -A - B) = P_0$, by multiplying its four displayed terms. The alternating term with no copy vanishes; terms with two or three copies vanish. Its three terms with one copy all equal $\phi(\mathbf{a}, \mathbf{b}, \mathrm{Im}\, M)$, since

$$\phi(\mathbf{a}, \mathrm{Im}\, M, -\mathbf{a} - \mathbf{b}) = \phi(\mathbf{a}, \mathbf{b}, \mathrm{Im}\, M), \quad \phi(\mathrm{Im}\, M, \mathbf{b}, -\mathbf{a} - \mathbf{b}) = \phi(\mathbf{a}, \mathbf{b}, \mathrm{Im}\, M).$$

Thus all one-copy terms together are $-M_0D + \langle \mathrm{Im}\, M, Z\rangle$. Finally the term with three copies is

$$\mathrm{Re}(M^3) = 4M_0^3 - 3M_0|M|^2, \qquad |M|^2 = \frac{\sum_i s_i^2}{18} + \frac{\sum_k d_k^2}{3}, \qquad M_0 = \frac{\sqrt{2}s_1}{6}.$$

Substitution proves (298)–(300) with their displayed factors. In particular it proves the absence of a $\sqrt{6}$ term without discarding any mixed term. The determinant formula uses the original isometry and its unchanged squared norm. □

11.2 Finite generators of both original lattices

Order the 24 original simple roots as

$$g_{5(i-1)+j+1} = a_{i,j} = e_{i,j} - e_{i,j+1} \quad (1 \le i \le 4,\ 0 \le j \le 4),$$

followed by

$$g_{21} = e_1 - e_2, \quad g_{22} = e_2 - e_3, \quad g_{23} = e_3 - e_4, \quad g_{24} = e_3 + e_4$$

in the original D_4 coordinates. All unspecified components are zero. Adjoin the five actual glue lifts

$$\begin{aligned} g_{25} &= (3\lambda, 3\lambda, 3\lambda, 3\lambda; 0), & g_{26} &= (3\lambda, 3\lambda, 0, 0; v_D),\\ g_{27} &= (3\lambda, 0, 3\lambda, 0; s_D), & g_{28} &= (2\lambda, 2\lambda, 2\lambda, 0; 0),\\ g_{29} &= (2\lambda, 4\lambda, 0, 2\lambda; 0). \end{aligned}$$

Then

$$N = \sum_{i=1}^{29} \mathbb{Z} g_i, \qquad H_i = (\rho, g_i) = \begin{cases} 1, & 1 \le i \le 24,\\ 30, 18, 18, 15, 20, & i = 25, 26, 27, 28, 29 \text{ respectively.} \end{cases} \tag{301}$$

Indeed the first 24 vectors span R, the last five generate the original glue, and subtraction of the corresponding glue lift leaves every element of N in R. The height calculation uses $(\rho_A, \lambda) = 5/2$, $(\rho_D, v_D) = (\rho_D, s_D) = 3$, and the simple-root heights one.

Define the following 30, fully specified neighbour generators:

$$h_1 = 6\alpha, \qquad h_i = g_i - H_i\alpha \quad (2 \le i \le 29), \qquad h_{30} = v = \rho/6 - \alpha. \tag{302}$$

Then $K = \sum_{i=1}^{29} \mathbb{Z} h_i$ and $\Lambda = \sum_{i=1}^{30} \mathbb{Z} h_i$. For if $X = \sum_i n_i g_i \in K$, then

$$X = \sum_{i=2}^{29} n_i(g_i - H_i\alpha) + \left(\sum_{i=1}^{29} n_i H_i\right)\alpha,$$

and the last coefficient is divisible by six. Conversely every displayed generator before h_{30} has height divisible by six. Adjoining h_{30} is exactly the original neighbour definition.

11.3 A complete finite calculation of the cubic value modules

For a lattice L and a homogeneous real cubic F, define its additive value module by

$$\mathcal{V}_F(L) = \sum_{X \in L} \mathbb{Z} F(X) \subset \mathbb{R}.$$

This definition retains every value and takes its generated additive group. It does not claim that every element of the group is one individual value of F.

Theorem 11.2 (Finite cubic-value generators, including negative integers)**.** *If $L = \sum_{i=1}^{m} \mathbb{Z} b_i$, allowing dependent generators, then $\mathcal{V}_F(L)$ is generated by the following finite list:*

$$\begin{aligned}
A_i &= F(b_i), && 1 \le i \le m, && (303)\\
D_{ij}^{+} &= F(b_i + b_j) - F(b_i) - F(b_j), && i < j, && (304)\\
D_{ij}^{-} &= F(b_i - b_j) - F(b_i) + F(b_j), && i < j, && (305)\\
D_{ijk} &= F(b_i + b_j + b_k) - F(b_i + b_j) - F(b_i + b_k) \\
&\quad - F(b_j + b_k) + F(b_i) + F(b_j) + F(b_k), && i < j < k. && (306)
\end{aligned}$$

Consequently (301) *gives exactly 4495 specified generators of $\mathcal{V}_F(N)$, and* (302) *gives exactly 4960 specified generators of $\mathcal{V}_F(\Lambda)$, for every fixed orthonormal octonion frame and its actual ϕ_{ijk}.*

Proof. Let B_F be the symmetric trilinear polarization, with $B_F(X, X, X) = F(X)$; explicitly its value is $1/6$ times the seven-term expression in (306), with its three inputs replacing the three generators. For every integer tuple n, expansion gives

$$F\left(\sum_i n_i b_i\right) = \sum_i n_i^3 A_i + \sum_{i<j} (n_i^2 n_j C_{ij} + n_i n_j^2 E_{ij}) + \sum_{i<j<k} n_i n_j n_k D_{ijk},$$

where $C_{ij} = 3B_F(b_i, b_i, b_j)$ and $E_{ij} = 3B_F(b_i, b_j, b_j)$. The two pair generators are $D_{ij}^{+} = C_{ij} + E_{ij}$ and $D_{ij}^{-} = -C_{ij} + E_{ij}$. Therefore their contribution is

$$\frac{n_i n_j (n_i + n_j)}{2} D_{ij}^{+} + \frac{n_i n_j (n_j - n_i)}{2} D_{ij}^{-}.$$

Both coefficients are integers, for all positive, zero, and negative integers n_i, n_j: if either integer is even the claim is immediate, and if both are odd their sum and difference are even. This proves that the finite list generates all values. Conversely each entry of the list is an integer linear combination of actual values on L. The proof does not use linear independence of the generators. Finally $29 + 2\binom{29}{2} + \binom{29}{3} = 4495$ and $30 + 2\binom{30}{2} + \binom{30}{3} = 4960$. □

For numerical arithmetic now specify the additional octonion frame choice completely. Use $\mathbb{O} = \mathbb{H} \oplus \mathbb{H}\ell$ in ordered coordinates (a, b), with

$$\begin{gathered}
(a, b)(c, d) = (ac - \overline{d}\, b,\ da + b\overline{c}),\\
u_0 = (1, 0), \quad u_1 = (i, 0), \quad u_2 = (j, 0), \quad u_3 = (k, 0), \qquad (307)\\
u_4 = (0, 1), \quad u_5 = (0, i), \quad u_6 = (0, j), \quad u_7 = (0, k).
\end{gathered}$$

Here $ij = k$, $jk = i$, $ki = j$, and the reversed products are negative. The positive oriented three-form triples are exactly

$$(123),\ (145),\ (176),\ (246),\ (257),\ (347),\ (365). \qquad (308)$$

Every ϕ_{ijk} is obtained from these by alternation, or is zero. This is a specified frame for the retained, previously frame-dependent isometry, without a change to any lattice coordinates or to the Albert matrix convention.

Theorem 11.3 (Exact value modules and their quotient of order twelve)**.** *For the frame* (307)*, the exact additive modules are*

$$\mathcal{V}_F(N) = \frac{1}{27}\mathbb{Z} \ \oplus \ \frac{\sqrt{2}}{54}\mathbb{Z} \ \oplus \ \frac{\sqrt{3}}{18}\mathbb{Z}, \tag{309}$$

$$\mathcal{V}_F(\Lambda) = \frac{1}{27}\mathbb{Z} \ \oplus \ \frac{\sqrt{2}}{216}\mathbb{Z} \ \oplus \ \frac{\sqrt{3}}{54}\mathbb{Z}. \tag{310}$$

In particular the inclusion of these two additive subgroups of $\mathbb{R}$ *has the exact quotient*

$$\mathcal{V}_F(\Lambda)/\mathcal{V}_F(N) \cong \mathbb{Z}/4 \oplus \mathbb{Z}/3 \cong \mathbb{Z}/12. \tag{311}$$

This is an inclusion of cubic value modules; it does not assert the false lattice inclusion $N \subset \Lambda$*.*

Proof. The polynomial coefficients in (298)–(300) are rational on both lattices for this exact multiplication table. The numbers $1, \sqrt{2}, \sqrt{3}$ are linearly independent over $\mathbb{Q}$. For example, isolating one radical in a rational relation and squaring either gives a rational multiple of another irrational radical, or a rational square equal to two, three, or 3/2, which is impossible by parity of prime exponents. Thus a value has a unique rational coefficient triple (P_0, P_2, P_3).

Apply the complete finite list in Theorem 11.2 to the displayed original generators. Exact substitution gives the following least common multiples of coefficient denominators. The denominators in each entry are ordered as (P_0, P_2, P_3), and a zero coefficient has denominator one:

	A_i	D^+_{ij}	D^-_{ij}	D_{ijk}
N	$(27, 27, 1)$	$(9, 18, 18)$	$(9, 18, 18)$	$(9, 18, 18)$.
Λ	$(27, 216, 27)$	$(27, 108, 54)$	$(27, 108, 54)$	$(9, 36, 54)$

(312)

This table is a finite rational calculation, with no bound on lattice points or heuristic sampling. The complete arithmetic certificate is the accompanying `verify_cubic.py`: its `residual_data` and `cubic` functions are exactly (291)–(300); it forms every $A_i, D^+_{ij}, D^-_{ij}, D_{ijk}$ for the 29 or 30 explicitly defined vectors, using rational arithmetic. The `cross` function executes the three cyclic positive and three negative entries for each of the seven triples (308). Hence the full finite calculation is specified by these equations, the finite index bounds in Theorem 11.2, and integer arithmetic. Its 4495 and 4960 entries verify the upper inclusions in (309)–(310).

Here are short certificates for the reverse inclusions. Coordinates in the next two tables are relative to the respective three proposed generators on the right sides of (309) and (310). In the first table all pair symbols use the g_i:

$F(g_1)$	-4	4	0
$F(g_2)$	-5	-4	0
$F(g_7)$	0	18	0
$F(g_{28})$	-80	52	0 .
$D^+_{21,28}$	0	-36	16
$D^+_{2,27}$	-27	27	3
$D^+_{2,25}$	-81	45	0

(313)

The determinants of the three columns obtained by transposing rows $(1,4,5)$, $(2,3,6)$, and $(2,7,6)$ are, respectively, $1792, -270, -1647$. Their greatest common divisor is one. In the second table all pair symbols use the h_i:

$$\begin{array}{r|rrr} F(h_2) & 14 & -272 & 0 \\ F(h_4) & 5 & -128 & 0 \\ F(h_{28}) & 13780 & -74672 & 0 \\ F(h_{30}) & 14 & -63 & -2 \\ D^{+}_{2,27} & 7425 & -60624 & -9 \end{array} . \tag{314}$$

Rows $(1,3,4)$, $(1,4,5)$, and $(2,4,5)$ give determinants $-5405504, 2315394, 1281267$, whose greatest common divisor is also one.

For clarity, the elementary integer argument using these determinants does not require a normal-form theorem. Let $W \subset \mathbb{Z}^3$ be the group generated by one displayed table. For any three of its columns forming a matrix C, the adjugate identity $C\,\mathrm{adj}(C) = \det(C)I$ implies $\det(C)\mathbb{Z}^3 \subset W$. Bezout's identity for the three displayed determinants, whose gcd is one, therefore gives $\mathbb{Z}^3 \subset W$. The opposite inclusion is immediate from the integer entries. Every table entry is an integer combination of actual values by its definition, proving the two reverse inclusions. Finally the coordinates of the inclusion of the first module in the second are $(a,b,c) \mapsto (a,4b,3c)$. Its cokernel is the displayed product of cyclic groups, with order twelve. This also proves the asserted exact quotient map. □

11.4 Frame dependence and exact symmetry transport

The finite-list theorem is valid for the actual coefficients of every orthonormal frame. The numerical modules (309)–(310) refer to the frame (307). This dependence has an exact comparison map. If $u'_a = Tu_a$ for a real isometry T fixing 1, then

$$\mathcal{L}_{u'} = (T,T,T)\mathcal{L}_u.$$

The symmetric part S of the trilinear tensor is unchanged, because T preserves the scalar coordinate and inner product. Writing $\mathbf{m} = \mathrm{Im}\, M$, its exact change is

$$F_{u'}(X) - F_u(X) = -3\big(\phi(T\mathbf{a}, T\mathbf{b}, T\mathbf{m}) - \phi(\mathbf{a},\mathbf{b},\mathbf{m})\big). \tag{315}$$

This follows from the complete expansion in the polynomial proof. If T is an octonion automorphism it preserves ϕ, so the cubic and its value modules are unchanged.

The dependence on a general orthonormal frame is substantive. Take the actual lattice vector with the only nonzero fibre coordinates

$$x_{1,1} = 1,\ x_{1,5} = -1, \quad x_{2,2} = 1,\ x_{2,4} = -1, \qquad d = (1,1,0,0).$$

It lies in $R \subset N$, has $A = u_1$, $B = u_2$, and $M = (u_4 + u_5)/\sqrt{3}$. In the Cayley frame its value is zero. Replace only

$$u'_3 = (2u_3 - \sqrt{5}u_4)/3, \qquad u'_4 = (\sqrt{5}u_3 + 2u_4)/3.$$

This is an exact orthonormal frame change; the same original lattice vector now has value $-\sqrt{15}/3$, because $\phi(u_1,u_2,u'_4) = \sqrt{5}/3$ and $\phi(u_1,u_2,u_5) = 0$. This value is outside $\mathbb{Q} + \sqrt{2}\mathbb{Q} + \sqrt{3}\mathbb{Q}$. Indeed the independent sign changes of $\sqrt{2}, \sqrt{3}$ are automorphisms of $\mathbb{Q}(\sqrt{2},\sqrt{3})$. If an element z in that field had $z^2 = 15$, each sign change would send it to z or $-z$. In the basis $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$ this makes z a rational multiple of a single basis vector. Its square would force one of $15, 15/2, 5, 5/2$ to be a

rational square, impossible by parity of its prime exponents. Thus the numerical specialization is not silently extended to an unspecified orthonormal frame.

Return to the explicit Cayley frame for the following numerical witnesses. Let $\sigma = \rho(289)$ act simultaneously by $e_{i,j} \mapsto e_{i,j+1 \bmod 6}$, and trivially on D_4. The exact isometry gives $\mathcal{L}\sigma^2 = c\mathcal{L}$ for $c(q_1, q_2, q_3) = (q_3, q_1, q_2)$. Cyclic real associativity therefore proves $F(\sigma^{2k}X) = F(X)$ for every real X. These maps preserve N by their original discriminant action. The stabilizer of this cubic within the six-element residue group is exactly its three-element subgroup: direct substitution at $\alpha = g_1$ gives

$$F(\alpha) = \frac{-4 + 2\sqrt{2}}{27}, \qquad F(\sigma\alpha) = \frac{-5 - 2\sqrt{2}}{27}, \tag{316}$$

which differ. All odd powers have the same effect on the cubic as σ, by its proved invariance under even powers.

The original reflection exchanging $e_{1,0}, e_{1,1}$ sends α to $-\alpha$, hence changes its nonzero cubic value to its negative. The original full glue-preserving lifts

$$\widehat{R}(X_1, X_2, X_3, X_4; d) = (X_2, X_1, X_3, -X_4; R_D d),$$
$$\widehat{H}(X_1, X_2, X_3, X_4; d) = (X_1, X_3, X_2, -X_4; H_D d).$$

also have explicit witnesses:

$$F(\widehat{R}\alpha) = 0, \qquad F(g_7) = \frac{\sqrt{2}}{3}, \qquad F(\widehat{H}g_7) = 0.$$

Their original matrices remain

$$R_D = \operatorname{diag}(-1, 1, 1, 1), \qquad H_D = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

These maps preserve the original lattice, as can be checked on its displayed generators. Both R_D and H_D preserve D_4: the former changes one sign, while the latter has integral coordinates on every even-sum integer vector and has output sum twice its first input coordinate. They are orthogonal involutions. Their marked discriminant actions fix v_D and exchange s_D, c_D, and exchange v_D, s_D while fixing c_D, respectively. On the binary glue generators, $\widehat{R}$ fixes the first two and sends the third to their second-plus-third sum; $\widehat{H}$ fixes the first and exchanges the second and third. On the ternary code generators $z_1 = (1, 1, 1, 0)$, $z_2 = (1, 2, 0, 1)$, the two signed component permutations act as $(z_1, z_2) \mapsto (z_1, 2z_2)$ and $(z_1, z_2) \mapsto (z_1, 2z_1 + 2z_2)$, respectively. Thus both preserve the full original glue and its inverse image. Every within-fibre coordinate permutation also preserves that inverse image, since its action on $\lambda + A_5$ is trivial: $Q\lambda - \lambda = Qe_0 - e_0 \in A_5$. These are concrete failures to preserve this cubic, without discarding their actual lattice actions or glue.

Negation preserves each lattice N and Λ, and its exact polynomial law is $F(-X) = -F(X)$. In the neighbour the nonzero witness is its displayed norm-four generator:

$$F(\rho) = 15\sqrt{2} + 16\sqrt{3}, \qquad F(v) = \frac{14}{27} - \frac{7\sqrt{2}}{24} - \frac{\sqrt{3}}{27}. \tag{317}$$

Both equalities follow by substituting the original coordinates of ρ and v into the proved polynomial.

Every original lattice isometry $T \in O(N)$, whether or not it preserves the cubic, has the exact replacement transport

$$\Gamma_T = \mathcal{L}T\mathcal{L}^{-1}, \qquad F^T(Y) = F(T^{-1}Y), \qquad F^T(TX) = F(X), \tag{318}$$

and

$$T\Lambda_\rho = \Lambda_{T\rho}, \qquad \mathcal{V}_{F^T}(\Lambda_{T\rho}) = \mathcal{V}_F(\Lambda_\rho), \qquad \mathcal{V}_{F^T}(N) = \mathcal{V}_F(N).$$

For the neighbour transport, write its exact elements as $x + k\rho/6$, where $x \in N$ and $(\rho, x) + k \equiv 0 \pmod 6$; application of T gives the identical congruence with $(T\rho, Tx)$. The inverse is T^{-1}. The value-module equalities then follow by the bijections and the displayed identity $F^T(TX) = F(X)$. In particular for $T = \sigma^{2k}$, one has $F^T = F$, so the same retained cubic has exactly the same value module on every neighbour $\Lambda_{\sigma^{2k}\rho}$ in this cyclic orbit. This statement names its target neighbour and does not assume the chosen ρ is fixed.

12 The residue cubic action and the common integral Leech correspondence

Retain the exact lattices $R = A^{\perp 4} \perp D \subset N$ and the binary and ternary codes from the index-72 construction. The four A coordinates remain the ordered six-element residue fibres. To avoid confusing a vector with a representation, write ϱ for the Weyl vector

$$\varrho_A = (5, 3, 1, -1, -3, -5)/2, \qquad \varrho_D = (3, 2, 1, 0), \qquad \varrho = \varrho_A^{\oplus 4} \perp \varrho_D.$$

Thus $\varrho \in N$ and $\varrho^2 = 84$. The original residue representation is still denoted by ρ. The specific element studied here is

$$r = \rho(361) = \rho(289)^2, \qquad r|_{A_i} = C^2, \quad Ce_j = e_{j+1 \bmod 6}, \qquad r|_D = I. \tag{RF1}$$

In particular $r^3 = I$. This is the actual residue coordinate permutation, with no conjugation or replacement.

For each $\sigma \in \Omega = O(N)\varrho$, choose an original root α_σ with $(\sigma, \alpha_\sigma) = 1$ and set

$$\begin{aligned} \ell_\sigma(x) &= (\sigma, x) \bmod 6, \qquad K_\sigma = \ker(\ell_\sigma : N \to \mathbb{Z}/6), \\ \Lambda_\sigma &= K_\sigma + \mathbb{Z}(\sigma/6 - \alpha_\sigma) \\ &= \{x + k\sigma/6 : x \in N,\ k \in \mathbb{Z},\ (\sigma, x) + k \equiv 0 \pmod 6\}. \end{aligned} \tag{RF2}$$

The preceding neighbor proof shows that the definition is independent of α_σ, every Λ_σ is the specified rootless even unimodular lattice, and $N \cap \Lambda_\sigma = K_\sigma$ with both indices six. The family considered in this section is precisely $\mathscr{F} = \{\Lambda_\sigma : \sigma \in \Omega\}$.

12.1 A necessary and sufficient equality criterion

Theorem 12.1 (Equality of the marked Weyl neighbors)**.** *For $\sigma, \tau \in \Omega$, the following three statements are equivalent:*

$$\Lambda_\sigma = \Lambda_\tau, \qquad K_\sigma = K_\tau, \qquad \tau - \epsilon\sigma \in 6N \text{ for some } \epsilon \in \{1, -1\}. \tag{RF3}$$

Proof. Equality of the two neighbors gives equality of the kernels by intersection with N. Each ℓ_σ is onto because $(\sigma, \alpha_\sigma) = 1$. If the kernels are equal, their induced isomorphisms $N/K_\sigma \to \mathbb{Z}/6$ differ by an automorphism of $\mathbb{Z}/6$. Its two units are $1, -1$. Therefore $(\tau - \epsilon\sigma, N) \subset 6\mathbb{Z}$. Unimodularity $N^\vee = N$ gives $(\tau - \epsilon\sigma)/6 \in N$. Conversely this congruence makes the two kernels equal.

It remains to show that this kernel criterion actually gives equality of the chosen unimodular extensions. The exact formula (RF2), after replacing k by $-k$, gives $\Lambda_{-\sigma} = \Lambda_\sigma$. We may consequently suppose $\tau = \sigma + 6t$ with $t \in N$. Both Weyl vectors have squared norm 84, so

$$0 = \tau^2 - \sigma^2 = 12(\sigma, t) + 36t^2, \qquad (\sigma, t) = -3t^2 \in 6\mathbb{Z}. \tag{RF4}$$

The last inclusion uses the actual evenness of N. For $x + k\tau/6$ in (RF2), put $x' = x + kt \in N$. Then $x + k\tau/6 = x' + k\sigma/6$, and

$$(\sigma, x') + k \equiv (\sigma, x) + k \equiv (\tau, x) + k \equiv 0 \pmod 6.$$

This proves $\Lambda_\tau \subset \Lambda_\sigma$. Interchanging σ, τ proves the reverse inclusion. Thus no additional unverified choice of an extension is being identified merely from equality of its common kernel. □

It follows in particular that the map

$$\Omega/\sim \;\longrightarrow\; \mathscr{F}, \qquad [\sigma] \longmapsto \Lambda_\sigma, \qquad \sigma \sim \tau \iff \tau \equiv \pm\sigma \pmod{6N}, \tag{RF5}$$

is a bijection. It is $O(N)$-equivariant: the original formula $T\Lambda_\sigma = \Lambda_{T\sigma}$ follows by substitution in (RF2), and T preserves the congruence relation since $T(N) = N$.

12.2 The exact tetracode obstruction to a fixed neighbor

Theorem 12.2 (No fixed member of this specified family)**.** *The actual residue element $r = \rho(361)$ fixes no lattice of $\mathscr{F}$. Every orbit of its action on $\mathscr{F}$ has exactly three members.*

Proof. Suppose first that $r\sigma \equiv -\sigma \pmod{6N}$. Applying r successively, using $r(N) = N$ and $r^3 = I$, gives $\sigma \equiv -\sigma \pmod{6N}$. Then $\sigma/3 \in N$, contradicting its pairing $1/3$ with the root $\alpha_\sigma \in N$. Thus (RF3) permits fixedness only if $t = (r\sigma - \sigma)/6 \in N$.

Every isometry of N preserves the original root lattice, permutes its four A_5 components, and acts in each of them by a signed coordinate permutation, as proved in the full automorphism theorem. Negation permutes the six entries of ϱ_A. Consequently each A block $s = (s_0, \dots, s_5)$ of every $\sigma \in \Omega$ is a permutation of the six distinct numbers $5/2, 3/2, 1/2, -1/2, -3/2, -5/2$. In this block put $\Delta_j = s_{j-2 \bmod 6} - s_j$. If the corresponding block $\Delta/6$ of t belongs to $A^\vee$, its coordinate differences are integral, so

$$\Delta_j \equiv c \pmod 6 \qquad \text{for every } j \tag{RF6}$$

for one integer c modulo six. On either three-cycle of the index permutation, the three Δ_j sum to zero. Hence $3c \equiv 0 \pmod 6$ and $c \in \{0, 2, 4\}$. Each Δ_j is an integer between -5 and 5. If $c = 0$, (RF6) forces all $\Delta_j = 0$, contrary to the distinctness of the entries on each three-cycle. Therefore $c = 2$ or 4.

In the original discriminant marking, a vector of the A class $t_i\lambda + A$ has common fractional part $-t_i/6$. The corresponding class of $\Delta/6$ is therefore $t_i = -c \pmod 6$, also in $\{2, 4\}$. The difference on the D block is exactly zero because $r|_D = I$. Thus membership of the full vector $(r\sigma - \sigma)/6$ in N would require a glue word

$$(t_1, t_2, t_3, t_4; 00) \in H, \qquad t_i \in \{2, 4\} \text{ for every } i. \tag{RF7}$$

In $H = \{(3u + 2z; y)\}$ this forces $u = 0$ and all four z_i nonzero. But the exact ternary code is

$$C_3 = \{(a + b, a + 2b, a, b) : a, b \in \mathbb{F}_3\}. \tag{RF8}$$

If exactly one parameter is zero, exactly one coordinate is zero. If both are nonzero, precisely one of $a + b, a + 2b$ is zero. Thus every nonzero word has weight three, and the zero word has weight zero. No word in (RF7) belongs to H. This contradicts $t \in N$.

Neither sign in (RF3) can occur. Since $r^3 = I$, every orbit therefore has cardinality three. □

The conclusion concerns exactly $\mathscr{F}$ and the action (RF1). The integral correspondence for these three-member orbits is constructed next; no assertion about other embeddings or constructions of a Leech lattice follows from this obstruction.

12.3 The canonical orbit and its common integral lattice

Put $\sigma_j = r^j\varrho$, $\Lambda_j = \Lambda_{\sigma_j}$ and $K_j = K_{\sigma_j}$, with j read modulo three. Keep the same original root $\alpha = e_0 - e_1$ in the first A block. Directly $(\sigma_j, \alpha) = 1$ for all three j. Define

$$v_j = \sigma_j/6 - \alpha, \quad v = v_0, \qquad q_1 = (\sigma_1 - \sigma_0)/6, \quad q_2 = (\sigma_2 - \sigma_0)/6. \tag{RF9}$$

The complete original coordinates are

$$\begin{aligned} q_1 &= \big((-2,-2,1,1,1,1)/3\big)^{\oplus 4} \perp 0, \\ q_2 &= \big((-1,-1,-1,-1,2,2)/3\big)^{\oplus 4} \perp 0. \end{aligned} \tag{RF10}$$

Their classes in $R^\vee/R$ are respectively $(4,4,4,4;00)$ and $(2,2,2,2;00)$. Neither is in H, while $3q_1, 3q_2 \in R$. In particular each $q_i + N$ has exact order three. Direct pairings give

$$q_1^2 = q_2^2 = 16/3, \quad (q_1,q_2) = 8/3, \quad (\varrho,q_1) = (\varrho,q_2) = -16, \qquad q_1 + q_2 \in R. \tag{RF11}$$

For $x \in N$, write its original glue class as $(3u + 2z; y)$, with $z = (a+b, a+2b, a, b) \in C_3$, and define the homomorphism

$$\beta : N \longrightarrow \mathbb{F}_3, \qquad \beta(x) = b = z_4. \tag{RF12}$$

This map is defined through the original quotient N/R. This definition uses the retained residue marking without changing it. The sum of the four z coordinates is also b modulo three. Write $\ell_j(x) = (\sigma_j, x) \bmod 6$. Then the exact three character formulas are

$$\ell_1(x) = \ell_0(x) + 4\beta(x), \qquad \ell_2(x) = \ell_0(x) + 2\beta(x) \quad \text{in } \mathbb{Z}/6. \tag{RF13}$$

The maps $b \mapsto 2b, 4b$ from $\mathbb{F}_3$ to $\mathbb{Z}/6$ are well-defined homomorphisms. To prove (RF13), the discriminant bilinear form gives

$$(q_1, x) \equiv \tfrac{5}{6}\, 4 \sum t_i \equiv \tfrac{1}{3} \sum t_i \equiv \tfrac{2}{3}\beta(x) \pmod{\mathbb{Z}}, \qquad t_i = 3u_i + 2z_i \pmod 6.$$

For q_2 the same calculation gives $(q_2, x) \equiv \beta(x)/3 \pmod{\mathbb{Z}}$. Multiplication by six and (RF9) prove both formulas.

Using the original dominant representatives $\omega_t = (1^t, 0^{6-t}) - \frac{t}{6}(1,\dots,1)$, put

$$u_* = (\omega_2, \omega_4, 0, \omega_2; 0) \in N. \tag{RF14}$$

Its ternary class is exactly the original generator $z_2 = (1,2,0,1)$. It has $u_*^2 = 4$, $(\varrho, u_*) = 12$ and $\beta(u_*) = 1$. The map

$$(\ell_0, \beta) : N \longrightarrow \mathbb{Z}/6 \oplus \mathbb{F}_3 \tag{RF15}$$

is onto: α maps to $(1,0)$ and u_* to $(0,1)$. Consequently

$$\begin{gathered} I = \ker(\ell_0, \beta) = K_0 \cap K_1 = K_1 \cap K_2 = K_2 \cap K_0 = \bigcap_j K_j, \\ [N : I] = 18, \qquad [K_j : I] = 3, \qquad \det I = 324. \end{gathered} \tag{RF16}$$

The kernel equalities follow from (RF13), because either $2b = 0$ or $4b = 0$ in $\mathbb{Z}/6$ forces $b = 0$ in $\mathbb{F}_3$. The action r permutes the K_j, so it preserves I. Also $3q_i \in I$: these are in R, hence have $\beta = 0$, and $(\varrho, 3q_i) = -48$ is divisible by six.

Define the full-rank lattice

$$J = I + \mathbb{Z}(3v). \tag{RF17}$$

It is contained in Λ_0. The element $6v = \varrho - 6\alpha$ lies in I: its ℓ_0 value is zero, and its glue has zero ternary part. The element $3v$ is not in N, because $(3v, \alpha) = -11/2$ is not an integer. Thus

$$N \cap J = I, \qquad [J : I] = 2, \qquad \det J = 324/4 = 81, \qquad [\Lambda_0 : J] = 9. \tag{RF18}$$

This lattice is positive definite, even, and rootless, because it is a sublattice of Λ_0.

All exact root corrections in the cyclic action are retained. Put $\kappa_j = \alpha - r^j\alpha \in R$. The vectors $r\alpha = e_2 - e_3$ and $r^2\alpha = e_4 - e_5$ are original simple roots with ϱ-pairing one. Hence $\kappa_j \in I$. The precise formulas, rather than literal identification of different generators, are

$$r^j v = v + q_j + \kappa_j \quad (j = 1, 2), \qquad r(3v) = 3v + 3q_1 + 3\kappa_1. \tag{RF19}$$

Since $rI = I$, $3q_1 \in I$ and $\kappa_1 \in I$, this proves $rJ \subset J$. Applying $r^3 = I$ gives equality $rJ = J$.

Theorem 12.3 (All three pairwise Leech intersections)**.** *The actual three-member orbit satisfies*

$$\Lambda_0 \cap \Lambda_1 = \Lambda_1 \cap \Lambda_2 = \Lambda_2 \cap \Lambda_0 = \bigcap_j \Lambda_j = J, \qquad [\Lambda_j : J] = 9. \tag{RF20}$$

The element r is an integral isometry of J, and induces isometries $\Lambda_j \to \Lambda_{j+1}$ with this common restriction.

Proof. First $J \subset \Lambda_0 \cap \Lambda_1$: I lies in both, and $3v = 3v_1 - 3q_1$ with $3q_1 \in I$. Suppose $y \in \Lambda_0 \cap \Lambda_1$. Using the same α, write $y = x_0 + k_0 v_0 = x_1 + k_1 v_1$, $x_j \in K_j$. The fractional part of (v_j, α) is $1/6$ for both j, so pairing with $\alpha \in N$ gives $k_0 \equiv k_1 \pmod 6$. Absorbing multiples of $6v_j \in K_j$, take the same $k \in \{0, \dots, 5\}$ in both expressions. Then $x_0 - x_1 = kq_1 \in N$. The exact order three of $q_1 + N$ forces $k = 0$ or 3. For these values $kq_1 \in I \subset K_1$, so $x_0 = x_1 + kq_1 \in K_0 \cap K_1 = I$. Thus $y = x_0 + kv \in J$, proving the first equality. Apply r and r^2, using $rJ = J$, to prove the other pairwise equalities. Their common index is nine by (RF18) and the isometries induced by r. □

For comparison with the original index-72 glue, let $R_0 = R \cap K_0$. Since $\beta|_R = 0$, (RF13) gives $R \cap K_j = R_0$ for every j, and $R \cap I = R_0$. The reduction map $K_0/R_0 \cong H$ identifies I/R_0 with the kernel of the original ternary coefficient b. Thus

$$[I : R_0] = 24, \qquad [J : R_0] = 48, \qquad [K_0 : R_0] = 72, \qquad [\Lambda_j : R_0] = 432. \tag{RF21}$$

All these are inclusions of the displayed original lattices.

12.4 The exact discriminant form and its cyclic three planes

Theorem 12.4 (The integral correspondence over $\mathbb{F}_3^4$)**.** *One has*

$$J^\vee = \Lambda_i + \Lambda_j \quad (i \neq j), \qquad J^\vee/J \cong \mathbb{F}_3^4. \tag{RF22}$$

In the ordered basis

$$e_1 = u_* + J, \quad e_2 = v + J, \quad e_3 = ru_* + J, \quad e_4 = rv + J, \tag{RF23}$$

the discriminant quadratic form is

$$q_J(a, b, c, d) = \frac{2}{3}(2ad + bc + bd) \pmod{2\mathbb{Z}}, \tag{RF24}$$

and three times its discriminant bilinear form has matrix

$$B = \begin{pmatrix} 0 & 0 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 \end{pmatrix} \quad \text{over } \mathbb{F}_3. \tag{RF25}$$

The three original Leech extensions are exactly the inverse images in $J^\vee$ of the isotropic planes

$$\begin{aligned} P_0 &= \{(a,b,0,0) : a,b \in \mathbb{F}_3\}, \\ P_1 &= \{(0,0,c,d) : c,d \in \mathbb{F}_3\}, \\ P_2 &= \{(a,b,a-b,b) : a,b \in \mathbb{F}_3\}. \end{aligned} \tag{RF26}$$

The induced action of the actual r is the matrix

$$\mathcal{R} = \begin{pmatrix} 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 2 \\ 1 & 0 & 2 & 2 \\ 0 & 1 & 0 & 2 \end{pmatrix}, \qquad \mathcal{R}^3 = I, \quad \mathcal{R}^T B \mathcal{R} = B, \quad \mathcal{R} P_j = P_{j+1}. \tag{RF27}$$

All entries in (RF23)–(RF27) use the original glue generator z_2, root α, pairing, and residue permutation.

Proof. Since $J \subset \Lambda_i = \Lambda_i^\vee$, each Λ_i lies in $J^\vee$. By (RF20), the two subgroups Λ_i/J and Λ_j/J have order nine and intersect trivially for $i \neq j$. Their sum has order 81, equal to $|J^\vee/J| = \det J = 81$, proving the first equality in (RF22).

The homomorphism $\beta : K_0/I \to \mathbb{F}_3$ is an isomorphism, with generator $u_* + I$. Hence Λ_0/J is generated by $u_* + J, v + J$. Both have order dividing three because $3u_* \in I$ and $3v \in J$. They are independent: if $au_* + bv \in J$, reduce first modulo N; the subgroup $J + N$ has the two cosets represented by $0, 3v$, whereas $v + N$ has order six. For $a, b \in \{0,1,2\}$ this forces $b = 0$. Then membership of au_* in $I = N \cap J$ forces $a = 0$. Thus $\Lambda_0/J \cong \mathbb{F}_3^2$ with the first two generators in (RF23). Applying r proves the corresponding statement for Λ_1/J and the last two generators. Their trivial intersection and total order 81 prove that (RF23) is indeed a basis of the full discriminant group.

Here are the full Euclidean Gram entries before any passage to a quotient:

$$\mathrm{Gram}(u_*, v, ru_*, rv) = \begin{pmatrix} 4 & 2 & -2 & -4/3 \\ 2 & 4 & -2/3 & -2/3 \\ -2 & -2/3 & 4 & 2 \\ -4/3 & -2/3 & 2 & 4 \end{pmatrix}. \tag{RF28}$$

For an explicit verification, each of the three nonzero A blocks of u_* pairs to $-2/3$ with its shifted block, giving $(u_*, ru_*) = -2$. One has $(u_*, \alpha) = (u_*, r\alpha) = (ru_*, \alpha) = 0$, $(\varrho, u_*) = 12$, $(\sigma_1, u_*) = -8$, and $(\sigma_2, u_*) = -4$, directly from (RF10) and (RF14). These give respectively $(u_*, v) = 2$, $(u_*, rv) = -4/3$, and $(v, ru_*) = -2/3$. Also $(\varrho, \sigma_1) = -12$, $(\varrho, r\alpha) = (\sigma_1, \alpha) = 1$, and $(\alpha, r\alpha) = 0$. Thus

$$(v, rv) = -12/36 - 1/6 - 1/6 = -2/3.$$

The four diagonal entries equal four, and $(ru_*, rv) = (u_*, v) = 2$ by the exact isometry r. This proves (RF28). Reducing its entries modulo $\mathbb{Z}$ gives (RF25). The diagonal and integral cross terms in the squared norm are even. Reducing the remaining cross terms modulo $2\mathbb{Z}$ gives exactly (RF24).

Changing any coordinate by three changes (RF24) by an even integer, so it is well-defined on the asserted field basis.

It remains to calculate the action without identifying a family permutation with a fixed Leech automorphism. By definition $re_1 = e_3$, $re_2 = e_4$. The vector $u_* + ru_* + r^2u_*$ lies in I: its β value is $1 + 1 + 1 = 0$, because r acts trivially on N/R, and its ℓ_0 value is $12 - 4 - 8 = 0$. Consequently

$$r^2u_* \equiv -u_* - ru_* \pmod{J}. \tag{RF29}$$

By (RF19),

$$v + rv + r^2v = 3v + q_1 + q_2 + \kappa_1 + \kappa_2.$$

The κ_j belong to I, and $3v \in J$. Furthermore $q_1 + q_2 - u_* + ru_* \in I$. Indeed $q_1 + q_2 \in R$, so its β value is zero; subtracting u_* and adding ru_* keeps that value zero. Its pairing with ϱ is $-32 - 12 - 4 = -48$, divisible by six. Thus

$$r^2v \equiv u_* - v - ru_* - rv \pmod{J}. \tag{RF30}$$

The four images in (RF29)–(RF30) give the four columns of $\mathcal{R}$ in (RF27). The matrix has cube the identity and preserves B either by multiplication, or by the already proved facts $r^3 = I$ and $rJ = J$ with its original isometric pairing.

The first two planes in (RF26) are respectively Λ_0/J and Λ_1/J by their proved bases. Their next image is the span of $(-1, 0, -1, 0)$ and $(1, -1, -1, -1)$, from (RF29)–(RF30). Solving their two linear parameters gives exactly $(a, b, a - b, b)$, proving the stated $P_2 = \Lambda_2/J$. Each plane is isotropic: the first two vanish in (RF24), and the third has $2ab + b(a-b) + b^2 = 3ab = 0$ in $\mathbb{F}_3$. They have nine elements and pairwise intersection zero, also following directly from (RF26). Finally each Λ_j is exactly the inverse image of its quotient plane because it contains J. This proves the claimed integral correspondence and all its maps, indices, pairings, and cyclic actions. □

12.5 The fixed fourth plane and its exact A_2^{12} extension

The same finite discriminant form has the additional plane

$$P_\infty = \{(a, b, -a + b, -b) : a, b \in \mathbb{F}_3\}. \tag{RF31}$$

Put

$$p = u_* - ru_*, \qquad q = v + ru_* - rv, \qquad M = J + \mathbb{Z}p + \mathbb{Z}q. \tag{RF32}$$

These are literal vectors in the original Euclidean space.

Theorem 12.5 (The fixed fourth extension)**.** *The plane P_∞ is isotropic and is fixed by the induced residue action. Its inverse image M is a positive definite even unimodular lattice with exactly 72 roots, of root system A_2^{12} and minimum squared norm two. The original residue element r is an integral isometry of M. It fixes four of these A_2 components pointwise and acts as a coordinate three-cycle on each of the other eight.*

The exact intersection and index relations are

$$\begin{gathered} M \cap \Lambda_j = J, \qquad [M : J] = [\Lambda_j : J] = 9, \\ M \cap N = L_0 := \{x \in N : \beta(x) = 0,\ (\varrho, x) \equiv 0 \pmod 2\}, \\ [M : L_0] = [N : L_0] = 6. \end{gathered} \tag{RF33}$$

If R_M denotes the lattice generated by all its roots, then

$$R_M \cong A_2^{\perp 12}, \qquad \det R_M = 3^{12} = 531441, \qquad [M : R_M] = 3^6 = 729. \tag{RF34}$$

Proof. Substitution of (RF31) in (RF24) gives the numerator $2a(-b)+b(-a+b)+b(-b) = -3ab = 0$ in $\mathbb{F}_3$. Thus P_∞ is isotropic. The matrix (RF27) maps its vector with parameters (a, b) to

$$(a + b, b, -a, -b), \tag{RF35}$$

which is again in P_∞, now with parameters $(a + b, b)$. The images of p, q in (RF23) are $(1, 0, -1, 0)$ and $(0, 1, 1, -1)$, a basis of P_∞. Consequently (RF32) is its exact inverse image under $J^\vee \to J^\vee/J$, and $[M : J] = 9$. Isotropy gives even norms on M; polarization gives integral pairings. Its determinant is $\det J/9^2 = 1$, and its positive definiteness and rank are inherited from the unchanged ambient real space. Equation (RF35), together with $rJ = J$, proves $rM = M$. Each of P_0, P_1, P_2 intersects P_∞ in zero, as follows by equating their coordinates in (RF26) and (RF31). Taking inverse images proves $M \cap \Lambda_j = J$.

For its relation with N, (RF19) gives the exact expression

$$q = ru_* - q_1 - \kappa_1. \tag{RF36}$$

Therefore

$$N + M = N + \mathbb{Z}(\varrho/2) + \mathbb{Z}q_1. \tag{RF37}$$

The two indicated cosets modulo N have orders two and three. For the first, pair with α; for the second, use (RF10). Subgroups of coprime orders intersect trivially, so the quotient in (RF37) has order six. Also $p \in N$ has $\beta(p) = 0$ and $(\varrho, p) = 12 - (-4) = 16$, hence image $(4, 0)$ in (RF15). Thus $p + I$ has order three. The vector $3q$ lies in J because $q + J$ has order three, and lies in N by (RF36) and $3q_1 \in N$. It is therefore in $N \cap J = I$. We also have $6v \in I$. Write an element of M using generators $I, 3v, p, q$ and reduce the coefficient of $3v$ modulo two and of q modulo three. If this element is in N, their two coprime-order cosets in (RF37) force both reduced coefficients to vanish. Hence

$$M \cap N = I + \mathbb{Z}p. \tag{RF38}$$

Its image in (RF15) is precisely $2(\mathbb{Z}/6) \oplus 0$, which proves the displayed formula for L_0 and $[N : L_0] = 6$. Since $\det N = \det M = 1$, the equal covolumes give $[M : L_0] = 6$, proving (RF33).

We next prove that the following root count is exhaustive. Every vector in $N + M$ has one of the six forms

$$x + a\varrho/2 - bq_1, \qquad x \in N,\ a \in \{0, 1\},\ b \in \{0, 1, 2\}. \tag{RF39}$$

When $a = 0$, the vector is in $R^\vee$. Every norm-two vector of this particular $R^\vee$ is already a root of R. Indeed, the A class minima are $0, 5/6, 4/3, 3/2, 4/3, 5/6$, and the three nonzero D classes have minimum one. In a nonzero class of $R^\vee/R$ with minimum at most two, the possible minima are exactly among

$$\{5/6, 1, 4/3, 3/2, 5/3, 11/6\}. \tag{RF40}$$

This exhausts them because at most two A classes can be nonzero under the bound; if two are nonzero, each has minimum $5/6$; with a nonzero D class there is at most one nonzero A class and its minimum must be $5/6$. No number in (RF40) is congruent to two modulo $2\mathbb{Z}$. All norms within a fixed discriminant class differ by an even integer, so no such class has a norm-two vector. This proves the claim about $R^\vee$.

All original roots have $\beta = 0$, so (RF33) says that those lying in M are exactly the ones with even ϱ height. In each original A block they are $e_i - e_j$ with $i \neq j$ and $i \equiv j \pmod 2$. These give two orthogonal A_2 systems per block, on the coordinate triples $\{0, 2, 4\}$ and $\{1, 3, 5\}$, and 48 roots in total. In D they are the eight roots $\pm e_1 \pm e_3,\ \pm e_2 \pm e_4$. They form four pairwise orthogonal A_1 systems. This accounts for exactly 56 roots with $a = 0$ in (RF39).

For $a = 1$, absorb the A dual vector $-bq_1$ into the marked A discriminant classes. If the original class of x is $(t_1, t_2, t_3, t_4; y) \in H$, the shifted class in each A is $t_i - 4b$. The exact half-Weyl-shift minima proved in the neighbor calculation are

$$\min_{w \in t\lambda + A} |w + \varrho_A/2|^2 = \begin{cases} 3/8 & t \equiv 0 \pmod 3, \\ 17/24 & t \not\equiv 0 \pmod 3, \end{cases} \qquad \min_{w+D=y} |w + \varrho_D/2|^2 = 1/2. \tag{RF41}$$

For completeness the first line follows by choosing centered quarter coordinates when $t = 0, 3$; for other t, the nearest twelfth coordinates have square sum $13/24$ and coordinate sum ± 1, and restoring zero sum costs $1/6$. The second line follows from the integer coordinate distances $(1/2, 0, 1/2, 0)$ and half-integer distances $(0, 1/2, 0, 1/2)$, with their ties meeting every D marking. Thus the total minimum in (RF39) with $a = 1$ is at least two. Equality requires $t_i - 4b \equiv 0 \pmod 3$ in every component. Since $t_i \equiv 2z_i \pmod 3$, this says $z_i = 2b$ for all four coordinates. For $b = 1, 2$ this is excluded by the exact code (RF8). For $b = 0$ it requires $z = 0$. The eight binary codewords and the two D ties then give exactly sixteen equality vectors $y = x + \varrho/2$. Every one belongs to M: the norm equation gives

$$2 = x^2 + (\varrho, x) + 21, \qquad (\varrho, x) = -19 - x^2, \tag{RF42}$$

which is odd because N is even. Also $\beta(x) = 0$ because $z = 0$. Therefore $x + 3\alpha \in L_0$ and $y = (x + 3\alpha) + 3v \in M$. This proves exhaustively that M has exactly $56 + 16 = 72$ roots.

Here are the full root coordinates and the resulting exact orthogonal decomposition. Let ζ_i be supported in the ith original A block with entries

$$\zeta = (1, -1, 1, -1, 1, -1)/4, \qquad \zeta^2 = 3/8.$$

In the following table every Z_y is an actual sum of those four mutually orthogonal vectors, and every δ_y is an original D root:

y	Z_y	δ_y
00	$\zeta_1 + \zeta_2 + \zeta_3 + \zeta_4$	$e_1 + e_3$
10	$-\zeta_1 - \zeta_2 + \zeta_3 + \zeta_4$	$e_1 - e_3$.
01	$-\zeta_1 + \zeta_2 - \zeta_3 + \zeta_4$	$e_2 + e_4$
11	$\zeta_1 - \zeta_2 - \zeta_3 + \zeta_4$	$e_2 - e_4$

(RF43)

The sixteen new roots are exactly

$$\{\epsilon Z_y + \eta \delta_y/2 : y \in \mathbb{F}_2^2,\ \epsilon, \eta \in \{1, -1\}\}. \tag{RF44}$$

To verify the signs from the original glue, the A half-shift minimizer for binary coordinate $u_i = 0$ is ζ_i and for $u_i = 1$ it is $-\zeta_i$. For $y = 00$ the two codewords are $0000, 1111$; for 10 they are $1100, 0011$; for 01 they are $1010, 0101$; and for 11 they are $0110, 1001$. These give respectively the four opposite pairs $\pm Z_y$ in (RF43). The two shifted D minimizers in these exact original markings are respectively $\pm(e_1 + e_3)/2, \pm(e_1 - e_3)/2, \pm(e_2 + e_4)/2, \pm(e_2 - e_4)/2$. This proves (RF44) with every sign and marking retained.

The four Z_y are mutually orthogonal and have squared norm $3/2$, as follows by dotting their four displayed sign rows. The δ_y are mutually orthogonal of squared norm two. The Z_y are orthogonal to the entire original D space, and to the eight old A_2 systems because their coordinates are constant on each parity triple. For each y, the six roots

$$\{\pm\delta_y\} \ \cup \ \{\epsilon Z_y + \eta \delta_y/2 : \epsilon, \eta = \pm 1\} \tag{RF45}$$

are an A_2 system: take simple roots δ_y and $Z_y - \delta_y/2$, whose Gram matrix is $\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$; their sum is $Z_y + \delta_y/2$. The four such systems are mutually orthogonal and orthogonal to the eight old systems. They exhaust the 72 proved roots. Thus their root lattice is $A_2^{\perp 12}$ of determinant 3^{12}, and unimodularity of M gives the index 3^6. This proves (RF34) and the Coxeter number three of this rooted Niemeier lattice directly from its twelve components.

Finally r fixes each ζ_i because it permutes coordinates within their parity triples; it fixes each δ_y because it is the identity on D. Hence it fixes the four systems (RF45) pointwise. On each of the eight old A_2 systems it cyclically permutes its three coordinate positions, giving its exact order-three Coxeter action. This completes the asserted classification of the fixed fourth extension and its precise relation to the three Leech lattices through J. □

13 The cubic on the common lattice and the fourth Niemeier extension

Retain the original coordinates, Cayley multiplication, F, the generators g_i and their heights H_i from Section 11, and the exact common lattice and fourth extension of Section 12. The symbol u below is the unchanged vector u_* in (RF32).

13.1 Integral generators of the common lattice and the fixed extension

Define the unchanged residue action by

$$(rX)_{i,j} = x_{i,j-2 \pmod 6}, \qquad (rX)_{D,k} = d_k.$$

The two primary parts of the original glue have the exact descriptions

$$H_2 = \{(\varepsilon_1+\varepsilon_2+\varepsilon_3, \varepsilon_1+\varepsilon_2, \varepsilon_1+\varepsilon_3, \varepsilon_1; \varepsilon_2, \varepsilon_3) : \varepsilon_i \in \mathbb{F}_2\},$$
$$H_3 = \{(t+s, t+2s, t, s; 00) : t, s \in \mathbb{F}_3\}.$$

In the original $(\mathbb{Z}/6)^4$ coordinates this means $k = 3w+2z$, with w the four binary entries of H_2 and z the four ternary entries of H_3. The binary D_4 marking is retained. Chinese remainders recover w, z uniquely, and the displayed parameterizations are injective. Consequently

$$\beta : N \longrightarrow \mathbb{F}_3, \qquad \beta(X) = z_4 = s$$

is a well-defined additive homomorphism. The actual lifts (301) satisfy

$$\beta(g_i) = 0 \quad (1 \le i \le 28), \qquad \beta(g_{29}) = 1.$$

Indeed roots have zero glue, the first three lifts are two-primary, and the last two have $(t, s) = (1, 0), (0, 1)$, respectively.

Set, with no change of representatives,

$$I = \{X \in N : (\rho, X) \equiv 0 \pmod 6,\ \beta(X) = 0\}, \qquad J = I + \mathbb{Z}(3v), \tag{319}$$
$$\omega_t = (\underbrace{1, \dots, 1}_{t}, 0, \dots, 0) - \frac{t}{6}(1,1,1,1,1,1), \qquad u = (\omega_2, \omega_4, 0, \omega_2; 0),$$
$$p = u - r(u), \qquad q = v + r(u) - r(v),$$
$$M = J + \mathbb{Z}p + \mathbb{Z}q. \tag{320}$$

Thus J and M are precisely the common-lattice and residue-fixed extension presentations being evaluated here; the following value-group proof needs no root count, lattice determinant, or classification label for M. For explicitness, the two extra vectors in the original five blocks are

$$p = ((1,1,-1,-1,0,0),(1,1,0,0,-1,-1),0,(1,1,-1,-1,0,0);0),$$
$$q = \tfrac{1}{3}((-2,4,4,-2,-2,-2),0,(2,2,-1,-1,-1,-1),(1,1,1,1,-2,-2);0).$$

The zero blocks have their original sizes six and four. These coordinates follow by substituting $v = \rho/6 - \alpha$ into (320); in particular the shifted α term in $r(v)$ is retained. Also $u \in N$, $\beta(u) = 1$, $(\rho, u) = 12$, and $(u, u) = 4$: its class is $(2,4,0,2;00)$, $(\rho_A, \omega_t) = t(6-t)/2$, and $(\omega_t, \omega_t) = t(6-t)/6$.

Theorem 13.1 (Integral generating lists with both kernel conditions)**.** *Define the following 32 ordered vectors:*

$$\begin{aligned} b_1 &= 6\alpha, & b_i &= g_i - H_i\alpha \quad (2 \le i \le 28), \\ b_{29} &= 3g_{29} - 60\alpha, & b_{30} &= 3v = \rho/2 - 3\alpha, \\ b_{31} &= p, & b_{32} &= q. \end{aligned} \tag{321}$$

Then the integral, not merely rational, equalities are

$$I = \sum_{i=1}^{29} \mathbb{Z} b_i, \qquad J = \sum_{i=1}^{30} \mathbb{Z} b_i, \qquad M = \sum_{i=1}^{32} \mathbb{Z} b_i.$$

Proof. Every one of the first 29 vectors has $\beta = 0$ and height divisible by six, by the exact heights and characters already calculated. Conversely take any integral expression $X = \sum_{i=1}^{29} n_i g_i$ for $X \in I$. Applying β shows $n_{29} = 3a$ for an integer a, regardless of which expression was chosen. The exact equality

$$X = \sum_{i=2}^{28} n_i(g_i - H_i\alpha) + a(3g_{29} - 60\alpha) + \left(n_1 + \sum_{i=2}^{28} n_i H_i + 60a\right)\alpha$$

has last coefficient (ρ, X), which is divisible by six. It therefore expresses X as an integer combination of the first 29 b_i. The reverse containment was proved for each generator. The last two equalities follow by adjoining exactly the vectors in (319) and (320). No assertion of linear independence is used. □

13.2 Complete finite certificates for the two new value groups

Theorem 13.2 (Equality of the common-lattice and fixed-extension value-generated groups)**.** *For the retained Cayley multiplication and the original ordered coordinates, the two additive cubic-value groups are exactly*

$$\mathcal{V}_F(J) = \mathcal{V}_F(M) = \frac{1}{27}\mathbb{Z} \oplus \frac{\sqrt{2}}{216}\mathbb{Z} \oplus \frac{\sqrt{3}}{18}\mathbb{Z}. \tag{322}$$

The equality concerns generated additive groups, not equality of individual-value sets $F(J)$ and $F(M)$.

Proof. Apply Theorem 11.2 to the first 30 and all 32 vectors in (321). The resulting list lengths are exactly

$$30 + 2\binom{30}{2} + \binom{30}{3} = 4960, \qquad 32 + 2\binom{32}{2} + \binom{32}{3} = 5984.$$

Write all pair and triple symbols in the next tables using these b_i, not the g_i or h_i. Substitution in the full polynomial gives the complete denominator certificate

	A_i	D^+_{ij}	D^-_{ij}	D_{ijk}
J	$(27, 216, 1)$	$(9, 36, 18)$	$(9, 36, 18)$	$(9, 36, 18)$
M	$(27, 216, 1)$	$(9, 36, 18)$	$(9, 36, 18)$	$(9, 36, 18)$

. (323)

As before each triple is the least common multiple of all reduced rational denominators in its full category, in the order (P_0, P_2, P_3); a zero has denominator one. The finite arithmetic is completely specified by (291)–(300), (321), and all indices in (303)–(306). The accompanying standard-library checker `check_jm_values.py`, using the included `cubic_engine.py`, executes every entry with exact fractions and asserts this table. It reconstructs all 72 marked glue classes, checks all 29 $\beta(g_i)$, and checks both kernel conditions on all the displayed I generators. These finite computations establish the upper containment in (322), including all mixed polarization terms involving p, q, not just their individual cubic values.

Here is a short certificate for the reverse containment, using only J. Coordinates of each row are taken in the ordered proposed value-group basis $(1/27, \sqrt{2}/216, \sqrt{3}/18)$:

$F(b_2)$	14	-272	0
$F(b_4)$	5	-128	0
$F(b_{27})$	23220	-111672	-90
$F(b_{28})$	13780	-74672	0
$F(b_{30})$	378	-1701	-18
$D^+_{2,27}$	7425	-60624	-3
$D^+_{11,21}$	36	48	-16

. (324)

Let C_1, C_2, C_3 have, respectively, columns obtained by transposing rows $(2, 3, 5)$, $(2, 5, 6)$, and $(1, 4, 7)$ of this table. Direct three-by-three determinants are

$$d_1 = -39859290, \qquad d_2 = 11531403, \qquad d_3 = -43244032.$$

The following explicit integer identity certifies their gcd one:

$$(-6882196252)d_1 + (-23788908901)d_2 + (-907)d_3 = 1. \tag{325}$$

Put $(c_1, c_2, c_3) = (-6882196252, -23788908901, -907)$. For each standard basis column $e_t \in \mathbb{Z}^3$ the exact integer reconstruction is

$$e_t = \sum_{j=1}^{3} C_j(c_j \operatorname{adj}(C_j)e_t). \tag{326}$$

All columns of every C_j represent integer combinations of actual cubic values on J; their integer coefficient vectors in (326) therefore give each of the three claimed generators as such a combination. This is an explicit reverse-containment morphism, not an index inference. The checker also expands every adjugate and verifies all three axis reconstructions, saving their integer coefficients in `EXACT_RESULTS.json`. Thus $\mathcal{V}_F(J)$ contains the right-hand side of (322). Since $J \subset M$, the same certificate proves its containment in $\mathcal{V}_F(M)$. The upper bounds already established finish both equalities. □

For additional exact checks on the unchanged representatives, the three individual values are

$$F(3v) = 14 - \frac{63}{8}\sqrt{2} - \sqrt{3}, \qquad F(p) = -2, \qquad F(q) = \frac{64}{27}.$$

They agree with the corresponding rows of the complete calculation; they are not substituted for the mixed-term certificate.

13.3 Exact inclusions, quotient maps, and the factor of the Albert determinant

Theorem 13.3 (The exact four-to-three chain of value groups)**.** *There is a strict chain of additive subgroups of the original real coefficient field*

$$\mathcal{V}_F(N) \subsetneq \mathcal{V}_F(J) = \mathcal{V}_F(M) \subsetneq \mathcal{V}_F(\Lambda).$$

The inclusion matrices in the ordered bases already displayed are

$$\mathcal{V}_F(N) \longrightarrow \mathcal{V}_F(J) = \mathcal{V}_F(M) : (a,b,c) \longmapsto (a,4b,c),$$

$$\mathcal{V}_F(J) = \mathcal{V}_F(M) \longrightarrow \mathcal{V}_F(\Lambda) : (a,b,c) \longmapsto (a,b,3c).$$

Their exact quotient maps are

$$\frac{a}{27} + \frac{b\sqrt{2}}{216} + \frac{c\sqrt{3}}{18} \longmapsto b \pmod 4,$$
$$\frac{a}{27} + \frac{b\sqrt{2}}{216} + \frac{c\sqrt{3}}{54} \longmapsto c \pmod 3.$$

In particular

$$\mathcal{V}_F(J)/\mathcal{V}_F(N) \cong \mathbb{Z}/4, \qquad \mathcal{V}_F(M)/\mathcal{V}_F(N) \cong \mathbb{Z}/4,$$
$$\mathcal{V}_F(\Lambda)/\mathcal{V}_F(J) \cong \mathbb{Z}/3, \qquad \mathcal{V}_F(\Lambda)/\mathcal{V}_F(M) \cong \mathbb{Z}/3,$$
$$\mathcal{V}_F(M)/\mathcal{V}_F(J) = 0.$$

Proof. The first basis comparison uses $\sqrt{2}/54 = 4(\sqrt{2}/216)$, with the other basis entries equal. The second uses $\sqrt{3}/18 = 3(\sqrt{3}/54)$, with its other entries equal. Uniqueness of rational radical coefficients proves that the displayed kernels are exactly the corresponding subgroups. Surjectivity is witnessed by $\sqrt{2}/216$ and $\sqrt{3}/54$, respectively, both actual elements of the proved additive groups. Their orders modulo the smaller groups are exactly four and three: the coefficient divisibility conditions are necessary and sufficient. This also proves strictness. The last quotient is zero by the equality of the two proved groups. In particular the composite inclusion is $(a,b,c) \mapsto (a,4b,3c)$, recovering the full quotient $\mathbb{Z}/4 \oplus \mathbb{Z}/3$, with explicit map

$$\frac{a}{27} + \frac{b\sqrt{2}}{216} + \frac{c\sqrt{3}}{54} \longmapsto (b \bmod 4, c \bmod 3).$$

Its identification with $\mathbb{Z}/12$ is $(\bar{b},\bar{c}) \mapsto 9b + 4c \pmod{12}$, whose inverse is reduction modulo four and modulo three. Every assertion follows from the proved cubic coefficient groups, not from an index of the original lattices. □

The factor two in the original Albert determinant is unchanged:

$$\det Q_{\mathbb{O}}(\lambda; \mathcal{L}X) = \lambda^3 - \lambda(X,X) + 2F(X).$$

Thus, if one instead asks for the additive group generated by the actual cubic term $2F$, it is exactly $2\mathcal{V}_F(L)$. For J, M this is

$$\frac{2}{27}\mathbb{Z} \oplus \frac{\sqrt{2}}{108}\mathbb{Z} \oplus \frac{\sqrt{3}}{9}\mathbb{Z}.$$

Multiplication by two is an injective additive map of the real coefficient field and restricts to the exact isomorphism $\mathcal{V}_F(L) \to \mathcal{V}_{2F}(L) = 2\mathcal{V}_F(L)$, $x \mapsto 2x$, with inverse $y \mapsto y/2$. It intertwines the same two inclusion matrices and the same quotients; no factor has been absorbed into F.

13.4 Reproduction and scope of the exact certificate

The accompanying evidence folder is

`supporting_materials/workbench/evidence/`
`period_lattice_continuation_20260906/jm_cubic/`

It contains both full source files `cubic_engine.py` and `check_jm_values.py`. Running the latter with Python's standard library performs the following finite proof certificate: all 20399 polarization entries for N, Λ, J, M, all displayed denominator and determinant assertions, the integer Bezout and adjugate axis reconstructions, and the exact inclusion coefficient comparisons. The output `EXACT_RESULTS.json` retains each selected column, determinant, Bezout coefficient, and every expanded integer linear combination reconstructing a value-group axis.

There is additionally an independent coefficient-identity check: the original 24 simple roots form a real basis of the original 24-dimensional space. The direct Cayley–Dickson implementation and the rational polynomial implementation are compared on the 24 basis vectors, their 552 ordered-by-index plus/minus pairs, and their 2024 three-element sums, a total of 2600 exact inputs. The finite polarization proof shows these evaluations determine every coefficient of a homogeneous cubic in those 24 variables. Consequently this is a full polynomial-identity certificate on the real ambient space, not sampling of lattice points. The field implementation retains all four rational coefficients in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, and verifies that the $\sqrt{6}$ coefficient is identically zero. A further 65 direct checks cover the displayed new generating lists and u, p, q.

This computation proves the stated additive value groups and their morphisms. It asserts neither equality of individual-value sets nor that the cubic group reconstructs a lattice or classifies all cubic-preserving lattice isometries.

14 A cyclic Wilson Leech lattice in the original triality space

14.1 The two retained metrics and the octonion convention

Let $V = \mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$ be the original ordered octonion space. Retain its metric and cubic

$$\begin{aligned} h_0(x,x) &= N(x_1) + N(x_2) + N(x_3), \\ \tau(x_1,x_2,x_3) = \mathrm{Re}((x_1x_2)x_3), &\qquad d(Q_{\mathbb{O}}(0;x)) = 2\tau(x_1,x_2,x_3), \end{aligned}$$

where

$$Q_{\mathbb{O}}(\lambda;x) = \begin{pmatrix} \lambda & x_3 & \bar{x}_2 \\ \bar{x}_3 & \lambda & x_1 \\ x_2 & \bar{x}_1 & \lambda \end{pmatrix}.$$

Wilson's norm on the same underlying ordered space is

$$q_{\mathrm{W}}(x) = \tfrac{1}{2} h_0(x,x).$$

The map

$$\iota : (V, q_{\mathrm{W}}) \longrightarrow (V, h_0), \qquad \iota(x_1,x_2,x_3) = (x_1/\sqrt{2}, x_2/\sqrt{2}, x_3/\sqrt{2}) \tag{327}$$

is an explicit isometry. Both metrics remain specified. In particular, for the original λ, which is not rescaled,

$$d(Q_{\mathbb{O}}(\lambda;\iota x)) = \lambda^3 - \lambda q_{\mathrm{W}}(x) + \frac{1}{\sqrt{2}}\tau(x_1,x_2,x_3). \tag{328}$$

Indeed the quadratic coordinates scale by $1/2$, while the trilinear term scales by $(1/\sqrt{2})^3 = 1/(2\sqrt{2})$ and its original coefficient is 2. Thus the off-diagonal cubic after ι is $1/(2\sqrt{2})$ times its value before ι.

For an explicit realization inside the same $\mathbb{O}$, choose a Cayley–Dickson basis $(1, \mathsf{i}, \mathsf{j}, \mathsf{k}, \ell, \mathsf{i}\ell, \mathsf{j}\ell, \mathsf{k}\ell)$, with

$$(a + b\ell)(c + d\ell) = (ac - \bar{d}b) + (da + b\bar{c})\ell, \qquad \mathsf{ij} = \mathsf{k}.$$

Wilson's ordered orthonormal basis is realized by

$$(i_\infty, i_0, i_1, i_2, i_3, i_4, i_5, i_6) = (1, \mathsf{i}, \mathsf{j}, \ell, \mathsf{k}, \mathsf{j}\ell, -\mathsf{k}\ell, \mathsf{i}\ell). \tag{329}$$

The seven oriented Fano triples are $(t, t+1, t+3)$ modulo 7: $i_t i_{t+1} = i_{t+3}$, with cyclic products within each triple and opposite signs on reversal. To verify (329), its seven corresponding products, in order $t = 0, \ldots, 6$, are

$$\mathsf{ij} = \mathsf{k}, \quad \mathsf{j}\ell = \mathsf{j}\ell, \quad \ell\mathsf{k} = -\mathsf{k}\ell, \quad \mathsf{k}(\mathsf{j}\ell) = \mathsf{i}\ell, \quad (\mathsf{j}\ell)(-\mathsf{k}\ell) = \mathsf{i}, \quad (-\mathsf{k}\ell)(\mathsf{i}\ell) = \mathsf{j}, \quad (\mathsf{i}\ell)\mathsf{i} = \ell.$$

Alternativity supplies cyclic products on each quaternionic line. The seven lines cover all pairs of different imaginary basis units; squares are -1 and 1 is the unit. This proves the algebra isomorphism on the full multiplication table, hence by bilinearity on all octonions. It also preserves conjugation and the norm. No conjugation of a coordinate of $Q_{\mathbb{O}}$ is performed.

14.2 The explicit eight-dimensional lattice and two sublattices

In the displayed Wilson basis write $e_0 = i_\infty, e_{t+1} = i_t$ and put

$$D_8 = \{(z_0, \ldots, z_7) \in \mathbb{Z}^8 : \textstyle\sum z_i \equiv 0 \pmod 2\},$$
$$s = \tfrac{1}{2}(-1 + i_0 + i_1 + i_2 + i_3 + i_4 + i_5 + i_6), \qquad L_{\mathrm{W}} = D_8 + \mathbb{Z}s.$$

This is Wilson's lattice L, renamed L_{W} to distinguish it from the project's original rank-24 Niemeier lattice. Since $2s \in D_8$ and $s \notin D_8$, it is exactly $D_8 \cup (s + D_8)$.

Lemma 14.1. *The lattice L_{W} is even and unimodular, has minimum norm 2, and has the explicit basis*

$$b_1 = s, \quad b_j = e_{j-1} - e_j \ (2 \le j \le 7), \quad b_8 = e_6 + e_7.$$

Its Gram matrix for the unchanged octonion norm is

$$G_8 = \begin{pmatrix} 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 2 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 2 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 2 & 0 \\ 1 & 0 & 0 & 0 & 0 & -1 & 0 & 2 \end{pmatrix}, \qquad \det G_8 = 1. \tag{330}$$

Its norm-two vectors are precisely the 112 vectors $\pm e_i \pm e_j$ with $i < j$ and the 128 vectors $\frac{1}{2}(\pm e_0 \pm \cdots \pm e_7)$ with an odd number of minus signs.

Proof. For $z \in D_8$, $N(z) \equiv \sum z_i = 0 \pmod 2$ and $h(s,z) = \frac{1}{2}\sum z_i - z_0 \in \mathbb{Z}$. As $N(s) = 2$, every norm in $D_8 + \mathbb{Z}s$ is even. The covolume of D_8 is 2, and its displayed extension has index 2, so L_{W} has covolume 1. The integral bilinear form follows by polarizing the even norm, and covolume one proves self-duality. The columns b_i all belong to L_{W}. Their determinant has absolute value one: expansion along the first coordinate gives $1/2$ times the determinant 2 of the indicated D_7 basis. They therefore form a basis. Their scalar products give the displayed matrix. Positivity and evenness give minimum at least 2, and $N(b_i) = 2$ attains it. Integer vectors of norm two have exactly two nonzero entries ± 1. Half-integer vectors have norm at least $8/4 = 2$, with equality exactly when all eight entries are $\pm 1/2$. Their membership in $s + D_8$ is exactly the stated odd-minus parity. □

Define, with the bars in the positions printed in Wilson,

$$\mathsf{M} = L_{\mathrm{W}}\bar{s}, \qquad \mathsf{N} = L_{\mathrm{W}}s.$$

All products are right products. Let B be the basis matrix with columns b_i. Right multiplication by s in this basis is the following integer matrix:

$$S = B^{-1}R_sB = \begin{pmatrix} 3 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \\ -2 & -1 & 0 & 1 & -1 & 1 & 0 & -2 \\ -4 & -1 & -1 & 1 & 0 & 0 & 1 & -3 \\ -6 & -1 & -1 & 0 & 0 & 1 & 0 & -4 \\ -8 & 0 & -1 & 0 & -1 & 1 & 1 & -5 \\ -10 & -1 & 0 & 0 & -1 & 0 & 1 & -5 \\ -5 & 0 & -1 & 1 & -1 & 0 & 0 & -2 \\ -7 & 0 & 0 & 0 & 0 & 0 & 0 & -4 \end{pmatrix}. \tag{331}$$

This is a direct multiplication-table calculation using (329); it can be checked by multiplying each column b_j by the displayed s. Set $T = -I_8 - S$. Since $s + \bar{s} = -1$, $s\bar{s} = 2$, and alternativity allows repeated s and $\bar{s} = -1 - s$, one has

$$T = B^{-1}R_{\bar{s}}B, \qquad S + T = -I_8, \qquad ST = TS = 2I_8, \qquad S^{\mathsf{T}}G_8S = T^{\mathsf{T}}G_8T = 2G_8. \tag{332}$$

Lemma 14.2. *Both* M, N *are sublattices of* L_{W} *of index* 16 *and minimum norm* 4*, and*

$$2L_{\mathrm{W}} \subset \mathsf{M}, \mathsf{N} \subset L_{\mathrm{W}}, \qquad \mathsf{M} + \mathsf{N} = L_{\mathrm{W}}, \qquad \mathsf{M} \cap \mathsf{N} = 2L_{\mathrm{W}}.$$

Their images $\overline{\mathsf{M}}, \overline{\mathsf{N}}$ *in* $\mathcal{E} = L_{\mathrm{W}}/2L_{\mathrm{W}}$ *are complementary four-dimensional* $\mathbb{F}_2$ *subspaces.*

Proof. The integer matrices S, T give the inclusions in L_{W}; $ST = 2I$ gives the inclusions of $2L_{\mathrm{W}}$. Norm composition shows that each right multiplication scales all squared lengths by $N(s) = N(\bar{s}) = 2$; thus its determinant has absolute value $(\sqrt{2})^8 = 16$, proving the indices and minimum norms. The identity $S + T = -I$ gives $\mathsf{M} + \mathsf{N} = L_{\mathrm{W}}$. Modulo $2L_{\mathrm{W}}$, each image has size $2^8/16 = 2^4$. Their sum is the eight-dimensional space $\mathcal{E}$, so their intersection is zero. This proves the asserted intersection upstairs. □

14.3 The binary glue code and its exact generators

Wilson's original lattice is exactly

$$\Lambda_{\mathrm{W}} = \{(x,y,z) \in L_{\mathrm{W}}^3 : x+y, x+z, y+z \in \mathsf{M}, \quad x+y+z \in \mathsf{N}\}. \tag{333}$$

It carries Wilson's norm q_{W}, as printed. Define the lattice inside the original triality metric by

$$\Lambda_{\mathrm{cyc}} = \iota(\Lambda_{\mathrm{W}}) \subset (V, h_0).$$

The simultaneous use of $\mathsf{M} = L_{\mathrm{W}}\bar{s}$ for pair sums and $\mathsf{N} = L_{\mathrm{W}}s$ for the total sum is essential.

Write any class in $\mathcal{E}$ uniquely as $m+n$, with $m \in \overline{\mathsf{M}}$ and $n \in \overline{\mathsf{N}}$. Then the image of (333) in $\mathcal{E}^3$ is the explicit binary code

$$\mathcal{C} = \{(m_1+n, m_2+n, m_1+m_2+n) : m_1, m_2 \in \overline{\mathsf{M}},\ n \in \overline{\mathsf{N}}\}. \tag{334}$$

Indeed the pair conditions say that the three N components are equal; the total-sum condition says that the sum of the three M components is zero. This proves both directions of the code description. Its dimension is $4+4+4=12$.

For a concrete lift, the first four columns of S and of T are bases of their images modulo 2. The determinant of the eight-column matrix $(T_1, T_2, T_3, T_4, S_1, S_2, S_3, S_4)$ is -7, so these columns are independent modulo 2; each of the two images already has dimension four. Put $m_j = b_j\bar{s}$, $n_j = b_j s$ for $1 \le j \le 4$. A complete set of 36 integer generators of Λ_{W} is

$$\begin{gathered}(2b_i, 0, 0),\ (0, 2b_i, 0),\ (0, 0, 2b_i) \quad (1 \le i \le 8),\\ (m_j, m_j, 0),\ (m_j, 0, m_j),\ (n_j, n_j, n_j) \quad (1 \le j \le 4).\end{gathered} \tag{335}$$

These vectors are all in (333). Their reductions span precisely (334); the first 24 generate its full kernel $(2L_{\mathrm{W}})^3$. This proves that they generate the entire lattice, not only a sublattice.

Theorem 14.3. *The explicitly generated lattice Λ_{W} with q_{W}, and therefore Λ_{cyc} with the unaltered h_0, is an even unimodular rank-24 lattice of minimum norm 4. In particular it is rootless. This is the actual Wilson Leech construction, through the explicit isometry* (327).

Proof. The code proves $[L_{\mathrm{W}}^3 : \Lambda_{\mathrm{W}}] = 2^{24-12} = 2^{12}$. Since L_{W} has covolume one for its original octonion metric, Λ_{W} has covolume 2^{12} for h_0. Changing to the separately recorded metric q_{W} multiplies volume in dimension 24 by $(1/\sqrt{2})^{24} = 2^{-12}$, so its covolume is exactly one.

For every triple of vectors in a real inner product space,

$$N(x)+N(y)+N(z) = N(x+y)+N(x+z)+N(y+z)-N(x+y+z).$$

For a vector of Λ_{W} the four terms on the right are norms in M or N, so each is divisible by 4: these are twice norms in the even lattice L_{W}. It follows that $q_{\mathrm{W}}(x,y,z)$ is even. Polarization of these even norms gives an integral bilinear form. An integral lattice of covolume one equals its dual, proving unimodularity.

If a nonzero vector had $q_{\mathrm{W}} = 2$, its three coordinate norms would sum to 4. Each nonzero coordinate in L_{W} has norm at least 2, so one coordinate is zero. The other two then lie in M, each of whose nonzero vectors has norm at least 4. At most one can be nonzero. That coordinate also lies in N by the total-sum condition, so it lies in $\mathsf{M} \cap \mathsf{N} = 2L_{\mathrm{W}}$ and has norm at least 8, a contradiction. Hence all nonzero norms are at least 4. The vectors $(2b_i, 0, 0)$ have $q_{\mathrm{W}} = 4$, so the minimum is exactly 4. The isometry ι transfers every statement to the original metric. □

14.4 Cyclic symmetry, the fixed lattice, and exact three-primary glue

Retain the canonical cycle $c(x,y,z) = (z,x,y)$. Denote the opposite cycle used in Wilson coordinates by $c_{\mathrm{W}}(x,y,z) = (y,z,x)$. Direct substitution gives

$$c_{\mathrm{W}} = c^{-1} = c^2, \qquad c_{\mathrm{W}}^2 = c, \qquad \langle c_{\mathrm{W}} \rangle = \langle c \rangle.$$

Both actions are retained with these directions; no action is renamed by silently reversing it. The defining three coordinate memberships, three pair conditions, and total-sum condition are all unchanged by c_{W}; hence both cycles preserve Λ_{W} and Λ_{cyc}. They preserve their metrics and the original cubic:

$$\mathrm{Re}((xy)z) = \mathrm{Re}((yz)x).$$

This follows from the octonion adjoint identities $h(ab,d) = h(b,\bar{a}d) = h(a,d\bar{b})$, which imply real associativity, and $\mathrm{Re}(ab) = \mathrm{Re}(ba)$. In the Albert algebra c_{W} is realized by conjugation with the *real* permutation matrix for $(1\,3\,2)$. It sends the coordinates (x_1, x_2, x_3) to (x_2, x_3, x_1) with no conjugations, and permutes the ordered diagonal frame accordingly. The inverse real permutation matrix, for $(1\,2\,3)$, realizes the canonical c and sends (x_1, x_2, x_3) to (x_3, x_1, x_2). Thus the fixed lattices of the two cycles are identical. For the retained residue isometry $\mathcal{L}r = c\mathcal{L}$, the opposite action satisfies $\mathcal{L}r^{-1} = c_{\mathrm{W}}\mathcal{L}$.

Theorem 14.4. *Before applying ι, the cyclic fixed lattice and its orthogonal complement in Λ_{W} are exactly*

$$F = \{(n,n,n) : n \in \mathsf{N}\}, \qquad C = \{(m_1, m_2, -m_1 - m_2) : m_1, m_2 \in \mathsf{M}\}.$$

Their ranks are 8 and 16. For the retained Wilson metric, and therefore after ι for the original triality metric, their Gram matrices in the displayed bases are

$$G_F = 3G_8, \qquad G_C = \begin{pmatrix} 2G_8 & G_8 \\ G_8 & 2G_8 \end{pmatrix}. \tag{336}$$

The fixed-lattice basis is $(b_j s, b_j s, b_j s)$, $1 \le j \le 8$; the coinvariant bases are $(b_j\bar{s}, 0, -b_j\bar{s})$ and $(0, b_j\bar{s}, -b_j\bar{s})$. In particular the fixed lattice has minimum norm 6 and determinant 3^8. The coinvariant lattice has determinant 3^8, and the map which negates its second L_{W} parameter identifies its displayed Gram matrix with the usual $A_2 \otimes E_8$ Gram matrix.

Proof. A fixed triple has the form (x,x,x). Its pair condition holds for every $x \in L_{\mathrm{W}}$ because $2L_{\mathrm{W}} \subset \mathsf{M}$. Its total condition is $3x \in \mathsf{N}$. As $L_{\mathrm{W}}/\mathsf{N}$ has exponent two, $3x \in \mathsf{N}$ is equivalent to $x \in \mathsf{N}$. This proves the first description. Orthogonality to the whole diagonal real eight-space says $x+y+z = 0$. The pair conditions are then exactly $x, y, z \in \mathsf{M}$, and the total condition is automatic. This proves the description of C.

Right multiplication by either s or $\bar{s}$ multiplies the octonion metric by 2. Thus

$$q_{\mathrm{W}}(as, as, as) = 3N(a),$$

and

$$q_{\mathrm{W}}(a\bar{s}, b\bar{s}, -(a+b)\bar{s}) = N(a) + N(b) + N(a+b) = 2N(a) + 2N(b) + 2h(a,b).$$

Polarization gives both full Gram matrices. Their determinants are

$$\det G_F = 3^8 \det G_8 = 3^8, \qquad \det G_C = \det \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}^8 (\det G_8)^2 = 3^8.$$

Minimum norm 6 for F follows from the exact factor 3 and the minimum 2 in L_{W}. The sign map $b \mapsto -b$ changes both off-diagonal blocks from G_8 to $-G_8$, proving the stated exact comparison to $A_2 \otimes E_8$. □

There is a complete integral comparison, including the glue:

$$0 \longrightarrow F \oplus C \longrightarrow \Lambda_{\mathrm{W}} \xrightarrow{\Sigma} \mathsf{N}/3\mathsf{N} \longrightarrow 0, \qquad \Sigma(x,y,z) = x+y+z \pmod{3\mathsf{N}}. \tag{337}$$

To prove surjectivity, for every $a \in L_{\mathrm{W}}$ the explicit vector

$$v(a) = (as, a, -a)$$

lies in Λ_{W} and has sum as. Its pair sums are $a(s+1) = -a\bar{s} \in \mathsf{M}$, $a(s-1) = -a\bar{s} - 2a \in \mathsf{M}$, and 0; its total lies in N. If a lattice vector has total $3n$, subtracting $(n,n,n) \in F$ leaves a vector of C. This proves the kernel assertion. Thus

$$\Lambda_{\mathrm{W}}/(F \oplus C) \cong \mathsf{N}/3\mathsf{N} \cong L_{\mathrm{W}}/3L_{\mathrm{W}} \cong (\mathbb{Z}/3)^8, \qquad [\Lambda_{\mathrm{W}} : F \oplus C] = 3^8.$$

All maps commute with c_{W} and with its inverse c; both cycles act trivially on the quotient.

The discriminant anti-isometry is also explicit. Use the quadratic discriminant convention $q_D(u+D) = h_D(u,u) \pmod{2\mathbb{Z}}$ for an even lattice D with its retained metric. The orthogonal projections of the lift $v(a)$ are

$$\begin{aligned} f(a) &= \tfrac{1}{3}(as, as, as), \\ \gamma(a) &= \left(\tfrac{2}{3}as,\ a - \tfrac{1}{3}as,\ -a - \tfrac{1}{3}as\right). \end{aligned} \tag{338}$$

They belong to F^* and C^*, respectively: their pairings with F or C are the pairings of the integral vector $v(a)$ with the same lattice, because the other projection is orthogonal. The classes $f(a)+F$ identify $L_{\mathrm{W}}/3L_{\mathrm{W}}$ with F^*/F. If $\gamma(a) \in C$, then $f(a) = v(a) - \gamma(a) \in \Lambda_{\mathrm{W}} \cap F_{\mathbb{R}} = F$, so $a \in 3L_{\mathrm{W}}$. Since C^*/C has order $\det G_C = 3^8$, the map

$$f(a) + F \longmapsto \gamma(a) + C$$

is an isomorphism of the two discriminant groups. Its quadratic sign is determined without an unspecified convention:

$$q_F(f(a)) = \tfrac{1}{3}N(a), \qquad q_C(\gamma(a)) = q_{\mathrm{W}}(v(a)) - \tfrac{1}{3}N(a) = 2N(a) - \tfrac{1}{3}N(a).$$

As $N(a)$ is even, their sum $2N(a)$ is zero modulo $2\mathbb{Z}$. This is the required anti-isometry. Equations (337) and (338) specify the overlattice itself, including its actual coset representatives.

14.5 Finite lattice symmetries and exact cubic tests

Independent sign changes of the three sectors preserve Λ_{W}. Changing the sign of x changes a pair sum by $-2x \in 2L_{\mathrm{W}} \subset \mathsf{M}$ and changes the total by $-2x \in 2L_{\mathrm{W}} \subset \mathsf{N}$. The same argument applies to each sector. All six sector permutations preserve the defining conditions too. Under signs $(\epsilon_1, \epsilon_2, \epsilon_3)$ the cubic τ is multiplied by $\epsilon_1\epsilon_2\epsilon_3$; hence exactly the even sign changes in this subgroup preserve it. Cyclic permutations preserve τ as already proved.

The odd permutation $(x,y,z) \mapsto (y,x,z)$ need not preserve it, even on the lattice itself. In Wilson's basis put

$$a = 2(1+i_0), \qquad b = 2(1+i_1), \qquad d = 2(i_2+i_3).$$

Every coordinate belongs to $2L_{\mathrm{W}}$, so $(a,b,d) \in \Lambda_{\mathrm{W}}$. The real part of $((1+i_0)(1+i_1))(i_2+i_3)$ is -1, because $i_0 i_1 = i_3$ and the i_3^2 term is the only real contribution. Reversing the first two factors changes this contribution to $+1$. Consequently

$$\tau(a,b,d) = -8, \qquad \tau(b,a,d) = 8. \tag{339}$$

After the specified scalar isometry the original determinant cubic has values $-4\sqrt{2}$ and $4\sqrt{2}$.

The simultaneous Fano-index operations

$$A : i_t \longmapsto i_{t+1}, \qquad B_0 : i_t \longmapsto i_{2t}, \qquad i_\infty \longmapsto i_\infty$$

are octonion algebra automorphisms, as the defining oriented Fano products show. They are orthogonal permutations preserving D_8 and the odd-minus half coset, and they fix the displayed s and $\bar{s}$. They consequently preserve $\mathsf{M}, \mathsf{N}, \Lambda_{\mathrm{W}}$. Applied simultaneously to all three sectors, they preserve τ. They satisfy $A^7 = B_0^3 = 1$ and $B_0 A B_0^{-1} = A^2$; their permutations are the distinct maps $t \mapsto 2^j t + k$ with $j = 0, 1, 2$ and $k \in \mathbb{F}_7$. Thus they form an explicitly realized $C_7 \rtimes C_3$ of order 21.

The even sign changes and the cycle c_{W} form $(\mathbb{Z}/2)^2 \rtimes C_3 \cong A_4$. They commute with the simultaneous Fano operations, and their intersection is trivial: a simultaneous Fano operation fixes the real unit in every sector and does not permute sectors. Hence an explicit finite group

$$A_4 \times (C_7 \rtimes C_3)$$

of order 252 preserves both the literal lattice and the original cubic. This is a proved subgroup; no assertion that it is the full cubic stabilizer is needed.

Wilson also gives the lattice symmetries

$$r_t(x, y, z) = (x, yi_t, zi_t) \qquad (t = 0, \ldots, 6).$$

For completeness their lattice action can be checked solely from the explicit finite matrices above. Let $U_t = B^{-1} R_{i_t} B$. The signed permutation multiplication table and (331) give

$$U_t \in GL_8(\mathbb{Z}), \qquad T^{-1}(U_t - I_8) \in M_8(\mathbb{Z}), \qquad \tfrac{1}{2}(U_t - I_8)T \in M_8(\mathbb{Z}) \quad (0 \le t \le 6). \tag{340}$$

These are finite exact matrix assertions, not assumptions about octonionic matrix associativity. The full multiplication rule and the exact rational algorithm checking every entry for all seven values are included in the accompanying `check_wilson.py`. Concretely, start with the 64 products $1 i_t = i_t 1 = i_t$, $i_t^2 = -1$, and the seven oriented Fano triples, form the eight columns $R_{i_t}(e_j)$, conjugate by the explicitly displayed B, and substitute the integer $T = -I_8 - S$. For each of the three matrices in (340) the reduced denominators of every entry are 1, for each $t = 0, 1, 2, 3, 4, 5, 6$. Also $\det U_t = 1$. This is a reproducible $7 \cdot 3 \cdot 64$-entry certificate. The first condition is also immediate from the signed permutation: right multiplication by i_t flips four coordinate signs, preserving the odd-minus parity of L_{W}.

The second matrix assertion says $L_{\mathrm{W}}(i_t - 1) \subset \mathsf{M}$, and the third says $\mathsf{M}(i_t - 1) \subset 2L_{\mathrm{W}}$. Thus $x + yi_t$ and $x + zi_t$ differ from the original pair sums by elements of M. The pair $(y + z)i_t$ remains in M, and the transformed total differs from $x + y + z$ by $(y + z)(i_t - 1) \in 2L_{\mathrm{W}} \subset \mathsf{N}$. All defining conditions are preserved. The inverse follows from $r_t^2(x, y, z) = (x, -y, -z)$ and the proved sign symmetries. Norm composition proves that r_t is an isometry.

Nevertheless none of these seven r_t preserves the original cubic. For $a_t = 2(1 + i_t)$, the triple (a_t, a_t, a_t) lies in $(2L_{\mathrm{W}})^3 \subset \Lambda_{\mathrm{W}}$. All products below occur in the associative real subalgebra generated by i_t, and give

$$\tau(a_t, a_t, a_t) = 8 \operatorname{Re}((1 + i_t)^3) = -16,$$

whereas $(1 + i_t)i_t = -1 + i_t$ and

$$\tau(a_t, a_t i_t, a_t i_t) = 8 \operatorname{Re}((1 + i_t)(-1 + i_t)^2) = 16.$$

Therefore the original determinant cubic on the isometric lattice Λ_{cyc} changes from $-8\sqrt{2}$ to $8\sqrt{2}$. This supplies actual lattice vectors as counterexamples, not only ambient real vectors outside the lattice.

14.6 Exact comparison with coordinate lattices and the marked lane

The construction has the explicit integral sandwich

$$(\sqrt{2}L_{\mathrm{W}})^3 \;\subset\; \Lambda_{\mathrm{cyc}} \;\subset\; (L_{\mathrm{W}}/\sqrt{2})^3, \qquad [\Lambda_{\mathrm{cyc}} : (\sqrt{2}L_{\mathrm{W}})^3] = [(L_{\mathrm{W}}/\sqrt{2})^3 : \Lambda_{\mathrm{cyc}}] = 2^{12}. \tag{341}$$

This is (334) transported by the exact isometry (327). The inclusions are literal inclusions in the fixed three octonion sectors, and each commutes with both c_{W} and the canonical $c = c_{\mathrm{W}}^{-1}$. Together with the explicit bases, the code, and the three-primary glue (338), it determines actual commensurability maps and cosets rather than only an abstract isometry class.

Under (329), the single negative column takes the Wilson odd-minus half roots to even-minus half roots in the current ordered Cayley–Dickson basis. The element s remains exactly

$$s = \tfrac{1}{2}(-1 + \mathsf{i} + \mathsf{j} + \mathsf{k} + \ell + \mathsf{i}\ell + \mathsf{j}\ell - \mathsf{k}\ell),$$

with coordinate vector $(-1,1,1,1,1,1,1,-1)/2$. The signed permutation has determinant $+1$ and all 64 basis products agree, as independently checked in the accompanying code. Thus the lattice, its cyclic action, and the triality cubic live in one specified multiplication convention.

14.7 Literal intersections with both retained lattice embeddings

Let $U = \mathrm{span}_{\mathbb{R}}(1, \mathsf{i}, \mathsf{j}, \mathsf{k})$ be the first four coordinates in the *current* ordered Cayley–Dickson basis, and let

$$D_4 = \{a = (a_0, a_1, a_2, a_3) \in \mathbb{Z}^4 : a_0 + a_1 + a_2 + a_3 \in 2\mathbb{Z}\} \subset U.$$

This quaternionic D_4 denotes the displayed four-coordinate lattice; the original Niemeier D summand below remains in its original place.

Let J_{W} denote the signed permutation matrix in (329). If $U_0 : \mathbb{R}^4 \to \mathbb{R}^8$ is the coordinate inclusion in the current basis, then

$$S^{-1}B^{-1}J_{\mathrm{W}}^{-1}U_0 = \begin{pmatrix} 1/2 & -1/2 & -1/2 & -1/2 \\ -1/2 & 0 & 0 & 1/2 \\ -1 & 1/2 & 0 & 1/2 \\ -3/2 & 1 & 1/2 & 1/2 \\ -2 & 1 & 1 & 1/2 \\ -5/2 & 3/2 & 1 & 1 \\ -5/4 & 3/4 & 3/4 & 3/4 \\ -7/4 & 3/4 & 3/4 & 3/4 \end{pmatrix}. \tag{342}$$

All matrices on the left have already been explicitly specified. Thus the equation is a direct rational multiplication certificate.

Lemma 14.5. *In this fixed coordinate frame,*

$$\mathsf{N} \cap U = 2D_4.$$

Proof. Since $\mathsf{N} \subset L_{\mathrm{W}}$, an element of $\mathsf{N} \cap U$ has integral first four coordinates: the half-integer coset of the current E_8 description has no zero coordinate among its last four. Write those integral coordinates as k_0, k_1, k_2, k_3. Membership in N is exactly integrality of the matrix (342) applied to k. Subtracting its last two rows gives $k_0/2$, so k_0 is even. The second row then forces k_3 even, the third forces k_1 even, and the fourth forces k_2 even. Write $k = 2a$. The first six rows are now integral. The seventh and eighth rows are integral precisely when $-5a_0 + 3a_1 + 3a_2 + 3a_3$ is even, equivalently $a_0 + a_1 + a_2 + a_3$ is even. Conversely these conditions make all eight rows integral. This proves both inclusions with no unstated lattice identification. □

We now retain the original lattices and their actual map $\mathcal{L}$. The original coordinates are $X = (x_{i,j}; d_k)$, $1 \le i \le 4$, $0 \le j \le 5$, $1 \le k \le 4$, with $\sum_j x_{i,j} = 0$. The root lattice is $R = A_5^{\perp 4} \perp D$, with its unchanged Euclidean metric. Its index-72 overlattice N is the inverse image of the original glue

$$(k_1, k_2, k_3, k_4; y) = \big(3u + 2(a+b, a+2b, a, b);\ (u_2+u_4, u_3+u_4)\big),$$

where $u \in \mathbb{F}_2^4$ has even weight and $a, b \in \mathbb{F}_3$; the first four entries are modulo 6. Here $k_i = -6x_{i,0} \pmod 6$ and y is the original $D^\vee/D$ marking, so $d = 0$ has $y = 00$. Retain the original neighbour

$$\rho_A = (5,3,1,-1,-3,-5)/2, \quad \rho_D = (3,2,1,0), \quad \rho = \rho_A^{\oplus 4} \perp \rho_D,$$

$$K = \{X \in N : (\rho, X) \in 6\mathbb{Z}\}, \qquad \alpha = (e_0 - e_1, 0, 0, 0; 0), \qquad \Lambda = K + \mathbb{Z}(\rho/6 - \alpha).$$

Thus $\rho \in N$, $\rho^2 = 84$, $(\rho, \alpha) = 1$. Every vector of N or Λ has rational original coordinates.

For exact use of the retained isometry, set

$$\sigma_i = \sum_{j=0}^{5} (-1)^j x_{i,j}, \qquad t_{i,p,j} = x_{i,p+2j} - \frac{1}{3}\sum_{h=0}^{2} x_{i,p+2h}.$$

Write $u_0 = 1, u_1 = \mathsf{i}, u_2 = \mathsf{j}, u_3 = \mathsf{k}$ and retain the remaining current Cayley–Dickson basis vectors $u_4, \dots, u_7$. The already specified map has the exact coordinate form

$$(\mathcal{L}X)_{j+1,a} = \begin{cases} t_{i,p,j} + \sqrt{2}\,\sigma_{a+1}/6, & 0 \le a \le 3, \\ t_{i,p,j} + \sqrt{3}\,d_{a-3}/3, & 4 \le a \le 7, \end{cases} \qquad a = 2(i-1) + p.$$

The placement of σ_{a+1} is retained, including its difference from the index i of the residual. No coordinate is reassigned.

Theorem 14.6. *The literal intersections in the original metric space are exactly*

$$\Lambda_{\mathrm{cyc}} \cap \mathcal{L}(N) = \Lambda_{\mathrm{cyc}} \cap \mathcal{L}(\Lambda) = \mathcal{I} := \{\sqrt{2}(a,a,a) : a \in D_4 \subset U\}. \tag{343}$$

In original Niemeier coordinates the inverse image of this common lattice is

$$\mathcal{L}^{-1}(\mathcal{I}) = \{x_{i,j} = (-1)^j a_{i-1},\ d = 0 :\ a \in D_4\} \subset K.$$

It has rank four, minimum norm 12*, and Gram matrix*

$$G_{\mathcal{I}} = 6\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & -1 \\ 0 & -1 & 2 & 0 \\ 0 & -1 & 0 & 2 \end{pmatrix}, \qquad \det G_{\mathcal{I}} = 4 \cdot 6^4 = 5184. \tag{344}$$

An explicit basis is $\sqrt{2}(\delta_j, \delta_j, \delta_j)$*, where* $\delta_1 = (1,-1,0,0)$*,* $\delta_2 = (0,1,-1,0)$*,* $\delta_3 = (0,0,1,-1)$*,* $\delta_4 = (0,0,1,1)$*.*

Proof. Each coordinate of Λ_{cyc} belongs to $\mathbb{Q}/\sqrt{2} = \sqrt{2}\mathbb{Q}$ in the current frame. For a rational original vector X, the first four coordinate formulas above therefore force $t_{i,p,j} = 0$ at the corresponding positions, by linear independence of $1, \sqrt{2}$ over $\mathbb{Q}$. The last four force both $t_{i,p,j} = 0$ and $d_{a-3} = 0$, by linear independence of $1, \sqrt{2}, \sqrt{3}$. For the latter independence, a relation with nonzero $\sqrt{3}$ coefficient would place $\sqrt{3}$ in $\mathbb{Q}(\sqrt{2})$; squaring $A + B\sqrt{2}$ forces $AB = 0$ and then would make 3

or 3/2 a rational square, a contradiction. All residuals therefore vanish and $d = 0$. The zero fibre sums give

$$x_{i,j} = (-1)^j \sigma_i/6.$$

First let $X \in N$. Its $A_5^\vee$ integral-difference conditions are exactly $k_i := \sigma_i/3 \in \mathbb{Z}$, yielding $x_{i,j} = (-1)^j k_i/2$. Their discriminant labels are $-3k_i = 3k_i \pmod 6$. Their three-primary parts are zero, so the injectivity of $(a, b) \mapsto (a + b, a + 2b, a, b)$ forces $a = b = 0$. Since $d = 0$, the binary condition is $u_2 + u_4 = u_3 + u_4 = 0$ and $\sum u_i = 0$. These equations give $u_1 = u_2 = u_3 = u_4$. Thus all four k_i have the same parity. Conversely these conditions satisfy every original glue requirement. The image is the diagonal triple

$$\mathcal{L}X = (k/\sqrt{2}, k/\sqrt{2}, k/\sqrt{2}), \qquad k \in U.$$

By the fixed-lattice description of Wilson's construction it belongs to Λ_{cyc} exactly when $k \in \mathsf{N}$. The previous lemma makes this equivalent to $k = 2a$, $a \in D_4$, which proves the first intersection equality.

For the second equality, a vector of Λ can be written $X = n + m\rho/6$ with $n \in N$, $m \in \mathbb{Z}$. The already proved necessary condition $d_X = 0$ gives $d_n = -m(3, 2, 1, 0)/6 \in D^\vee$. All coordinates of a vector of $D^\vee$ are integral or all half-integral; the fourth coordinate is zero, so all are integral. Its third coordinate forces $6 \mid m$, and hence $X \in N$. In fact $\Lambda \cap N = K$: in $k + m(\rho/6 - \alpha) \in N$ the pairing with the original root α forces $6 \mid m$, and $(\rho, \rho - 6\alpha) = 84 - 6 = 78 \in 6\mathbb{Z}$. Thus an intersection vector belongs to K. For the diagonal-fibre vectors just found,

$$(\rho, X) = \frac{3}{2} \sum_{i=1}^{4} k_i.$$

Consequently the condition $X \in K$ is $\sum k_i \equiv 0 \pmod 4$. Every $k = 2a$ with $a \in D_4$ satisfies it, proving equality of the two intersections and the displayed inverse image.

Finally $h_0(\sqrt{2}(a, a, a), \sqrt{2}(b, b, b)) = 6h(a, b)$. The four δ_j form a basis of D_4, since their determinant has absolute value two and D_4 has index two in $\mathbb{Z}^4$. Their scalar products give (344). The least nonzero norm in D_4 is two: a norm-one integer vector has odd coordinate sum, while $(1, -1, 0, 0)$ has norm two. This proves the exact minimum and determinant. □

The two slice indices are also literal. Define the common four-dimensional real subspace $\mathcal{U} = \{(u, u, u) : u \in U\}$. The proof gives

$$\mathcal{L}(N) \cap \mathcal{U} = \{(k, k, k)/\sqrt{2} : k \in P\}, \quad P = 2\mathbb{Z}^4 + \mathbb{Z}(1, 1, 1, 1),$$

$$\mathcal{L}(\Lambda) \cap \mathcal{U} = \{(k, k, k)/\sqrt{2} : k \in P_0\}, \quad P_0 = \{k \in P : \textstyle\sum k_i \equiv 0 \pmod 4\} = 2D_4 + \mathbb{Z}(1, 1, 1, 1).$$

The first proof does not need Wilson membership to obtain these slice equations. The lattice P has basis $2e_0, 2e_1, 2e_2, (1, 1, 1, 1)$ and covolume 8. The lattice $2D_4$ has covolume $2^4 \cdot 2 = 32$, so $[P : 2D_4] = 4$. The class of $(1, 1, 1, 1)$ has exact order two modulo $2D_4$, yielding $[P_0 : 2D_4] = 2$. Therefore

$$[\mathcal{L}(N) \cap \mathcal{U} : \mathcal{I}] = 4, \qquad [\mathcal{L}(\Lambda) \cap \mathcal{U} : \mathcal{I}] = 2.$$

The rank-four intersection, strictly below rank 24, also proves that these full lattices are not commensurable *in the retained embedding*: a common finite-index subgroup would have full rank. Their exact existing relation is the common lattice with the explicit embeddings, inverse map, metrics, and indices just proved.

14.8 The cubic value group on the literal common lattice

Retain $F(y) = \tau(y_1, y_2, y_3)$, so the Albert off-diagonal determinant is $2F$. For $y = \sqrt{2}(a, a, a) \in \mathcal{I}$ write $a = a_0 + \mathbf{a}$ in the same quaternionic U. Since $\mathbf{a}^2 = -N(\mathbf{a})$ and a_0 is real, direct multiplication inside this associative subalgebra gives

$$\mathrm{Re}(a^3) = a_0^3 - 3a_0N(\mathbf{a}) = 4a_0^3 - 3a_0N(a).$$

Every scalar and factor is therefore

$$F(\sqrt{2}(a, a, a)) = 2\sqrt{2}(4a_0^3 - 3a_0N(a)). \tag{345}$$

For $a \in D_4$, $N(a)$ is even, because $N(a) \equiv \sum a_i = 0 \pmod 2$. The integer in parentheses is therefore even, so every value belongs to $4\sqrt{2}\mathbb{Z}$. For the actual lattice vector $a = (1, 1, 0, 0)$ it equals $4 - 3 \cdot 2 = -2$, giving $F = -4\sqrt{2}$. Consequently the *additive subgroup generated by all values* is exactly

$$\langle F(\mathcal{I})\rangle_{\mathbb{Z}} = 4\sqrt{2}\mathbb{Z}, \qquad \langle d(Q_{\mathbb{O}}(0; \mathcal{I}))\rangle_{\mathbb{Z}} = 8\sqrt{2}\mathbb{Z}.$$

This assertion concerns the generated additive groups; it does not claim that every element of either group is an individually attained cubic value. For an explicit distinction, put $P(a) = a_0(a_0^2 - 3(a_1^2 + a_2^2 + a_3^2))$. If $3 \mid P(a)$, reduction modulo three forces $3 \mid a_0$, and substitution $a_0 = 3b$ gives $P(a) = 9b(3b^2 - a_1^2 - a_2^2 - a_3^2)$. Thus $3 \mid P(a)$ implies $9 \mid P(a)$. The value $P(a) = 6$ is impossible, so $F = 12\sqrt{2}$ is not attained, although it belongs to the proved additive value group.

14.9 Primary source and exact reproducibility data

Wilson's octonionic construction is the source of the pair and total conditions [31, Sections 2–4, pp. 2–6]. His page 2 uses the odd-minus lattice L and $R = \bar{L}$; page 3 gives pair sums in $L\bar{s}$, the total in Ls, and the metric equal to one half of the sum of octonion norms. The displayed basis map and (327) retain these conjugations and the exact scale in the present coordinates. The distinct coordinate change on Wilson's page 6 is not substituted for this isometry.

The complete integer-arithmetic program and its output are supplied in the following directory (the two displayed lines form one path):

```
supporting_materials/workbench/evidence/
lattice_foundation_20260906/wilson/
```

The files are `check_wilson.py` and `EXACT_CHECKS.json`. They record the generating-lattice index 4096, Gram determinant 1, even diagonal, integral cyclic action, the fixed and coinvariant Gram matrices, and the index 6561 of their sum. The same program checks all 64 multiplication-table products for (329) and the stated exact cubic tests. The proofs above establish rootlessness and the global descriptions; these conclusions are not inferred from a finite search.

15 The exact cubic image of the original common lattice

Retain the coordinates a_0, a_1, a_2, a_3 and the original lattice

$$D_4 = \{a \in \mathbb{Z}^4 : a_0 + a_1 + a_2 + a_3 \in 2\mathbb{Z}\}. \tag{D4I1}$$

The polynomial already used in section 14.8 is

$$P(a) = a_0(a_0^2 - 3(a_1^2 + a_2^2 + a_3^2)). \tag{D4I2}$$

Here and throughout this section F is the original triality cubic $F = \tau$, with its ordered octonionic arguments as in sections 14 and 14.8. In particular, on the literal common lattice of theorem 14.6,

$$\mathcal{I} = \{\sqrt{2}(a,a,a) : a \in D_4\}, \qquad F(\sqrt{2}(a,a,a)) = 2\sqrt{2}\,P(a). \tag{D4I3}$$

No coordinate permutation, scaling of P, or alteration of the lattice is made. The witnesses below are actual points of this original lattice. The image proof is elementary and requires no external arithmetic existence theorem.

15.1 Necessary restrictions in the original coordinates

For every integer b, one has $b^2 \equiv b \pmod 2$. Consequently, for $a \in D_4$,

$$a_0^2 + a_1^2 + a_2^2 + a_3^2 \equiv a_0 + a_1 + a_2 + a_3 \equiv 0 \pmod 2.$$

Writing $s = a_1^2 + a_2^2 + a_3^2$, the factor $a_0^2 - 3s$ is congruent to $a_0^2 + s \equiv 0 \pmod 2$. Thus $P(a)$ is even.

Modulo three the original polynomial satisfies $P(a) \equiv a_0^3$. If $3 \mid P(a)$, the field $\mathbb{Z}/3\mathbb{Z}$ has no nonzero element with zero cube, so $3 \mid a_0$. Substitution of $a_0 = 3b$ into the original formula gives

$$P(a) = 3b(9b^2 - 3s) = 9b(3b^2 - s). \tag{D4I4}$$

Therefore

$$P(a) \in 2\mathbb{Z}, \qquad 3 \mid P(a) \implies 9 \mid P(a). \tag{D4I5}$$

15.2 Three explicit affine lines and their exact values

Define the common direction

$$v = (-3, 1, 1, 1). \tag{D4I6}$$

Its coordinate sum is zero, and direct substitution gives

$$P(v) = -3\big(9 - 3(1+1+1)\big) = 0.$$

For every $r \in \mathbb{Z}$ and $\varepsilon \in \{-1, 1\}$, put

$$L_\varepsilon(r) = (-3r - \varepsilon, r, r, r + \varepsilon) = rv + (-\varepsilon, 0, 0, \varepsilon). \tag{D4I7}$$

Its coordinate sum is $-3r-\varepsilon+r+r+r+\varepsilon = 0$, so it belongs to D_4. The original last-three-coordinate square sum and first-coordinate square are respectively

$$s = r^2 + r^2 + (r+\varepsilon)^2 = 3r^2 + 2r\varepsilon + 1, \qquad (-3r - \varepsilon)^2 = 9r^2 + 6r\varepsilon + 1.$$

Thus the factor in the original cubic is exactly

$$a_0^2 - 3s = (9r^2 + 6r\varepsilon + 1) - (9r^2 + 6r\varepsilon + 3) = -2,$$

and therefore

$$P(L_\varepsilon(r)) = (-3r - \varepsilon)(-2) = 6r + 2\varepsilon. \tag{D4I8}$$

For every $h \in \mathbb{Z}$, define the third line

$$L_0(h) = (-3h, h+1, h-1, h) = hv + (0, 1, -1, 0). \tag{D4I9}$$

Its coordinate sum is zero, so it also belongs to D_4. Here

$$s = (h+1)^2 + (h-1)^2 + h^2 = 3h^2 + 2,$$

and consequently

$$P(L_0(h)) = (-3h)(9h^2 - 3(3h^2+2)) = (-3h)(-6) = 18h. \tag{D4I10}$$

These identities hold for all signed integer parameters. In particular, $L_0(0) = (0, 1, -1, 0)$ has cubic value zero, and the zero vector also has cubic value zero.

15.3 The global image and a proved reconstruction map

Set

$$S = \{n \in \mathbb{Z} : n \in 2\mathbb{Z} \text{ and } (3 \nmid n \text{ or } 9 \mid n)\}. \tag{D4I11}$$

Theorem 15.1 (Complete cubic image and explicit section)**.** *The original polynomial has the exact image*

$$P(D_4) = 18\mathbb{Z} \ \cup \ (6\mathbb{Z}+2) \ \cup \ (6\mathbb{Z}-2) = S. \tag{D4I12}$$

Equivalently,

$$P(D_4) = \{n \in \mathbb{Z} : n \bmod 18 \in \{0, 2, 4, 8, 10, 14, 16\}\}. \tag{D4I13}$$

The following map is an actual section $\sigma : S \to D_4$:

$$\sigma(n) = \begin{cases} (-3h, h+1, h-1, h), & n = 18h, \\ (-3r-\varepsilon, r, r, r+\varepsilon), & n = 2(3r+\varepsilon), \quad \varepsilon \in \{-1, 1\}. \end{cases} \tag{D4I14}$$

Its displayed parameters are uniquely determined in their respective cases, and $P \circ \sigma = \mathrm{id}_S$.

Proof. Equation (D4I5) proves $P(D_4) \subseteq S$. For the reverse inclusion let $n \in S$. If $3 \mid n$, then $9 \mid n$, and evenness gives $n = 18h$ for a unique $h \in \mathbb{Z}$. Equation (D4I10) gives $P(L_0(h)) = n$.

If $3 \nmid n$, write $n = 2q$. Then $3 \nmid q$, so there is a unique $\varepsilon \in \{-1, 1\}$ with $q \equiv \varepsilon \pmod 3$. The number $r = (q - \varepsilon)/3$ is consequently a unique integer. Equation (D4I8) gives

$$P(L_\varepsilon(r)) = 6r + 2\varepsilon = 2q = n.$$

The two displayed cases are disjoint and cover S, and both witnesses lie in D_4 by section 15.2. These calculations prove the asserted section identity and both inclusions in (D4I12). The first arithmetic progression contributes residue zero modulo eighteen; the second contributes $2, 8, 14$; and the third contributes $16, 4, 10$. This proves (D4I13). □

The classification includes every positive, zero, and negative value. The full excluded set is the union of the odd integers with $18\mathbb{Z}+6$ and $18\mathbb{Z}+12$. Let

$$A = \{a \in \mathbb{Z}^4 : a_0 + a_1 + a_2 + a_3 = 0\} \tag{D4I15}$$

be the literal sum-zero sublattice in the original coordinates. The three witnesses lie in $A \subseteq D_4$, so the two inclusions $S \subseteq P(A) \subseteq P(D_4) = S$ prove

$$P(A) = P(D_4) = S. \tag{D4I16}$$

The image of the original cubic $F = \tau$ on the common lattice is therefore exactly

$$\begin{aligned} F(\mathcal{I}) &= 2\sqrt{2}\,S \\ &= 4\sqrt{2}\big((9\mathbb{Z}) \cup (3\mathbb{Z}+1) \cup (3\mathbb{Z}-1)\big). \end{aligned} \tag{D4I17}$$

Because the original off-diagonal Albert determinant is $2F$, its corresponding image is $4\sqrt{2}\,S$. The additive group generated by that determinant image is $8\sqrt{2}\mathbb{Z}$, while the additive group generated by $F(\mathcal{I})$ is $4\sqrt{2}\mathbb{Z}$. Indeed every P value is even, and the first two lines at parameter zero attain $P = 2$ and $P = -2$. Thus the generated additive group of the P values is exactly $2\mathbb{Z}$; applying the two original scalar factors proves both group assertions without identifying an image set with its generated group.

15.4 Exact local images with explicit local sections

For a prime p, let

$$D_{4,p} = \{a \in \mathbb{Z}_p^4 : a_0 + a_1 + a_2 + a_3 \in 2\mathbb{Z}_p\}. \tag{D4I18}$$

This is precisely $D_4 \otimes_{\mathbb{Z}} \mathbb{Z}_p$ in the original coordinates. To verify the equality directly, use the integer basis

$$\begin{aligned} b_0 &= (1,-1,0,0), \quad b_1 = (0,1,-1,0), \\ b_2 &= (0,0,1,-1), \quad b_3 = (0,0,1,1). \end{aligned} \tag{D4I19}$$

For any a in the displayed definition of $D_{4,p}$, the unique coefficients of these basis vectors are

$$\begin{aligned} c_0 &= a_0, & c_1 &= a_0 + a_1, \\ c_3 &= (a_0 + a_1 + a_2 + a_3)/2, & c_2 &= c_3 - a_3. \end{aligned} \tag{D4I20}$$

All belong to $\mathbb{Z}_p$, and substitution gives the original four coordinates. Conversely a $\mathbb{Z}_p$-linear combination of the four vectors has coordinate sum $2c_3$. These two direct calculations prove the tensor-product identification, including at $p = 2$. Taking integral coefficients gives the same basis description of the original D_4.

The three line identities of section 15.2 hold over $\mathbb{Z}_p$, since they are integer polynomial identities. Each line stays in the sum-zero submodule of $D_{4,p}$.

Theorem 15.2 (All local images and the exact local-to-global statement)**.** *For the unchanged polynomial P,*

$$\begin{aligned} P(D_{4,2}) &= 2\mathbb{Z}_2, \\ P(D_{4,3}) &= \mathbb{Z}_3^{\times} \cup 9\mathbb{Z}_3, \\ P(D_{4,p}) &= \mathbb{Z}_p \qquad (p \geq 5). \end{aligned} \tag{D4I21}$$

An integer n belongs to every local image if and only if $n \in S$, and hence if and only if $n \in P(D_4)$. The global section is exactly (D4I14)*; explicit local sections are given in the proof.*

Proof. For $p = 2$, the parity calculation of section 15.1, reduced modulo two, gives $P(D_{4,2}) \subseteq 2\mathbb{Z}_2$. Conversely, given an arbitrary $n = 2q \in 2\mathbb{Z}_2$, put

$$r = (q-1)/3 \in \mathbb{Z}_2.$$

Division by three is valid since three is a unit in $\mathbb{Z}_2$. Then $L_1(r) \in D_{4,2}$ and

$$P(L_1(r)) = 6r + 2 = 2q = n.$$

Thus $n \mapsto L_1((n/2-1)/3)$ is an explicit section on $2\mathbb{Z}_2$, and proves the first equality.

For $p = 3$, reduction of the original polynomial modulo three and the substitution $a_0 = 3b$ in (D4I4) prove that any value divisible by three belongs to $9\mathbb{Z}_3$. Every unit $n \in \mathbb{Z}_3^\times$ is congruent modulo three to exactly one of 2ε, with $\varepsilon \in \{-1, 1\}$. Set

$$r = (n - 2\varepsilon)/6 \in \mathbb{Z}_3.$$

This quotient is integral because the numerator is divisible by three and two is a unit. The line $L_\varepsilon(r)$ then has P value $6r + 2\varepsilon = n$. Every $n \in 9\mathbb{Z}_3$ can be written uniquely as $18h$ with $h = n/18 \in \mathbb{Z}_3$, again because two is a unit. The third line has $P(L_0(h)) = 18h = n$. These two disjoint cases give the explicit local section

$$n \longmapsto \begin{cases} L_\varepsilon((n-2\varepsilon)/6), & n \in \mathbb{Z}_3^\times, \quad n \equiv 2\varepsilon \pmod 3, \\ L_0(n/18), & n \in 9\mathbb{Z}_3. \end{cases} \tag{D4I22}$$

They prove the second equality in (D4I21).

For every prime $p \geq 5$, six is a unit. Given any $n \in \mathbb{Z}_p$, choose $r = (n-2)/6 \in \mathbb{Z}_p$. The point $L_1(r)$ belongs to $D_{4,p}$ and has $P(L_1(r)) = 6r + 2 = n$. Thus $n \mapsto L_1((n-2)/6)$ is a section on all of $\mathbb{Z}_p$, proving the third equality.

For integers n, the first local condition is precisely evenness; the second is precisely $3 \nmid n$ or $9 \mid n$; and the other primes impose no condition. Membership in every local image is therefore exactly $n \in S$. The explicit integral section in theorem 15.1 proves that those conditions suffice for global integral representability. This proves the local-to-global assertion without approximation or inference from a finite modular calculation. □

15.5 The exact factorization and quotient metric on the sum-zero slice

Define the integral map into the original sublattice A of (D4I15) by

$$\Phi : \mathbb{Z}^3 \longrightarrow A, \qquad \Phi(w, u, v) = (-3w - u - v, w + u, w + v, w). \tag{D4I23}$$

Its coordinate sum is zero. For $a \in A$, its proposed inverse is

$$\Phi^{-1}(a) = (a_3, a_1 - a_3, a_2 - a_3). \tag{D4I24}$$

Substitution returns the original second, third, and fourth coordinates immediately; the first becomes $-a_1 - a_2 - a_3 = a_0$ by the defining equation of A. Substituting Φ into the inverse returns (w, u, v). These two equalities prove that Φ is an exact integral-module isomorphism.

For $a = (a_0, y, z, w) \in A$, put $T = y + z + w = -a_0$. Direct expansion of the original polynomial gives

$$\begin{aligned} P(a) &= -T(T^2 - 3(y^2 + z^2 + w^2)) \\ &= 2T(y^2 + z^2 + w^2 - yz - yw - zw) \\ &= T((y-z)^2 + (y-w)^2 + (z-w)^2). \end{aligned} \tag{D4I25}$$

Under Φ, one has $T = 3w + u + v$, and the three differences are $u - v, u, v$. Consequently

$$P(\Phi(w, u, v)) = 2(3w + u + v)(u^2 - uv + v^2). \tag{D4I26}$$

This identity explains the three original lines without changing their coordinates. On $L_\varepsilon(r)$, the quotient differences are $(u, v) = (-\varepsilon, -\varepsilon)$, so $u^2 - uv + v^2 = 1$; on $L_0(h)$, they are $(u, v) = (1, -1)$, so $u^2 - uv + v^2 = 3$. The original first coordinates remain $-3r - \varepsilon$ and $-3h$, respectively.

The exact quotient map is

$$\chi : A \longrightarrow \mathbb{Z}^2, \qquad \chi(a) = (a_1 - a_3, a_2 - a_3). \tag{D4I27}$$

It is onto because $\chi\Phi(0,u,v) = (u,v)$. An element lies in its kernel exactly when its last three coordinates all equal a common integer w; its first coordinate is then $-3w$. Thus $\ker\chi = \mathbb{Z}(-3,1,1,1)$. The section $(u,v) \mapsto \Phi(0,u,v)$ proves the split exact sequence

$$0 \longrightarrow \mathbb{Z}(-3,1,1,1) \longrightarrow A \xrightarrow{\chi} \mathbb{Z}^2 \longrightarrow 0. \tag{D4I28}$$

For $\omega = (-1+\sqrt{-3})/2$, one has $\omega + \overline{\omega} = -1$ and $\omega\overline{\omega} = 1$. Direct multiplication therefore proves

$$(u + v\omega)(u + v\overline{\omega}) = u^2 - uv + v^2. \tag{D4I29}$$

The quadratic factor in the original cubic is exactly this Eisenstein norm, with the preceding coefficient two unchanged.

The original Euclidean metric on these same coordinates is, by expansion,

$$\|\Phi(w,u,v)\|^2 = 12w^2 + 8w(u+v) + 2(u^2 + uv + v^2). \tag{D4I30}$$

Write $d = (-3,1,1,1)$ for the direction when used in the metric calculation. Its squared norm is twelve, and the exact pairing is

$$\langle \Phi(w,u,v), d\rangle = 12w + 4(u+v). \tag{D4I31}$$

Hence orthogonal projection away from the real line $\mathbb{R}d$ is given in the original coordinates by

$$\begin{aligned} \pi\Phi(w,u,v) &= \Phi(w,u,v) - \left(w + \frac{u+v}{3}\right) d \\ &= \left(0, \frac{2u-v}{3}, \frac{2v-u}{3}, -\frac{u+v}{3}\right). \end{aligned} \tag{D4I32}$$

This depends only on the quotient coordinate $\chi(a) = (u,v)$. Its squared norm is

$$\begin{aligned} \|\pi\Phi(w,u,v)\|^2 &= \frac{(2u-v)^2 + (2v-u)^2 + (u+v)^2}{9} \\ &= \frac{2}{3}(u^2 - uv + v^2). \end{aligned} \tag{D4I33}$$

The quotient metric on $A/\mathbb{Z}d$, realized by this exact projection, thus has Gram matrix

$$\frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \tag{D4I34}$$

in the displayed quotient coordinates (u,v). The retained common-lattice map $a \mapsto \sqrt{2}(a,a,a)$ multiplies every squared norm by six, as proved in theorem 14.6. Consequently the corresponding quotient Gram matrix on the image of A in $\mathcal{I}$ is

$$\begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}. \tag{D4I35}$$

This is the exact quotient-metric calculation on the indicated rank-three slice. It neither identifies that slice with the full D_4 nor changes the original common-lattice metric.

16 The marked Niemeier lattice, its twelve theta indices, and the umbral shadow

The original objects are

$$A = \{(x_0, \dots, x_5) \in \mathbb{Z}^6 : \sum_j x_j = 0\}, \qquad D = \{d \in \mathbb{Z}^4 : \sum_i d_i \equiv 0 \pmod 2\}, \qquad R = A^{\perp 4} \perp D.$$

The metrics are the displayed Euclidean metrics. The four copies of A retain their original fibre order $f_i 289^j$ for $0 \le j < 6$, with fibre labels $(00, 10, 01, 11)$. Their discriminant generator is $\lambda = (5, -1, -1, -1, -1, -1)/6$. The discriminant of D retains $10 \mapsto v = (1, 0, 0, 0)$, $01 \mapsto s = (1, 1, 1, 1)/2$, and $11 \mapsto c = (-1, 1, 1, 1)/2$. Consequently the original norm-valued quadratic refinement, with codomain $\mathbb{Q}/2\mathbb{Z}$, is

$$q_R(k_1, k_2, k_3, k_4; y) = \frac{5}{6} \sum_i k_i^2 + \mathbf{1}_{y \neq 0} \pmod{2\mathbb{Z}}.$$

The original glue and Niemeier lattice are

$$\begin{aligned} E &= \{u \in \mathbb{F}_2^4 : \sum_i u_i = 0\}, \qquad \tau(u) = (u_2 + u_4, u_3 + u_4), \\ j(a, b) &= (a + b, a + 2b, a, b), \qquad C_3 = j(\mathbb{F}_3^2), \\ H &= \{(3u + 2j(a, b); \tau(u)) : u \in E,\ (a, b) \in \mathbb{F}_3^2\}, \\ N &= \{x \in R^\vee : x + R \in H\}. \end{aligned}$$

Thus $|H| = 8 \cdot 9 = 72$ and $N/R = H$, with the root system $X = A_5^4 D_4$ proved in the retained lattice calculation. In particular, no norm-two vectors are introduced by the nonzero glue classes: their original minima are four or six. We use this proved literal root system, rather than a rank-only identification, throughout the construction below.

16.1 The primary prescription and its complete coefficient matrix

Write $e(t) = \exp(2\pi i t)$ and $q = e(\tau)$ for $\tau \in \mathbb{H}$. Cheng–Duncan–Harvey, arXiv:1307.5793v2, Section 3.2 [24, Sections 3.2–3.3], source labels `thetexp`, `transf_theta`, and `transf_theta_2`, use

$$\theta_{m,r}(\tau, z) = \sum_{k \equiv r \ (2m)} q^{k^2/(4m)} e(kz), \qquad r \in \mathbb{Z}/(2m).$$

Their Section 3.3, source label `def:OmegaMatrices`, defines

$$\Omega_m(n)_{r,s} = \begin{cases} 1, & r + s \equiv 0 \pmod{2n}, \quad r - s \equiv 0 \pmod{2m/n}, \\ 0, & \text{otherwise}, \end{cases} \qquad n \mid m.$$

Their ADE table, source label `ADE1`, assigns

$$\Omega^{A_{m-1}} = \Omega_m(1), \qquad \Omega^{D_{m/2+1}} = \Omega_m(1) + \Omega_m(m/2),$$

and extends by addition over components with the same Coxeter number. These are the primary authors' prescribed maps. The following calculation specializes and proves their values in the present marked coordinates; it does not assert a new ADE classification.

The six-cycle $e_j \mapsto e_{j+1 \bmod 6}$ on an A fibre has characteristic polynomial $(x^6-1)/(x-1)$ on its augmentation space. Its order is six. In D, take the original simple roots $\alpha_1 = e_1 - e_2$, $\alpha_2 = e_2 - e_3$, $\alpha_3 = e_3 - e_4$, $\alpha_4 = e_3 + e_4$. Their reflection product $s_{\alpha_1} s_{\alpha_2} s_{\alpha_3} s_{\alpha_4}$ sends

$$e_1 \mapsto e_2, \quad e_2 \mapsto e_3, \quad e_3 \mapsto -e_1, \quad e_4 \mapsto -e_4.$$

It has order six and characteristic polynomial

$$(x^3+1)(x+1) = (x^2-x+1)(x+1)^2 = \frac{(x^2-1)(x^6-1)}{(x-1)(x^3-1)}.$$

Thus the original five components have Coxeter number $h = m = 6$, and the primary theta prescription has precisely $2h = 12$ indices and denominator $4h = 24$ in its exponents. The root rank remains $4 \cdot 5 + 4 = 24$; the glue index remains 72.

Put $J = \Omega_6(3)$. Its two congruences are

$$r + s \equiv 0 \pmod 6, \qquad r - s \equiv 0 \pmod 4.$$

The simultaneous solutions modulo twelve consist of the single value $s \equiv 5r \pmod{12}$: this value satisfies both congruences, and the difference of any two solutions is divisible by both six and four, hence by twelve. Therefore

$$\boxed{\Omega^X = 5I_{12} + J, \qquad J_{r,s} = \mathbf{1}_{s \equiv 5r \ (12)}.}$$

In the full original index order $0, 1, \ldots, 11$ this is

$$\Omega^X = \begin{pmatrix}
6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 5 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 5 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 5 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 5 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 6 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 5 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 5 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 6 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 5 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 5
\end{pmatrix}.$$

The unary series and the actual shadow are

$$S_{6,r}(\tau) = \frac{1}{2\pi \mathrm{i}} \frac{\partial \theta_{6,r}}{\partial z}\Big|_{z=0} = \sum_{k \equiv r \ (12)} k q^{k^2/24}, \qquad S^X = \Omega^X S_6.$$

All series and their derivatives converge normally on compact subsets of their stated domains, by Gaussian decay. Substitution $k \mapsto -k$ proves $S_{6,-r} = -S_{6,r}$, so $S_{6,0} = S_{6,6} = 0$. Let $\mathcal{A} : \mathbb{C}^5 \to \mathbb{C}^{12}$ insert zero in positions zero and six and put $(\mathcal{A}u)_{12-r} = -u_r$ for $1 \le r \le 5$. This retains, and explicitly reconstructs, all twelve components. Then

$$\Omega^X \mathcal{A} = \mathcal{A} M^X, \qquad \boxed{M^X = \begin{pmatrix}
5 & 0 & 0 & 0 & 1 \\
0 & 4 & 0 & 0 & 0 \\
0 & 0 & 6 & 0 & 0 \\
0 & 0 & 0 & 4 & 0 \\
1 & 0 & 0 & 0 & 5
\end{pmatrix}.}$$

Indeed its (r,s) entry is $\Omega^X_{r,s} - \Omega^X_{r,12-s}$; direct substitution gives the matrix. In particular the fourth A-type multiplicity and the full D-type contribution have both been retained. The diagonal entries $(5,4,6,4,5)$ are exactly the Coxeter exponent multiplicities: four copies of $(1,2,3,4,5)$ and one copy of $(1,3,3,5)$.

16.2 Quadratic refinements and the finite theta transformations

Let $L = \mathbb{Z}\ell$ with $(\ell,\ell) = 12$. Its dual is $L^\vee = \mathbb{Z}\ell/12$. Hence

$$A_L = L^\vee/L = \mathbb{Z}/12, \qquad q_L(r) = r^2/12 \pmod{2\mathbb{Z}}, \qquad Q_L(r) = q_L(r)/2 = r^2/24 \pmod{\mathbb{Z}},$$

and the associated bilinear form is $B_L(r,s) = rs/12 \pmod{\mathbb{Z}}$. The symbols q_L and Q_L deliberately have different codomains and retain the factor two.

Define the unitary Fourier matrices and reflection by

$$(F_\pm)_{r,s} = 12^{-1/2} e(\pm rs/12), \qquad R_{r,s} = \mathbf{1}_{s \equiv -r \ (12)}.$$

Finite geometric-series summation gives

$$F_\pm^2 = R, \qquad F_- = RF_+ = F_+R, \qquad F_+F_- = I.$$

The substitution $k \mapsto k+12$ shows that $e(k^2/24)$ depends only on $r = k \bmod 12$. Thus, with $D_{r,s} = e(r^2/24)\delta_{r,s}$,

$$\theta_6(\tau+1, z) = D\theta_6(\tau, z).$$

Poisson summation of the Gaussian gives

$$\theta_6(-1/\tau, z/\tau) = \sqrt{-\mathrm{i}\tau}\, e(6z^2/\tau) F_- \theta_6(\tau, z),$$
$$\theta_6(-1/\tau, -z/\tau) = \sqrt{-\mathrm{i}\tau}\, e(6z^2/\tau) F_+ \theta_6(\tau, z).$$

Here $\sqrt{-\mathrm{i}\tau}$ is the holomorphic square root positive when τ is positive imaginary. For completeness, apply the Fourier transform convention $\widehat{f}(u) = \int_{\mathbb{R}} f(x) e(-ux)\, dx$ to $f(x) = \exp(\pi \mathrm{i}\tau x^2/12 + 2\pi \mathrm{i} z x)$. Completing the square evaluates its transform; the real positive Gaussian integral is obtained by squaring the integral and using polar coordinates, and holomorphic continuation gives the expression for $\Im\tau > 0$. Periodizing $f(r+12x)$, its n-th Fourier coefficient is $12^{-1} e(nr/12) \widehat{f}(n/12)$. Summing these coefficients at zero and grouping n by its residue modulo twelve gives the first displayed transformation. Absolute Gaussian convergence justifies the integration, grouping, and Fourier evaluation. Replacing z by $-z$ and using $\theta_{6,-r}(\tau, z) = \theta_{6,r}(\tau, -z)$ gives the second. The second, with its plus-sign Fourier matrix, is precisely CDH's convention; the first uses the usual generator $S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ with elliptic argument z/τ.

The matrices carrying the finite multiplier for this first convention are

$$U(T) = D, \qquad U(S) = e(-1/8) F_-.$$

The square-root weight factor has not been included in U. Differentiation of the theta transformation at $z = 0$ gives

$$S_6(-1/\tau) = \tau\sqrt{-\mathrm{i}\tau} F_- S_6(\tau), \qquad S_6(\tau+1) = D S_6(\tau).$$

Equivalently, CDH's reflected convention gives $S_6(-1/\tau) = -\tau\sqrt{-\mathrm{i}\tau} F_+ S_6(\tau)$. The equality follows from $RS_6 = -S_6$. Thus the unary series has weight $3/2$ with the same finite multiplier; theta has weight $1/2$.

The permutation J preserves Q_L because $((5r)^2 - r^2)/24 = r^2 \in \mathbb{Z}$. Since $5^{-1} = 5$ modulo twelve,

$$(JF_\pm)_{r,s} = 12^{-1/2} e(\pm 5rs/12) = (F_\pm J)_{r,s}.$$

Therefore Ω^X commutes with both finite multiplier matrices. The shadow transformation follows by applying this proved intertwiner. Its component series vanish at infinity since every nonzero summand has strictly positive exponent; the modular transformation shows the same cuspidal behavior at every cusp of the full modular group.

There is also an exact comparison with the original A5 discriminant, without replacing its form. Reduction

$$p : \mathbb{Z}/12 \longrightarrow \mathbb{Z}/6, \qquad p(r) = r \bmod 6$$

has kernel $\{0, 6\}$ and satisfies

$$q_A(p(r)) = \frac{5}{6} r^2 = 10 q_L(r) \pmod{2\mathbb{Z}}.$$

Thus it is a surjective quadratic similitude with precisely multiplier ten, and its bilinear forms satisfy the same multiplier relation. It is not asserted to be an isometry. Applying p to four coordinates and the identity to the retained D discriminant gives

$$\widetilde{D} = (\mathbb{Z}/12)^4 \oplus \mathbb{F}_2^2 \longrightarrow D_R, \qquad (r; y) \longmapsto (p(r_1), \dots, p(r_4); y).$$

The pullback quadratic refinement is exactly $10 \sum_i q_L(r_i) + \mathbf{1}_{y \neq 0}$, its radical is $\{0, 6\}^4 \oplus 0$, and its quotient by that radical is the original (D_R, q_R). To prove the radical claim, pairing with each coordinate generator forces $5r_i/6 \in \mathbb{Z}$, hence $6 \mid r_i$; the nondegenerate D form forces $y = 0$. The vectors so obtained have quadratic value 30 in each nonzero coordinate, hence zero modulo two. The inverse image of the original H has order $16 \cdot 72 = 1152$ and is isotropic for this pullback form. This gives the exact relation of the theta indices to all original glue coordinates, with the form and multiplier specified on both sides.

16.3 The full Fourier determinant and the retained order-twelve character

Let $\chi : \mathrm{SL}_2(\mathbb{Z}) \to \mathbb{Z}/12$ be the retained marking

$$\chi(T) = -1, \qquad \chi(S) = 3.$$

These values respect the presentation $S^4 = I$, $(ST)^3 = S^2$: in the abelian target the required relations are $4 \cdot 3 = 0$ and $3(3-1) = 2 \cdot 3$ modulo twelve. They therefore define a character, and its value at T makes it onto. We now calculate the determinant of the entire twelve-dimensional finite theta representation, rather than that of a reduced component list.

Put $\zeta = e(1/12)$. The Vandermonde determinant gives

$$\det F_+ = 12^{-6} \prod_{0 \le j < k \le 11} (\zeta^k - \zeta^j).$$

For every such pair,

$$\zeta^k - \zeta^j = 2\mathrm{i} \sin\!\left(\frac{\pi(k-j)}{12}\right) \exp\!\left(\frac{\pi \mathrm{i}(j+k)}{12}\right),$$

whose sine is strictly positive. There are $\binom{12}{2} = 66$ pairs, and each index occurs in eleven of them, so

$$\sum_{j<k} (j+k) = 11 \sum_{j=0}^{11} j = 11 \cdot 66 = 726.$$

The phase of the determinant is consequently

$$\mathrm{i}^{66}\exp(726\pi\mathrm{i}/12) = (-1)\mathrm{i} = -\mathrm{i}.$$

Its remaining factor is positive. Unitarity, proved by the finite geometric sum above, makes its absolute value one. It follows exactly that

$$\boxed{\det F_+ = -\mathrm{i}, \qquad \det F_- = \mathrm{i}.}$$

In particular retaining CDH's reflection changes the Fourier determinant by $\det R = (-1)^5 = -1$; it cannot be dropped.

The sum of the squares of all twelve indices is

$$\sum_{r=0}^{11} r^2 = \frac{11\cdot 12\cdot 23}{6} = 506.$$

Consequently

$$\boxed{\det U(T) = e(506/24) = e(1/12), \qquad \det U(S) = e(-12/8)\det F_- = -\mathrm{i} = e(-3/12).}$$

For an explicit check that the finite matrices have the relevant metaplectic relations, let $G = \sum_{r=0}^{11} e(r^2/24)$. Listing the squares modulo twenty-four gives

$$G = 1 + 4e(1/24) + 2e(1/6) + 2e(3/8) + 2e(2/3) + e(1/2).$$

The first and last terms cancel, as do the terms with arguments $1/6$ and $2/3$. The elementary sine and cosine values at $\pi/12$ and $3\pi/4$ give

$$4e(1/24) + 2e(3/8) = \sqrt{6}(1+\mathrm{i}) = \sqrt{12}\,e(1/8).$$

Completing the square in the finite sum proves

$$(F_-D)^2 = e(1/8)D^{-1}F_-, \qquad (F_-D)^3 = e(1/8)R.$$

For the first equality, its (r,s) entry before evaluation is $12^{-1}e(s^2/24)\sum_t e(t^2/24 - t(r+s)/12)$; shifting t by $r+s$ produces exactly the stated Gauss sum and matrix entry. The second follows by multiplication and $F_-^2 = R$. It follows that

$$(U(S)U(T))^3 = U(S)^2 = -\mathrm{i}R, \qquad U(S)^4 = -I_{12}.$$

Taking determinants kills this metaplectic central sign because the dimension twelve is even. The determinant therefore descends to an ordinary character of $\mathrm{SL}_2(\mathbb{Z})$. Its two computed generator values, together with generation by S, T, prove

$$\boxed{\det U(\gamma) = e(-\chi(\gamma)/12).}$$

The dual determinant is its inverse and is exactly $e(\chi(\gamma)/12)$, with the retained signs $T \mapsto e(-1/12)$, $S \mapsto \mathrm{i}$.

This is a statement about the determinant multiplier, with its domain and codomain specified. It is not a statement that a vector of weight $1/2$ has scalar weight one after taking a determinant. The separate Dedekind-eta identities

$$\eta(\tau+1) = e(1/24)\eta(\tau), \qquad \eta(-1/\tau) = \sqrt{-\mathrm{i}\tau}\,\eta(\tau)$$

show that η^2 has weight one and multiplier $\det U$, whereas η^{-2} has weight minus one and multiplier $(\det U)^{-1}$. Indeed $\eta^2(S\tau) = (-\mathrm{i})\tau\eta^2(\tau)$ and $\eta^{-2}(S\tau) = \mathrm{i}\tau^{-1}\eta^{-2}(\tau)$. The integer-weight factors $(c\tau+d)^{\pm 1}$ obey the cocycle identity by matrix multiplication, so the two generator formulas prove the statements for every γ. At $-I$ the character and odd integer weight each contribute a minus sign, whose product is one.

16.4 The residue Fourier map and what the Weyl quotient retains

The original marking is $\kappa(289^j) = j \pmod 6$, with the full ordered residue list

$$(289^j)_{j=0}^{5} = (1, 289, 361, 169, 121, 529) \pmod{840}.$$

On the six-dimensional complex coordinate carrier define

$$E_2 : \mathbb{C}[S_{840}] \longrightarrow \mathbb{C}[\mathbb{Z}/12], \qquad E_2(e_{289^j}) = e_{2j},$$

and on a separately indexed $\mathbb{C}^6$ define

$$B(e_k) = \frac{e_k + e_{k+6}}{\sqrt{2}}, \qquad 0 \le k < 6.$$

Both are injective and preserve the indicated coordinate Hermitian products. Retain the original negative-sign normalized Fourier matrix $F_6^- = (6^{-1/2} e(-kj/6))_{k,j}$. Entry-by-entry calculation gives

$$\boxed{F_- E_2 = B F_6^-.}$$

Indeed the (r, j) entry of the left side is $12^{-1/2} e(-rj/6)$, and the right side has the same value, using r mod 6. Thus the original unnormalized sample transform is carried into this formula by its exact factor $\sqrt{6}$. Translation $j \mapsto j + 1$ intertwines under E_2 with translation $r \mapsto r + 2$ modulo twelve. This gives a literal Fourier and translation map, not an identification of different quadratic forms. In particular $Q_L(6) = 3/2 = 1/2 \pmod{\mathbb{Z}}$, so the quotient $\mathbb{Z}/12 \to \mathbb{Z}/6$ does not carry Q_L itself to a quadratic form on the quotient. Its exact multiplier-ten comparison with the original A5 form was proved above.

For the original lattice action, $\rho(289)$ is the same six-cycle in each of the four A coordinate sets and is the identity on D. Each coordinate transposition of A is reflection in the corresponding root $e_i - e_j$, so the full residue action is in $W(R) = W(A)^4 \times W(D)$. It acts trivially on every A discriminant because $Q_\sigma \lambda - \lambda = e_{\sigma(0)} - e_0 \in A$, and it fixes the D discriminant. Consequently

$$S_{840} \xrightarrow{\rho} O(N) \longrightarrow O(N)/W(R)$$

is the trivial homomorphism. This proves exactly which residue-group action is killed, without discarding its already constructed Fourier map.

The original marked quotient is

$$G = O(N)/W(R) \cong \mathrm{GL}_2(\mathbb{F}_3).$$

Here its actual coordinate map uses $\ell_1 = (1, 1)$, $\ell_2 = (1, 2)$, $\ell_3 = (1, 0)$, $\ell_4 = (0, 1)$. For $A \in \mathrm{GL}_2(\mathbb{F}_3)$ determine the unique signs and permutation by

$$\ell_i A = \varepsilon_i \ell_{p(i)}, \qquad (P_p u)_i = u_{p(i)}, \qquad B_p \tau(u) = \tau(P_p u).$$

The four row lines exhaust the projective line, so the signs and permutation exist uniquely. The binary equation defines B_p uniquely since τ is onto. Then $(k; y) \mapsto (\varepsilon_i k_{p(i)}; B_p y)$ preserves precisely the original two glue codes, and its ternary restriction is $M_g j = jA$. Conversely a marked glue stabilizer gives this same A. Multiplication of these identities proves the group isomorphism, with the original row/column convention retained.

To compare its chosen signed lifts with CDH's simple-root permutation section, let $Q_{\mathrm{rev}} e_j = e_{5-j}$ in an A fibre. When $\varepsilon_i = -1$, replace the representative $-I$ by $-Q_{\mathrm{rev}}$; their quotient is $Q_{\mathrm{rev}} \in W(A)$

and $-Q_{\mathrm{rev}}(e_j - e_{j+1}) = e_{4-j} - e_{5-j}$. Thus the resulting map permutes the original simple roots by diagram reversal. When $\varepsilon_i = 1$ no change is needed. For D, put $\Delta = (\alpha_1\ \alpha_2\ \alpha_3\ \alpha_4)$. Associate the original classes v, s, c with outer nodes $1, 4, 3$, respectively; their fundamental weights are e_1, $(1,1,1,1)/2$, and $(1,1,1,-1)/2$, the last of which differs from the retained c by $(1,0,0,-1) \in D$. For B_p, let P_{B_p} permute these outer nodes exactly as B_p permutes v, s, c and fix node two. Then

$$d_{B_p} = \Delta P_{B_p} \Delta^{-1}$$

is the exact original-coordinate diagram representative. Its orthogonality follows because permuting the outer nodes preserves the original Gram matrix. It preserves D, and its action on fundamental weights proves its discriminant action is exactly B_p. Its quotient with the originally chosen D lift has trivial discriminant action, hence belongs to the proved Weyl kernel $W(D)$. This specifies the actual change of section on every summand.

Define the following four class functions using those retained maps:

$$a_g = \#\{i : p(i) = i\}, \quad b_g = \sum_{p(i)=i} \varepsilon_i, \quad s_g = \mathrm{sgn}(B_p|_{\{v,s,c\}}), \quad c_g = \#\,\mathrm{Fix}(B_p|_{\{v,s,c\}}) - 1.$$

The first two are the traces on the four-component permutation and signed permutation representations. The last is the trace on the two-dimensional sum-zero subspace of the three outer-node permutation representation, and s_g is its sign character. Thus these are precisely CDH's $(\bar{\chi}^{X_A}, \chi^{X_A}, \chi^{X_D}, \check{\chi}^{X_D})$; the remaining $\bar{\chi}^{X_D}$ is one because the single D component is fixed. The comparison of sections above proves this identification for the actual marked quotient.

CDH Section 5.1 [24, Section 5.1] defines the full twisted coefficient matrix by

$$(\Omega_g^{X_A})_{r,s} = \big(b_g \mathbf{1}_{r \text{ even}} + a_g \mathbf{1}_{r \text{ odd}}\big)\delta_{r,s},$$

and, for its exceptional $m = 6$ D4 case, by

$$(\Omega_g^{X_D})_{r,s} = \begin{cases} c_g\delta_{r,s}, & r \equiv 0,3 \pmod 6, \\ \delta_{r,s} + s_g J_{r,s}, & r \equiv 1,5 \pmod 6, \\ s_g\delta_{r,s} + s_g J_{r,s}, & r \equiv 2,4 \pmod 6. \end{cases}$$

All deltas in this formula have indices modulo twelve. The exact maps from the marked glue quotient to the prescribed twisted shadows are

$$g \longmapsto \Omega_g^X = \Omega_g^{X_A} + \Omega_g^{X_D}, \qquad S_g^X = \Omega_g^X S_6.$$

They are class-function constructions from traces of representations; the assignment $g \mapsto \Omega_g^X$ is not asserted to be a group representation. On the odd five-component presentation the explicit matrix is

$$\boxed{M_g^X = \begin{pmatrix} a_g+1 & 0 & 0 & 0 & s_g \\ 0 & b_g & 0 & 0 & 0 \\ 0 & 0 & a_g+c_g & 0 & 0 \\ 0 & 0 & 0 & b_g & 0 \\ s_g & 0 & 0 & 0 & a_g+1 \end{pmatrix}.}$$

This follows by the same signed component reconstruction $\mathcal{A}$: in rows two and four the two D terms cancel, in row three the contribution is c_g, and rows one and five pair with one another. For

reference, all possible character tuples are

$[g]$	a_g	b_g	s_g	c_g	number of elements
$1A$	4	4	1	2	1
$2A$	4	-4	1	2	1
$2B$	2	0	-1	0	12
$4A$	0	0	1	2	6
$3A$	1	1	1	-1	8
$6A$	1	-1	1	-1	8
$8A, 8B$	0	0	-1	0	12

with the CDH class names of their table `tab:chars:eul:6`. The formulas apply to each actual A, so their validity does not depend on a choice of conjugacy-class names. The finite checker obtains the table from all 48 invertible matrices over $\mathbb{F}_3$ and the original row forms.

The distinguished residue action has quotient $g = 1$ and therefore produces the untwisted shadow, while a product $\rho(r)L_g$ produces the same S_g^X for every r. The four-component signed action and the D4 triality quotient survive and enter the displayed matrix. The central element $-I \in \mathrm{GL}_2(\mathbb{F}_3)$, for example, changes b_g from four to minus four while fixing a_g, s_g, c_g; it changes the even-index odd-space entries with exactly that sign.

The already marked composite

$$\mathrm{SL}_2(\mathbb{Z}) \xrightarrow{\chi} \mathbb{Z}/12 \longrightarrow \mathbb{Z}/6 \xrightarrow{j\mapsto 289^j} S_{840}$$

is onto, has kernel $\chi^{-1}(\{0,6\})$, and sends T to $289^5 = 529$. Following it by the Weyl quotient kills it, by the proved residue calculation. Following χ into the dual determinant multiplier retains its complete order twelve, by the determinant theorem. These are explicit different outgoing morphisms from the same marking, including their kernels and signs.

16.5 The shadow-to-Fricke map with the original scalar seventy-two

The Coxeter calculation above gives the original formal frame exponent

$$\pi_X = -5[1] + [2] - [3] + 5[6].$$

Let $\mathcal{E}(\sum a_d[d]) = \prod_d \eta(d\tau)^{a_d}$. It is a group homomorphism from addition of formal exponents to multiplication of nonzero holomorphic functions, by the exponent laws. The retained scalar function is

$$\boxed{T_6(\tau) = \mathcal{E}(-\pi_X) = \frac{\eta(\tau)^5\eta(3\tau)}{\eta(2\tau)\eta(6\tau)^5}.}$$

The minus sign on π_X is essential: $\mathcal{E}(\pi_X) = T_6^{-1}$.

Here is a direct coefficient map from the actual shadow matrix to this eta function. Its diagonal difference vector is $\alpha = (5, 4, 6, 4, 5)$, and set $\alpha_0 = 0$ modulo six. Define the Lambert-series map at this marked matrix by

$$\mathscr{L}(\Omega^X)(\tau) = 1 + \sum_{n\geq 1} \alpha_{n \bmod 6} \frac{nq^n}{1-q^n}.$$

Convergence is absolute and locally uniform since $|q| < 1$ and α is bounded. The coefficient of q^N is exactly $\sum_{d|N} d\alpha_{d \bmod 6}$; thus this specifies every coefficient, not only its initial terms. For all positive integers n the retained multiplicities satisfy

$$\alpha_{n \bmod 6} = 5 - \mathbf{1}_{2|n} + \mathbf{1}_{3|n} - 5\mathbf{1}_{6|n}.$$

Checking the six residues proves this identity. The product definition $\eta(\tau) = q^{1/24}\prod_{n\geq 1}(1-q^n)$ and termwise logarithmic differentiation then give

$$\boxed{f^X := \mathscr{L}(\Omega^X) = -q\frac{d}{dq}\log T_6 = 5\lambda_6 + \lambda_2 - \lambda_3,}$$

where the original functions on the right are

$$\lambda_d = q\frac{d}{dq}\log\frac{\eta(d\tau)}{\eta(\tau)} = \frac{d-1}{24} + \sum_{n\geq 1}\sigma_1(n)(q^n - dq^{dn}).$$

The constant is $[5(6-1)+(2-1)-(3-1)]/24 = 1$, and the other coefficients agree by the six-residue identity. This proves the shadow-to-eta calculation directly for every coefficient.

The precise primary connecting map is CDH Section 4.2 [24, Section 4.2]:

$$\sigma^X(\tau,z) = \sum_{r\ (12)} \overline{S_r^X(\tau)}\theta_{6,r}(\tau,z), \qquad \mathscr{S}_{1,1}(\sigma^X) = f^X.$$

Its source labels are `eqn:sigmaX`, `def:weight2`, `weight2:Aseries`, and `weight2_EZ`. The displayed Lambert map is its explicit specialization at this root-system shadow. It verifies the resulting function independently of the general Shimura-lift theorem. In particular the shadow and the retained Fricke function are connected by this complete coefficient map.

The original modular subgroup here is $\Gamma_0(6)$, and $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(6)] = 12$. To compute the index, reduction modulo six gives $\mathrm{SL}_2(\mathbb{Z}/6) \cong \mathrm{SL}_2(\mathbb{F}_2)\times\mathrm{SL}_2(\mathbb{F}_3)$ of order $6\cdot 24 = 144$. Reduction is onto: elementary upper and lower unipotent matrices generate over each field, and CRT choices of their entries realize each factor while acting trivially on the other. The upper triangular subgroup modulo six has $\varphi(6)\cdot 6 = 12$ elements (choose its first unit diagonal entry and its upper-right entry; the other diagonal entry is its inverse). Its inverse image is $\Gamma_0(6)$, proving index $144/12 = 12$. This modular-group index and the twelve theta indices have their specified constructions; neither replaces the lattice index seventy-two.

Under the exact Fricke transformation $W_6\tau = -1/(6\tau)$, the eta inversion formula sends each $\eta(dW_6\tau)$ to $\sqrt{-\mathrm{i}(6/d)\tau}\,\eta((6/d)\tau)$. The retained exponent vector at $d = 1,2,3,6$ is $(5,-1,1,-5)$; its reversed vector is its negative and its exponent sum is zero. The τ factors therefore cancel. Since each $6/d$ is positive, the consistent square-root branch gives the positive scalar

$$\prod_{d|6}(6/d)^{a_d/2} = 6^{5/2}3^{-1/2}2^{1/2} = 72.$$

Consequently

$$\boxed{T_6(W_6\tau) = 72/T_6(\tau).}$$

For the weight-two slash action of the determinant-six representative,

$$(f|_2 W_6)(\tau) = (6\tau^2)^{-1}f(-1/(6\tau)).$$

Differentiating the last identity and using $d(W_6\tau)/d\tau = (6\tau^2)^{-1}$ gives

$$\boxed{f^X|_2 W_6 = -f^X.}$$

On the Jacobi side, the primary Eichler–Zagier operator $\mathcal{W}_6(6)$ acts on theta by $\Omega_6(6) = R$. This is verified by its defining congruences, $r+s = 0 \pmod{12}$, $r-s = 0 \pmod 2$. On the odd shadow

and on the corresponding skew-Jacobi form its eigenvalue is minus one. The specialized coefficient map just proved carries that negative eigenline to the negative Fricke eigenline of f^X. This proves an exact equivariant linear comparison on these lines; it does not evaluate an $\mathrm{SL}_2(\mathbb{Z})$ character on the determinant-six matrix.

Finally, the original sheet coordinates remain

$$J_6 = T_6 + 5 + 72/T_6, \qquad Y_6 = T_6 - 72/T_6.$$

Direct substitution and differentiation prove

$$J_6(W_6\tau) = J_6(\tau), \quad Y_6(W_6\tau) = -Y_6(\tau), \quad Y_6^2 = (J_6-5)^2-288, \quad dJ_6 = Y_6\, dT_6/T_6 = -2\pi\mathrm{i}Y_6 f^X\, d\tau.$$

Thus the original scalar 72, the additive shift 5, the branch constant 288, and the differential factor $-2\pi\mathrm{i}$ all survive the complete shadow-to-sheet calculation.

16.6 Established scope and reproducibility

The constructed maps are the ADE coefficient specialization, the exact finite-theta intertwiner, the multiplier determinant and its dual, the quadratic similitude to the original discriminants, the marked glue-group trace prescription, the original six-to-twelve Fourier map, and the coefficient-level shadow-to-Fricke map. Their kernels, weights, signs, and constants were specified and proved above. They establish concrete relations even where a direct isometry or an identical group action is absent. They do not construct a new graded moonshine module or a new moonshine theorem. The name "umbral shadow" here refers to precisely the CDH prescription, with its source provenance retained.

The standard-library program `check_shadow.py` uses exact arithmetic in $\mathbb{Q}[z]/(z^8-z^4+1)$ with $z = e(1/24)$. It verifies both full and odd matrices, the Fourier determinant by the Vandermonde product, the Gauss sum, the character signs, the Fourier embedding, all 48 original-code character records, the first 240 Lambert coefficients, and the squared positive Fricke scalar 72^2. The proofs above establish the unbounded identities independently of those finite checks.

References

[1] J. W. Milnor, *Topology from the Differentiable Viewpoint*, University Press of Virginia, 1965, §7, Theorem B and Lemma 4, pp. 42–50. `https://www.maths.ed.ac.uk/~aar/papers/milnortop.pdf`.

[2] L. Alpöge, with the disclosed Fable/Claude contribution, *The* $(3,4,\infty)$ *modular family of 2-tori, completed at its three special points, is a complex structure on* S^6, together with the citable canonical audit and source bundle, Zenodo record 22238779 (2026), doi:10.5281/zenodo.22238779. Local audit locators are below `supporting_materials/workbench/research/`; the one-copy quotient and framing discussion are in `cusp_sphere_route_strengthening.tex` and `cusp_s12_carrier_homotopy.tex`.

[3] K. Conrad, $SL_2(\mathbb{Z})$, author-hosted expository notes, Theorem 1.1, pp. 1–2; Theorem C.1, pp. 18–19, author's PDF, accessed 5 September 2026.

[4] A. Borel and J.-P. Serre, *Groupes de Lie et puissances réduites de Steenrod*, Amer. J. Math. **75** (1953), no. 3, 409–448; Sections 12 and 15, doi:10.2307/2372495.

[5] A. Hatcher, *Spectral Sequences in Algebraic Topology*, Chapter 5 draft, Section 5.1; Theorem 5.41(b), p. 573, and the suspension calculation, pp. 577–578, author-hosted chapter PDF.

[6] A. Hatcher, *Algebraic Topology*, Cambridge University Press, Cambridge, 2002; the opening low-dimensional homotopy-group table, p. 339, author's electronic edition.

[7] D. Husemoller, *Fibre Bundles*, third edition, Graduate Texts in Mathematics, vol. 20, Springer, New York, 1994; Chapter 17, Section 7, Theorem 7.2, pp. 274–275, doi:10.1007/978-1-4757-2261-1.

[8] D. C. Isaksen, G. Wang, and Z. Xu, *Stable homotopy groups of spheres: from dimension 0 to 90*, Publ. Math. IHÉS **137** (2023), 107–243; Table 1, p. 113, doi:10.1007/s10240-023-00139-1.

[9] Y. Komori, K. Matsumoto, and H. Tsumura, *The Theory of Zeta-Functions of Root Systems*, Springer Monographs in Mathematics, Springer, Singapore, 2023; Section 10.1, pp. 213 and 216, doi:10.1007/978-981-99-0910-0.

[10] A.-L. Mare and M. Willems, *Topology of the octonionic flag manifold*, Münster J. Math. **6** (2013), 483–523, published article; arXiv:0809.4318.

[11] H. Toda, *Composition Methods in Homotopy Groups of Spheres*, Annals of Mathematics Studies **49**, Princeton University Press, Princeton, NJ, 1962; Proposition 3.1, doi:10.1515/9781400882625.

[12] L. Alpöge, with the disclosed Fable/Claude contribution, *The* $(3, 4, \infty)$ *modular family of* 2*-tori, completed at its three special points, is a complex structure on* S^6, canonical audit and reconstruction source bundle, Zenodo record 22238779 (2026), original manuscript §3.7 and §§4.1–4.5. `https://doi.org/10.5281/zenodo.22238779`.

[13] *Real* S^{12} *and* S^{24} *Routes from Alpöge's Candidate* $(3, 4, \infty)$ *Two-Torus Fibration*, companion research draft in the preceding canonical source bundle; native quotient, localized cusp map, and framed-cusp calculation. Current proof-bearing edition: `https://www.overleaf.com/read/rtmyqxyrzprn`.

[14] J. W. Milnor, *Topology from the Differentiable Viewpoint*, University Press of Virginia, 1965, §7, Theorem B and Lemma 4. `https://www.maths.ed.ac.uk/~aar/papers/milnortop.pdf`.

[15] L. S. Pontryagin, *Homotopy classification of mappings of an* $(n+2)$*-dimensional sphere on an* n*-dimensional one*, Dokl. Akad. Nauk SSSR **70** (1950), 957–959. `https://www.sas.rochester.edu/mth/sites/doug-ravenel/otherpapers/pont3.pdf`.

[16] L. S. Pontryagin, *Smooth manifolds and their applications in homotopy theory*, Trudy Mat. Inst. Steklov **45** (1955); English translation, Amer. Math. Soc. Transl. (2) **11** (1959). `https://www.sas.rochester.edu/mth/sites/doug-ravenel/otherpapers/pont4.pdf`.

[17] D. Johnson, *Spin structures and quadratic forms on surfaces*, J. London Math. Soc. (2) **22** (1980), 365–373.

[18] A. Debray and S. Gunningham, *The Arf–Brown TQFT of pin-minus surfaces*, arXiv:1803.11183, especially §3.1.1 and Proposition 5.1. `https://adebray.github.io/papers/arf_brown.pdf`.

[19] A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002; group table p. 339, Corollary 4.24, p. 360, and pp. 384–385. https://pi.math.cornell.edu/~hatcher/AT/AT.pdf.

[20] *The retained-angle cusp quotient has nonzero top-cell class*, complete derivative-framing theorem and its based Pontryagin–Thom proof, 5 September 2026; companion to the canonical cusp/sphere-route draft. Current cumulative source collection: https://www.overleaf.com/read/rtmyqxyrzprn.

[21] *The higher torus wrap: obstruction, exact local lift, and the Jordan–triality rung*, canonical higher-rung research draft, section on peripheral modulus and the mod–24 boundary, 5 September 2026. Current cumulative source collection: https://www.overleaf.com/read/rtmyqxyrzprn.

[22] A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002, Corollary 4.24, p. 360, stable homotopy groups and the integral relation $\eta^3 = 12\nu$, pp. 384–385, and Proposition 4.56 with proof on p. 388. https://pi.math.cornell.edu/~hatcher/AT/AT.pdf.

[23] I. Yokota, *Exceptional Lie groups*, 2009; Sections 1.10, 1.14, 1.16 and Theorem 2.7.1, https://arxiv.org/abs/0902.0431.

[24] M. C. N. Cheng, J. F. R. Duncan, and J. A. Harvey, *Umbral Moonshine and Niemeier Lattices*, Research in the Mathematical Sciences **1** (2014), Article 3; Section 2.4, definition of the Weyl quotient and Table 3, https://arxiv.org/abs/1307.5793v2.

[25] A.-L. Mare and M. Willems, *Topology of the octonionic flag manifold*, Münster Journal of Mathematics **6** (2013), 483–523; Definition 2.2, Propositions 2.4 and 2.7, pp. 491–492, and equations (5.4)–(5.6), pp. 510–511, https://www.uni-muenster.de/FB10/mjm/vol_6/mjm_vol_6_14.pdf; https://arxiv.org/abs/0809.4318.

[26] The Clankers, *Residual divisor shells, support-homogeneous signs, and the $A_5^4D_4$ Niemeier refinement of the modulo-840 Erdős–Straus datum*, April 2026, maintained-source snapshot of 5 August 2026; theorems thm:reduced-shell, thm:isotropic, thm:root-preservation, and thm:two-channel. The exact source title, statement excerpts, snapshot digest, and counting correction are included in the accompanying *ES–Niemeier source provenance* [28].

[27] *Exact support–glue–Fricke transport*, theorem thm:exact-support-glue-Fricke-transport in the maintained Erdős–Straus Foundation source, inspected 5 September 2026. The exact earlier-glue generators and transporter, with their original equation labels, are reproduced in the accompanying *ES–Niemeier source provenance* [28].

[28] *ES–Niemeier source provenance*, companion document to *The modulo-840 lattice, triality, and exact shell maps*, 5 September 2026; ES_NIEMEIER_PROVENANCE.md, included with this module. It supplies exact passages and labels from the maintained residual note, the support–glue theorem, and the canonical *Higher torus wrapping and the octonionic 24-dimensional continuation.*

[29] J. H. Conway and N. J. A. Sloane, *Twenty-three constructions for the Leech lattice*, Proc. R. Soc. Lond. A **381** (1982), 275–283, pp. 277–280, https://doi.org/10.1098/rspa.1982.0071.

[30] R. E. Borcherds, *The Leech lattice*, Proc. R. Soc. Lond. A **398** (1985), 365–376, Sections 6–7, https://math.berkeley.edu/~reb/papers/leech/leech.pdf.

[31] R. A. Wilson, *Octonions and the Leech lattice*, Journal of Algebra **322** (2009), 2186–2190, https://doi.org/10.1016/j.jalgebra.2009.03.021. Author version submitted 18 December 2008, revised 20 March 2009, Sections 2–4, preprint pp. 2–6, https://webspace.maths.qmul.ac.uk/r.a.wilson/pubs_files/octoLeech1rev.pdf.